

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON DC**

**Before the Honorable Clark S. Cheney
Chief Administrative Law Judge**

In the Matter of

**CERTAIN SEMICONDUCTOR DEVICES,
PRODUCTS CONTAINING SAME, AND
COMPONENTS THEREOF**

Investigation No. 337-TA-1500

COMPLAINANT'S OPENING CLAIM CONSTRUCTION BRIEF

TABLE OF CONTENTS

I.	APPLICABLE LAW	1
II.	LEVEL OF SKILL IN THE ART	3
III.	ASSERTED PATENTS.....	3
A.	The '546 and '167 Patents	3
B.	The '235 Patent.....	4
C.	The '983 Patent	5
IV.	DISPUTED TERMS.....	6
A.	'546 Patent/'167 Patent.....	6
1.	“layer” / “first liner layer” / “refractory metal layer” ('546 patent at claims 1-14; '167 patent at claims 1, 3, 4).....	6
2.	“intermix region” ('546 patent at claims 1, 2, 6, 8-9, 13; '167 patent at claims 1, 2, 6).....	9
B.	'235 Patent	10
1.	“from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide” ('235 patent at claims 1, 10).....	10
C.	'983 Patent	12
1.	“adjacent to” ('983 patent at claims 1, 5, 13, 17, 20)	12
2.	“the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate” ('983 patent at claims 1, 13, 20).....	14
3.	“the drift isolation region is disposed under the gate” ('983 patent at claims 4, 16).....	16
4.	“the channel having a channel length that is controlled by ion implantation” ('983 patent at claims 7, 18)	16
5.	“edge of the gate” ('983 patent at claims 1, 2, 6, 13-15, 20)	17
V.	CONCLUSION.....	18

TABLE OF AUTHORITIES

CASES

<i>Accentra, Inc. v. Staples, Inc.</i> , 500 F. App'x 922 (Fed. Cir. 2013)	13, 14
<i>Avid Tech., Inv. v. Harmonic, Inc.</i> , 812 F.3d 1040 (Fed. Cir. 2016)	8
<i>Barrette Outdoor Living, Inc. v. Fortress Iron, LP</i> , 156 F.4th 1353 (Fed. Cir. 2025)	2
<i>BASF Corp. v. Johnson Matthey Inc.</i> , 875 F.3d 1360 (Fed. Cir. 2017)	2
<i>CCS Fitness, Inc. v. Brunswick Corp.</i> , 288 F.3d 1359 (Fed. Cir. 2002)	2
<i>Certain Child Car Seats</i> , Inv. 337-TA-1459, Order No. 13 (Feb. 27, 2026)	8
<i>Certain Icemaking Machines and Components Thereof</i> , Inv. 337-TA-1369, Order No. 10 (Jan. 4, 2024)	12
<i>Cox Commc'ns, Inc. v. Sprint Commc'ns Co.</i> , 838 F.3d 1224 (Fed. Cir. 2016)	9, 13
<i>Fisher-Price, Inc. v. Graco Children's Prods., Inc.</i> , 154 F. App'x. 903 (Fed. Cir. 2005)	6
<i>Housey Pharms., Inc. v. Astrazeneca UK Ltd.</i> , 366 F.3d 1348 (Fed. Cir. 2004)	2
<i>InterDigital Commc'ns, LLC v. Int'l Trade Comm'n</i> , 690 F.3d 1318 (Fed. Cir. 2012)	8, 16
<i>K-2 Corp. v. Salomon S.A.</i> , 191 F.3d 1356 (Fed. Cir. 1999)	1, 9, 16
<i>MBO Lab'ys, Inc. v. Becton, Dickinson & Co.</i> , 474 F.3d 1323 (Fed. Cir. 2007)	13
<i>Nautilus, Inc. v. Biosig Instruments, Inc.</i> , 572 U.S. 898 (2014)	1, 2, 7
<i>Phillips v. AWH Corp.</i> , 415 F.3d 1303 (Fed. Cir. 2005)	1, 2, 11

<i>Power-One, Inc. v. Artesyn Techs., Inc.</i> , 599 F.3d 1343 (Fed. Cir. 2010)	13
<i>Sonix Tech. Co. v. Publ’ns Int’l, Ltd.</i> , 844 F.3d 1370 (Fed. Cir. 2017)	2, 6
<i>Summit 6, LLC v. Samsung Elecs. Co.</i> , 802 F.3d 1283 (Fed. Cir. 2015)	11
<i>Thorner v. Sony Computer Ent. Am. LLC</i> , 669 F.3d 1362 (Fed. Cir. 2012)	1, 2, 16
<i>Toshiba Corp. v. Imation Corp.</i> , 681 F.3d 1358 (Fed. Cir. 2012)	8
<i>Vitronics Corp. v. Conception, Inc.</i> , 90 F.3d 1576 (Fed. Cir. 1996)	8

OTHER AUTHORITIES

35 U.S.C. § 112(b)	2
--------------------------	---

Pursuant to the Procedural Schedule (Order No. 6) and the Revised Ground Rules (Order No. 4), Complainant GlobalFoundries U.S. Inc. (“GlobalFoundries”) hereby submits its Opening Claim Construction Brief.

Respondents’ claim construction positions follow a familiar pattern. For five of the disputed terms, Respondents contend that ordinary, well-understood words—such as “layer” and “adjacent”—cannot be understood at all. For the remaining terms, Respondents ask the Court to rewrite the claim language. Both approaches are legal error.

The indefiniteness contentions fail at the threshold. Respondents must prove indefiniteness by clear and convincing evidence, *Nautilus, Inc. v. Biosig Instruments, Inc.*, 572 U.S. 898, 912 n.10 (2014), and a claim is definite if it “inform[s] those skilled in the art about the scope of the invention with reasonable certainty,” *id.* at 910. Every challenged term is recited in the claims and described in the specification with more than sufficient context to reasonably apprise a POSA of the claim scope.

Respondents’ affirmative constructions fare no better. Each departs from the words the patentees chose without the lexicography or disavowal required to justify a departure. *Thorner v. Sony Computer Ent. Am. LLC*, 669 F.3d 1362, 1365-66 (Fed. Cir. 2012). And each would inject ambiguity, read out disclosed embodiments, or render other claim language superfluous. “Courts do not rewrite claims; instead, we give effect to the terms chosen by the patentee.” *K-2 Corp. v. Salomon S.A.*, 191 F.3d 1356, 1364 (Fed. Cir. 1999).

Each disputed term should be given its plain and ordinary meaning.

I. APPLICABLE LAW

In construing patent claims, courts consider intrinsic evidence to the patent, which includes the claims, the specification, and the prosecution history, as well as extrinsic evidence. *See Phillips v. AWH Corp.*, 415 F.3d 1303, 1314 (Fed. Cir. 2005) (en banc). “[T]here is a ‘heavy presumption’

that the language in the claim carries its ordinary and customary meaning.” *Housey Pharms., Inc. v. Astrazeneca UK Ltd.*, 366 F.3d 1348, 1352 (Fed. Cir. 2004) (quoting *CCS Fitness, Inc. v. Brunswick Corp.*, 288 F.3d 1359, 1366 (Fed. Cir. 2002)). “[T]he ordinary and customary meaning of a claim term is the meaning that the term would have to a person of ordinary skill in the art in question at the time of the invention, i.e., as of the effective filing date of the patent application.” *Phillips*, 415 F.3d at 1313.

“There are only two exceptions to th[e] general rule” that the words of a claim are generally given their ordinary and customary meaning: “1) when a patentee sets out a definition and acts as his own lexicographer, or 2) when the patentee disavows the full scope of a claim term either in the specification or during prosecution.” *Thorner*, 669 F.3d at 1365. Courts should not “confin[e] the claims to” specific embodiments, even when “a patent describes only a single embodiment.” *Phillips*, 415 F.3d at 1323.

A patent specification “shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.” 35 U.S.C. § 112(b). This requires patent claims to “inform those skilled in the art about the scope of the invention with reasonable certainty.” *Nautilus*, 572 U.S. 898, 910. “[T]he definiteness requirement must take into account the inherent limitations of language.” *Id.* at 909. But even if a claim is broad, breadth is not indefiniteness. *BASF Corp. v. Johnson Matthey Inc.*, 875 F.3d 1360, 1367 (Fed. Cir. 2017). “[A]n inventor ‘need not define his invention with mathematical precision in order to comply with the definiteness requirement.’” *Barrette Outdoor Living, Inc. v. Fortress Iron, LP*, 156 F.4th 1353, 1367 (Fed. Cir. 2025). The patent challenger has the burden to prove indefiniteness by clear and convincing evidence. *Sonix Tech. Co. v. Publ’ns Int’l, Ltd.*, 844 F.3d 1370, 1377 (Fed. Cir. 2017).

II. LEVEL OF SKILL IN THE ART

A person of ordinary skill in the art (“POSA”) at the time of the inventions would have had a bachelor’s degree in electrical engineering, materials science, physics, chemistry, or a related field, together with at least two to three years of experience in semiconductor device design, fabrication, or process integration, or equivalent education and experience. Ex. 1 (Liu Decl.) ¶¶ 23–26. The Court need not resolve any dispute regarding the level of ordinary skill to construe the disputed terms: under any party’s articulation, each term should be given its plain and ordinary meaning.

III. ASSERTED PATENTS

A. The ’546 and ’167 Patents

U.S. Patent No. 9,865,546 (“the ’546 patent”) (Ex. 2) and U.S. Patent No. 10,707,167 (“the ’167 patent”) (Ex. 3) share a common specification and are both titled “Contacts to Semiconductor Substrate and Methods of Forming Same.”

The patents “relate[] to contacts over a semiconductor substrate and methods of forming the same,” and “[m]ore specifically, . . . to forming an intermix region at an interface of a bottom portion of contacts and an upper portion of the semiconductor substrate.” ’546 patent, 1:8-13. According to the patent, in a p-type field effect transistor, “performance with a Ti-based silicide liner is a challenge for 14/10 nm silicon-on-insulator (14/10 SOI) technologies.” *Id.*, 1:35-37. Although “a bilayer of NiPt and Ti has been employed to coat contact openings over the source and drain,” “this bilayer requires deposition of a relatively thick metal layer,” and “[a]s semiconductor devices get smaller, requiring a thick bilayer in the contact causes contact opening fill issues, such as an incomplete contact fill.” *Id.*, 1:35-44.

The disclosed inventions “utilize an intermix concept as a template for formation of a contact to a semiconductor substrate of a semiconductor structure,” ’546 patent, 2:35-38, and

“involve[] depositing a first liner layer in a contact opening such that an intermix region is formed at an interface of the bottom portion of the contact opening and an upper portion of the semiconductor substrate thereunder, e.g., source and drain,” *id.*, 2:43-48. Because “[u]nreacted (non-diffused) portions of the first liner layer may be removed prior to completion of fabricating the contact,” the disclosed methods “do not increase the deposited metal thickness in the contact openings,” and “contact opening fill issues are avoided and overall contact resistance is improved.” *Id.*, 2:48-53. The patent expressly describes the “intermix region” as “having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate.” *Id.*, 2:16-19.

B. The '235 Patent

U.S. Patent No. 8,330,235 (“the '235 patent”) (Ex. 4) is titled “Method to Reduce MOL Damage on NiSi.” The patent “relates to semiconductor devices with improved nickel silicide layers” and “is particularly applicable in fabricating transistors with improved device performance by avoiding nickel silicide degradation.” '235 patent, 1:11-15.

The specification explains that during “overetch” processes, “underlying NiSi layers are damaged, causing degradation of the NiSi.” *Id.*, 2:54-57. “The degradation can be morphological, wherein mouse bites or missing NiSi is observed, . . . or electrical, wherein the resistivity is increased,” and “[e]ither or both types of degradation ultimately result in poor device parameters such as higher sheet resistance and external resistance.” *Id.*, 2:57-62.

The '235 patent addresses that problem by “formulating the NiSi layer with a platinum composition gradient with the amount of platinum increasing in a direction toward its upper surface, i.e., in a direction away from the source/drain regions and the gate electrode, thereby shielding the NiSi layer from physical damage during post NiSi processing as when removing an overlying stress liner by etching.” '235 patent, 5:3-11. “During annealing, platinum diffuses to

form a composition gradient with the amount of platinum increasing in a direction away from the substrate and gate electrode.” *Id.*, 5:17-20. This “concentration gradient of platinum effectively shields the NiSi layer from physical damage during subsequent processing, e.g., etching, while concomitantly maintaining the integrity of the NiSi layer.” *Id.*, 5:17-23. In the disclosed embodiment, “[t]he resulting NiSi layer 401 has a composition gradient of platinum from about 8 wt. % to about 12 wt. %, such as about 10 wt. %, at the silicon interface 403 to about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %, at the opposite surface.” *Id.*, 7:6-10.

C. The '983 Patent

U.S. Patent No. 8,507,983 (“the '983 patent”) (Ex. 5) is titled “High Voltage Device.” The specification explains that “Lateral Double-Diffused (LD) transistors have been widely employed in high voltage applications,” and that “[f]or high performance LD transistors, low drain-to-source on-resistance (Rdson) is desired to minimize its power dissipation when it is turned on, as well as high breakdown voltage to maximize its voltage capability.” '983 patent, 1:16-22. “To achieve low Rdson, the channel of the high LD transistor should be as short as possible.” *Id.*

Conventional processing made short channels difficult to achieve: “process overlay issues make it difficult for alignment of the HV channel well, which defines the channel length,” and “the thin gate electrodes used prevent them serving as a hard mask, further exacerbating process control in forming the channel well.” '983 patent, 1:31-38. “From the foregoing discussion, it is desirable to provide reliable HV devices formed with short channel lengths to have low Rdson.” *Id.*, 1:39-41.

The '983 patent discloses a device that “includes a doped channel well disposed in the substrate adjacent to a first edge of the gate,” where “[t]he first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate,” and “[t]he first edge of the gate and channel edge defines an effective channel length of the device” that “is self-aligned to the

gate.” ’983 patent, Abstract. “A doped drift well adjacent to a second edge of the gate is also included.” *Id.* The specification further teaches that “[t]he effective channel length, in one embodiment, is controlled by ion implantation which is self-aligned to the gate edge,” and “may be controlled by channel or body implant conditions selected to satisfy threshold voltage and breakdown voltage requirements”—for example, “shorter effective channel length may be achieved by decreasing tilt angle, implant energy or dosage and vice-versa.” ’983 patent, 5:57-67. “With the channel well being self-aligned to the gate edge of the implant, process overlay issues are avoided.” *Id.*, 6:1-2.

IV. DISPUTED TERMS

A. ’546 Patent/’167 Patent

1. “layer” / “first liner layer” / “refractory metal layer” (’546 patent at claims 1-14; ’167 patent at claims 1, 3, 4)

Complainant’s Proposed Construction	Respondents’ Proposed Construction
Plain and ordinary meaning	Indefinite
Not indefinite	If not indefinite, “a separate and defined [first liner / refractory metal] coating of material”

Respondents contend that “layer,” “first liner layer,” and “refractory metal layer” are indefinite. They are not. Respondents bear the burden of proving indefiniteness by clear and convincing evidence. *Sonix Tech.*, 844 F.3d at 1377. The term “layer” is one of the most basic and widely used words in the semiconductor art. Indeed, Respondents use the terms “layer” and “multi-metal layer” on their own website to describe their processes. Ex. 6 (Tower Webpage). Terms that are “readily understood by both the expert and the layperson” are sufficiently definite. *Fisher-Price, Inc. v. Graco Children’s Prods., Inc.*, 154 F. App’x. 903, 907 (Fed. Cir. 2005). A POSA would have no difficulty reasonably ascertaining the scope of “layer,” “first liner layer,”

and “refractory metal layer” in view of the guidance provided by the claims and specification. Liu Decl. ¶¶ 33–38.

The claim language confirms these “layer” terms are not indefinite. Claim 1 of the ’546 patent recites “depositing a first liner layer to conformally coat the contact opening” and “depositing a refractory metal layer over the first intermix region.” ’546 patent at cl. 1. Claim 1 of the ’167 patent recites “a first liner layer substantially coating a sidewall of the contact opening” and “a refractory metal layer substantially coating the first liner layer in the contact opening.” ’167 patent at cl. 1. The claims thus tell a POSA where each layer is located, its relationship with other structures (and other layers), and what it separates. That is far more than “reasonable certainty” requires. *Nautilus*, 572 U.S. at 910.

The specification and the dependent claims provide even more guidance. The specification identifies exemplary materials from which the first liner layer may be formed (“at least one of: NiPt, Ni, Ti, TiSi, Co, Er, Pt, and Yb”) and exemplary processes of how layers are formed (“using ionized PVD or radio frequency PVD (RFPVD) or other directional energetic thin film deposition techniques”). ’546 patent, 5:60-65; *see also* cls. 3, 5. The specification likewise identifies exemplary materials for the “refractory metal layer” (“W, Ti, Ta or Mo employed directly as a ‘metal’; a refractory metal employed as a constituent of a ‘compound’ thereof, e.g., TiN, TaN, etc., or even as a constituent of an ‘alloy’ with another metal, e.g., titanium-tungsten”) and teaches that such a “layer” may be a “bilayer.” ’546 patent, 7:3-8; *see also* cl. 4; ’167 patent at cls. 3, 4. A POSA reading these descriptions in the specification and claims would reasonably ascertain the scope of what a “layer” comprises.

Respondents’ alternative construction “a separate and defined [first liner / refractory metal] coating of material” should be rejected. It is black-letter law that any departure from the ordinary meaning of a claim term must rest on lexicography or “clear and unmistakable” disclaimer. *Avid*

Tech., Inv. v. Harmonic, Inc., 812 F.3d 1040, 1045 (Fed. Cir. 2016); *see also Toshiba Corp. v. Imation Corp.*, 681 F.3d 1358, 1369 (Fed. Cir. 2012); *InterDigital Commc'ns, LLC v. Int'l Trade Comm'n*, 690 F.3d 1318, 1324 (Fed. Cir. 2012). There is none here. The words “separate” and “defined” appear nowhere in the claims, and Respondents identify no lexicography and no disavowal that could justify importing them.

Respondents’ “separate and defined coating of material” requirement would also read out disclosed embodiments, a result that “is rarely, if ever, correct.” *Vitronics Corp. v. Conceptronic, Inc.*, 90 F.3d 1576, 1583 (Fed. Cir. 1996); *see also Certain Child Car Seats*, Inv. 337-TA-1459, Order No. 13, at 13 (Feb. 27, 2026) (declining to adopt a claim construction “inconsistent with at least one express embodiment”). The specification teaches a “dielectric layer 120” that “may include a single layer or more than two separate layers.” ’546 patent, 4:56-62. The specification similarly discloses that the “refractory metal layer” may be a “Ti/TiN bilayer.” *Id.*, 7:45-46. And the specification teaches that the first liner layer is deposited such that “a first intermix region” is “formed at an interface of the bottom portion of each contact opening 130a, 130b and the upper portion of each of source 124a and drain 124b”—that is, the liner material intermixes with the adjoining substrate rather than remaining wholly “separate” from it. ’546 patent, 5:65-6:2. Respondents’ proposed limitation of “layer” to “a separate and defined . . . coating of material” would exclude each of those disclosed “layers.”

If anything, Respondents’ proposed construction needlessly injects uncertainty into the claims. Respondents’ proposed construction supplies no metric for how “separate” a coating must be from what, and no guidance as to what would make a coating sufficiently “defined.” Nor does it provide any guidance as to what a “coating of material” comprises, or how such a coating differs from a “layer.” Replacing a well-understood claim term with undefined words amounts to an

improper rewrite of the claim language. *K-2 Corp.*, 191 F.3d at 1364 (“Courts do not rewrite claims; instead, we give effect to the terms chosen by the patentee.”).

The terms “layer,” “first liner layer,” and “refractory metal layer” are not indefinite and should be given their plain and ordinary meaning. No construction is required.

2. “intermix region” (’546 patent at claims 1, 2, 6, 8-9, 13; ’167 patent at claims 1, 2, 6)

Complainant’s Proposed Construction	Respondents’ Proposed Construction
Plain and ordinary meaning	Indefinite
Not indefinite	

Respondents contend that “intermix region” is indefinite, and they offer no alternative construction at all. Respondents cannot carry their clear-and-convincing burden. *Cox Commc’ns, Inc. v. Sprint Commc’ns Co.*, 838 F.3d 1224, 1228 (Fed. Cir. 2016). A POSA reading the claims in light of the specification would readily understand an “intermix region” to be a region in which materials from different layers have intermixed. Liu Decl. ¶ 40.

The claims themselves identify what an intermix region is, where it is located, and what it contains. Liu Decl. ¶¶ 39–42. Claim 1 of the ’546 patent recites “causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate.” ’546 patent at cl. 1. Claim 1 of the ’167 patent recites “an intermix region at an upper surface of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate.” ’167 patent at cl. 1. The claims thus specify the structure, location, and material composition of the claimed “intermix region,” providing adequate objective boundaries and reasonably apprising POSAs of claim scope. And the dependent claims provide even more guidance, reciting that “the intermix region includes a titanium-based silicide” and that “the

intermix region has a thickness of approximately 3 nanometers to approximately 5 nanometers.”

’167 patent at cls. 2, 6; *see also* ’546 patent at cls. 2, 6, 9, 13.

The specification confirms “intermix region” is not indefinite. It explains that intermix regions “originate from the high energy of the incoming sputtered atoms (from first liner layer 132) which are thus pushed into the Si lattice of source 124a and drain 124b,” so that “[t]he first few metal atoms, arriving with relatively high energy, can be forced into the interstitial sites of the Si lattice to create a thin region (intermix regions 134a, 134b) of mixed Si and metal on the order of 3-5 nm in thickness.” ’546 patent, 6:3-19. The intermix region accordingly “include[s] a mixture of the material used for first liner layer 132 and a material used for the respective semiconductor substrate 110 thereunder.” *Id.*, 6:6-9. The specification even provides exemplary compositions of the “intermix region”: the region “may have a relatively high Si content (greater than 50% Si in atomic composition),” *id.*, 6:19-21, and may comprise “a NiPt silicide,” “a NiPtTi silicide,” “a Ni silicide,” or “a NiTi silicide,” *id.*, 7:45-58.

In short, the claims and specification here provide the “significant guidance” needed for a POSA to understand the scope of “intermix region” with reasonable certainty. The term is not indefinite and should be given its plain and ordinary meaning.

B. ’235 Patent

1. “from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide” (’235 patent at claims 1, 10)

Complainant’s Proposed Construction	Respondents’ Proposed Construction
Plain and ordinary meaning	Indefinite
Not indefinite	

Respondents again contend that this phrase is indefinite and offer no alternative construction. Respondents cannot carry their clear-and-convincing burden.

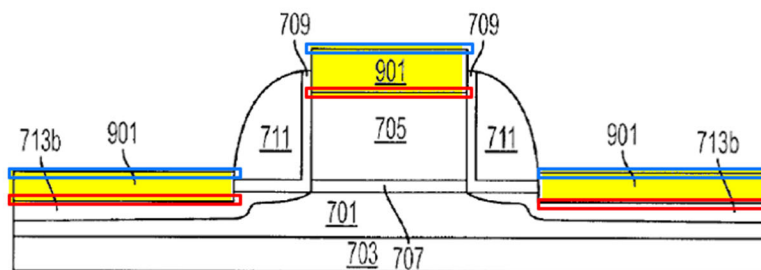
Claim 1 recites a “silicon substrate” with “a nickel silicide layer on the gate electrode and source/drain regions.” The claim further recites that the “nickel silicide layer comprises platinum having a composition gradient, *from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide*, with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode.” ’235 patent at cl. 1; *see also id.* at cl. 10. A POSA would understand that the disputed term identifies the endpoints of a platinum composition gradient, namely, from the bottom of the nickel silicide (a nickel silicide surface “at a silicon interface”) to the top of the silicide (“an opposite surface of the nickel silicide”). Liu Decl. ¶¶ 43–48. This is “little more than the application of the widely accepted meaning of commonly understood words.” *Phillips*, 415 F.3d at 1314; *see also Summit 6, LLC v. Samsung Elecs. Co.*, 802 F.3d 1283, 1291 (Fed. Cir. 2015) (“‘Being provided to’ is comprised of commonly used terms; each is used in common parlance and has no special meaning in the art. Because the plain and ordinary meaning of the disputed claim language is clear, the district court did not err by declining to construe the claim term.”). Indeed, it was the Examiner who suggested revisions to the claim that resulted in the claim language “from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide.” Ex. 7 (’235 File History) at 6 (7/26/2012 Applicant Arguments). The claims were subsequently allowed with that language. *Id.* at 19–20 (Notice of Allowance and Reasons for Allowance).

The specification provides even more guidance, stating:

The resulting NiSi layers 901 have a composition gradient of platinum from about 8 wt. % to about 12 wt. %, such as about 10 wt. %, *at the bottom surface that contacts the silicon gate electrodes 705 and source/drain regions 713b*, to about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %, *at the opposite surface*.

’235 patent at 8:43–48. The claimed “surfaces” for the platinum gradient as described are further shown in Figure 9, which depicts a nickel silicide 901 (yellow) with a “bottom surface that contacts

the silicon gate electrodes 705 and source/drain regions 713b” (red box) and an “opposite surface” (blue box).



The specification further discloses that the platinum concentration gradient of the NiSi layer is “increasing in amount toward the upper surface, i.e., increasing in amount in a direction away from the P-channel and N-channel transistors” ’235 patent, 3:59-67; *id.* at 5:3-11 (“formulating the NiSi layer with a platinum composition gradient with the amount of platinum increasing in a direction toward its upper surface, i.e., in a direction away from the source/drain regions and the gate electrode.”); *see also id.* at 7:4-12. And the specification provides examples of why these endpoints matter: increasing the platinum composition between the two claimed surfaces results in “shielding the NiSi layer from physical damage during post NiSi processing as when removing an overlying stress liner by etching.” *Id.* at 5:3-11. Where the intrinsic record supplies this degree of guidance, an indefiniteness challenge fails. *See Certain Icemaking Machines and Components Thereof*, Inv. 337-TA-1369, Order No. 10, at 18 (Jan. 4, 2024) (finding an indefiniteness contention “undercut by the significant guidance that the claims, when read in light of the specification, provide regarding the scope of the claimed” limitation).

The phrase “from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide” is not indefinite and should be given its plain and ordinary meaning.

C. ’983 Patent

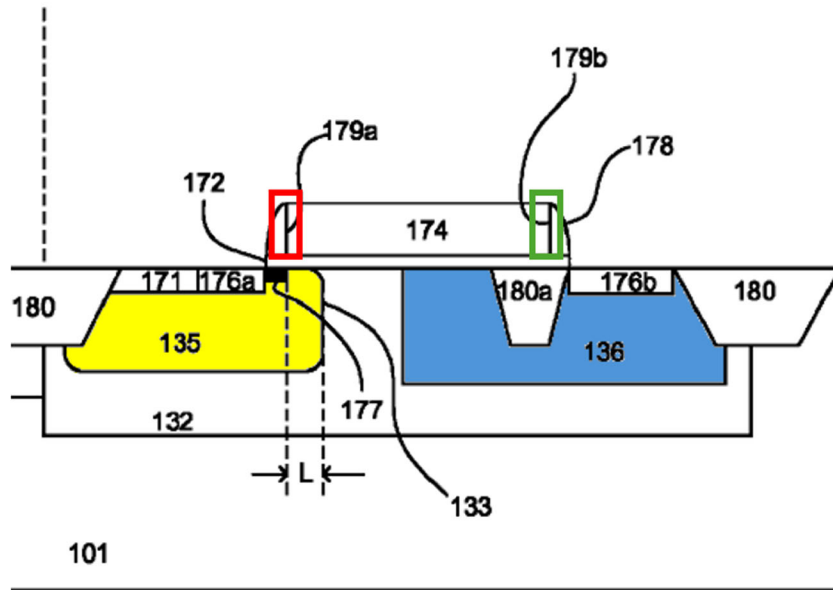
1. “adjacent to” (’983 patent at claims 1, 5, 13, 17, 20)

Complainant's Proposed Construction	Respondents' Proposed Construction
Plain and ordinary meaning	Indefinite
Not indefinite	

Respondents contend that “adjacent to” is indefinite. It is not. The plain and ordinary meaning of “adjacent” is “situated near or next.” Ex. 8 (2004 Merriam-Webster Dictionary) at 4. Indeed, courts routinely construe “adjacent” with no difficulty. *MBO Lab’ys, Inc. v. Becton, Dickinson & Co.*, 474 F.3d 1323, 1333 (Fed. Cir. 2007) (construing “adjacent” to mean “next to” and rejecting a proposed construction of “contiguous or connected”). Liu Decl. ¶¶ 49–52.

The meaning of terms like “adjacent” is “dictated by context.” See *Accentra, Inc. v. Staples, Inc.*, 500 F. App’x 922, 930-31 (Fed. Cir. 2013) (reversing indefiniteness finding because “[n]ear” is a word whose meaning is dictated by context”); *Power-One, Inc. v. Artesyn Techs., Inc.*, 599 F.3d 1343, 1348-50 (Fed. Cir. 2010) (rejecting indefiniteness challenge to “near”). Respondents bear the clear-and-convincing burden, *Cox Commc’ns*, 838 F.3d at 1228, and cannot carry it here.

The claims here supply context expressly. For example, claim 1 recites a “gate” with a “first edge” and a “second edge.” The term “adjacent” is used to specify that a “doped channel well” and a “heavily doped region” are “adjacent to a first edge of the gate” on one hand, while a “doped drift well” is “adjacent to a second edge of the gate.” ’983 patent, cl. 1. The specification similarly describes a “doped channel well” that is “adjacent to a first edge of the gate” on one hand and a “doped drift well” that is “adjacent to a second edge of the gate.” ’983 patent, 1:45-55, 1:58-66, 2:4-13. Figure 1 confirms that spatial relationship, with a channel well 135 (yellow) next to or near a first edge of the gate 179a (red) and a drift well 136 (blue) next to or near the second edge of the gate 179b (green):



In short, each use of “adjacent” in the claims and the specification identifies both structures and the reference feature, e.g., a first or second edge of the gate. With those guideposts, a POSA would readily understand the claim scope with reasonable certainty. “Adjacent to” is not indefinite and should be given its plain and ordinary meaning.

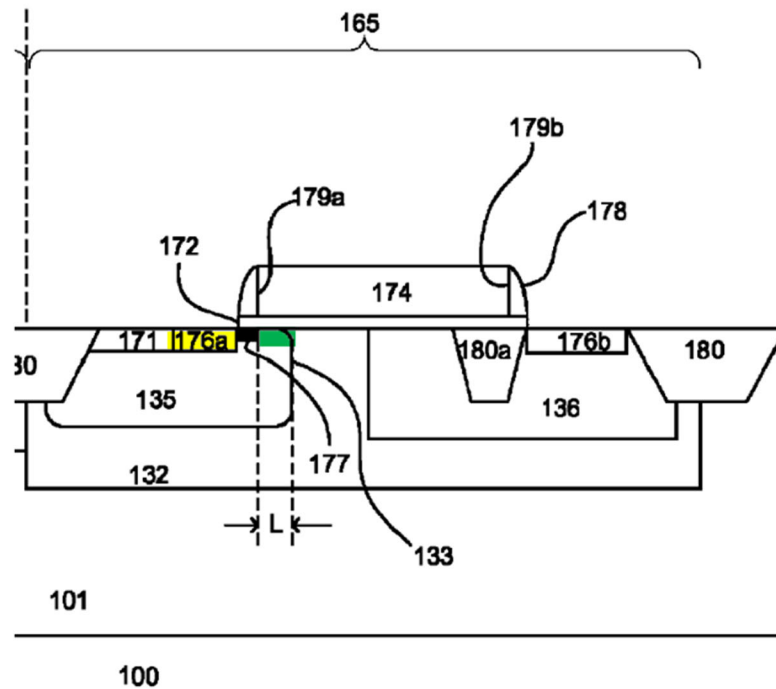
2. “the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate” (’983 patent at claims 1, 13, 20)

Complainant’s Proposed Construction	Respondents’ Proposed Construction
Plain and ordinary meaning	Indefinite
Not indefinite	

Respondents contend that this phrase is indefinite. Once again, they offer no alternative construction and cannot carry their clear-and-convincing burden. Regardless, a skilled artisan would reasonably understand the scope of this limitation in view of the claim language and specification, which confirm that the channel and the first heavily doped region adjacent to the first edge of the gate are offset from one another. Liu Decl. ¶¶ 54-57.

As with “adjacent to,” the meaning of this term is “dictated by context.” *Accentra, Inc.*, 500 F.App’x 922 at 930–931. The intrinsic record again provides that context. Claim 1 recites

that “the channel is displaced from a first heavily doped region adjacent to the first edge of the gate.” The specification describes this relationship, disclosing a channel with “channel length L” (green) and a “source diffusion region 176a” (yellow) with a “source extension region 177” in between. ’983 patent, 5:25-32, 5:52-56. As shown below, the channel (green) is displaced or offset from heavily doped region 176a (yellow), due to the intervening region 177.



The dependent claims remove any doubt about the claimed displacement. Claim 2 recites a device “further comprising spacer on the first edge of the gate, wherein the spacer defines the displacement of the channel from the first heavily doped region.” ’983 patent at cl. 2; *see also id.* at cl. 14. The intrinsic record therefore not only uses the term but recites an exemplary structure that establishes the displacement.

To the extent Respondents’ challenge rests on the phrase “adjacent to,” it fails for the reasons given above. Regardless, “the significant guidance that the claims, when read in light of the specification, provide regarding the scope of the claimed” term refutes Respondents’ position.

3. “the drift isolation region is disposed under the gate” (’983 patent at claims 4, 16)

Complainant’s Proposed Construction	Respondents’ Proposed Construction
Plain and ordinary meaning	“the drift isolation region is positioned at least in part under the gate”

This phrase requires no construction beyond its plain and ordinary meaning. Respondents’ proposed construction replaces “disposed” with “positioned at least in part”: the drift isolation region is ~~disposed~~ **positioned at least in part** under the gate. There is no lexicography or disavowal that justifies this rewrite of the claims. *Thorner*, 669 F.3d at 1365-66; *InterDigital*, 690 F.3d at 1324. “Courts do not rewrite claims; instead, we give effect to the terms chosen by the patentee.” *K-2 Corp.*, 191 F.3d at 1364. Respondents’ proposed construction should be rejected and this term should be given its plain and ordinary meaning.

The specification independently forecloses Respondents’ construction. It describes the claimed arrangement and then separately identifies a partial arrangement as an alternative: “As shown, the drift isolation region 180a is disposed in the drift well 136 under the gate,” but “[d]isposing the drift well at other positions, *such as partially underlying the gate* may also useful.” ’983 patent, 5:22-24. The specification thus treats a drift isolation region “disposed . . . under the gate” and one “partially underlying the gate” as distinct configurations. Respondents’ construction would collapse that distinction, inviting legal error.

4. “the channel having a channel length that is controlled by ion implantation” (’983 patent at claims 7, 18)

Complainant’s Proposed Construction	Respondents’ Proposed Construction
Plain and ordinary meaning	“the channel having a channel length that is formed by implanting ions into the substrate to form a channel well”

This term requires no construction beyond its plain and ordinary meaning. Respondents again propose rewriting the claim language, replacing “controlled by ion implantation” with

“formed by implanting ions into the substrate to form a channel well.” There is again no lexicography or disclaimer to support Respondents’ proposed rewrite.

The claim language and specification refute Respondents’ proposed construction. The claim language uses the term “controlled by,” not “formed by.” The specification uses both terms separately: while components (such as the “channel”) are “formed,” dimensions of such components (such as the “channel length”) are “controlled.” ’983 patent, 8:56-57 (“the substrate is implanted with dopants *to form* a channel well 135.”), 5:57-59 (“The effective channel length, in one embodiment, is *controlled by* ion implantation”). It makes little sense to “form” a “channel length” as Respondents propose.

5. “edge of the gate” (’983 patent at claims 1, 2, 6, 13-15, 20)

Complainant’s Proposed Construction	Respondents’ Proposed Construction
Plain and ordinary meaning	“edge of the gate electrode over the gate dielectric”

This term requires no construction beyond its plain and ordinary meaning. Respondents proposes replacing “gate” with “gate electrode” and adding the words “over the gate dielectric,” so that “edge of the gate” becomes “edge of the gate electrode over the gate dielectric.” Respondents are incorrect. The specification describes “a gate with first and second edges 179a-b or sidewalls.” ’983 patent, 4:46-51. A POSA would thus readily understand that an “edge of the gate” simply refers to a sidewall of the gate. No further construction is needed.

Respondents’ proposed construction errs in at least two respects. First, Respondents equate the “gate” with the “gate electrode.” But each asserted independent claim recites “a gate, the gate including a gate electrode over a gate dielectric,” and the specification likewise teaches that “[t]he gate includes a HV gate electrode 174.” ’983 patent, 4:46-51. The “gate” is therefore broader than the “gate electrode” it includes, and Respondents’ attempt to substitute the component for the whole should be rejected. Second, Respondents’ proposed construction adds the phrase “over the

gate dielectric.” There is no disclaimer or lexicography to support limiting the “edge of the gate” to only portions “over the gate dielectric,” further confirming that Respondents’ proposed construction is error.

V. CONCLUSION

For the foregoing reasons, GlobalFoundries respectfully requests that the Court reject Respondents’ proposed constructions and indefiniteness contentions and hold that each disputed term should be given its plain and ordinary meaning.

Word Count Certificate

Pursuant to Ground Rule 1.7, the undersigned certifies that this Opening Claim Construction Brief complies with the word limitation set forth in the Ground Rule 7.2. The certified word count includes footnotes and excludes caption, tables of contents or authorities, words in figures, the word count certificate, signature block, and the certificate of service.

Certified word count: 5157.

Dated: August 25, 2026

Respectfully submitted,

/s/ Bert C. Reiser

Clement Naples
Cleary Gottlieb Steen & Hamilton LLP
One Liberty Plaza
New York, NY 10006
Telephone: (212) 225-2516
Facsimile: (212) 225-3999
cnaples@cgsh.com

Bert C. Reiser
Sami Al-Marzoog
Cleary Gottlieb Steen & Hamilton LLP
2112 Pennsylvania Ave, N.W.
Washington, DC 20037
Telephone: (202) 974-1500
Facsimile: (202) 974-1999
breiser@cgsh.com
salmarzoog@cgsh.com

Thomas W. Yeh
Cleary Gottlieb Steen & Hamilton LLP
650 California St, Suite 2400
San Francisco, CA 94108
Telephone: (415) 796 4400
Facsimile: (415) 796-4499
tyeh@cgsh.com

Email: team-globalfoundries-itc-cgshonly@cgsh.com

*Counsel for Complainant
GlobalFoundries U.S. Inc.*

CERTIFICATE OF SERVICE

It is hereby certified that copies of the foregoing **COMPLAINANT'S OPENING CLAIM CONSTRUCTION BRIEF** were served on August 25, 2026 as follows:

The Honorable Lisa R. Barton Secretary to the Commission U.S. International Trade Commission 500 E Street, SW, Room 112 Washington, DC 20436	BY EDIS
The Honorable Clark S. Cheney Chief Administrative Law Judge U.S. International Trade Commission 500 E Street, SW, Room 317 Washington, DC 20436	By Email and Box: Cheney1500@usitc.gov
Stephen R. Smith COOLEY LLP 1299 Pennsylvania Ave., N.W. Suite 700 Washington, D.C. 20004 <i>Counsel for Respondents Tower Semiconductor Ltd., Tower Partners Semiconductor, Ltd., Tower Semiconductor Italy S.r.l., Tower US Holdings, Inc., Tower Semiconductor San Antonio, Inc., Tower Semiconductor NPB Holdings, Inc., Tower Semiconductor Newport Beach, Inc., and Newport Fab LLC</i>	By Email: TowerITC@cooley.com

/s/ Erika J. Weinstein

Erika J. Weinstein

CLEARY GOTTlieb STEEN & HAMILTON LLP

EXHIBIT 1

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON DC**

**Before the Honorable Clark S. Cheney
Chief Administrative Law Judge**

In the Matter of

**CERTAIN SEMICONDUCTOR DEVICES,
PRODUCTS CONTAINING SAME, AND
COMPONENTS THEREOF**

Investigation No. 337-TA-1500

**DECLARATION OF DAVID LIU, PH.D. IN SUPPORT OF COMPLAINANT
GLOBALFOUNDRIES' OPENING CLAIM CONSTRUCTION BRIEF**

TABLE OF CONTENTS

I.	INTRODUCTION	2
II.	BACKGROUND AND EDUCATION	3
III.	MATERIALS CONSIDERED AND COMPENSATION	5
IV.	LEGAL PRINCIPLES	6
A.	Person of Ordinary Skill in the Art	6
B.	Claim Construction	7
V.	THE '546/'167 PATENT'S "LAYER"/ "FIRST LINER LAYER" / "REFRACTORY METAL LAYER"	8
VI.	THE '546/'167 PATENT'S "INTERMIX REGION"	10
VII.	THE '235 PATENT'S "FROM A NICKEL SILICIDE SURFACE AT A SILICON INTERFACE TO AN OPPOSITE SURFACE OF THE NICKEL SILICIDE"	12
VIII.	THE '983 PATENT'S "ADJACENT TO"	14
IX.	THE '983 PATENT'S "THE CHANNEL IS DISPLACED FROM [A/THE] FIRST HEAVILY DOPED REGION ADJACENT TO THE FIRST EDGE OF THE GATE"	16
X.	CONCLUSION.....	17

I. INTRODUCTION

1. I have been retained as an expert by Complainant GlobalFoundries U.S. Inc. (“Complainant” or “GlobalFoundries”) in connection with the above-captioned action to provide my analysis and conclusions on certain technical aspects of this dispute.

2. I have personal knowledge of the matters set forth in this declaration and, if called upon to do so, would testify to such matters in court.

3. In this declaration, I provide opinions relating to U.S. Patent Nos. 8,330,235 (“the ’235 patent”), 8,507,983 (“the ’983 patent”), 9,865,546 (“the ’546 patent”) and 10,707,167 (“the ’167 patent”), including summaries of these patents and opinions relating to terms recited in the claims of these patents.

4. I understand that these patents, together with U.S. Patents No. 9,093,425 (the “’425 patent”) and 10,062,748 (the “’748 patent”) are the asserted patents in the above-captioned proceeding (the “Asserted Patents”). I have not been asked to form an opinion on any claim terms of the ’425 or the ’748 patents and as of the date of this declaration I have not formed such an opinion.

5. I understand that Tower Semiconductor, Inc. (“Respondent” or “Tower”) asserts that certain terms of the Asserted Patents are indefinite. These terms include (1) “layer”/“first liner layer”/“refractory metal layer” and (2) “intermix region” from the ’546 and ’167 patents; (3) “from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide from the ’235 patent; and (4) “adjacent to” and (5) “the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate” from the ’983 patent.

6. In my opinion, each of the terms above would have had a meaning understood by a POSITA with reasonable certainty, and therefore are not indefinite.

II. BACKGROUND AND EDUCATION

7. I received a Bachelor of Science degree in Electrical Engineering from the University of California, Berkeley, in 1983. I received a Master of Science and Ph.D. degrees in Electrical Engineering from Stanford University in 1985 and 1989, respectively. My Ph.D. dissertation, entitled "Physics and Technology of Novel Conductivity Modulated Power MOSFET" was done under the supervision of Prof. James D. Plummer, who is well known in the field of integrated circuit technology, especially in the area of power devices.

8. From 1989 to 1992, I was a member of technical staff at Texas Instruments, Inc. At Texas Instruments, my job responsibilities included process integration, device modeling, high-voltage CMOS process integration, and investigating novel source-side injection mechanisms for Flash EPROM channel hot-electron programming.

9. From 1992 to 1995, I was a member of the technical staff at Advanced Micro Devices, Inc. (AMD) where I was a key contributor in optimizing Flash cell and periphery devices in AMD's CMOS-based 0.5um and 0.35um Flash EPROM technology. My job responsibilities at AMD also included process integration, device modeling, and development of triple-well process technology for accommodate x-decoder transistors and high voltage transistors for negative gate erase operation. While at AMD, I was awarded a Spotlight Award for developing a method of manufacturing a self-aligned source (SAS) etch for a NOR flash memory.

10. In 2000, I co-founded Progressant Technologies in Fremont, California. Progressant Technologies developed IP for negative differential resistance transistor technology and was eventually acquired by Synopsys, Inc.

11. From 2000 to 2004, I was a Senior Manager at Xilinx, Incorporated, where I was responsible for developing nonvolatile memory process technology for flash and CPLD product applications, as well as advanced CMOS process technology (specifically 75nm CMOS

technology node, a half node version between 90nm and 65nm). The flash memory products I developed at Xilinx are used as the configuration memory for the FPGA, with the configuration interface between them being the JTAG interface.

12. From 2004 to 2007, I was a Senior Scientist at Maxim Integrated Products where I was responsible for developing Embedded Non-volatile Memory process technology for Power Management product applications.

13. Since 2007, I have served as a technical consultant where I have provided expert advice regarding Flash memory technology, CMOS process technology, and semiconductor device physics.

14. I have 20 years of experience as an engineer and engineering manager/director of Complementary Metal Oxide Semiconductor (CMOS) technology development. During my career, I have worked at some of the leading technology companies in the world, such as Texas Instruments, Advanced Micro Devices, Altera Corporation (now a subsidiary of Intel Corp.), and Xilinx. At these companies, my work focused on various aspects of CMOS and semiconductor technology.

15. I hold over 99 U.S. patents, a large number of which are directed to logic CMOS processes technology and high voltage CMOS process technology. The vast majority of my patents are in the area of CMOS process technology and memory architecture. As such, I am intimately familiar with the implementation of integrated process flows that essentially are repetitive sequences of modules, each composed of a layer patterning step followed by selective deposition and etching of various semiconductor materials, with the area of selectivity dictated by the layer patterning.

16. I have also authored several technical papers that have been published in well-respected, peer-reviewed journals, such as the IEEE Electron Device Letters, the IEEE Journal of Solid-State Circuits, and the IEEE Transactions on Electron Devices.

17. As an example, I worked on a new conductivity-modulated Power MOSFET that features a buried minority-carrier injector, implemented at the bottom of a filled trench, to enhance the current conduction capability of a vertical Power DMOS. This work was published in the IEEE Transactions on Electron Devices. In my experience, there is a continual pursuit toward the reduction of on-resistance for a Power MOSFET device with a given voltage and current handling capability.

18. My opinions are based on my years of education, research, and experience, as well as my study of relevant materials. In forming my opinions, I have also considered the materials identified in this declaration and in the claim construction brief.

19. My qualifications generally are set forth in my curriculum vitae, which is submitted as Appendix A.

III. MATERIALS CONSIDERED AND COMPENSATION

20. I am being compensated for my time at my ordinary hourly rate of \$600/hr. I have no financial interest in the outcome of this case, and my compensation is not dependent on the contents of this declaration, the substance of my analysis, conclusions, or testimony that I may give, or this case's outcome.

21. In connection with rendering my opinions I have reviewed the '235 patent, the '983 patent, the '546 patent, and the '167 patent, and their prosecution histories. I have also reviewed the below materials.

Exhibit #	Description
2	'546 patent
3	'167 patent
4	'235 patent
5	'983 patent
6	“RF-SOI & RF-CMOS Platforms,” Tower Semiconductor, https://towersemi.com/technology/rf-and-hpa/rf-soi-rf-cmos-platforms/ (last visited August 25, 2026)
7	Excerpts of the File History of the '235 Patent
8	<i>The Merriam-Webster Dictionary</i> , 2004, definition of “adjacent”

IV. LEGAL PRINCIPLES

22. I am not a lawyer. GlobalFoundries’s counsel has instructed me as to the applicable legal standards relevant to this declaration.

A. Person of Ordinary Skill in the Art

23. I understand that assessments of the meanings of claim terms are made from the perspective of a person of ordinary skill in the art (“POSA”) at the time of the priority date for the patent being discussed. For the purposes of this declaration, I have been asked to assume that the applicable priority date for the patents I address is: February 5, 2009 for the '235 patent; February 23, 2009 for the '983 patent; June 3, 2015 for the '546 patent and for the '167 patent.

24. I understand that factors considered in determining the ordinary level of skill in a field of art include: the level of education and experience of persons working in the field; the types of problems encountered in the field; the teachings of the prior art regarding solutions to such problems; and the sophistication of the technology at the time of the alleged invention. I understand that a person of ordinary skill in the art (“POSA”) is not a specific real individual, but rather is a

hypothetical individual having the qualities reflected by the factors above and knowledge of all relevant prior art references.

25. In my opinion, a person of ordinary skill in the art at the time of the inventions would have had a bachelor's degree in electrical engineering, materials science, physics, chemistry, or a related field, together with at least two to three years of experience in semiconductor device design, fabrication, or process integration, or equivalent education and experience.

26. I would have qualified as a person of ordinary skill in the art with respect to each patent I address in this declaration as of their respective assumed priority dates. My opinions are provided from the perspective that a person of ordinary skill in the art would have had as of those dates.

B. Claim Construction

27. I understand that the Court has not construed the claims of the patents I address in this declaration, and the parties are currently briefing claim construction issues.

28. I understand that the purpose of claim construction is to determine the scope and meaning of the patent claims from the perspective of a person of ordinary skill in the art in light of the evidence.

29. I understand that the claims of a patent, its specification, and its file history are considered "intrinsic evidence," and that other evidence, such as dictionaries and expert opinions, is considered "extrinsic evidence."

30. I understand that intrinsic evidence is the most important source for determining the meaning of a claim term, and that extrinsic evidence also may be useful when considered in the context of the intrinsic evidence but cannot be used to vary from or contradict how the term is used in the intrinsic evidence.

31. I understand that the plain meaning of a claim term is the ordinary and customary meaning of the term to a person of ordinary skill in the art at the time of the alleged invention, in the context of the patent. I understand that a claim term should be given its plain and ordinary meaning unless the intrinsic record conveys a different or inconsistent use of the term.

32. I have been informed that a patent claim is indefinite when it fails to inform, with reasonable certainty, those skilled in the art about the scope of the invention. I have also been informed that a party challenging a claim as indefinite must prove it by clear and convincing evidence.

V. THE '546/'167 PATENT'S "LAYER"/ "FIRST LINER LAYER" / "REFRACTORY METAL LAYER"

GlobalFoundries' Proposal	Tower's Proposal
The terms are not indefinite.	Indefinite
The terms should be construed in accordance with their plain and ordinary meaning.	Or, if not indefinite: "a separate and defined [first liner / refractory metal] coating of material"

33. The word "layer" appears throughout claims 1-14 of the '546 patent and claims 1, 3, and 4 of the '167 patent. For example, claim 1 of the '546 patent recites "[a] method for forming a contact in a dielectric layer to a semiconductor substrate" including "forming a contact opening in the dielectric layer over the semiconductor substrate," "depositing a first liner layer to conformally coat the contact opening," and "depositing a refractory metal layer over the first intermix region."

34. The term "layer" is one of the most basic and widely used words in the semiconductor industry. The '546 and '167 claims use the term consistently with its well-known usage in the art. In my opinion, a POSA would have no difficulty reasonably ascertaining the scope of "layer," "first liner layer," and "refractory metal layer" when read in view of the claims and

specification. The claim language informs a POSA of each layer's location and its relationship with other structures. For example, claim 1 of the '546 patent recites "depositing a first liner layer to conformally coat the contact opening" and "depositing a refractory metal layer over the first intermix region." '546 patent at cl. 1. Claim 1 of the '167 patent recites "a first liner layer substantially coating a sidewall of the contact opening" and "a refractory metal layer substantially coating the first liner layer in the contact opening." '167 patent at cl. 1. The requirements that, e.g., the "first liner layer" either "conformally coat the contact opening" or "substantially coat[] a sidewall of the contact opening" inform a POSA of specific structure of the liner layer in relation to the contact opening.

35. The specification identifies exemplary materials from which the first liner layer may be formed ("at least one of: NiPt, Ni, Ti, TiSi, Co, Er, Pt, and Yb") and exemplary processes of how layers are formed ("using ionized PVD or radio frequency PVD (RFPVD) or other directional energetic thin film deposition techniques"). '546 patent, 5:60-65; *see also* cls. 3, 5. The specification likewise identifies exemplary materials for the "refractory metal layer" ("W, Ti, Ta or Mo employed directly as a 'metal'; a refractory metal employed as a constituent of a 'compound' thereof, e.g., TiN, TaN, etc., or even as a constituent of an 'alloy' with another metal, e.g., titanium-tungsten") and teaches that such a "layer" may be a "bilayer." '546 patent, 7:3-8; *see also* cl. 4; '167 patent at cls. 3, 4. A POSA reading these descriptions in the specification and claims would reasonably ascertain the scope of what a "layer" comprises. Additionally, I have reviewed a webpage from Tower that uses the term "layer" in the context of "multi-metal layer CMOS" and "Cu or Al layers." Ex. 6 at 1.

36. Based on the plain language of the terms, the knowledge of a POSA, the claim language, and the specification, it is my opinion that, a POSA would have understood term “layer” without further construction, under its plain and ordinary meaning.

37. This plain meaning would not require a “separate and defined coating of material” that Tower proposes. For example, a POSA would not understand how “separate” a coating must be from other parts in order to be sufficiently “defined.” Nor would a POSA understand what a “coating of material,” as Tower proposes, comprises or how such a coating differs from a “layer.”

38. Further, Tower’s proposal appears to exclude embodiments disclosed in the specification of the ’546 and ’167 patents. The specification teaches a “dielectric layer 120” that “may include a single layer or more than two separate layers.” ’546 patent, 4:56-62. The specification similarly discloses that the “refractory metal layer” may be a “Ti/TiN bilayer.” *Id.*, 7:45-46. The specification further teaches that the first liner layer is deposited such that “a first intermix region” is “formed at an interface of the bottom portion of each contact opening 130a, 130b and the upper portion of each of source 124a and drain 124b”—that is, the liner material intermixes with the adjoining substrate rather than remaining wholly “separate” from it. ’546 patent, 5:65-6:2. Respondents’ proposed limitation of “layer” to “a separate and defined . . . coating of material” would exclude each of those disclosed “layers.”

VI. THE ’546/’167 PATENT’S “INTERMIX REGION”

GlobalFoundries’ Proposal	Tower’s Proposal
The term is not indefinite. The term should be construed in accordance with its plain and ordinary meaning.	Indefinite

39. The claims 1, 2, 6, 8-9, and 13 of the ’546 patent, and claims 1, 2, and 6 of the ’167 patent recite an “intermix region.” For example, claim 1 of the ’546 patent recites “causing a

portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first *intermix region* at the upper portion of the semiconductor substrate.” Similarly, claim 1 of the ’167 patent recites “an intermix region at an upper surface of the semiconductor substrate in the contact opening.”

40. In my opinion, the “intermix region” is not indefinite. A POSITA would have reasonable certainty about the existence and nature of an “intermix region” not only for the clarity of the term itself, but also the additional context provided by the claims, and specifically would understand the claimed “intermix region” to be a region in which materials from different layers have intermixed. Specifically, the claims identify what an intermix region is, where it is located, and what it contains. For instance, claim 1 of the ’167 patent recites “the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate.” ’167 patent at cl. 1. Similarly, as discussed above, claim 1 of the ’546 patent recites “causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate.” ’546 patent at cl. 1. The claims thus specify the structure, location, and material composition of the claimed “intermix region.” A POSA would have understood from the claims the objective boundaries of the claimed “intermix region” and have been reasonably apprised of the claim scope. And the dependent claims provide even more guidance, reciting that “the intermix region includes a titanium-based silicide” and that “the intermix region has a thickness of approximately 3 nanometers to approximately 5 nanometers.” ’167 patent at cls. 2, 6; *see also* ’546 patent at cls. 2, 6, 9, 13.

41. Further, a POSA would understand from the specification the scope of the claimed “intermix region.” The specification explains that intermix regions “originate from the high energy

of the incoming sputtered atoms (from first liner layer 132) which are thus pushed into the Si lattice of source 124a and drain 124b,” so that “[t]he first few metal atoms, arriving with relatively high energy, can be forced into the interstitial sites of the Si lattice to create a thin region (intermix regions 134a, 134b) of mixed Si and metal on the order of 3-5 nm in thickness.” ’546 patent, 6:3-19. The intermix region in embodiments accordingly “include a mixture of the material used for first liner layer 132 and a material used for the respective semiconductor substrate 110 thereunder.” *Id.*, 6:6-9. The specification even provides exemplary compositions of the “intermix region”: the region “may have a relatively high Si content (greater than 50% Si in atomic composition),” *id.*, 6:19-21, and may comprise “a NiPt silicide,” “a NiPtTi silicide,” “a Ni silicide,” or “a NiTi silicide,” *id.*, 7:45-58.

42. In view of the foregoing, it is my opinion that a POSA would have understood the term “intermix region” with reasonable certainty in light of the claim language and the specification.

VII. THE ’235 PATENT’S “FROM A NICKEL SILICIDE SURFACE AT A SILICON INTERFACE TO AN OPPOSITE SURFACE OF THE NICKEL SILICIDE”

GlobalFoundries’ Proposal	Tower’s Proposal
The term is not indefinite.	Indefinite
The term should be construed in accordance with its plain and ordinary meaning.	

43. Claims 1 and 10 of the ’235 patent recite a “nickel silicide layer compris[ing] platinum having a composition gradient, *from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide*, with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode....”

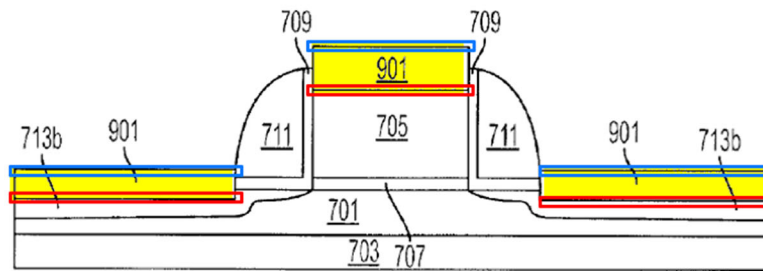
44. In my opinion, a POSA would understand this claim with reasonable certainty, and therefore the claim is not indefinite. In my opinion, the plain language of the claim provides more than reasonable certainty to a POSA about the endpoints of the platinum composition gradient, namely, from the bottom of the nickel silicide (a nickel silicide surface “at a silicon interface”) to the top of the silicide (“an opposite surface of the nickel silicide”).

45. Further, the specification also provides guidance that supports my understanding of the claim term:

The resulting NiSi layers 901 have a composition gradient of platinum from about 8 wt. % to about 12 wt. %, such as about 10 wt. %, ***at the bottom surface that contacts the silicon gate electrodes 705 and source/drain regions 713b***, to about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %, ***at the opposite surface***.

'235 patent at 8:43-48. The claimed “surfaces” for the platinum gradient as described are further shown in Figure 9, which depicts a nickel silicide 901 (yellow) with two surfaces: a “bottom surface that contacts the silicon gate electrodes 705 and source/drain regions 713b” (red box) and an “opposite surface” (blue box). In my opinion, the specification thus provides reasonable guidance about the claimed “surface at a silicon interface” and “opposite surface” of the nickel silicide.

46. The specification provides even more guidance confirming the scope of this term. The specification discloses that the platinum concentration gradient of the NiSi layer is “increasing in amount toward the upper surface, i.e., increasing in amount in a direction away from the P-channel and N-channel transistors” '235 patent, 3:59-67; *id.* at 5:3-11 (“formulating the NiSi layer with a platinum composition gradient with the amount of platinum increasing in a direction toward its upper surface, i.e., in a direction away from the source/drain regions and the gate electrode.”); *see also id.* at 7:4-12. And the specification even explains why these endpoints matter:



increasing the platinum composition between the two claimed surfaces results in “shielding the NiSi layer from physical damage during post NiSi processing as when removing an overlying stress liner by etching.” *Id.* at 5:3-11.

47. I also note that during prosecution, it was the Examiner who suggested revisions to the claim that resulted in the claim language “from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide.” Ex. 7 at 6 (7/26/2012 Applicant Arguments). The claims were subsequently allowed with that language. Ex. 7 at 19-20 (Notice of Allowance and Reasons for Allowance).

48. Based on the foregoing, it is my opinion that a POSA would be able to reasonably ascertain the scope of the claim with the term “from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide,” that the term is not indefinite.

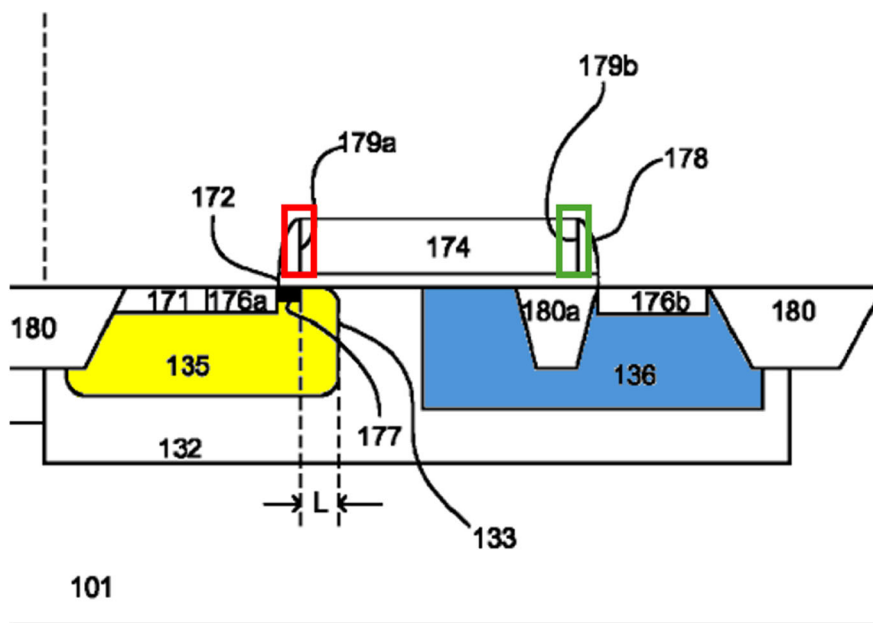
VIII. THE '983 PATENT'S “ADJACENT TO”

GlobalFoundries' Proposal	Tower's Proposal
The term is not indefinite.	Indefinite
The term should be construed in accordance with its plain and ordinary meaning.	

49. Claim 1, 5, 13, 17 and 20 of the '983 patent recite the patent term “adjacent to.” For example, claim 1 recites “a doped channel well disposed in the substrate adjacent to a first edge of the gate” and “a doped drift well adjacent to a second edge of the gate.”

50. In my opinion, a POSA would have been able to reasonably ascertain the meaning of “adjacent to.” “Adjacent” is a commonly used word and means “near” or “next to.” Ex. 8 at 4. The claims and specification provide further guidance.

51. For example, claim 1 recites a “gate” with a “first edge” and a “second edge.” The term “adjacent” is used to specify that a “doped channel well” and a “heavily doped region” are “adjacent to a first edge of the gate” while a “doped drift well” is “adjacent to a second edge of the gate.” ’983 patent, cl. 1. The specification similarly and consistently describes a “doped channel well” that is “adjacent to a first edge of the gate” and a “doped drift well” that is “adjacent to a second edge of the gate.” ’983 patent, 1:45-55, 1:58-66, 2:4-13. And Figure 1 confirms that spatial relationship, with a channel well 135 (yellow) in a source region next to or near a first edge of the gate 179a (red) and a drift well 136 (blue) in a drain region next to or near the second edge of the gate 179b (green):



52. In my opinion, each use of “adjacent” in the claims and the specification identifies both structures and the reference feature, e.g., a first or second edge of the gate. With the

guideposts laid out in the specification, a POSA would readily understand the claim scope with reasonable certainty.

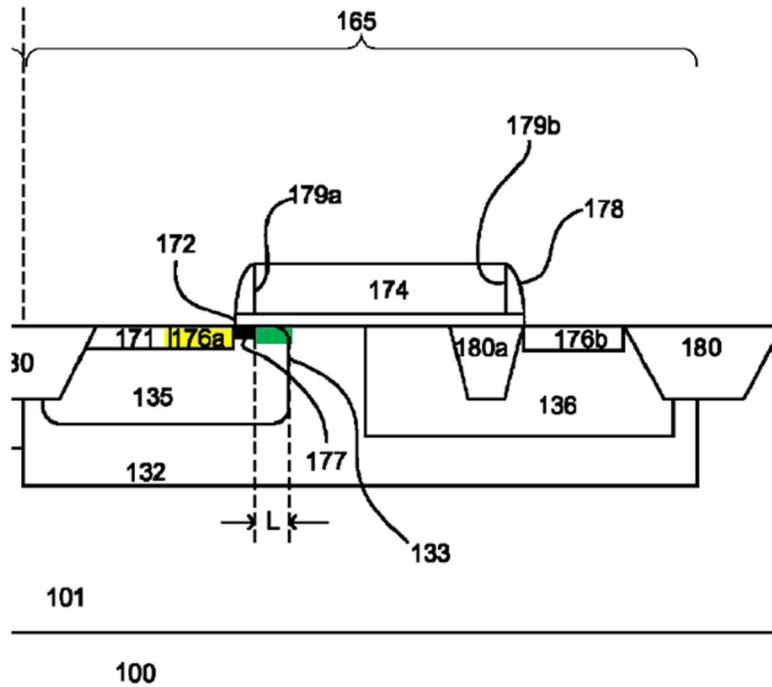
IX. THE '983 PATENT'S "THE CHANNEL IS DISPLACED FROM [A/THE] FIRST HEAVILY DOPED REGION ADJACENT TO THE FIRST EDGE OF THE GATE"

GlobalFoundries' Proposal	Tower's Proposal
The term is not indefinite. The term should be construed in accordance with its plain and ordinary meaning.	Indefinite

53. Claims 1, 13, and 20 of the '983 patent recite "the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate."

54. In my opinion, this phrase describes a straightforward structural relationship: the channel is offset from the first heavily doped region, which is positioned adjacent to the gate's first edge. "Displaced" is a common term in the art, and the plain and ordinary meaning of "displaced" is offset, as would be well understood by a POSA when reading the claims in view of the intrinsic record.

55. Further, in my opinion, the specification confirms this. Claim 1 recites that "the channel is displaced from a first heavily doped region adjacent to the first edge of the gate." The specification describes this, disclosing a channel with "channel length L" (green) and a "source diffusion region 176a" (yellow) with a "source extension region 177" in between as shown below. '983 patent, 5:25-32, 5:52-56. Accordingly, in the illustrated embodiment, the channel (green) is displaced or offset from heavily doped region 176a (yellow), due to the source extension region 177.



56. Claim 2, a dependent claim, further recites a device “further comprising spacer on the first edge of the gate, wherein the spacer defines the displacement of the channel from the first heavily doped region.” ’983 patent at cl. 2; *see also id.* at cl. 14. In my opinion, this recitation in claim 2 provides an example of the structure that establishes the displacement, providing further guidance to a POSITA.

57. In light of the foregoing, it is my opinion that a POSA would have determined the claim scope with a reasonable certainty.

X. CONCLUSION

58. My opinions in this declaration are based on information I have considered to date. I reserve the right to revise or supplement my opinions as the case progresses, as additional information comes to light, and/or in response to evidence or testimony presented by Respondent

I declare under penalty of perjury that the foregoing contains my true and correct opinions.

Executed in Arlington, Virginia on August 25, 2026

A handwritten signature in black ink, appearing to read 'David Liu', is positioned above a horizontal line.

David Liu, Ph.D.

APPENDIX A

DAVID KUAN-YU LIU, Ph.D.

davidkyliu@comcast.net

PROFESSIONAL PROFILE

- Technical expert with over 20 years of industrial experience in the development of advanced CMOS devices and process technology, with special emphasis on non-volatile semiconductor memory devices and process technology, and high-voltage semiconductor devices, with additional 18 years of experience serving as expert consultant.
 - Named inventor on 99 US patents.
- Consultant and expert witness for past 18 years in semiconductor intellectual property evaluation and patent litigation, with deposition experience and successful ITC trial testifying experience.
- Successful track record in transitioning products from development phase to high-volume manufacturing phase, both as an individual contributor and as manager/director.
- Co-founder of an LLC in the development and licensing of intellectual property with emphasis on embedded flash memory technology.
- Co-founder of a semiconductor memory start-up company (Progressant Technologies, Inc.); participated in its successful sale to Synopsys, Inc.
- Fluent in Mandarin Chinese (native language) and English; over 10 years experience as real-time translator of church sermons and proficient in written translation.

EDUCATION

Stanford University, Palo Alto, California

Ph.D., Electrical Engineering (1989)

Thesis: Physics and Technology of Novel Conductivity-Modulated Power MOSFET's

M.S., Electrical Engineering (1985)

Coursework in integrated-circuit technology, semiconductor device physics, computer modeling of semiconductor devices and fabrication processes, and solid-state physics

University of California, Berkeley, California

B.S., Electrical Engineering (1983)

Certificate of Distinction (awarded annually to the top EECS graduate)

EXPERT WITNESS WORK - CASE EXPERIENCE:

Case: **Himax Technologies Limited v. Foxconn Fukuyama Technologies,**
Jurisdiction: Japan Commercial Arbitration Association (JCAA), Case No. 25-03
Work Product: Expert consulting work relating to fabless-foundry arbitration.
Nature of Case: Arbitration hearing
Counsel: Retained by Anderson Mori & Tomotsune (Contact: Andrew Halliwell)
Date: 03/26 – Present

Case: **Tower Semiconductor Ltd. v. GlobalFoundries, Inc.**
Jurisdiction: Investigation No. 337-TA-1500
United States International Trade Commission
Work Product: Expert consulting work relating to LDMOS process technologies,
Counsel: Retained by Cleary Gottlieb Steen & Hamilton (LLP Contact: Thomas Yeh)
Date: 04/26 – Present

Case: **Advanced Memory Technologies, Inc. v. Micron Technology, Inc.**
Jurisdiction: Civil Action No. 1:25-cv-01036 (W.D. Tex.)
Work Product: Expert consulting work relating to Booster Circuit. Patent analysis and prior art review toward preparation of EPR petition support and draft of declaration.
Nature of Case: EPR.
Counsel: Retained by Fish & Richardson (Contact: Tim Riffe)
Date: 11/25 – Present

Case: **Apple v. WeCrevention, Inc.**
Jurisdiction: Post-Grant Patent Matters Adverse to WeCrevention, Inc.
Work Product: Expert consulting work relating to High Speed Chip Memory Module. Patent analysis and prior art review toward preparation of IPR petition support and draft of declaration.
Nature of Case: IPR.
Counsel: Retained by Fish & Richardson (Contact: Dr. Hyun Jin In)
Date: 11/25 – Present

Case: **Monterey Research LLC v. Renesas Electronics Corp.**
Jurisdiction: Civil Action No. 2:24-cv-00238-JRG (E.D. Tex.)
Work Product: Expert consulting work relating to Charge Trapping Nonvolatile Memory. Patent analysis and prior art review toward preparation of IPR petition support and draft of declaration.
Nature of Case: Civil action and IPR.
Counsel: Retained by Morrison Foerster (Contact: Yuka Teraguchi)
Date: 03/25 – 07/26

Case: **Palisade Techs., LLP v. Micron Tech., Inc.**
Jurisdiction: Civil Action N.D. Cal, No. 7:24-cv-00262 (W.D. Tex.)
Work Product: Expert consulting work relating to 3D NAND Flash Memory Technology. Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration on 2 patents.
Nature of Case: Civil action and IPR.
Counsel: Retained by O'Melveny & Myers LLP (Contact: John Kappos)
Date: 12/24 – 01/26

Case: **Yangtze Memory Technologies Company, Ltd. vs. Micron Technology, Inc.**
Jurisdiction: Civil Action N.D. Cal, No. 3:223-cv-05792-RFL
Easter District of Texas
Work Product: Expert consulting work relating to 3D NAND Flash Memory Technology. Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration on 4 patents. ***Deposed on 3 petitions that have been instituted.***
Nature of Case: Civil action and IPR.
Counsel: Retained by Orrick, Herrington & Sutcliffe (Contact: Jason Lang)
Date: 03/24 – Present

Case: **Force MOS Technology Co., Ltd. vs. Inergy Technology, Inc.**
Jurisdiction: Civil Action E.D. TX, No. 2:22-cv-00460-JRG
Easter District of Texas
Work Product: Expert consulting work relating to Vertical Trench DMOS process technology. Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration on 2 patents. Declaration for claim construction on 3 patents. ***Deposed on Claim Construction Declaration, and deposed on declarations for IPR petitions, and deposed on reply declaration.*** Continuing work on claim construction and technical consulting.
Nature of Case: Civil action and IPR.
Counsel: Retained by Baker McKenzie (Contact: McKenzie Martin)
Date: 10/23 – 10/25

Case: **GlobalFoundries U.S. vs. International Business Machines**
Jurisdiction: 03348-KMK-AEK (S.D.N.Y. 2023)
Work Product: Expert consulting work relating to trade secrets involving advanced CMOS process and device technology (Nanosheet technology). Reviewed evidence and summarized findings.
Nature of Case: Civil action.
Counsel: Retained by Latham & Watkins (Contact: Sami Al-Marzoog)
Date: 10/23 – 2/25

Case: **Element Capital. vs. Beijing BOE Display Technology Co., Ltd.**
Jurisdiction: Civil Action No. 2:22-cv-00118-JRG
Eastern District of Texas
Work Product: Expert consulting work relating to OLED technology. Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration on 3 patents.
Nature of Case: Civil action.
Counsel: Retained by Fish & Richardson (Contact: Linhong Zhang)
Date: 4/23 – 12/23

Case: **Sonrai Memory Ltd. vs. Micron Technology, Inc.,**
Jurisdiction: Civil Action No. 6:22-cv-000855-FB
Western District of Texas
Work Product: Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration resulting in institution of the petition and ***deposed on supplemental declaration.***
Nature of Case: Civil action.
Counsel: Retained by Fish & Richardson (Contact: Tim Riffe)
Date: 12/22 – 7/24

Case: **Daedalus Prime LLC vs. Samsung Electronics Co.,**
Jurisdiction: Civil Action E.D. TX, Marshall Div., No. 2:22-cv-353
Eastern District of Texas, Marshall Div., and
Investigation No. 337-TA-1336
Work Product: Expert consulting work relating to FINFET process technology. Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration on 2 patents. Source code database viewing (GDS layout and process flow documents) for supporting of non-infringement position.
Nature of Case: Civil action and ITC Investigation on Certain Semiconductor Devices, Mobile Devices Containing the Same and Components Thereof..
Counsel: Retained by Fish & Richardson (Contact: Kevin Su)
Date: 10/22 – 4/23

Case: **Netlist, Inc. vs. Samsung Electronics Co. Ltd. et al.,**
Jurisdiction: Civil Action No. 2:21-cv-00463
Eastern District of Texas
Work Product: Expert consulting work relating to memory modules.
Nature of Case: Civil action.
Counsel: Retained by Fish & Richardson (Contact: Tony Nguyen)
Date: 8/22 – 12/22

Case: **Vervain, LLC vs. Micron Technology & related IPR Petition**
Jurisdiction: Action No. 6:21-cv-00487-ADA
Western District of Texas
Work Product: Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration on 4 patents, all 4 petitions instituted, and ***deposed on both initial declaration and reply declaration on all 4 patents.***
Counsel: Retained by Orrick, Herrington & Sutcliffe LLP (Contact: Jared Bobrow)
Date: 8/21 – 2/23

Case: **Monterey Research vs. Qualcomm PTAB Matter**
Work Product: Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration.
Counsel: Retained by Jones Day (Contact: David Cochran)
Date: 12/21 – 2/22

Case: **Semiconductor Energy Laboratory Co., OLED Patent Matters**
Jurisdiction: Action No. 8:21-cv-00554-JAK-ADS
Central District of California
Work Product: Expert for infringement on 3 patents relating to OLED driver circuit and layout, responsibilities include source code viewing, GDS layers viewing, and claim charts review.
Nature of Case: Civil action.
Counsel: Retained by Fish & Richardson (Contact: Bryan Basso)
Date: 9/21 – 4/22

Case: **Apple Inc. IPR Petition for U.S. Patents 6,622,108 & 6,807,505**
Work Product: Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration.
Counsel: Retained by Erise IP (Contact: Paul Hart)
Date: 8/21 – 9/21

Case: **Texas Instruments IPR Petition**
Work Product: Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration.
Counsel: Retained by Fish & Richardson (Contact: Josh Griswold)
Date: 9/20 – 4/21

Case: ***Micron Technology vs. UMC/JinHua***
Jurisdiction: Case No. 4:17-CV-06932-MMC
United States District Court for the Northern District of California
Work Product: Testifying expert, trade secret documents review, and declaration submission.
Nature of Case: Trade secret contention of Micron confidential documents.
Counsel: Retained by Jones Day (Contact: Randy Kay)
Date: 3/17 – 12/23

Case: ***SK Hynix vs. BitMicro***
Jurisdiction: Investigation No. 337-TA-1097
United States International Trade Commission
Work Product: Testifying expert for non-infringement on 2 patents, issued expert rebuttal report, ***deposition taken***, issued witness statement.
Nature of Case: ITC infringement case concerning module stacking of SDRAM.
Counsel: Retained by Fish Richardson (Contact: Joseph Colaanni)
Date: 11/18 – 4/19

Case: ***Rohm Company IPR Petition***
Work Product: Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration.
Counsel: Retained by Fish & Richardson (Contact: Matt Colvin)
Date: 4/18 – 8/18

Case: ***Apple Inc., vs. Qualcomm Incorporated***
Jurisdiction: Case No. 3:17-CV-02402
United States District Court for the Southern District of California
Work Product: Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration, ***2 depositions taken***.
Counsel: Fish & Richardson (Contact: Tony Nguyen)
Date: 1/18 – 5/19

Case: ***Macronix vs. Toshiba***
Jurisdiction: Investigation No. 337-TA-1046
United States International Trade Commission
Work Product: Testifying expert for infringement and non-invalidity on 2 patents, claim construction analysis, technical tutorial, database viewing for evidence gathering, issuing expert report and expert rebuttal report, ***deposition taken for 2 patents***, issued witness statement and rebuttal witness statement, ***testified in ITC court on both infringement and invalidity***.
Nature of Case: ITC infringement case concerning advanced NAND Flash memory.
Counsel: Retained by Fish Richardson (Contact: David Barken)
Date: 4/17 – 10/18

Case: *Novatek Microelectronics Corp. IPR Petition*
Work Product: Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration.
Counsel: Retained by Kroub, Silbersher & Kolmykov PLLC (Contact: Sergey Kolmykov)
Date: 2/17 – 4/17

Case: *Taiwan Semiconductor Manufacturing Company IPR Petition*
Work Product: Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration.
Counsel: Retained by Fish & Richardson (Contact: John Goetz)
Date: 7/15 – 4/17

Case: *Triune Systems, LLC vs. Wayne Chen, et al.,*
Jurisdiction: Case No. 296-03209-2013,
District Court of Collin County, TX
Work Product: Testifying expert, database viewing, expert report, *video-taped testimony*
Nature of Case: Defending against trade secret contention.
Counsel: Retained by Gordon & Rees LLP (Contact: Robert Bragalone)
Date: 5/15 – 9/15

Case: *Macronix vs. Spansion*
Jurisdiction: Investigation No. 337-TA-922
United States International Trade Commission
Work Product: Testifying expert, claim construction analysis, technical tutorial, database viewing for evidence gathering, expert report, expert rebuttal report.
Nature of Case: ITC infringement case concerning Flash memory cell
Counsel: Retained by Fish Richardson (Contact: David Barken)
Date: 9/14 – 1/15

Case: *Macronix vs. Spansion*
Jurisdiction: Investigation No. 337-TA-909
United States International Trade Commission
Work Product: Database viewing for evidence gathering, support technical content of expert report, support technical content against rebuttal report.
Nature of Case: ITC infringement case concerning Flash memory cell
Counsel: Retained by Winston & Strawn (Contact Michael Murray)
Date: 5/14 – 1/15

Case: ***Keranos, LLC v. Analog Devices et. al.***
Jurisdiction: Civil Case No. 2:10-CV-00207
United States District Court for the Eastern District of Texas
Work Product: Claim Chart Generation and Claim Construction
Nature of Case: Patent infringement case concerning split gate Flash memory cell
Counsel: Retained by Agility IP (Contact Michelle Breit)
Date: 2/09 – 2/14

Case: ***Silicon Storage Technology, Inc. v. Xicor, LLC***
Jurisdiction: Civil Case No. CV 10-01515
United States District Court for the Northern District of California
Work Product: Expert report and declarations
Nature of Case: Matter concerning United States Reissue Patent RE38,370 on the process of flash memory cell.
Counsel: Retained by Shore Chan Bragalone LLP (Contact Patrick Conroy)
Date: 6/11 – 10/11

Case: ***Fast Memory Erase, LLC v. Spansion et.al.***
Jurisdiction: Civil Case No. 3:10CV-481
United States District Court for the Northern District of Texas
Work Product: Declarations
Nature of Case: Defending Motion for Attorney Fees.
Counsel: Retained by Shore Chan Bragalone LLP (Contact Patrick Conroy)
Date: 2/10 – 6/10

Case: ***Fast Memory Erase, LLC v. Spansion et.al.***
Jurisdiction: Civil Case No. 3-08cv0977M
United States District Court for the Northern District of Texas
Work Product: Testifying expert: issuing expert report, declarations, and ***deposition taken.***
Nature of Case: Patent infringement case concerning minimizing source current during erase of Flash memory cell.
Counsel: Retained by Shore Chan Bragalone LLP (Contact Patrick Conroy)
Date: 2/09 – 5/10

PROFESSIONAL EXPERIENCE:

10/07 to present: **Technical Consultant/Expert Witness**

Serving as a technical consultant and advisor in the area of semiconductor device and process technologies. Expert witness in ITC cases and district court cases, with successful testifying experience in trial hearing. Depositions taken at least 12 times for both litigation cases and IPR petitions. Experienced in source code viewings of circuit schematics and GDS layouts; seasoned in reverse engineering of integrated circuits, as well as techniques for de-processing and analyzing semiconductor chips. Responsibilities including gathering and reviewing of evidence, issuing declaration on claim construction, giving technical tutorial, issuing expert report and expert rebuttal report. Technical consultant for evaluating numerous flash memory and CMOS image sensor patent portfolios for licensing and acquisition consideration. Assessed qualitative merits of each patent portfolio and identified potential infringers and potential licensees. Generated numerous claim charts against potential infringers based on published patents, technical papers and datasheets, and reverse engineering reports.

10/07 to 4/11: **Jonker, LLC**

Inventor and Partner. Developed an intellectual property (IP) portfolio (7 U.S. patents granted and 3 U.S. patent applications pending) on zero-cost CMOS-logic-compatible embedded flash memory. Successfully negotiated the acquisition of this IP portfolio by a major semiconductor company.

8/04 to 10/07: **Maxim Integrated Products, San Jose, California**

Senior Scientist. Responsible for developing embedded non-volatile memory process technology for battery management products. Duties included supervising the development of integrated process flows at the 0.35 μ m and 0.18 μ m CMOS technology nodes. Analyzed and solved process-related reliability issues.

5/00 to 8/04: **Xilinx, Incorporated, San Jose, California**

Senior Manager. Responsible for developing non-volatile memory process technology for flash and CPLD products, as well as advanced CMOS process technology (75nm node). Duties included supervising the development of integrated non-volatile memory process flows at multiple CMOS technology nodes to meet product application requirements. Analyzed and solved process-related reliability issues.

2/00 to 5/00 **Progressant Technologies, Fremont, California**

Co-Founder. Facilitated IP development and sale of the company to Synopsys, Inc.

9/98 to 2/00

Programmable Silicon Solutions, Sunnyvale, California

Director of Process Engineering. Responsible for developing a fully logic compatible embedded flash memory technology for WSMC (a semiconductor foundry). Developed a new integrated process flow to implement a proprietary flash cell in a high-performance logic process, supervised testchip and product tapeout, process optimization, device optimization, and product yield enhancement. Applied interpersonal skills to align the goals of the foundry technology development group to that of PSS. Participated in the strategic planning of a technology roadmap and foundry strategy for the company.

9/97 to 9/98

AMIC Technology, Santa Clara, California

Director of Flash Technology. Responsible for developing 0.35μm ETOX-based flash memory technology. Prepared the business plan and technology roadmap. Developed a new integrated process flow, supervised testchip and product tapeout, process optimization, device optimization, and product yield enhancement.

5/96 to 9/97

Altera Corp., San Jose, California

Device Engineering Manager. Responsible for technology development, process integration, and foundry interface for 0.35μm generation of logic and EEPROM programmable logic device technology. Duties included project schedule planning, supervision of testchip and product tapeout, process optimization, and product yield enhancement.

7/95 to 5/96

Information Storage Devices, San Jose, California

Technology Development Manager/Program Manager. Led a project team of 10 in technology development, process integration, and first product introduction, with a new foundry. Duties included project schedule planning, supervision of testchip and product tapeout, process optimization, resolution of sorting issues and yield enhancement.

5/92 to 7/95

Advanced Micro Devices, Sunnyvale, California

Member of Technical Staff. Led the development of low-energy/DINOR-type flash EPROM technology. Key individual contributor in optimizing the process and design of flash cell and periphery devices in AMD's 0.5μm and 0.35μm flash EPROM technologies. Duties included process integration, device modeling, and development of triple-well technology and high-voltage transistors for negative gate erase operation.

1/89 to 4/92

Texas Instruments, Dallas, Texas

Member of Technical Staff. Research and development of 16Mb generation of flash EPROM technology, as well as a new generation of antifuse-based FPGAs. Duties included process integration, device modeling, high-voltage CMOS process integration, and investigation of a novel source-side injection mechanism for EPROM channel hot-electron programming.

AWARDS/HONORS

U.C. Berkeley Certificate of Distinction Award, 1983

AEA Faculty Development Fellowship, 1983-1985

AMD Spotlight Award, 1995

PATENTS

1. U.S. Patent 5,106,773, "Programmable Gate Array and Methods for Its Fabrication" (with K.-L. Chen and H. Tigelaar), issued April 21, 1992.
2. U.S. Patent 5,166,557, "Gate Array with Built-in Programmable Circuitry" (with K.-L. Chen), issued November 24, 1992.
3. U.S. Patent 5,202,576, "Asymmetrical Non-Volatile Memory Cell, Arrays, and Methods for Fabricating the Same" (with M. Wong), issued April 13, 1993.
4. U.S. Patent 5,219,782, "Sublithographic Antifuse and method for Manufacturing" (with K.-L. Chen), issued June 15, 1993.
5. U.S. Patent 5,250,464, "Method of Making a Low Capacitance, Low Resistance Sidewall Antifuse Structure" (with M. Wong), issued October 5, 1993.
6. U.S. Patent 5,264,384, "Method of Making a Non-Volatile Memory Cell" (with C. Kaya), issued November 23, 1993.
7. U.S. Patent 5,300,803, "Source Side Injection Non-volatile Memory Cell," issued April 5, 1994.
8. U.S. Patent 5,365,105, "Sidewall Antifuse Structure and Method for Making," issued November 15, 1994.
9. U.S. Patent 5,371,402, "Low Capacitance, Low Resistance Sidewall Antifuse Structure and Process" (with M. Wong), issued December, 1994
10. U.S. Patent 5,395,797, "Antifuse Structure and Method of Fabrication," issued March 7, 1995.
11. U.S. Patent 5,470,773, "Method of Protecting a Stacked Gate Edge in a Semiconductor Device from Self Aligned Source (SAS) Etch" (with Y. Sun and C. Chang), issued November 28, 1995.
12. U.S. Patent 5,482,880, "Non-Volatile Memory and Fabrication Method," (with C. Kaya), issued January 9 1996.
13. U.S. Patent 5,517,443, "Method and System for Protecting a Stacked Gate Edge in a Semiconductor Device from Self Aligned Source (SAS) Etch in a Semiconductor Device," issued May 14, 1996.
14. U.S. Patent 5,521,867, "Adjustable Threshold Voltage Conversion Circuit" (with Jian Chen), issued May 28, 1996.
15. U.S. Patent 5,534,455, "Method and System for Protecting a Stacked Gate Edge in a Semiconductor Device from Self Aligned Source (SAS) Etch in a Semiconductor Device," issued July 9, 1996.

16. U.S. Patent 5,541,875, "High Energy Buried Layer Implant To Provide A Low Resistance P-Well in A Flash EPROM Array" (with Jian Chen), issued May 28, 1996.
17. U.S. Patent 5,590,076, "Channel Hot-Carrier Page Write," issued December 31, 1996.
18. U.S. Patent 5,596,531, "Method for Decreasing the Discharge Time of A Flash Memory Cell," issued January 21, 1997.
19. U.S. Patent 5,612,914, "Asymmetrical Non-volatile Memory Cell, Arrays and Methods for Fabricating Same" (with Man Wong), issued May 18, 1997.
20. U.S. Patent 5,624,859, "Method for Providing Device Isolation and Off-State Leakage Current for a Semiconductor Device," issued April 29, 1997.
21. U.S. Patent 5,625,220, "Sublithographic Antifuse," (with K.L. Chen), issued April 29, 1997.
22. U.S. Patent 5,646,430, "Non-volatile Memory Cell Having Lightly-Doped Source Region" (with C. Kaya), Issued July 8, 1997.
23. U.S. Patent 5,650,964, "Method of Inhibiting Degradation of Ultra Short Channel Charge-carrying Devices during Discharge," (with Jian Chen), Issued July 22, 1997.
24. U.S. Patent 5,652,155, "Method for Making Semiconductor circuit Including Non-ESD Transistors with Reduced Degradation Due to an Impurity Implant," Issued July 27, 1997.
25. U.S. Patent 5,656,509, "Method and Test Structure for Determining Gouging in a Flash EPROM Cell During SAS Etch," Issued July 29, 1997
26. U.S. Patent 5,661,059, "Boron Penetration to Suppress Short Channel Effect in P-channel Device," Issued August 26, 1997.
27. U.S. Patent 5,674,764, "Method of Making Asymmetrical Non-volatile Memory Cell," Issued October 7, 1997.
28. U.S. Patent 5,693,972, "Method and System for Protecting a Stacked Gate Edge in a Semiconductor Device from Self-Aligned Source (SAS) Etch in a Semiconductor Device," Issued December 2, 1997.
29. U.S. Patent 5,751,631, "Flash memory cell and a new method for sensing the content of the new memory cell," Issued May 12, 1998
30. U.S. Patent 5,789,295, "Method of Eliminating or Reducing Poly1 Oxidation at Stacked Gate Edge in Flash EPROM Process," Issued August 4, 1998.
31. U.S. Patent 5,814,854, "Highly scalable FLASH EEPROM cell," Issued September 29, 1998.
32. U.S. Patent 5,814,864, "Semiconductor Circuit Including Non-ESD Transistors with Reduced Degradation Due to an Impurity Implant," Issued September 29, 1998.

33. U.S. Patent 5,912,836, "Circuit for Detecting Both Charge Gain and Charge Loss Properties in a Non-Volatile Memory Array," Issued June 15, 1999.
34. U.S. Patent 5,930,174, "Circuit and Method for Erasing Flash Memory Array," Issued July 27, 1999.
35. U.S. Patent 5,981,994, "Method and Semiconductor circuit for Maintaining Integrity of Field Threshold Voltage Requirements," Issued November 9, 1999.
36. U.S. Patent 5,995,418, "Circuit and Method for Erasing Flash Memory Array," Issued November 30, 1999.
37. U.S. Patent 6,026,017, "Compact nonvolatile memory," Issued February 15, 2000.
38. U.S. Patent 6,027,974, "Nonvolatile memory," Issued February 22, 2000.
39. U.S. Patent 6,088,263, "Non-volatile memory using substrate electrons," Issued July 11, 2000.
40. U.S. Patent 6,091,636, "Flash memory cell and a new method for sensing the content of the new memory cell," Issued July 18, 2000.
41. U.S. Patent 6,127,225, "Memory cell having implanted region formed between select and sense transistors," Issued October 3, 2000.
42. U.S. Patent 6,159,800, "Method of forming memory cell," Issued December 12, 2000.
43. U.S. Patent 6,188,604, "Flash memory cell & array with improved pre-program and erase characteristics," Issued February 13, 2001.
44. U.S. Patent 6,185,133, "Flash EPROM using junction hot hole injection for erase," Issued February 6, 2001.
45. U.S. Patent 6,252,799, "Device with embedded Flash and EERPOM memories," Issued June 26, 2001.
46. U.S. Patent 6,326,265, "Device with Embedded Flash and EPROM Memories," Issued December 4, 2001.
47. U.S. Patent 6,417,550, "High voltage MOS devices with high gated-diode breakdown voltage and punch-through voltage," Issued July 9, 2002.
48. U.S. Patent 6,479,862, "Charge trapping device and method for implementing a transistor having a negative differential resistance mode," Issued November 12, 2002.
49. U.S. Patent 6,512,274, "CMOS-process compatible, tunable NDR (negative differential resistance) device and method of operating same," Issued January 28, 2003.
50. U.S. Patent 6,596,617, "CMOS compatible process for making a tunable negative differential resistance (NDR) device," Issued July 22, 2003.

51. U.S. Patent 6,624,026, "Nonvolatile Memory," Issued September 23, 2003.
52. U.S. Patent 6,680,245, "Method for making both a negative differential resistance (NDR) device and a non-NDR device using a common MOS process," Issued January 20, 2004.
53. U.S. Patent 6,686,631, "Negative differential resistance (NDR) device and method of operating same," Issued February 3, 2004.
54. U.S. Patent 6,693,027, "Method for configuring a device to include a negative differential resistance (NDR) characteristic," Issued February 17, 2004.
55. U.S. Patent 6,700,155, "Charge trapping device and method for implementing a transistor having a configurable threshold," Issued March 2, 2004.
56. U.S. Patent 6,711,063, "EEPROM memory cell array architecture for substantially eliminating leakage," Issued March 23, 2004.
57. U.S. Patent 6,835,979, "Nonvolatile Memory," Issued December 28, 2004.
58. U.S. Patent 6,969,894, "Variable threshold semiconductor device and method of operating same," Issued November 29, 2005.
59. U.S. Patent 6,972,234, "High Voltage MOS Devices with High Gated-Diode Breakdown Voltage and Punch-through Voltage," Issued December 6, 2005.
60. U.S. Patent 6,972,465, "CMOS process compatible, tunable negative differential resistance (NDR) device and method of operating same," Issued December 6, 2005.
61. U.S. Patent 7,067,873, "Charge Trapping Device," Issued June 27, 2006.
62. U.S. Patent 7,091,077, "Method of directionally trimming polysilicon width," Issued August 15, 2006.
63. U.S. Patent 7,109,078, "CMOS compatible process for making a charge trapping device," Issued September 19, 2006.
64. U.S. Patent 7,301,194, "Shrinkable and highly coupled double poly EEPROM with Inverter," Issued November 27, 2007.
65. U.S. Patent 7,436,710, "EEPROM memory device with cell having NMOS in a P pocket as a control gate, PMOS program/erase transistor, and PMOS access transistor in a common well," Issued October 14, 2008.
66. U.S. Patent 7,535,758, "One or Multiple-Times Programming Device," Issued May 19, 2009.
67. U.S. Patent 7,782,668, "Integrated circuit embedded with Non-volatile One-Time-Programmable and Multiple-Time Programmable memory," Issued August 24, 2010.

68. U.S. Patent 7,787,295, "Integrated circuit embedded with Non-volatile Multiple-Time Programmable memory having variable coupling," Issued August 31, 2010.
69. U.S. Patent 7,787,304, "Method of making integrated circuit embedded with Non-volatile One-Time-Programmable and Multiple-Time Programmable memory," Issued August 31, 2010
70. U.S. Patent 7,787,309, "Method of operating integrated circuit embedded with Non-volatile One-Time Programmable and Multiple-Time Programmable memory," Issued August 31, 2010.
71. U.S. Patent 7,791,955, "Method of erasing a block of memory cells," Issued September 7, 2010.
72. U.S. Patent 7,835,184, "EEPROM memory cell with first-dopant-type control gate transistor, and second-dopant type program/erase and access transistors formed in common well," Issued November 16, 2010
73. U.S. Patent 7,835,186, "Method of programming a selected memory cell," Issued November 16, 2010.
74. U.S. Patent 7,852,672, "Integrated Circuit embedded with Non-Volatile programmable memory having variable coupling," Issued December 14, 2010
75. U.S. Patent 7,876,615, "Method of operating integrated circuit embedded with Non-Volatile programmable memory having variable coupling related application data," Issued January 25, 2011
76. U.S. Patent 7,920,426, "Non-volatile memory programmable through areal capacitive coupling," Issued April 5, 2011
77. U.S. Patent 7,944,750, "Multi-programmable Non-Volatile Memory Cell," Issued May 17, 2011
78. U.S. Patent 8,203,861, "Non-Volatile One-Time Programmable and Multiple-Time Programmable Device," Issued June 19, 2012
79. U.S. Patent 8,208,299, "Integrated Circuit Embedded with Non-Volatile Programmable Memory Having Variable Coupling and Separate Read/Write Paths," Issued June 26, 2012
80. U.S. Patent 8,300,470, "Two Terminal Programmable Hot Channel Electron Non-Volatile Memory," Issued October 30, 2012
81. U.S. Patent 8,305,805, "Common Drain Non-Volatile Multiple-Time Programmable Memory," Issued November 6, 2012
82. U.S. Patent 8,325,519, "Method of Operating Integrated Circuit Embedded with Non-Volatile Programmable Memory having Variable Coupling Ratio," Issued December 4, 2012.

83. U.S. Patent 8,580,622, "Method of Making Integrated Circuit Embedded with Non-Volatile Programmable Memory having Variable Coupling," Issued November 12, 2013.
84. U.S. Patent 8,582,342, "Non-Volatile One-Time-Programmable and Multiple-Time Programmable Memory Configuration Circuit," Issued November 12, 2013.
85. U.S. Patent 8,599,612, "Method of Operating Integrated Circuit Embedded with Non-Volatile Programmable Memory having Variable Coupling Related Application data," Issued December 3, 2013.
86. U.S. Patent 8,705,263, "Non-Volatile One-Time-Programmable and Multiple-Time Programmable Memory Configuration Circuit," Issued April 22, 2014.
87. U.S. Patent 8,988,103, "Capacitively Coupled Logic Gate," Issued March 24, 2015.
88. U.S. Patent 9,064,192, "Automatic Valid Vote Count Storage using Secure Embedded Non-Volatile Memory," Issued June 23, 2015.
89. U.S. Patent 9,224,739, "Method of Making Integrated Circuit Embedded with Non-Volatile Programmable Memory having Variable Coupling," Issued December 29, 2015.
90. U.S. Patent 9,305,931, "Zero Cost NVM Cell Using High Voltage Devices in Analog Process," Issued April 5, 2016.
91. U.S. Patent 9,324,438, "Method of Operating Incrementally Programmable Non-Volatile Memory," Issued April 26, 2016.
92. U.S. Patent 9,558,813, "Method of Operating Incrementally Programmable Non-Volatile Memory," Issued January 31, 2017.
93. U.S. Patent 9,570,161, "Method of Operating Incrementally Programmable Non-Volatile Memory," Issued February 14, 2017.
94. U.S. Patent 9,570,175, "Incrementally Programmable Non-Volatile Memory," Issued February 14, 2017.
95. U.S. Patent 10,061,738, "Ephemeral Peripheral Device," Issued August 28, 2018.
96. U.S. Patent 10,115,467, "One time Accessible (OTA) Non-Volatile Memory," Issued October 30, 2018.
97. U.S. Patent 10,198,889, "Automatic Valid Vote Count Storage Using Secure Embedded Non-Volatile Memory," Issued February 5, 2019
98. U.S. Patent 10,559,700, "Zero Cost ANVM Cell Using High Voltage Devices in Analog Process," Issued February 11, 2020.
99. U.S. Patent 10,629,607, "Semiconductor CMOS Non-Volatile Memory Device," Issued April 21, 2020

PUBLICATIONS

1. S.D. Leeke, D. K. Y. Liu and J. P. McVittie, "Plasma mode trench etching with direct hydrocarbon injection," *Materials Research Society Symposium Proceedings*, Vol. 68, pp. 21-27, 1986.
2. D. K. Y. Liu and J. D. Plummer, "A novel trench-injector power device with low ON-resistance and high switching speed," *IEEE Electron Device Letters*, Vol. 9, No. 7, pp. 321-323, 1988.
3. D. M. Boisvert, D. K. Y. Liu and J. D. Plummer, "Circuit approaches to increasing IGBT switching speed," *IEEE Journal of Solid-State Circuits*, Vol. 23, No. 5, pp. 1276-1279, 1988.
4. D. K. Y. Liu and J. D. Plummer, "Device physics and optimization of conductivity-modulated power MOSFET's," *IEEE Transactions on Electron Devices*, Vol. 35, No. 12, pp. 2457-2458, 1988.
5. D. K. Y. Liu, K.-L. Chen, H. Tigelaar, J. Paterson and S. O. Chen, "Scaled dielectric antifuse structure for field-programmable gate array applications," *IEEE Electron Device Letters*, Vol. 12, No. 4, pp. 151-153, 1991.
6. M. Wong, D. K. Y. Liu, M. M. Moslehi and D. W. Reed, "Preoxidation treatment using HCl/HF vapor," *IEEE Electron Device Letters*, Vol. 12, No. 8, pp. 425-426, 1991.
7. D. K. Y. Liu, C. Kaya, M. Wong, J. Paterson and P. Shah, "Optimization of a source-side injection FAMOS cell for flash EPROM applications," *International Electron Devices Meeting Technical Digest*, pp. 315-318, 1991.
8. K.-L. Chen, D. K. Y. Liu, G. Misium, W. M. Gosney, S.-J. Wang, J. Camp and H. Tigelaar, "A sublithographic antifuse structure for field-programmable gate array applications," *IEEE Electron Device Letters*, Vol. 13, No. 1, pp. 53-55, 1992.
9. K. T. San, C. Kaya, D. K. Y. Liu, T. P. Ma, and P. Shah, "A new technique for determining capacitive coupling coefficients in flash EPROMs," *IEEE Electron Device Letters*, Vol. 13, No. 6, pp. 328-331, 1992.
10. C. Kaya, D. K. Y. Liu, J. Paterson and P. Shah, "Buried source-side injection (BSSI) for flash EPROM programming," *IEEE Electron Device Letters*, Vol. 13, No. 9, pp. 465-467, 1992.
11. M. Wong, D. K. Y. Liu and S. S.-W. Huang, "Analysis of the subthreshold slope and the linear transconductance techniques for the extraction of the capacitance coupling coefficients of floating-gate devices," *IEEE Electron Device Letters*, Vol. 13, No. 11, pp. 566-568, 1992.
12. D. K. Y. Liu and J. D. Plummer, "Design and analysis of a new conductivity-modulated power MOSFET," *IEEE Transactions on Electron Devices*, Vol. 40, No. 2, pp. 428-438, 1993.
13. J. Z. Peng, S. Haddad, H. Fang, C. Chang, S. Longcor, B. Ho, Y. Sun, D. Liu, Y. Tang, J. Hsu, S. Luan, J. Lien, "Flash EPROM endurance simulation using physics-based models," In *International Electron Devices Meeting Technical Digest*, pp. 295-298, 1994.

14. J. Chen, J. Hsu, S. Luan, Y. Tang, D. Liu, S. Haddad, C. Chang, S. Longcor, J. Lien, "Short Channel enhanced degradation during discharge of Flash EEPROM memory Cell," *International Electron Devices Meeting Technical Digest*, pp. 331-334, 1995.
15. V. H. Chan and D. Liu, "An Enhanced Erase Mechanism During Channel Fowler-Nordheim Tunneling in Flash EPROM Memory Devices," *IEEE Electron Devices Letters*, Vol. 20, No. 3, pp. 140-142, 1999.

EXHIBIT 2



US009865546B2

(12) **United States Patent**
Alptekin et al.

(10) **Patent No.:** **US 9,865,546 B2**

(45) **Date of Patent:** **Jan. 9, 2018**

(54) **CONTACTS TO SEMICONDUCTOR
SUBSTRATE AND METHODS OF FORMING
SAME**

21/76846; H01L 21/76879; H01L
21/76883; H01L 23/5283; H01L
23/53266; H01L 23/53261

See application file for complete search history.

(71) Applicant: **GLOBALFOUNDRIES INC.**, Grand
Cayman, KY (US)

(56) **References Cited**

(72) Inventors: **Emre Alptekin**, Wappingers Falls, NY
(US); **Nicolas L. Breil**, San Jose, CA
(US); **Christian Lavoie**, Pleasantville,
NY (US); **Ahmet S. Ozcan**,
Chappaqua, NY (US); **Kathryn T.
Schonenberg**, Wappingers Falls, NY
(US)

U.S. PATENT DOCUMENTS

8,278,200 B2 10/2012 Lavoie et al.
2012/0161324 A1 * 6/2012 Heinrich H01L 21/28518
257/757
2014/0027883 A1 * 1/2014 Hu H01L 21/743
257/534

(73) Assignee: **GLOBALFOUNDRIES INC.**, Grand
Cayman (KY)

OTHER PUBLICATIONS

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 30 days.

Filed Application titled "Metal Semiconductor Alloy Contact Resis-
tance Improvement," U.S. Appl. No. 14/172,525, filed Feb. 4, 2014,
37 pages.

* cited by examiner

(21) Appl. No.: **14/729,298**

(22) Filed: **Jun. 3, 2015**

(65) **Prior Publication Data**

US 2016/0358860 A1 Dec. 8, 2016

(51) **Int. Cl.**

H01L 23/532 (2006.01)

H01L 23/528 (2006.01)

H01L 21/768 (2006.01)

H01L 23/485 (2006.01)

(52) **U.S. Cl.**

CPC **H01L 23/53266** (2013.01); **H01L 21/7684**
(2013.01); **H01L 21/76843** (2013.01); **H01L**
21/76846 (2013.01); **H01L 21/76879**
(2013.01); **H01L 21/76883** (2013.01); **H01L**
23/485 (2013.01); **H01L 23/53223** (2013.01);
H01L 23/53238 (2013.01)

(58) **Field of Classification Search**

CPC H01L 21/7684; H01L 21/76843; H01L

Primary Examiner — Fernando L Toledo

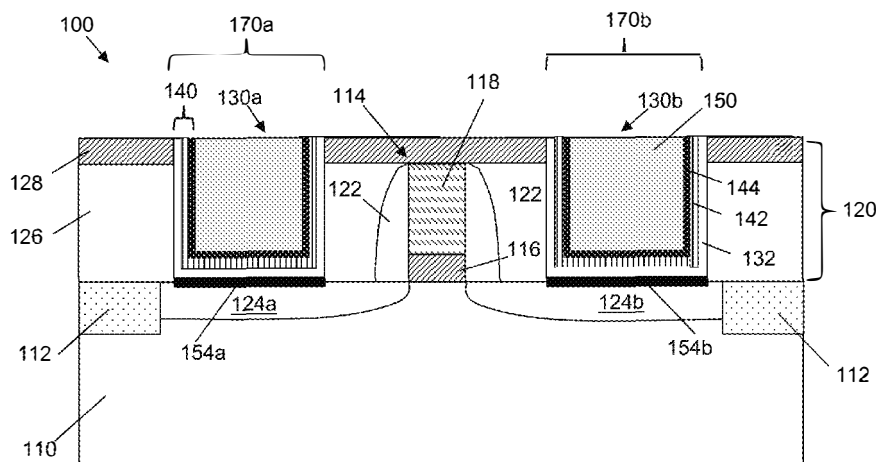
Assistant Examiner — Lawrence-Linh T Nguyen

(74) *Attorney, Agent, or Firm* — Yuanmin Cai; Hoffman
Warnick LLC

(57) **ABSTRACT**

An aspect of the invention includes a method for forming a
contact in a dielectric layer over a semiconductor substrate.
The method may comprise: forming a contact opening in a
dielectric layer over the semiconductor substrate to expose
an upper portion of the semiconductor substrate; depositing
a first liner layer to conformally coat the contact opening;
causing a portion of the first liner layer to diffuse into the
upper portion of the semiconductor substrate to form a first
intermix region at the upper portion of the semiconductor
substrate; depositing a refractory metal layer over the first
intermix region; and depositing a metal in the contact
opening thereby forming the contact.

14 Claims, 5 Drawing Sheets



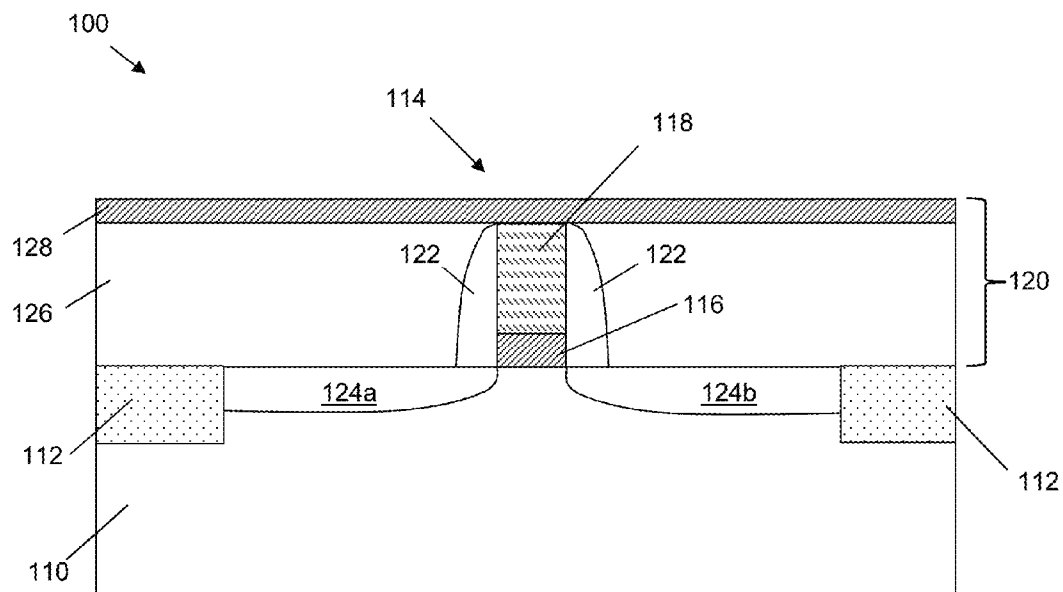


FIG. 1

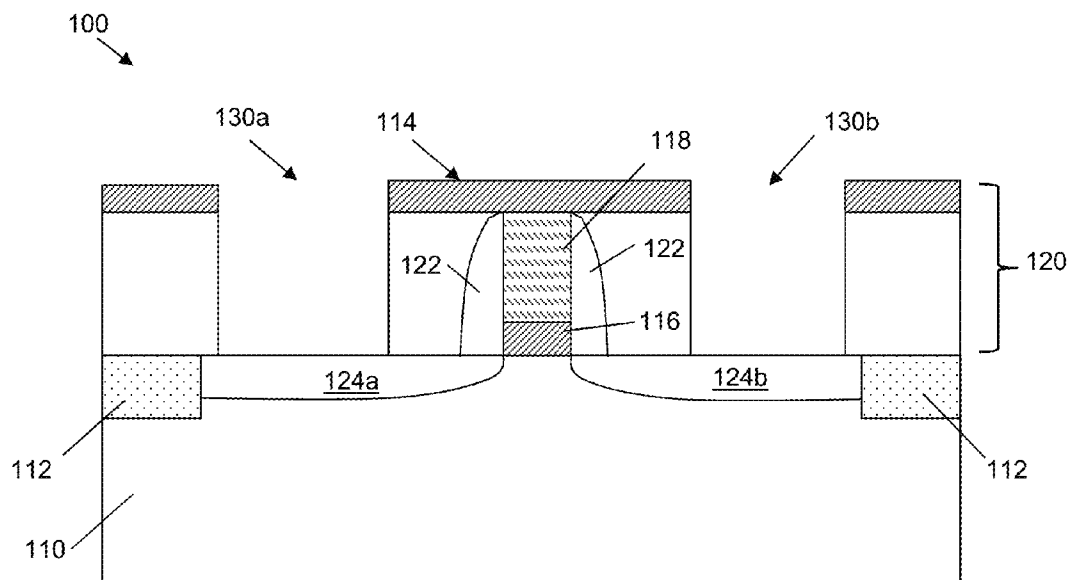


FIG. 2

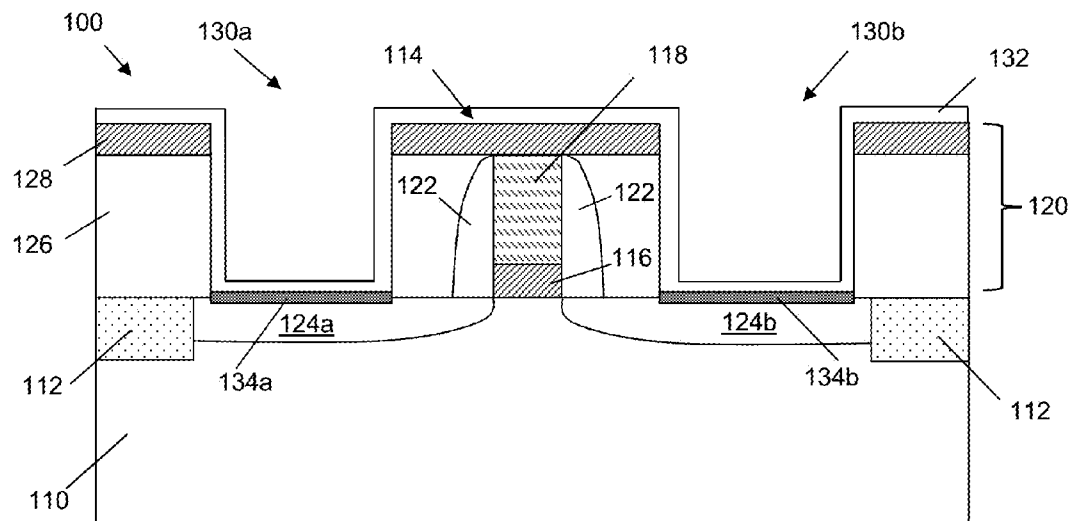


FIG. 3

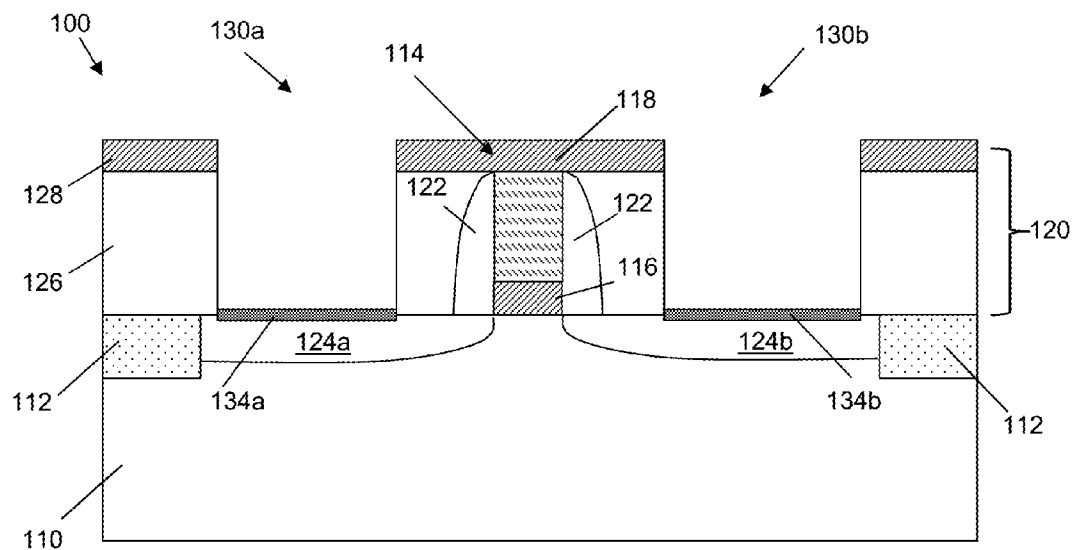


FIG. 4

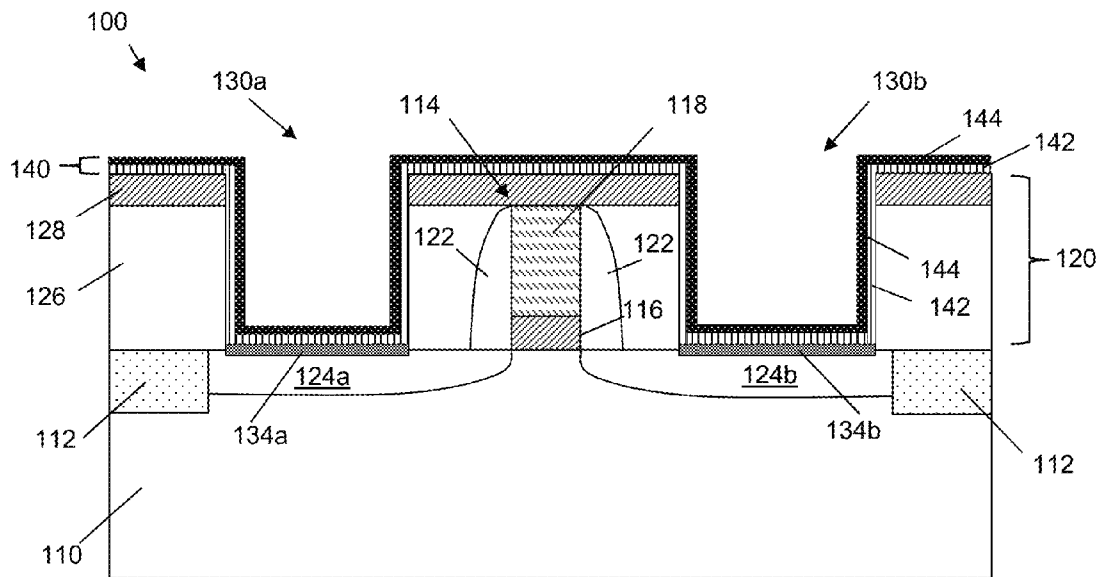


FIG. 5

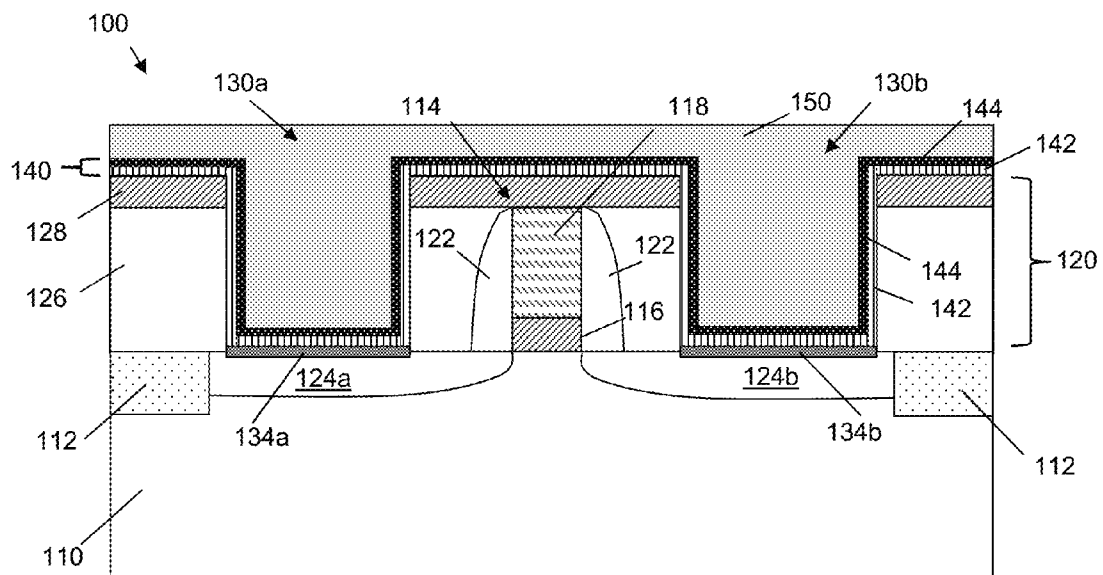


FIG. 6

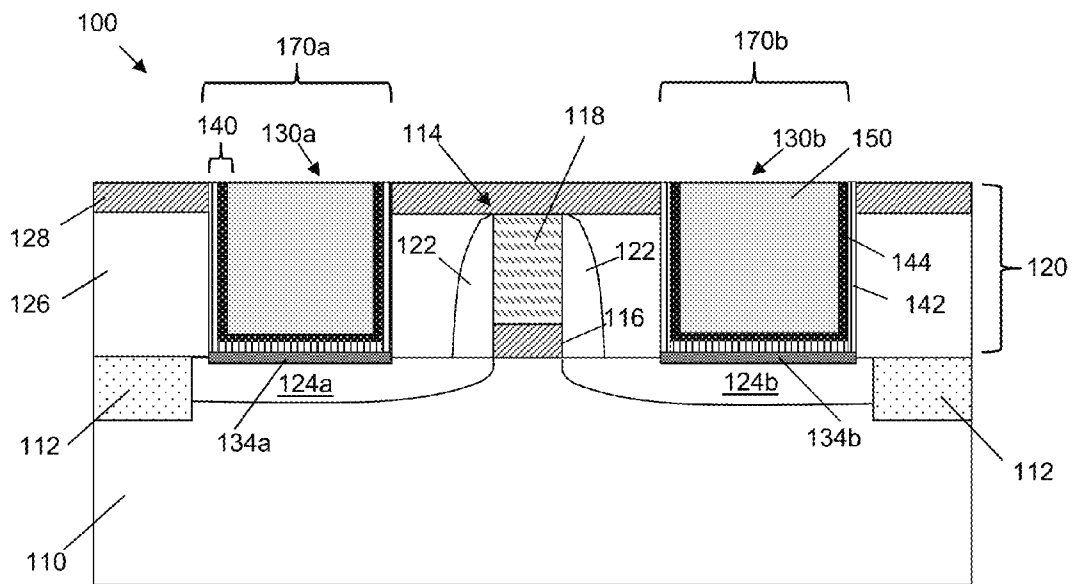


FIG. 7

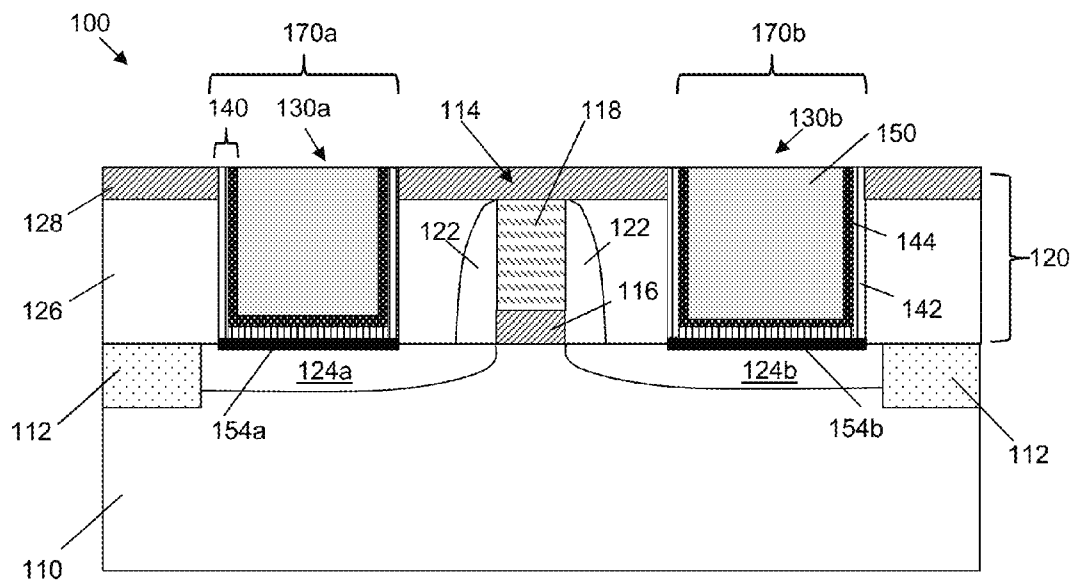


FIG. 8

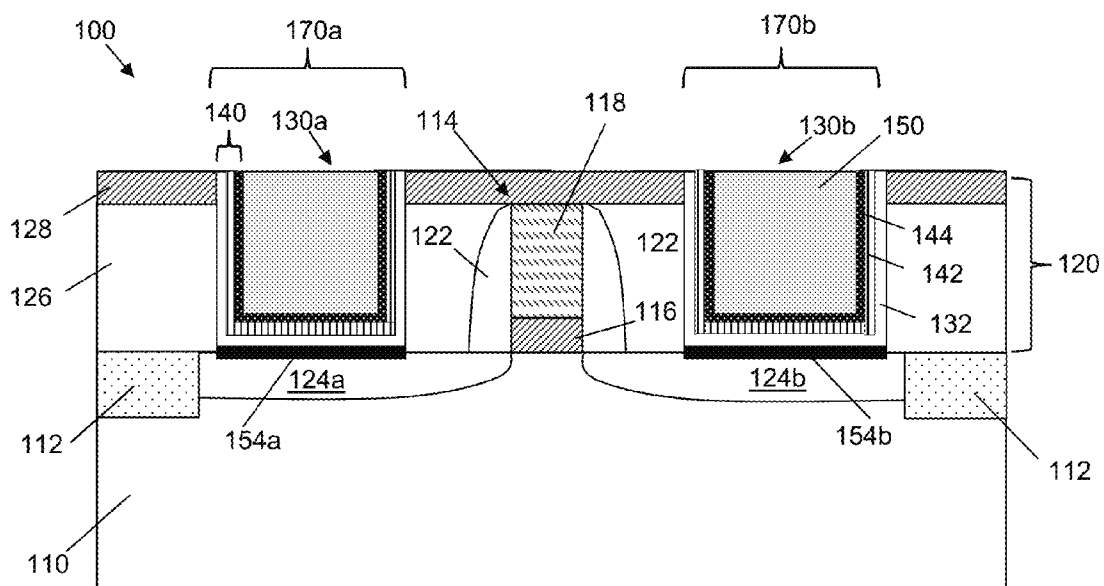


FIG. 9

1

CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME

BACKGROUND

Technical Field

The present invention relates to contacts over a semiconductor substrate and methods of forming the same. More specifically, the present invention relates to forming an intermix region at an interface of a bottom portion of contacts and an upper portion of the semiconductor substrate, respectively.

Related Art

State of the art field effect transistors (FETs) can be fabricated by depositing a gate conductor over a gate dielectric and a semiconductor substrate. Generally, the FET fabrication process implements lithography and etching processes to define the gate structures. After providing the gate structures, source/drain extensions may be formed into a portion of the semiconductor substrate and on both sides of each gate structure by ion implantation. Sometimes this implant is performed using a spacer to create a specific distance between the gate structure and the implanted junction. Source and drain may be doped with a p-type or n-type dopant. The term "p-type" refers to the addition of impurities to an intrinsic semiconductor that creates deficiencies of valence electrons. "N-type" refers to the addition of impurities that contributes free electrons to an intrinsic semiconductor. After formation of the transistor, a set of contact openings may be created to expose the source and drain. The contact openings may be coated with a Ti-based silicide liner. Subsequently, contact openings may be filled with a metal material, such as W.

In a p-type FET (PFET), performance with a Ti-based silicide liner is a challenge for 14/10 nm silicon-on-insulator (14/10 SOI) technologies. In order to improve workfunction of Ti-based silicide in the source and drain of a PFET, a bilayer of NiPt and Ti has been employed to coat contact openings over the source and drain. However, this bilayer requires deposition of a relatively thick metal layer. As semiconductor devices get smaller, requiring a thick bilayer in the contact causes contact opening fill issues, such as an incomplete contact fill.

SUMMARY

A first aspect of the invention includes a method for forming a contact in a dielectric layer to a semiconductor substrate. The method comprises: forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate; depositing a first liner layer to conformally coat the contact opening; causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate; depositing a refractory metal layer over the first intermix region; and depositing a metal in the contact opening thereby forming the contact.

A method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain. The method comprises: forming a contact opening in the dielectric layer to expose an upper portion of each of the source and the drain; depositing a first liner layer to conformally coat each contact opening and causing a portion of the first liner layer to diffuse into the upper portion of each

2

of the source and the drain to form a first intermix region at the upper portion of the each of the source and the drain; depositing a refractory metal layer over the first intermix region; and depositing a metal to substantially coat the refractory metal layer and substantially fill each of the openings thereby forming the set of contacts.

A third aspect of the invention includes a semiconductor structure including a contact to a semiconductor substrate. The contact comprises: a contact opening in a dielectric layer over the semiconductor substrate; a first liner layer substantially coating a sidewall of the contact opening; a refractory metal layer substantially coating the first liner layer in the contact opening; a metal substantially filling the contact opening; and an intermix region at an upper portion of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate.

BRIEF DESCRIPTION OF THE DRAWINGS

The embodiments of this invention will be described in detail, with reference to the following figures, wherein like designations denote like elements, and wherein:

FIGS. 1-8 show cross-sectional views of a semiconductor structure undergoing processing according to embodiments of the invention.

FIG. 9 shows a cross-sectional view of another embodiment of the semiconductor structure according to embodiments of the invention.

DETAILED DESCRIPTION

Aspects of the present invention utilize an intermix concept as a template for formation of a contact to a semiconductor substrate of a semiconductor structure. In some embodiments, the semiconductor structure may be a transistor, e.g., PFET or n-type FET (NFET) having a gate, a source, and a drain as is known in the art. In other embodiments, semiconductor structure may be a wire, a FinFet, or a hetero-bipolar junction transistor (HBT) or other silicon based microelectronic devices. One aspect of the present invention involves depositing a first liner layer in a contact opening such that an intermix region is formed at an interface of the bottom portion of the contact opening and an upper portion of the semiconductor substrate thereunder, e.g., source and drain. Unreacted (non-diffused) portions of the first liner layer may be removed prior to completion of fabricating the contact. Therefore, aspects of the present invention do not increase the deposited metal thickness in the contact openings. Thus, contact opening fill issues are avoided and overall contact resistance is improved.

Referring now to FIG. 1, a semiconductor structure 100 is shown including a semiconductor substrate 110, an isolation region 112, and a gate structure 114 within a dielectric layer 120 over substrate 112. Substrate 110 may be employed as a semiconductor layer of a semiconductor-on-insulator substrate (not shown), or a bulk substrate (not shown) as understood in the art. Substrate 110 may include any now known or later developed semiconductor material including but not limited to silicon, germanium, silicon germanium, silicon carbide, and those consisting essentially of one or more III-V compound semiconductors having a composition defined by the formula $Al_{X1}Ga_{X2}In_{X3}As_{Y1}P_{Y2}N_{Y3}Sb_{Y4}$, where X1, X2, X3, Y1, Y2, Y3, and Y4 represent relative proportions, each greater than or equal to zero and X1+X2+

3

$X_3+Y_1+Y_2+Y_3+Y_4=1$ (1 being the total relative mole quantity). Other suitable substrates include II-VI compound semiconductors having a composition $Zn_{A1}Cd_{A2}Se_{B1}Te_{B2}$, where A1, A2, B1, and B2 are relative proportions each greater than or equal to zero and $A_1+A_2+B_1+B_2=1$ (1 being a total mole quantity). Furthermore, a portion or entire substrate may be strained. Substrate 110 may be doped, undoped or contain doped and undoped regions therein.

In some embodiments of the invention, substrate 110 may include at least one isolation region 112 therein. Isolation region 112 may be a trench isolation region or a field oxide isolation region as is known in the art. A trench isolation region can be formed utilizing a conventional trench isolation process well known to those skilled in the art. For example, lithography, etching and filling of the trench with a trench dielectric such as an oxide may be used in forming the trench isolation region. Optionally, a liner may be formed in the trench prior to trench fill. After the trench is filled, a densification step may be performed as well as a planarization process. A field oxide isolation region may be formed utilizing a so-called local oxidation of silicon process. The at least one isolation region 112 may provide isolation between neighboring gate structures, typically required when the neighboring gate structures have opposite conductivities, i.e., n-type transistors and p-type transistors. As such, the at least one isolation region 112 can separate, for example, a first device region in which an n-type transistor device can be formed and second device region in which a p-type transistor device can be formed.

Further, FIG. 1 illustrates that a gate 114 may be formed on substrate 110. Gate structure 114 may include a gate dielectric material 116 and a gate conductor 118. Gate structure 114 may be used to control output current of the semiconducting device through electrical or magnetic fields. Gate dielectric 116 may include, but is not limited to: HfO_2 , ZrO_2 , La_2O_3 , Al_2O_3 , TiO_2 , $SrTiO_3$, $LaAlO_3$, Y_2O_3 , HfO_xN_y , ZrO_xN_y , $La_2O_xN_y$, $Al_2O_xN_y$, TiO_xN_y , $SrTiO_xN_y$, $LaAlO_xN_y$, $Y_2O_xN_y$, $SiON$, SiN_x , a silicate thereof, and an alloy thereof. Each value of x is independently from 0.5 to 3 and each value of y is independently from 0 to 2. In some embodiments, a multilayered gate dielectric structure comprising different gate dielectric materials, e.g., silicon dioxide, and a dielectric metal oxide can be formed and used as gate dielectric material 116. Gate conductor 118 may include, but is not limited to: doped polysilicon, doped silicon germanium, an elemental metal (e.g., tungsten, titanium, tantalum, aluminum, nickel, ruthenium, palladium and platinum), an alloy of at least two elemental metals, an elemental metal nitride (e.g., tungsten nitride, aluminum nitride, and titanium nitride), or multilayered combinations thereof.

Gate structure 114 may be formed by depositing gate dielectric 116 on substrate 110. As used herein, "depositing" may include any now known or later developed techniques appropriate for the material to be deposited including but are not limited to, for example: chemical vapor deposition (CVD), low-pressure CVD (LPCVD), plasma-enhanced CVD (PECVD), semi-atmosphere CVD (SACVD) and high density plasma CVD (HDPCVD), rapid thermal CVD (RTCVD), ultra-high vacuum CVD (UHVCVD), limited reaction processing CVD (LRPCVD), metalorganic CVD (MOCVD), sputtering deposition, ion beam deposition, electron beam deposition, laser assisted deposition, thermal oxidation, thermal nitridation, spin-on methods, physical vapor deposition (PVD), atomic layer deposition (ALD), chemical oxidation, molecular beam epitaxy (MBE), plating, evaporation. Additionally, subsequent to the deposition

4

of gate dielectric 116, gate conductor 118 can be deposited on gate dielectric 116. The precise thickness of gate dielectric 116 and gate conductor 118 may vary widely with the intended application.

Gate dielectric 116 and gate conductor 118 can be patterned into a functional gate structure 114. Gate structure 114 can be formed by lithography and etching. Lithography can include forming a photoresist (not shown) on the top-most surface of gate conductor 118 exposing the photoresist to a desired pattern of radiation, and then developing the exposed photoresist with a resist developer to provide a patterned photoresist atop the gate conductor 118. At least one etch is then employed which transfers the pattern from the patterned photoresist into the various materials. As used herein, "etching" may include any now known or later developed techniques appropriate for the material to be etched including but not limited to, for example: isotropic etching, anisotropic etching, plasma etching, sputter etching, ion beam etching, reactive-ion beam etching and reactive-ion etching (RIE).

Still referring to FIG. 1, a spacer 122 may be formed on each sidewall of gate 114. Spacers 122 may be formed by depositing a spacer material and etching the spacer material. Spacers 122 may include a spacer dielectric material as is known in the art, such as, but not limited to: a dielectric oxide, dielectric nitride, and/or dielectric oxynitride. Additionally, a source 124a and drain 124b may be formed in a portion of substrate 110 on either side of gate 114. In some embodiments, a source extension region and a drain extension region (not separately shown) may be formed prior to forming spacer 122 utilizing an extension ion implantation process. As such, a portion of each of the source extension region and the drain extension region would be located beneath the spacer 122. After forming the spacers 122, source 124a and drain 124b are formed utilizing a source/drain ion implantation process. An activation anneal may follow the implantation processes. Source 124a (including the corresponding source extension region) and drain 124b (including the corresponding drain extension region) may be doped with a p-type or n-type dopant. The term "p-type" refers to the addition of impurities to an intrinsic semiconductor that creates deficiencies of valence electrons. In a silicon-containing substrate, examples of p-type dopants, i.e., impurities, include, but are not limited to, boron, aluminum, gallium and indium. "N-type" refers to the addition of impurities that contributes free electrons to an intrinsic semiconductor. In a silicon-containing substrate, examples of n-type dopants, i.e., impurities, include, but are not limited to, antimony, arsenic and phosphorous. The concentration of dopants within source 124a (and the corresponding source extension region) and drain 124b (and the corresponding drain extension region) can be within ranges known in the art.

Still referring to FIG. 1, a dielectric layer 120 may be formed such that dielectric layer 120 substantially surrounds gate 114. In one embodiment of the invention, formation of dielectric layer 120 may include deposition of a dielectric material 126 and a dielectric cap material 128. While shown as two separate layers, it is to be understood that dielectric layer 120 is not limited to only two layers. Rather, in other embodiments of the invention, dielectric layer 120 may include a single layer or more than two separate layers. Dielectric material 126 may include, but is not limited to: silicon nitride (Si_3N_4), silicon oxide (SiO_2), fluorinated SiO_2 (FSG), hydrogenated silicon oxycarbide ($SiCOH$), porous $SiCOH$, boro-phospho-silicate glass (BPSG), silsesquioxanes, carbon (C) doped oxides (i.e., organosilicates) that

5

include atoms of silicon (Si), carbon (C), oxygen (O), and/or hydrogen (H), thermosetting polyarylene ethers, SiLK (a polyarylene ether available from Dow Chemical Corporation), a spin-on silicon-carbon containing polymer material available from JSR Corporation, other low dielectric constant (<3.9) material, or layers thereof. The upper surface of gate 114 is exposed immediately after depositing dielectric material 126. In some embodiments, a planarization process or an etch back process follows the deposition of the dielectric material 126. As is illustrated, dielectric material 126 may have an upper surface that is coplanar with an upper surface of gate 114. Dielectric material 126 may have a bottom surface that contacts an upper surface of source 124a, an upper surface of the drain 124b, and an upper surface of insulation regions 112. The thickness of the dielectric material 126 that can be employed in the present application may vary depending on the type of dielectric material employed as well as the method that was employed in forming the same. In one embodiment, dielectric material 126 has a thickness from 80 nm to 500 nm. Other thicknesses that are greater or lesser than the range provided above can also be used for dielectric material 126.

Subsequently, dielectric cap material 128 may be deposited on dielectric material 126. Dielectric cap material 128 may include an insulator material that differs from dielectric material 126. Examples of insulator materials that can be used as dielectric cap material 128 include, for example, silicon dioxide, silicon nitride and silicon oxynitride. Dielectric cap material 128 may have a thickness from 1 nm to 20 nm. Other thicknesses can also be used for the thickness of the dielectric cap material 128.

As shown in FIG. 2, a plurality of contact openings 130a, 130b may be formed in a dielectric layer 120 to substrate 110. That is, contact openings 130a, 130b may be formed through dielectric cap material 128 and dielectric material 126 to expose source 124a and drain 124b. For example, contact opening 130a exposes source 124a, while contact opening 130b exposes drain 124b. Contact openings 130a, 130b can be formed by lithography and etching. Portions of dielectric material 126 and dielectric cap material 128 remain on the upper surfaces of gate 114. As is shown in the drawings of the present application, a portion of dielectric cap material 128 that is located on the upper surface gate structure 114 may extend beyond the edges of gate structure 114 and beyond an outermost edge of each spacer 122. Further, dielectric cap material 128 and dielectric material 126 that are present above isolation regions 112 extend beyond the outermost edges of isolation regions 112. In some embodiments, plurality of openings 130a, 130b have a width (measured from one exposed sidewall of one portion of dielectric material 126 to another exposed sidewall of another portion of dielectric material 126) of from 10 nm-50 nm. Other widths that are greater or lesser than the aforementioned widths can also be used in the present application.

As shown in FIG. 3, a first liner layer 132 may be deposited to conformally coat each contact opening 130a, 130b and substantially cover dielectric cap material 128. First liner layer 132 may be deposited using ionized PVD or radio frequency PVD (RFPVD) or other directional energetic thin film deposition techniques such as Ion beam assisted deposition, performed at room temperature or above. First liner layer 132 may include, but is not limited to, at least one of: NiPt, Ni, Ti, TiSi, Co, Er, Pt, and Yb. First liner layer 132 may be deposited such that a first intermix region 134a, 134b is formed at an interface of the bottom

6

portion of each contact opening 130a, 130b and the upper portion of each of source 124a and drain 124b.

First intermix regions 134a, 134b may originate from the high energy of the incoming sputtered atoms (from first liner layer 132) which are thus pushed into the Si lattice of source 124a and drain 124b. Therefore, first intermix regions 134a, 134b may include a mixture of the material used for first liner layer 132 and a material used for the respective semiconductor substrate 110 thereunder. In the embodiment shown in FIG. 3, first intermix regions 134a, 134b may have a composition including a material used for first liner layer 132 and materials used for the respective source and drain 124a, 124b thereunder. While the Si lattice is a dense way to pack Si atoms, it has a relatively open structure due to its bonding being covalent. The first few metal atoms, arriving with relatively high energy, can be forced into the interstitial sites of the Si lattice to create a thin region (intermix regions 134a, 134b) of mixed Si and metal on the order of 3-5 nm in thickness, for example. In some cases, first intermix region 134a, 134b may have a relatively high Si content (greater than 50% Si in atomic composition). In other embodiments, the compositions of first intermix regions 134a, 134b may vary depending on one or more factors including, but not limited to: the type of material being deposited, the composition of source 124a and drain 124b, the energy of the atoms of the material being deposited, and/or the conditions of the deposition technique (temperature, pressure, reaction, atmosphere, etc.). In some embodiments, it may be desirable to select a target composition of first intermix regions 134a, 134b in view of one or more criteria, e.g., etch selectivity. First intermix region 134a, 134b may be amorphous and a subsequent anneal may crystallize intermix region 134a, 134b into a lower resistance layer as will be described herein.

First intermix regions 134a, 134b allow for formation of a very thin contact layer that is normally very difficult to control with standard sputtering tools. First intermix regions 134a, 134b are much more conformal on Si structures that are not planar. Additionally, upon annealing of first intermix regions 134a, 134b (which will be described further herein), very little if any metal diffusion is needed to form the low resistance contact layer which is often a source of defects in small devices. Encroachment of silicide under spacers 122 and encroachment on defective silicon areas may be substantially reduced or eliminated by this intermixing concept, thus increasing the yield of semiconductor structures or devices.

Referring now to FIG. 4, another step of the method according to embodiments of the invention is shown. In this step, first liner layer 132 is removed from each contact opening 130a, 130b as well as from the upper surface of dielectric cap material 128. That is, the remaining portion of first liner layer 132 which does not diffuse into the respective source 124a and drain 124b is removed. First liner layer 132 may be removed by etching using one of the various etching techniques previously described. In some embodiments, first liner layer 132 may be removed by an aqua regia etch process. Aqua regia etching is a wet etch process and involves providing a solution of hydrochloric and nitric acids to the surfaces to be etched. First liner layer 132 may be removed such that intermix regions 134a, 134b are exposed at the bottom portion of contact openings 130a, 130b.

As shown in FIG. 5, another aspect of the method according to embodiments of the invention is shown to include depositing a refractory metal layer 140 to conformally coat each contact opening 130a, 130b and substan-

tially cover dielectric cap material **128** and first intermix regions **134a**, **134b** at the bottom portion of each contact opening **130a**, **130b**. Refractory metal layer **140** may include, but is not limited to: W, Ti, Ta or Mo employed directly as a “metal”; a refractory metal employed as a constituent of a “compound” thereof, e.g., TiN, TaN, etc.; or even as a constituent of an “alloy” with another metal, e.g., titanium-tungsten. In some embodiments, as shown in FIG. 5, refractory metal layer **140** may include a bilayer of a first refractory material **142** and a second refractory material **144** such that first refractory metal material **142** substantially separates first liner layer **132** from second refractory metal layer **144**. First and second refractory metal materials **142**, **144** may include any of the refractory metal materials previously mentioned. For example, in some embodiments, first refractory metal material **142** may include Ti and second refractory metal material **144** may include TiN. In this example, first refractory metal material **142** may be deposited using RFPVD and second refractory metal material **144** may be deposited using ALD. While shown as a bilayer, it is to be understood that refractory metal layer **140** may include any number of layers of refractory metal materials without departing from aspects of the invention.

Referring now to FIG. 6, a metal **150** may be deposited over dielectric cap material **128** and within each contact opening **130a**, **130b** to substantially coat refractory metal layer **140** therein and substantially fill contact openings **130a**, **130b**. Alternatively, a plating process that fills each contact opening **130a**, **130b** from the bottom upwards can be used. In an embodiment of the invention where refractory metal layer **140** includes a bilayer, second refractory metal material **144** may substantially separate first refractory metal material **142** from metal layer **150**. Metal layer **150** may include, but is not limited to: W, Co, Cu, and Al. As shown in FIG. 7, semiconductor structure **100** may undergo a planarization process such as, for example, chemical mechanical polishing (CMP) to create contacts **170a**, **170b** within contact openings **130a**, **130b**. During the planarization process, portions of refractory metal layer **140** and metal layer **150** that extend beyond the upper surface of dielectric cap material **128** are removed such that refractory metal layer **140** and metal layer **150** within contact openings **130a**, **130b** are coplanar with the upper surface of dielectric cap material **128**.

In one example, contacts **170a** and **170b** may include NiPt as first liner layer **132**, a Ti/TiN bilayer as refractory metal layer **140**, and W as metal **150**. In this embodiment, first intermix regions **134a**, **134b** may include a NiPt silicide and second intermix regions **154a**, **154b** may include a NiPtTi silicide. However, it is to be understood that other combinations of materials may be used to create contacts **170a**, **170b** without departing from aspects of the invention. For example, in another embodiment, first liner layer **132** may include Ni, refractory metal layer **140** may include a Ti/TiN bilayer, and metal **150** may include W. In this embodiment, first intermix regions **134a**, **134b** may include a Ni silicide and second intermix regions **154a**, **154b** may include a NiTi silicide.

As shown in FIG. 8, a laser anneal process may be performed such that a second intermix region **154a**, **154b** is formed at the interface of the bottom portion of contacts **170a**, **170b** and upper portion of source **124a** and drain **124b**. Second intermix regions **154a**, **154b** may have a thickness of 3-5 nm. The laser anneal process may transform the composition of first intermix region **134a**, **134b** (FIGS. 3-7) into a second composition resulting in second intermix region **154a**, **154b** in place of first intermix region **134a**,

134b. That is, the laser anneal process may crystalize intermix regions **134a**, **134b** into a lower resistance layer, i.e. a second intermix region **154a**, **154b**. Second intermix regions **154a**, **154b** may have compositions including a mixture of the materials used for source **124a** and drain **124b**, first liner layer **132**, and refractory metal layer **140**. In an embodiment where Ti is used as refractory metal layer **140**, second intermix region **154a**, **154b** may be a silicide, such as but not limited to: TiSi, NiSi, or NiPtTiSi. That is, during the laser anneal process, refractory metal **140** may diffuse into first intermix regions **134a**, **134b** (FIGS. 3-7) to create second intermix regions **154a**, **154b**. The laser anneal process may be performed at temperature of approximately 700° C.-1100° C. for between approximately 0.1 milliseconds to approximately 1 millisecond. More particularly, the laser anneal process may be performed at approximately 800° C. for approximately 0.25 milliseconds. In some embodiments, the anneal temperature and duration may be dependent on the materials being used. In one embodiment, the anneal can utilize a laser having an energy from 500 watts to 2000 watts. In another embodiment, the anneal can be performed utilizing a raster scan beam size of from 1 cm long to 100 microns wide. In one example, the anneal can be performed utilizing a 10.6 micron CO₂ laser. In order to promote the efficiency of the CO₂ laser coupling with the semiconductor substrate, a laser with a shorter wavelength (e.g., InGaAs laser at 880 nm) can be used in conjunction with the CO₂ laser.

Referring now to FIG. 9, another embodiment of the invention is shown. In this embodiment, first liner layer **132** is not removed prior to depositing refractory metal layer **140**. In this embodiment, first liner layer **132** is deposited as described with respect to FIG. 3, such that the deposition of first liner layer **132** creates first intermix region **134a**, **134b**. However, after deposition of first liner layer **132**, refractory metal layer **140** is deposited as described with respect to FIG. 5. While refractory metal layer **140** is shown as a bilayer, refractory metal layer **140** may include any number of layers of refractory metal materials, as previously discussed. Further, this embodiment also includes etching refractory metal layer **140** from the upper surface of dielectric cap layer **128** such that a portion of refractory metal layer **140** remains substantially coating the sidewalls of contact openings **130a**, **130b** and first intermix regions **134a**, **134b** at the bottom portion of contact openings **130a**, **130b** as described with respect to FIG. 6. Additionally, a metal layer **150** may be deposited and polished as described with respect to FIGS. 6-7. Lastly, this embodiment may also undergo the laser anneal process to create second intermix regions **154a**, **154b** as described with respect to FIG. 8. Second intermix regions **154a**, **154b** may include a mixture of the materials used for source and drain regions **124a**, **124b**, first liner layer **132**, and refractory metal layer **140**. This embodiment allows for a higher quantity of first liner layer **132** in contacts **170a**, **170b** due to the remaining portions of first liner layer **132** on sidewalls of contact openings **130a**, **130b**. Additionally, this embodiment results in a simpler process as the step of removing liner layer **132** is not performed, and therefore selectivity may not be a problem.

Contacts **170a**, **170b** of this embodiment may include NiPt as first liner layer **132**, a Ti/TiN bilayer for refractory metal layer **140**, and W as metal **150**. In this embodiment, first intermix regions **134a**, **134b** may include NiPt silicide and second intermix regions **154a**, **154b** may include a NiPtTi silicide. Alternatively, first liner layer **132** may include Ti, refractory metal **140** may include TiN, and metal

150 may include W. In this embodiment, first intermix region 134a, 134b may include TiSi and second intermix region 154a, 154b may include TiSi with a higher concentration of Ti.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

The descriptions of the various embodiments of the present invention have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

We claim:

1. A method for forming a contact in a dielectric layer to a semiconductor substrate, the method comprising:
 - forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate;
 - depositing a first liner layer to conformally coat the contact opening
 - causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate;
 - depositing a refractory metal layer over the first intermix region; and
 - depositing a metal in the contact opening thereby forming the contact.
2. The method of claim 1, further comprising:
 - removing a remaining non-diffused portion of the first liner layer to expose the first intermix region at the upper portion of the semiconductor substrate prior to depositing the refractory metal layer.
3. The method of claim 1, wherein the depositing the first liner layer includes depositing at least one of: nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb).
4. The method of claim 1, wherein depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN).
5. The method of claim 1, wherein depositing the first liner layer includes performing physical vapor deposition.

6. The method of claim 1, further comprising:
 - polishing the metal; and
 - performing a laser anneal of the opening such that a second intermix region is formed at the upper portion of the semiconductor substrate and a bottom surface of the contact, wherein the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer.

7. The method of claim 1, wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal in the opening includes depositing tungsten (W).

8. A method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain, the method comprising:

- forming a contact opening in the dielectric layer to expose an upper-portion of each of the source and the drain;
- depositing a first liner layer to conformally coat each contact opening and causing a portion of the first liner layer to diffuse into the upper portion of each of the source and the drain to form a first intermix region at the upper portion of the each of the source and the drain;
- depositing a refractory metal layer over the first intermix region; and
- depositing a metal to substantially coat the refractory metal layer and substantially fill each of the openings thereby forming the set of contacts.

9. The method of claim 8, further comprising:

- removing a remaining non-diffused portion of the first liner layer to expose the intermix region prior to depositing the refractory metal layer.

10. The method of claim 8, wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb).

11. The method of claim 8, wherein the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN).

12. The method of claim 8, wherein the depositing the first liner layer includes performing physical vapor deposition.

13. The method of claim 8, further comprising:
 - polishing the metal in each of the openings; and
 - performing a laser anneal of each of the openings such that a second intermix region is formed at the upper portion of the respective source and drain thereunder, wherein the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer.

14. The method of claim 13, wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal in the opening includes depositing tungsten (W).

* * * * *

EXHIBIT 3



US010707167B2

(12) **United States Patent**
Alptekin et al.

(10) **Patent No.:** **US 10,707,167 B2**

(45) **Date of Patent:** **Jul. 7, 2020**

(54) **CONTACTS TO SEMICONDUCTOR
SUBSTRATE AND METHODS OF FORMING
SAME**

(71) Applicant: **GLOBALFOUNDRIES INC.**, Grand
Cayman (KY)

(72) Inventors: **Emre Alptekin**, Wappingers Falls, NY
(US); **Nicolas L. Breil**, San Jose, CA
(US); **Christian Lavoie**, Pleasantville,
NY (US); **Ahmet S. Ozcan**,
Chappaqua, NY (US); **Kathryn T.
Schonenberg**, Wappingers Falls, NY
(US)

(73) Assignee: **GLOBALFOUNDRIES INC.**, Grand
Cayman (KY)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 362 days.

(21) Appl. No.: **15/826,939**

(22) Filed: **Nov. 30, 2017**

(65) **Prior Publication Data**

US 2018/0090447 A1 Mar. 29, 2018

Related U.S. Application Data

(62) Division of application No. 14/729,298, filed on Jun.
3, 2015, now Pat. No. 9,865,546.

(51) **Int. Cl.**
H01L 23/532 (2006.01)
H01L 21/768 (2006.01)
H01L 23/485 (2006.01)
H01L 21/285 (2006.01)

(52) **U.S. Cl.**
CPC .. **H01L 23/53266** (2013.01); **H01L 21/28518**
(2013.01); **H01L 21/7684** (2013.01); **H01L**
21/76843 (2013.01); **H01L 21/76846**

(2013.01); **H01L 21/76855** (2013.01); **H01L**
21/76879 (2013.01); **H01L 21/76883**
(2013.01); **H01L 23/485** (2013.01); **H01L**
21/2855 (2013.01); **H01L 23/53223** (2013.01);
H01L 23/53238 (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

8,278,200 B2	10/2012	Lavoie et al.
2012/0161324 A1	6/2012	Heinrich et al.
2014/0027883 A1	1/2014	Hu
2015/0220791 A1	8/2015	Wu et al.
2015/0221740 A1 *	8/2015	Alptekin H01L 29/456 257/288

* cited by examiner

Primary Examiner — Fernando L Toledo

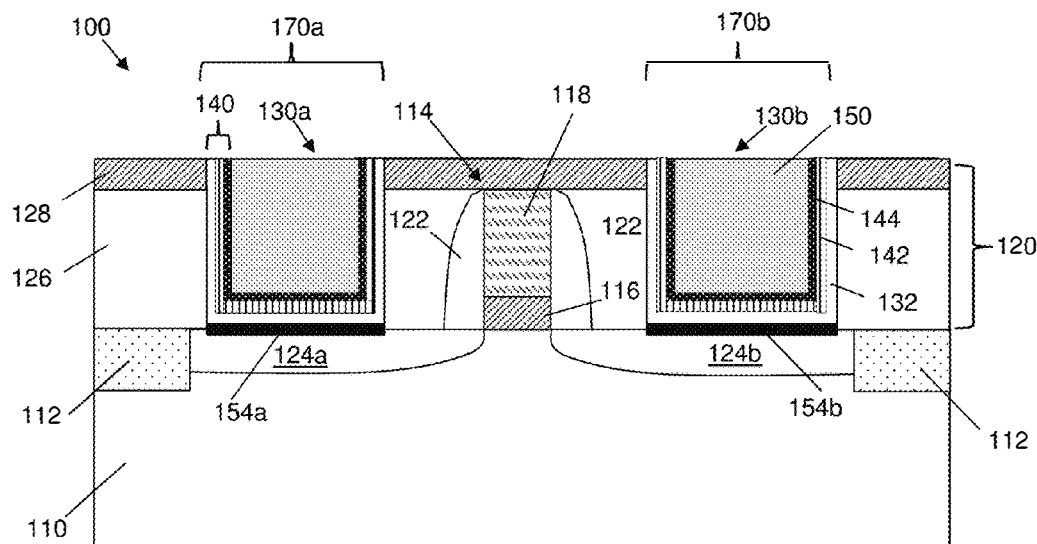
Assistant Examiner — Lawrence-Lin T Nguyen

(74) *Attorney, Agent, or Firm* — Anthony Canale;
Hoffman Warnick LLC

(57) **ABSTRACT**

An aspect of the invention includes a method for forming a contact in a dielectric layer over a semiconductor substrate. The method may comprise: forming a contact opening in a dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate; depositing a first liner layer to conformally coat the contact opening; causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate; depositing a refractory metal layer over the first intermix region; and depositing a metal in the contact opening thereby forming the contact.

6 Claims, 5 Drawing Sheets



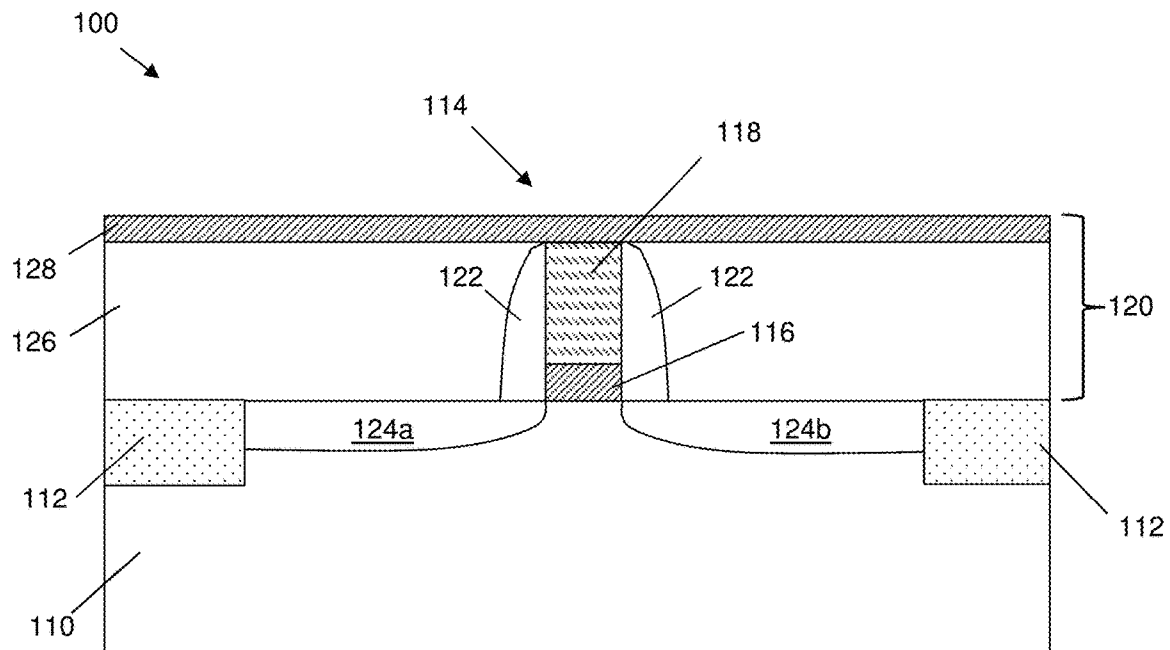


FIG. 1

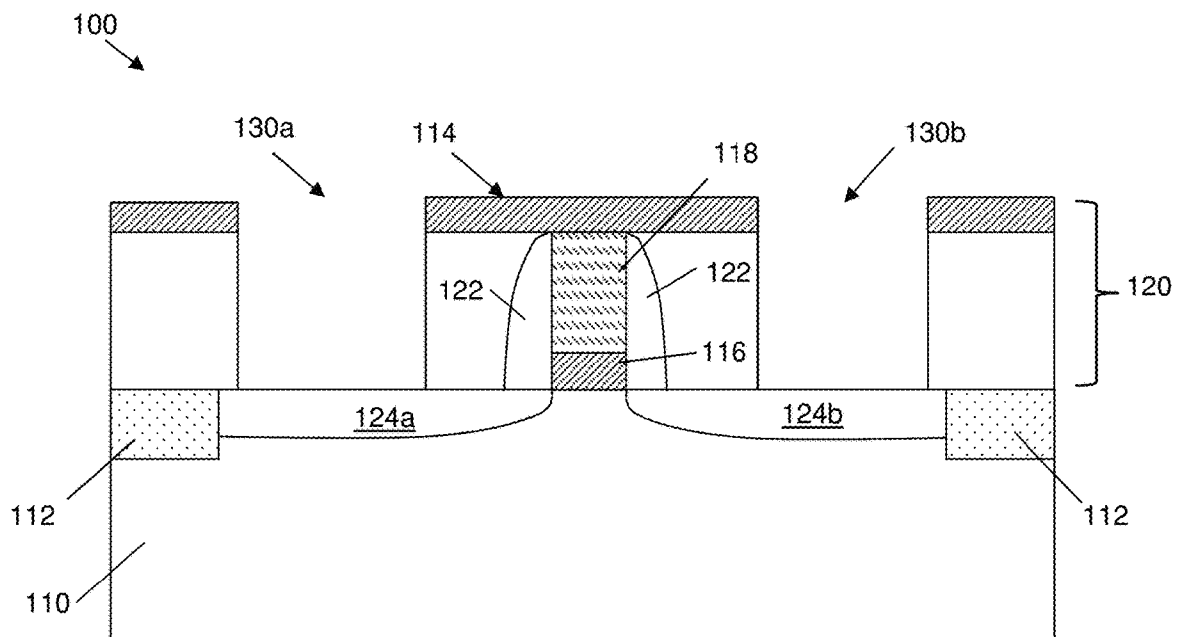


FIG. 2

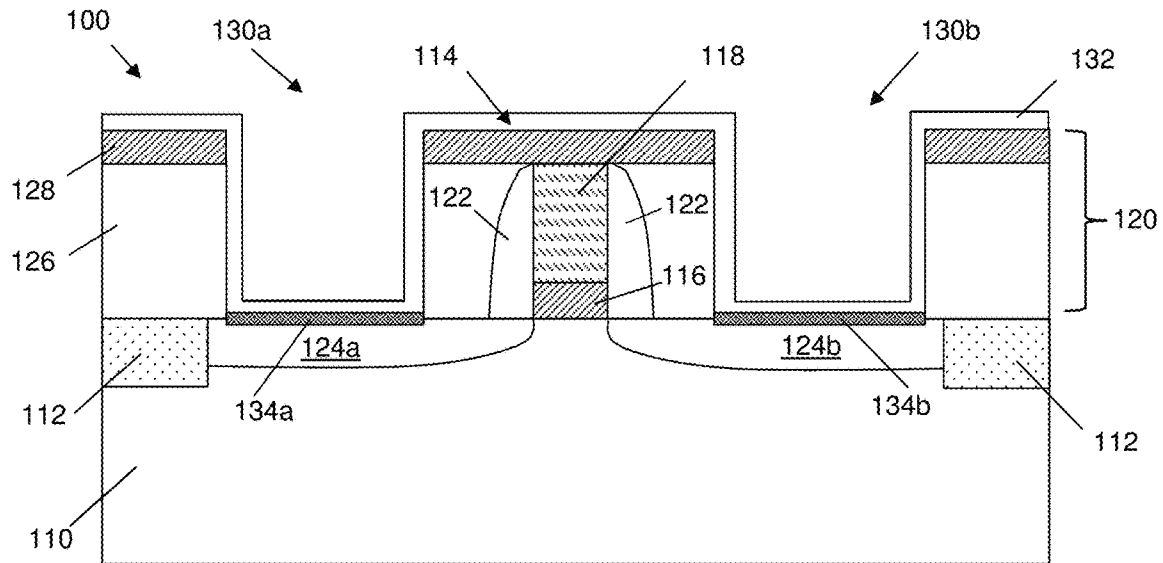


FIG. 3

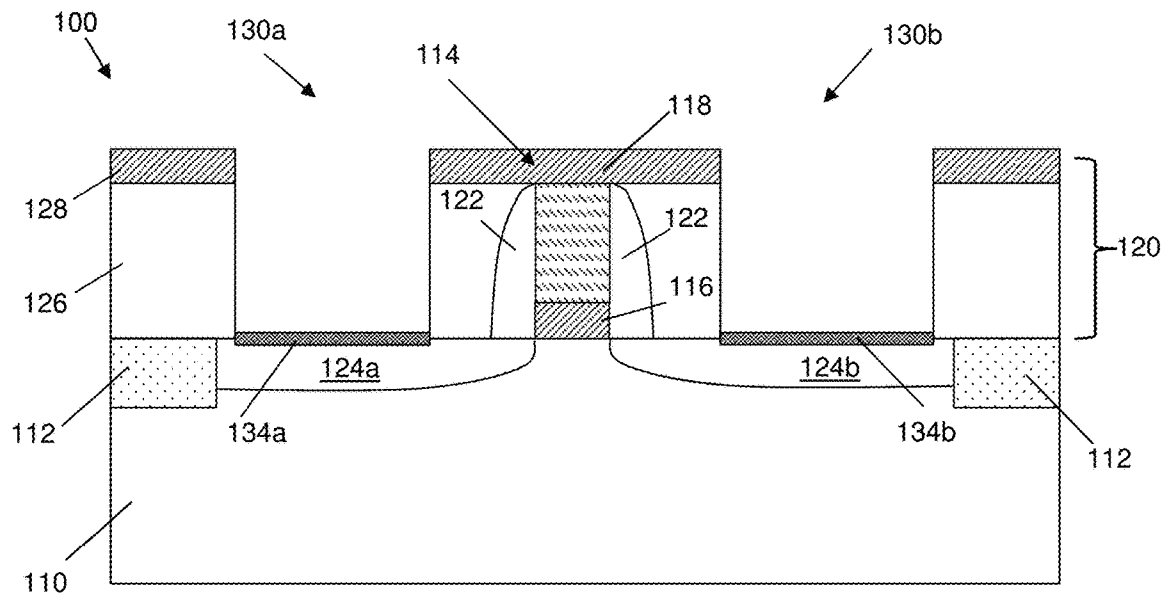


FIG. 4

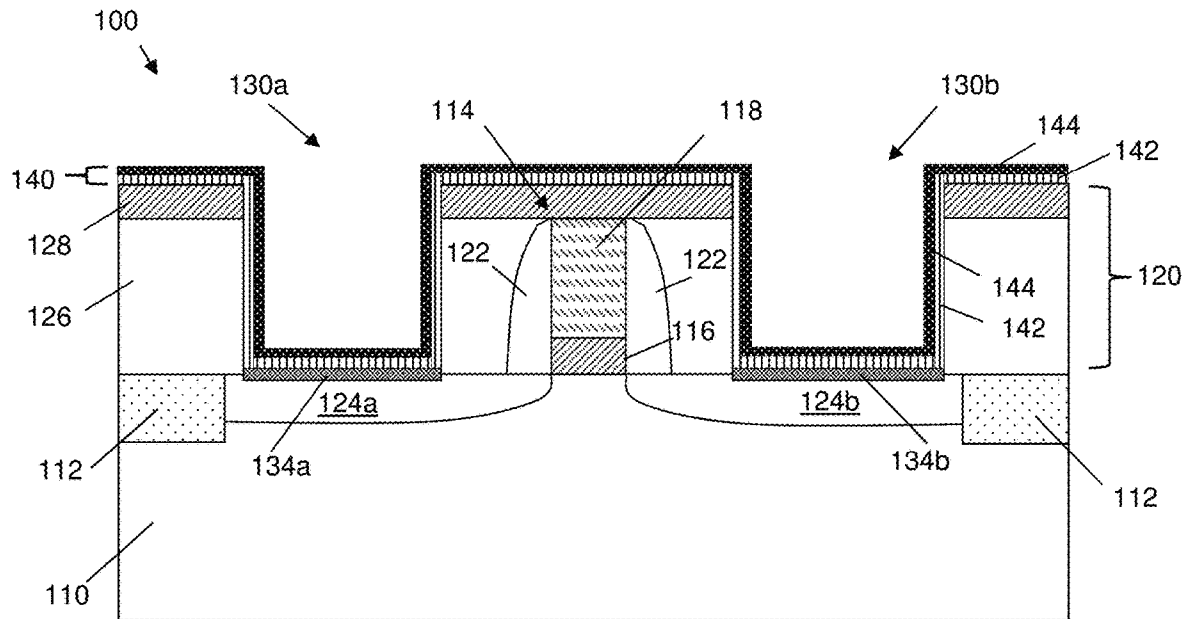


FIG. 5

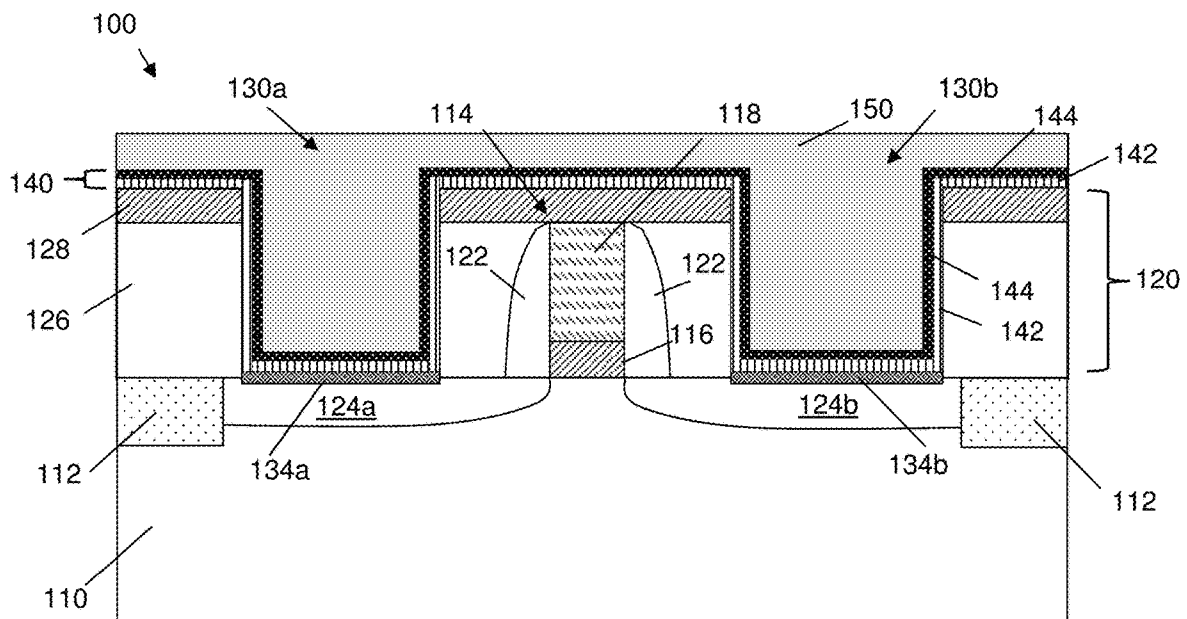


FIG. 6

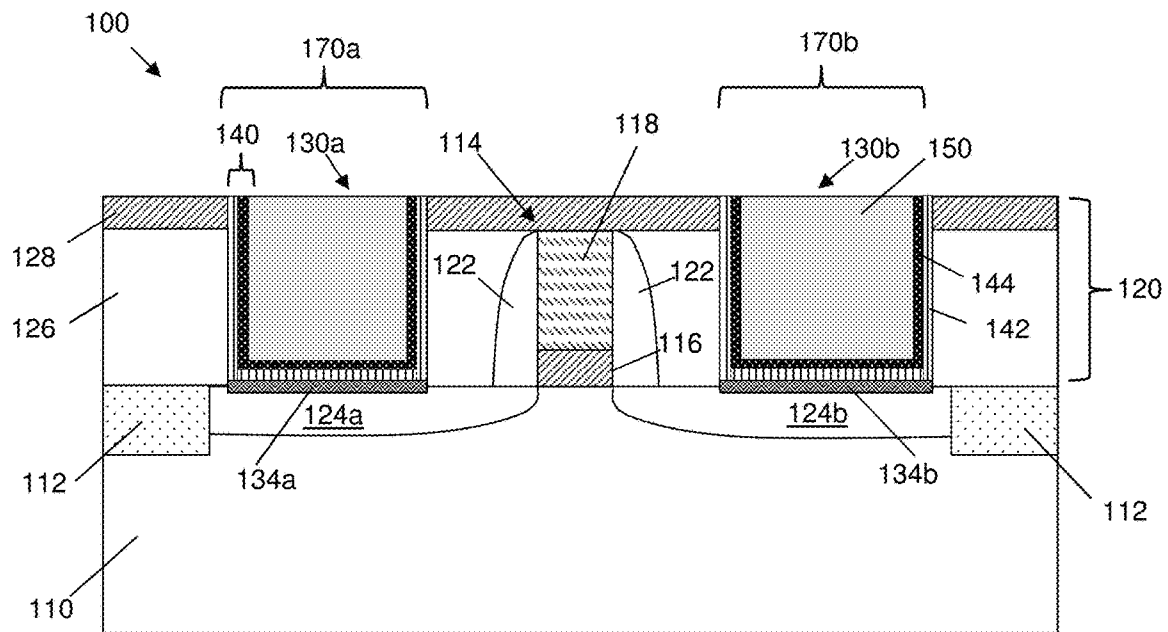


FIG. 7

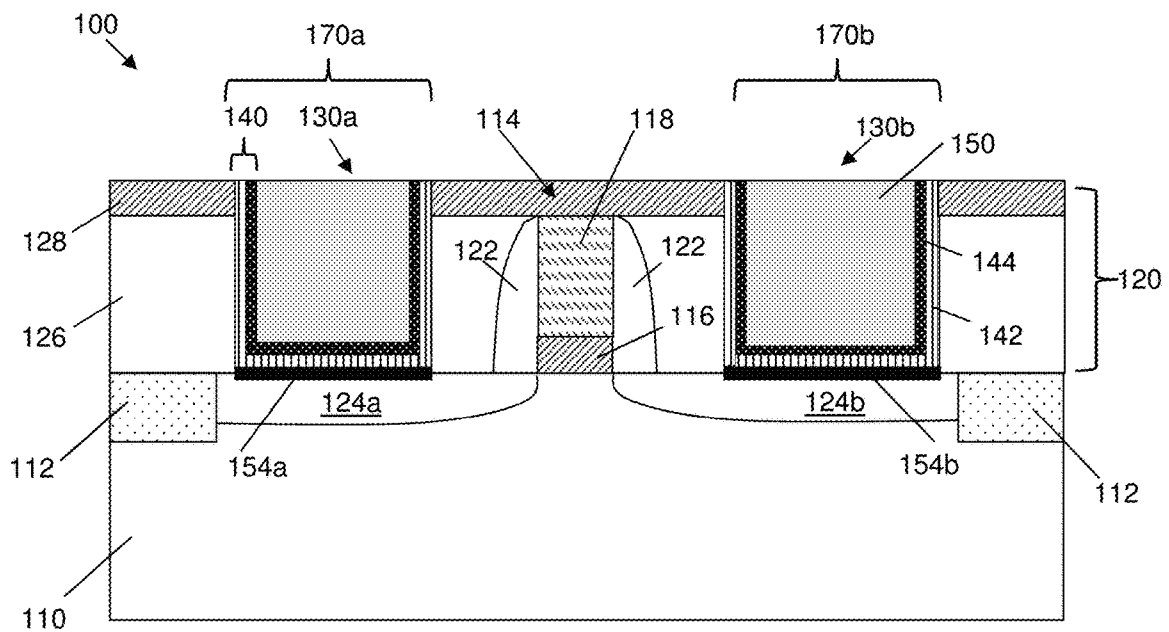


FIG. 8

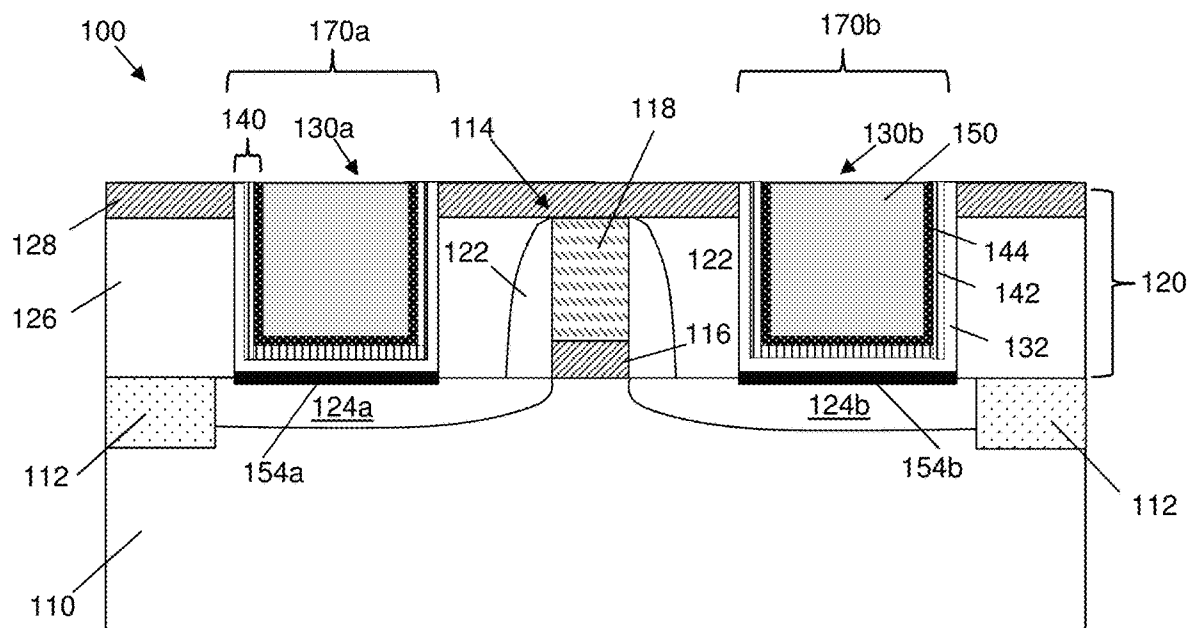


FIG. 9

1

CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME

BACKGROUND

Technical Field

The present invention relates to contacts over a semiconductor substrate and methods of forming the same. More specifically, the present invention relates to forming an intermix region at an interface of a bottom portion of contacts and an upper portion of the semiconductor substrate, respectively.

Related Art

State of the art field effect transistors (FETs) can be fabricated by depositing a gate conductor over a gate dielectric and a semiconductor substrate. Generally, the FET fabrication process implements lithography and etching processes to define the gate structures. After providing the gate structures, source/drain extensions may be formed into a portion of the semiconductor substrate and on both sides of each gate structure by ion implantation. Sometimes this implant is performed using a spacer to create a specific distance between the gate structure and the implanted junction. Source and drain may be doped with a p-type or n-type dopant. The term "p-type" refers to the addition of impurities to an intrinsic semiconductor that creates deficiencies of valence electrons. "N-type" refers to the addition of impurities that contributes free electrons to an intrinsic semiconductor. After formation of the transistor, a set of contact openings may be created to expose the source and drain. The contact openings may be coated with a Ti-based silicide liner. Subsequently, contact openings may be filled with a metal material, such as W.

In a p-type FET (PFET), performance with a Ti-based silicide liner is a challenge for 14/10 nm silicon-on-insulator (14/10 SOI) technologies. In order to improve workfunction of Ti-based silicide in the source and drain of a PFET, a bilayer of NiPt and Ti has been employed to coat contact openings over the source and drain. However, this bilayer requires deposition of a relatively thick metal layer. As semiconductor devices get smaller, requiring a thick bilayer in the contact causes contact opening fill issues, such as an incomplete contact fill.

SUMMARY

A first aspect of the invention includes a method for forming a contact in a dielectric layer to a semiconductor substrate. The method comprises: forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate; a first liner layer to conformally coat the contact opening; causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate; depositing a refractory metal layer over the first intermix region; and depositing a metal in the contact opening thereby forming the contact.

A method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain. The method comprises: forming a contact opening in the dielectric layer to expose an upper portion of each of the

2

source and the drain; depositing a first liner layer to conformally coat each contact opening and causing a portion of the first liner layer to diffuse into the upper portion of each of the source and the drain to form a first intermix region at the upper portion of the each of the source and the drain; depositing a refractory metal layer over the first intermix region; and depositing a metal to substantially coat the refractory metal layer and substantially fill each of the openings thereby forming the set of contacts.

A third aspect of the invention includes a semiconductor structure including a contact to a semiconductor substrate. The contact comprises: a contact opening in a dielectric layer over the semiconductor substrate; a first liner layer substantially coating a sidewall of the contact opening; a refractory metal layer substantially coating the first liner layer in the contact opening; a metal substantially filling the contact opening; and an intermix region at an upper portion of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate.

BRIEF DESCRIPTION OF THE DRAWINGS

The embodiments of this invention will be described in detail, with reference to the following figures, wherein like designations denote like elements, and wherein:

FIGS. 1-8 show cross-sectional views of a semiconductor structure undergoing processing according to embodiments of the invention.

FIG. 9 shows a cross-sectional view of another embodiment of the semiconductor structure according to embodiments of the invention.

DETAILED DESCRIPTION

Aspects of the present invention utilize an intermix concept as a template for formation of a contact to a semiconductor substrate of a semiconductor structure. In some embodiments, the semiconductor structure may be a transistor, e.g., PFET or n-type FET (NFET) having a gate, a source, and a drain as is known in the art. In other embodiments, semiconductor structure may be a wire, a FinFet, or a hetero-bipolar junction transistor (HBT) or other silicon based microelectronic devices. One aspect of the present invention involves depositing a first liner layer in a contact opening such that an intermix region is formed at an interface of the bottom portion of the contact opening and an upper portion of the semiconductor substrate thereunder, e.g., source and drain. Unreacted (non-diffused) portions of the first liner layer may be removed prior to completion of fabricating the contact. Therefore, aspects of the present invention do not increase the deposited metal thickness in the contact openings. Thus, contact opening fill issues are avoided and overall contact resistance is improved.

Referring now to FIG. 1, a semiconductor structure **100** is shown including a semiconductor substrate **110**, an isolation region **112**, and a gate structure **114** within a dielectric layer **120** over substrate **112**. Substrate **112** may be employed as a semiconductor layer of a semiconductor-on-insulator substrate (not shown), or a bulk substrate (not shown) as understood in the art. Substrate **112** may include any now known or later developed semiconductor material including but not limited to silicon, germanium, silicon germanium, silicon carbide, and those consisting essentially of one or more III-V compound semiconductors having a composition

defined by the formula $\text{Al}_{X1}\text{Ga}_{X2}\text{In}_{X3}\text{As}_{Y1}\text{P}_{Y2}\text{N}_{Y3}\text{Sb}_{Y4}$, where X1, X2, X3, Y1, Y2, Y3, and Y4 represent relative proportions, each greater than or equal to zero and $X1+X2+X3+Y1+Y2+Y3+Y4=1$ (1 being the total relative mole quantity). Other suitable substrates include II-VI compound semiconductors having a composition $\text{Zn}_{A1}\text{Cd}_{A2}\text{Se}_{B1}\text{Te}_{B2}$, where A1, A2, B1, and B2 are relative proportions each greater than or equal to zero and $A1+A2+B1+B2=1$ (1 being a total mole quantity). Furthermore, a portion or entire substrate may be strained. Substrate 112 may be doped, undoped or contain doped and undoped regions therein.

In some embodiments of the invention, substrate 110 may include at least one isolation region 112 therein. Isolation region 112 may be a trench isolation region or a field oxide isolation region as is known in the art. A trench isolation region can be formed utilizing a conventional trench isolation process well known to those skilled in the art. For example, lithography, etching and filling of the trench with a trench dielectric such as an oxide may be used in forming the trench isolation region. Optionally, a liner may be formed in the trench prior to trench fill. After the trench is filled, a densification step may be performed as well as a planarization process. A field oxide isolation region may be formed utilizing a so-called local oxidation of silicon process. The at least one isolation region 112 may provide isolation between neighboring gate structures, typically required when the neighboring gate structures have opposite conductivities, i.e., n-type transistors and p-type transistors. As such, the at least one isolation region 112 can separate, for example, a first device region in which an n-type transistor device can be formed and second device region in which a p-type transistor device can be formed.

Further, FIG. 1 illustrates that a gate 114 may be formed on substrate 112. Gate structure 114 may include a gate dielectric material 116 and a gate conductor 118. Gate structure 114 may be used to control output current of the semiconducting device through electrical or magnetic fields. Gate dielectric 116 may include, but is not limited to: HfO_2 , ZrO_2 , La_2O_3 , Al_2O_3 , TiO_2 , SrTiO_3 , LaAlO_3 , Y_2O_3 , HfO_xN_y , ZrO_xN_y , $\text{La}_2\text{O}_x\text{N}_y$, $\text{Al}_2\text{O}_x\text{N}_y$, TiO_xN_y , SrTiO_xN_y , LaAlO_xN_y , $\text{Y}_2\text{O}_x\text{N}_y$, SiO_n , SiN_x , a silicate thereof, and an alloy thereof. Each value of x is independently from 0.5 to 3 and each value of y is independently from 0 to 2. In some embodiments, a multilayered gate dielectric structure comprising different gate dielectric materials, e.g., silicon dioxide, and a dielectric metal oxide can be formed and used as gate dielectric material 116. Gate conductor 118 may include, but is not limited to: doped polysilicon, doped silicon germanium, an elemental metal (e.g., tungsten, titanium, tantalum, aluminum, nickel, ruthenium, palladium and platinum), an alloy of at least two elemental metals, an elemental metal nitride (e.g., tungsten nitride, aluminum nitride, and titanium nitride), or multilayered combinations thereof.

Gate structure 114 may be formed by depositing gate dielectric 116 on substrate 112. As used herein, "depositing" may include any now known or later developed techniques appropriate for the material to be deposited including but are not limited to, for example: chemical vapor deposition (CVD), low-pressure CVD (LPCVD), plasma-enhanced CVD (PECVD), semi-atmosphere CVD (SACVD) and high density plasma CVD (HDPCVD), rapid thermal CVD (RTCVD), ultra-high vacuum CVD (UHVCVD), limited reaction processing CVD (LRPCVD), metalorganic CVD (MOCVD), sputtering deposition, ion beam deposition, electron beam deposition, laser assisted deposition, thermal oxidation, thermal nitridation, spin-on methods, physical vapor deposition (PVD), atomic layer deposition (ALD),

chemical oxidation, molecular beam epitaxy (MBE), plating, evaporation. Additionally, subsequent to the deposition of gate dielectric 116, gate conductor 118 can be deposited on gate dielectric 116. The precise thickness of gate dielectric 116 and gate conductor 118 may vary widely with the intended application.

Gate dielectric 116 and gate conductor 118 can be patterned into a functional gate structure 114. Gate structure 114 can be formed by lithography and etching. Lithography can include forming a photoresist (not shown) on the top-most surface of gate conductor 118 exposing the photoresist to a desired pattern of radiation, and then developing the exposed photoresist with a resist developer to provide a patterned photoresist atop the gate conductor 118. At least one etch is then employed which transfers the pattern from the patterned photoresist into the various materials. As used herein, "etching" may include any now known or later developed techniques appropriate for the material to be etched including but not limited to, for example: isotropic etching, anisotropic etching, plasma etching, sputter etching, ion beam etching, reactive-ion beam etching and reactive-ion etching (RIE).

Still referring to FIG. 1, a spacer 122 may be formed on each sidewall of gate 114. Spacers 122 may be formed by depositing a spacer material and etching the spacer material. Spacers 122 may include a spacer dielectric material as is known in the art, such as, but not limited to: a dielectric oxide, dielectric nitride, and/or dielectric oxynitride. Additionally, a source 124a and drain 124b may be formed in a portion of substrate 110 on either side of gate 114. In some embodiments, a source extension region and a drain extension region (not separately shown) may be formed prior to forming spacer 122 utilizing an extension ion implantation process. As such, a portion of each of the source extension region and the drain extension region would be located beneath the spacer 122. After forming the spacers 122, source 124a and drain 124b are formed utilizing a source/drain ion implantation process. An activation anneal may follow the implantation processes. Source 124a (including the corresponding source extension region) and drain 124b (including the corresponding drain extension region) may be doped with a p-type or n-type dopant. The term "p-type" refers to the addition of impurities to an intrinsic semiconductor that creates deficiencies of valence electrons. In a silicon-containing substrate, examples of p-type dopants, i.e., impurities, include, but are not limited to, boron, aluminum, gallium and indium. "N-type" refers to the addition of impurities that contributes free electrons to an intrinsic semiconductor. In a silicon-containing substrate, examples of n-type dopants, i.e., impurities, include, but are not limited to, antimony, arsenic and phosphorus. The concentration of dopants within source 124a (and the corresponding source extension region) and drain 124b (and the corresponding drain extension region) can be within ranges known in the art.

Still referring to FIG. 1, a dielectric layer 120 may be formed such that dielectric layer 120 substantially surrounds gate 114. In one embodiment of the invention, formation of dielectric layer 120 may include deposition of a dielectric material 126 and a dielectric cap material 128. While shown as two separate layers, it is to be understood that dielectric layer 120 is not limited to only two layers. Rather, in other embodiments of the invention, dielectric layer 120 may include a single layer or more than two separate layers. Dielectric material 126 may include, but is not limited to: silicon nitride (Si_3N_4), silicon oxide (SiO_2), fluorinated SiO_2 (FSG), hydrogenated silicon oxycarbide (SiCOH), porous

SiCOH, boro-phospho-silicate glass (BPSG), silsesquioxanes, carbon (C) doped oxides (i.e., organosilicates) that include atoms of silicon (Si), carbon (C), oxygen (O), and/or hydrogen (H), thermosetting polyarylene ethers, SiLK (a polyarylene ether available from Dow Chemical Corporation), a spin-on silicon-carbon containing polymer material available from JSR Corporation, other low dielectric constant (<3.9) material, or layers thereof. The upper surface of gate 114 is exposed immediately after depositing dielectric material 126. In some embodiments, a planarization process or an etch back process follows the deposition of the dielectric material 126. As is illustrated, dielectric material 126 may have an upper surface that is coplanar with an upper surface of gate 114. Dielectric material 126 may have a bottom surface that contacts an upper surface of source 124a, an upper surface of the drain 124b, and an upper surface of isolation regions 112. The thickness of the dielectric material 126 that can be employed in the present application may vary depending on the type of dielectric material employed as well as the method that was employed in forming the same. In one embodiment, dielectric material 126 has a thickness from 80 nm to 500 nm. Other thicknesses that are greater or lesser than the range provided above can also be used for dielectric material 126.

Subsequently, dielectric cap material 128 may be deposited on dielectric material 126. Dielectric cap material 128 may include an insulator material that differs from dielectric material 126. Examples of insulator materials that can be used as dielectric cap material 128 include, for example, silicon dioxide, silicon nitride and silicon oxynitride. Dielectric cap material 128 may have a thickness from 1 nm to 20 nm. Other thicknesses can also be used for the thickness of the dielectric cap material 128.

As shown in FIG. 2, a plurality of contact openings 130a, 130b may be formed in a dielectric layer 120 to substrate 110. That is, contact openings 130a, 130b may be formed through dielectric cap material 128 and dielectric material 126 to expose source 124a and drain 124b. For example, contact opening 130a exposes source 124a, while contact opening 130b exposes drain 124b. Contact openings 130a, 130b can be formed by lithography and etching. Portions of dielectric material 126 and dielectric cap material 128 remain on the upper surfaces of gate 114. As is shown in the drawings of the present application, a portion of dielectric cap material 128 that is located on the upper surface gate structure 114 may extend beyond the edges of gate structure 114 and beyond an outermost edge of each spacer 122. Further, dielectric cap material 128 and dielectric material 126 that are present above isolation regions 112 extend beyond the outermost edges of isolation regions 112. In some embodiments, plurality of openings 130a, 130b have a width (measured from one exposed sidewall of one portion of dielectric material 126 to another exposed sidewall of another portion of dielectric material 126) of from 10 nm-50 nm. Other widths that are greater or lesser than the aforementioned widths can also be used in the present application.

As shown in FIG. 3, a first liner layer 132 may be deposited to conformally coat each contact opening 130a, 130b and substantially cover dielectric cap material 128. First liner layer 132 may be deposited using ionized PVD or radio frequency PVD (RFPVD) or other directional energetic thin film deposition techniques such as Ion beam assisted deposition, performed at room temperature or above. First liner layer 132 may include, but is not limited to, at least one of: NiPt, Ni, Ti, TiSi, Co, Er, Pt, and Yb. First liner layer 132 may be deposited such that a first intermix

region 134a, 134b is formed at an interface of the bottom portion of each contact opening 130a, 130b and the upper portion of each of source 124a and drain 124b.

First intermix regions 134a, 134b may originate from the high energy of the incoming sputtered atoms (from first liner layer 132) which are thus pushed into the Si lattice of source 124a and drain 124b. Therefore, first intermix regions 134a, 134b may include a mixture of the material used for first liner layer 132 and a material used for the respective semiconductor substrate 110 thereunder. In the embodiment shown in FIG. 3, first intermix regions 134a, 134b may have a composition including a material used for first liner layer 132 and materials used for the respective source and drain 124a, 124b thereunder. While the Si lattice is a dense way to pack Si atoms, it has a relatively open structure due to its bonding being covalent. The first few metal atoms, arriving with relatively high energy, can be forced into the interstitial sites of the Si lattice to create a thin region (intermix regions 134a, 134b) of mixed Si and metal on the order of 3-5 nm in thickness, for example. In some cases, first intermix region 134a, 134b may have a relatively high Si content (greater than 50% Si in atomic composition). In other embodiments, the compositions of first intermix regions 134a, 134b may vary depending on one or more factors including, but not limited to: the type of material being deposited, the composition of source 124a and drain 124b, the energy of the atoms of the material being deposited, and/or the conditions of the deposition technique (temperature, pressure, reaction, atmosphere, etc.). In some embodiments, it may be desirable to select a target composition of first intermix regions 134a, 134b in view of one or more criteria, e.g., etch selectivity. First intermix region 134a, 134b may be amorphous and a subsequent anneal may crystallize intermix region 134a, 134b into a lower resistance layer as will be described herein.

First intermix regions 134a, 134b allow for formation of a very thin contact layer that is normally very difficult to control with standard sputtering tools. First intermix regions 134a, 134b are much more conformal on Si structures that are not planar. Additionally, upon annealing of first intermix regions 134a, 134b (which will be described further herein), very little if any metal diffusion is needed to form the low resistance contact layer which is often a source of defects in small devices. Encroachment of silicide under spacers 122 and encroachment on defective silicon areas may be substantially reduced or eliminated by this intermixing concept, thus increasing the yield of semiconductor structures or devices.

Referring now to FIG. 4, another step of the method according to embodiments of the invention is shown. In this step, first liner layer 132 is removed from each contact opening 130a, 130b as well as from the upper surface of dielectric cap material 128. That is, the remaining portion of first liner layer 132 which does not diffuse into the respective source 124a and drain 124b is removed. First liner layer 132 may be removed by etching using one of the various etching techniques previously described. In some embodiments, first liner layer 132 may be removed by an aqua regia etch process. Aqua regia etching is a wet etch process and involves providing a solution of hydrochloric and nitric acids to the surfaces to be etched. First liner layer 132 may be removed such that intermix regions 134a, 134b are exposed at the bottom portion of contact openings 130a, 130b.

As shown in FIG. 5, another aspect of the method according to embodiments of the invention is shown to include depositing a refractory metal layer 140 to conform

mally coat each contact opening **130a**, **130b** and substantially cover dielectric cap material **128** and first intermix regions **134a**, **134b** at the bottom portion of each contact opening **130a**, **130b**. Refractory metal layer **140** may include, but is not limited to: W, Ti, Ta or Mo employed directly as a “metal”; a refractory metal employed as a constituent of a “compound” thereof, e.g., TiN, TaN, etc.; or even as a constituent of an “alloy” with another metal, e.g., titanium-tungsten. In some embodiments, as shown in FIG. 5, refractory metal layer **140** may include a bilayer of a first refractory material **142** and a second refractory material **144** such that first refractory metal material **142** substantially separates first liner layer **132** from second refractory metal layer **144**. First and second refractory metal materials **142**, **144** may include any of the refractory metal materials previously mentioned. For example, in some embodiments, first refractory metal material **142** may include Ti and second refractory metal material **144** may include TiN. In this example, first refractory metal material **142** may be deposited using RFPVD and second refractory metal material **144** may be deposited using ALD. While shown as a bilayer, it is to be understood that refractory metal layer **140** may include any number of layers of refractory metal materials without departing from aspects of the invention.

Referring now to FIG. 6, a metal **150** may be deposited over dielectric cap material **128** and within each contact opening **130a**, **130b** to substantially coat refractory metal layer **140** therein and substantially fill contact openings **130a**, **130b**. Alternatively, a plating process that fills each contact opening **130a**, **130b** from the bottom upwards can be used. In an embodiment of the invention where refractory metal layer **140** includes a bilayer, second refractory metal material **144** may substantially separate first refractory metal material **142** from metal layer **150**. Metal layer **150** may include, but is not limited to: W, Co, Cu, and Al. As shown in FIG. 7, semiconductor structure **100** may undergo a planarization process such as, for example, chemical mechanical polishing (CMP) to create contacts **170a**, **170b** within contact openings **130a**, **130b**. During the planarization process, portions of refractory metal layer **140** and metal layer **150** that extend beyond the upper surface of dielectric cap material **128** are removed such that refractory metal layer **140** and metal layer **150** within contact openings **130a**, **130b** are coplanar with the upper surface of dielectric cap material **128**.

In one example, contacts **170a** and **170b** may include NiPt as first liner layer **132**, a Ti/TiN bilayer as refractory metal layer **140**, and W as metal **150**. In this embodiment, first intermix regions **134a**, **134b** may include a NiPt silicide and second intermix regions **154a**, **154b** may include a NiPtTi silicide. However, it is to be understood that other combinations of materials may be used to create contacts **170a**, **170b** without departing from aspects of the invention. For example, in another embodiment, first liner layer **132** may include Ni, refractory metal layer **140** may include a Ti/TiN bilayer, and metal **150** may include W. In this embodiment, first intermix regions **134a**, **134b** may include a Ni silicide and second intermix regions **154a**, **154b** may include a NiTi silicide.

As shown in FIG. 8, a laser anneal process may be performed such that a second intermix region **154a**, **154b** is formed at the interface of the bottom portion of contacts **170a**, **170b** and upper portion of source **124a** and drain **124b**. Second intermix regions **154a**, **154b** may have a thickness of 3-5 nm. The laser anneal process may transform the composition of first intermix region **134a**, **134b** (FIGS. 3-7) into a second composition resulting in second intermix

region **154a**, **154b** in place of first intermix region **134a**, **134b**. That is, the laser anneal process may crystallize intermix regions **134a**, **134b** into a lower resistance layer, i.e. a second intermix region **154a**, **154b**. Second intermix regions **154a**, **154b** may have compositions including a mixture of the materials used for source **124a** and drain **124b**, first liner layer **132**, and refractory metal layer **140**. In an embodiment where Ti is used as refractory metal layer **140**, second intermix region **154a**, **154b** may be a silicide, such as but not limited to: TiSi, NiSi, or NiPtTiSi. That is, during the laser anneal process, refractory metal **140** may diffuse into first intermix regions **134a**, **134b** (FIGS. 3-7) to create second intermix regions **154a**, **154b**. The laser anneal process may be performed at temperature of approximately 700° C.-1100° C. for between approximately 0.1 milliseconds to approximately 1 millisecond. More particularly, the laser anneal process may be performed at approximately 800° C. for approximately 0.25 milliseconds. In some embodiments, the anneal temperature and duration may be dependent on the materials being used. In one embodiment, the anneal can utilize a laser having an energy from 500 watts to 2000 watts. In another embodiment, the anneal can be performed utilizing a raster scan beam size of from 1 cm long to 100 microns wide. In one example, the anneal can be performed utilizing a 10.6 micron CO₂ laser. In order to promote the efficiency of the CO₂ laser coupling with the semiconductor substrate, a laser with a shorter wavelength (e.g., InGaAs laser at 880 nm) can be used in conjunction with the CO₂ laser.

Referring now to FIG. 9, another embodiment of the invention is shown. In this embodiment, first liner layer **132** is not removed prior to depositing refractory metal layer **140**. In this embodiment, first liner layer **132** is deposited as described with respect to FIG. 3, such that the deposition of first liner layer **132** creates first intermix region **134a**, **134b**. However, after deposition of first liner layer **132**, refractory metal layer **140** is deposited as described with respect to FIG. 5. While refractory metal layer **140** is shown as a bilayer, refractory metal layer **140** may include any number of layers of refractory metal materials, as previously discussed. Further, this embodiment also includes etching refractory metal layer **140** from the upper surface of dielectric cap layer **128** such that a portion of refractory metal layer **140** remains substantially coating the sidewalls of contact openings **130a**, **130b** and first intermix regions **134a**, **134b** at the bottom portion of contact openings **130a**, **130b** as described with respect to FIG. 6. Additionally, a metal layer **150** may be deposited and polished as described with respect to FIGS. 6-7. Lastly, this embodiment may also undergo the laser anneal process to create second intermix regions **154a**, **154b** as described with respect to FIG. 8. Second intermix regions **154a**, **154b** may include a mixture of the materials used for source and drain regions **124a**, **124b**, first liner layer **132**, and refractory metal layer **140**. This embodiment allows for a higher quantity of first liner layer **132** in contacts **170a**, **170b** due to the remaining portions of first liner layer **132** on sidewalls of contact openings **130a**, **130b**. Additionally, this embodiment results in a simpler process as the step of removing liner layer **132** is not performed, and therefore selectivity may not be a problem.

Contacts **170a**, **170b** of this embodiment may include NiPt as first liner layer **132**, a Ti/TiN bilayer for refractory metal layer **140**, and W as metal **150**. In this embodiment, first intermix regions **134a**, **134b** may include NiPt silicide and second intermix regions **154a**, **154b** may include a NiPtTi silicide. Alternatively, first liner layer **132** may

include Ti, refractory metal **140** may include TiN, and metal **150** may include W. In this embodiment, first intermix region **134a**, **134b** may include TiSi and second intermix region **154a**, **154b** may include TiSi with a higher concentration of Ti.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

The descriptions of the various embodiments of the present invention have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

We claim:

1. A semiconductor structure including a contact to a semiconductor substrate, the contact comprising:
 - a contact opening in a dielectric layer over the semiconductor substrate;

a first liner layer substantially coating a sidewall of the contact opening;

a refractory metal layer substantially coating the first liner layer in the contact opening;

a metal substantially filling the contact opening; and

an intermix region at an upper surface of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate;

wherein the refractory metal layer includes a multi-layer having at least a first refractory metal material and a second refractory metal material, the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal.

2. The contact of claim 1, wherein the intermix region includes a titanium-based silicide.

3. The contact of claim 1, wherein the first liner layer includes titanium (Ti) and the refractory metal layer includes titanium nitride (TiN).

4. The contact of claim 1, wherein the first liner layer includes nickel platinum (NiPt), the first refractory metal material includes titanium (Ti) and the second refractory metal material includes titanium nitride (TiN).

5. The contact of claim 1, wherein the metal includes tungsten (W).

6. The contact of claim 1, wherein the intermix region has a thickness of approximately 3 nanometers to approximately 5 nanometers.

* * * * *

EXHIBIT 4



US008330235B2

(12) **United States Patent**
Ramani et al.

(10) **Patent No.:** **US 8,330,235 B2**
(45) **Date of Patent:** **Dec. 11, 2012**

(54) **METHOD TO REDUCE MOL DAMAGE ON NISI**

(75) Inventors: **Karthik Ramani**, Hopewell Junction, NY (US); **Paul R. Besser**, Suunyvale, CA (US)

(73) Assignee: **Globalfoundries Inc.**, Grand Cayman (KY)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **13/096,511**

(22) Filed: **Apr. 28, 2011**

(65) **Prior Publication Data**

US 2011/0198670 A1 Aug. 18, 2011

Related U.S. Application Data

(62) Division of application No. 12/366,378, filed on Feb. 5, 2009, now Pat. No. 7,994,038.

(51) **Int. Cl.**
H01L 29/78 (2006.01)

(52) **U.S. Cl.** **257/413**; 257/383; 257/E29.156

(58) **Field of Classification Search** 257/369, 257/383, 412, E29.139, E29.242, 382, 388, 257/413, E29.156, 377

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,841,449 B1 1/2005 Bertrand et al.
7,768,072 B2 * 8/2010 Tsai et al. 257/368
2005/0095799 A1 * 5/2005 Wang et al. 438/303
2005/0247926 A1 11/2005 Sun et al.
2007/0020929 A1 * 1/2007 Purtell et al. 438/682

2007/0034967 A1 * 2/2007 Nayfeh et al. 257/369
2007/0093047 A1 * 4/2007 Okuno et al. 438/592
2008/0090369 A1 * 4/2008 Akiyama et al. 438/308
2008/0237750 A1 * 10/2008 Tsai et al. 257/412
2008/0274611 A1 * 11/2008 Cabral et al. 438/653
2009/0028744 A1 * 1/2009 Hui et al. 420/456
2010/0256405 A1 * 10/2010 Dussarrat et al. 556/32

OTHER PUBLICATIONS

Lee et al. "Fully Silicided Ni1-xPtxSi Metal Gate Electrode for p-MOSFETS", Electrochem. Solid-State Lett., vol. 8, Issue 7, pp. G156-G159, May 12, 2005.*

* cited by examiner

Primary Examiner — Matthew Landau

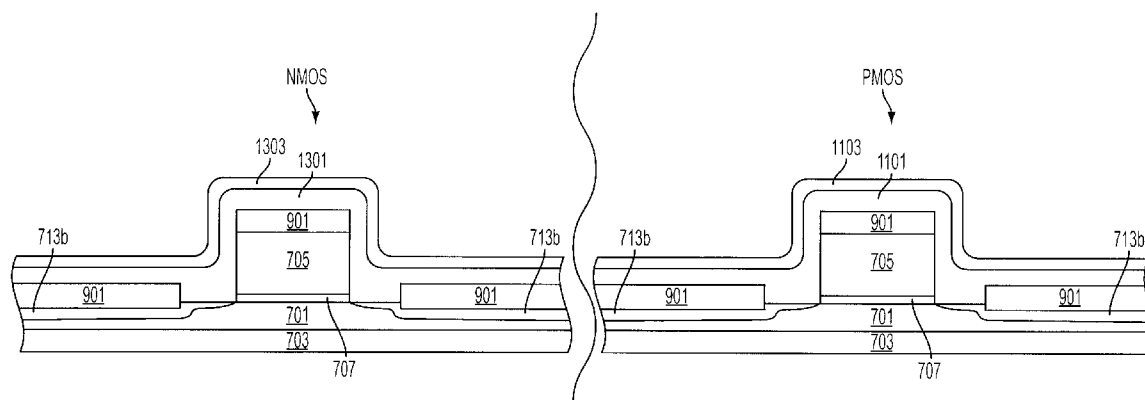
Assistant Examiner — Joseph C Nicely

(74) *Attorney, Agent, or Firm* — Ditthavong Mori & Steiner, P.C.

(57) **ABSTRACT**

Transistor devices are formed with nickel silicide layers formulated to prevent degradation upon removal of overlying stress liners. Embodiments include transistors with nickel silicide layers having a platinum composition gradient increasing in platinum content toward the upper surfaces thereof, i.e., increasing in platinum in a direction away from the gate electrode and source/drain regions. Embodiments include forming a first layer of nickel having a first amount of platinum and forming, on the first layer of nickel, a second layer of nickel having a second amount of platinum, the second weight percent of platinum being greater than the first weight percent. The layers of nickel are then annealed to form a nickel silicide layer having the platinum composition gradient increasing in platinum toward the upper surface. The platinum concentration gradient protects the nickel silicide layer during subsequent processing, as during etching to remove overlying stress liners, thereby avoiding a decrease in device performance.

20 Claims, 13 Drawing Sheets



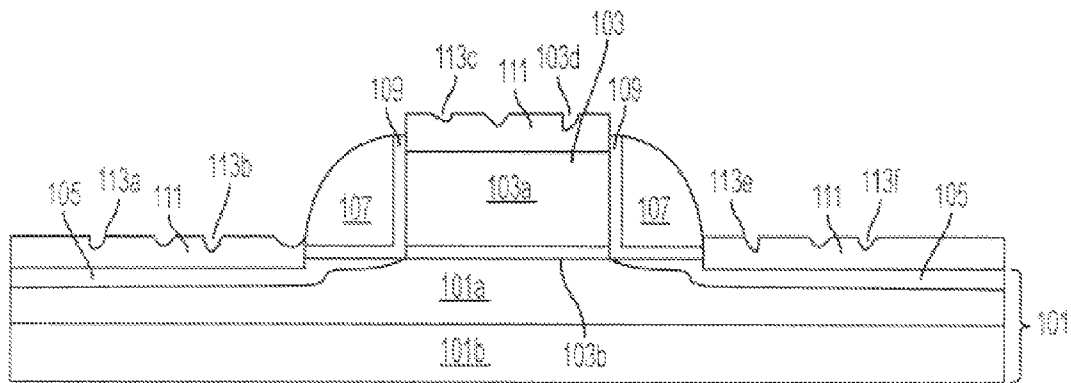


FIG. 1
(PRIOR ART)

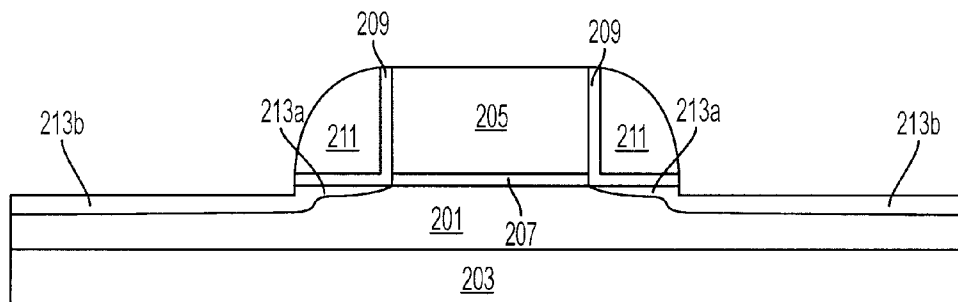


FIG. 2

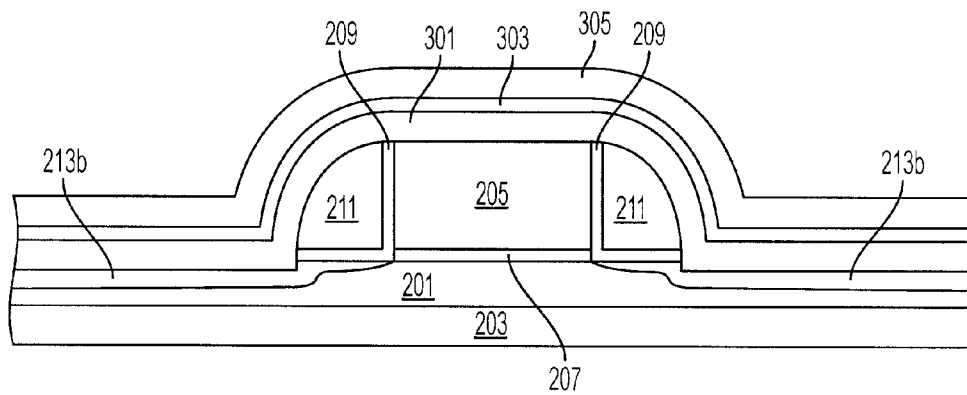


FIG. 3

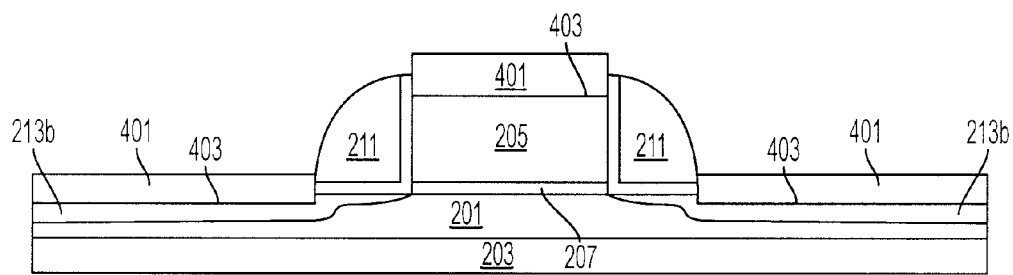


FIG. 4

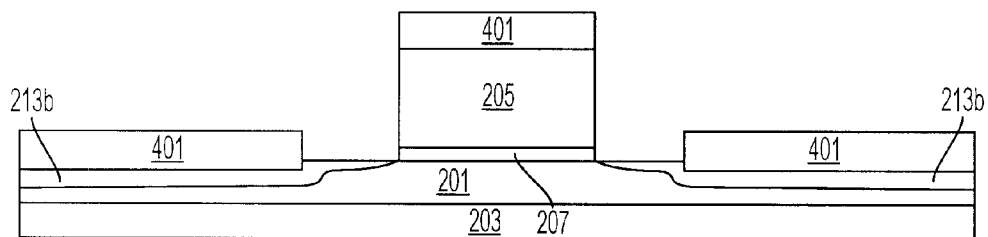


FIG. 5

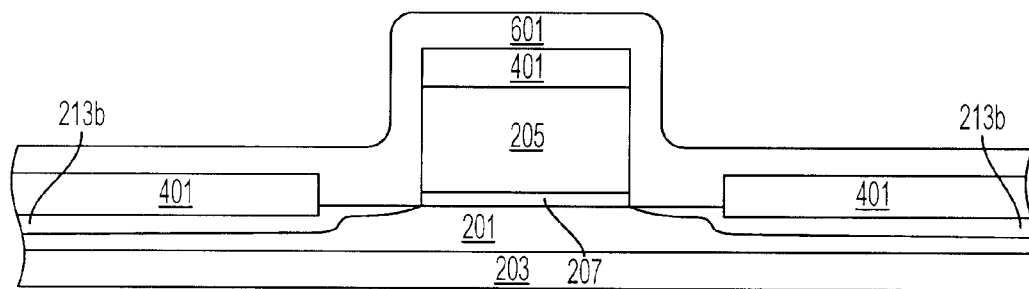


FIG. 6

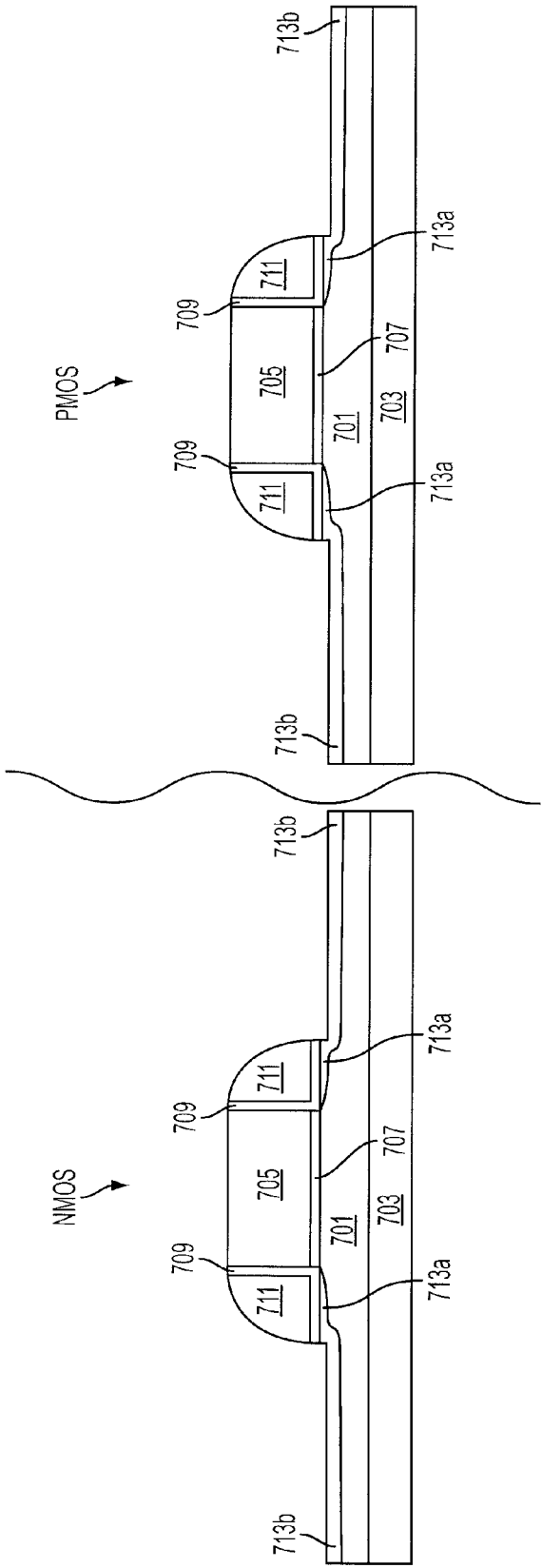


FIG. 7

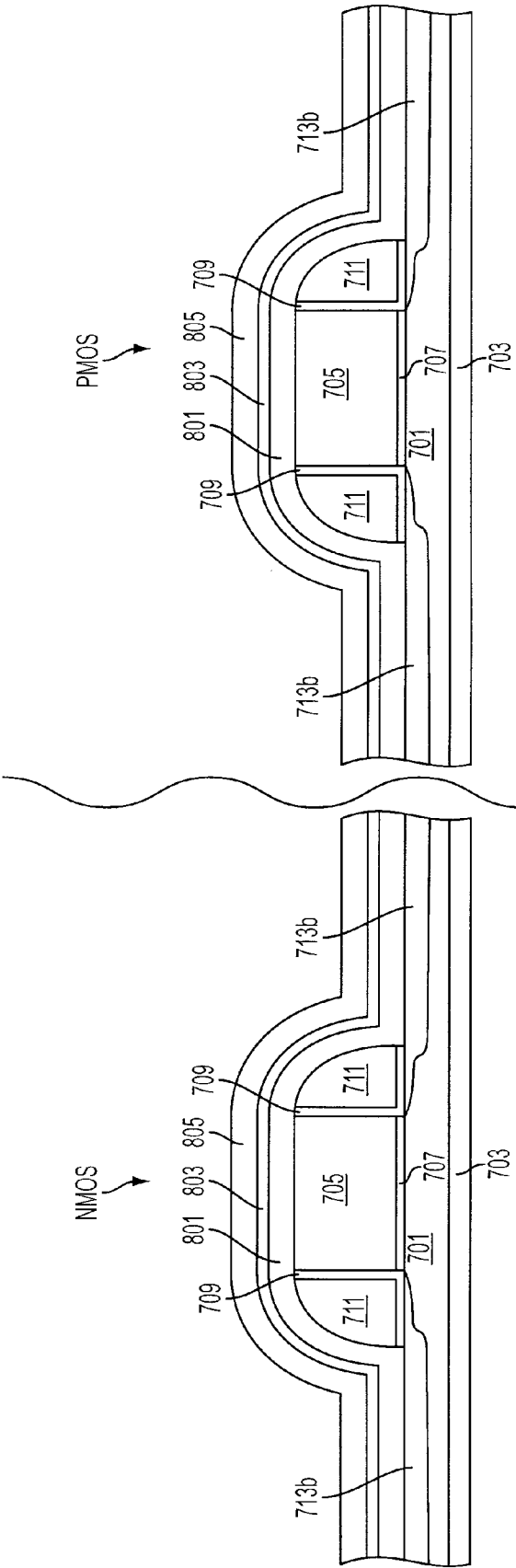


FIG. 8

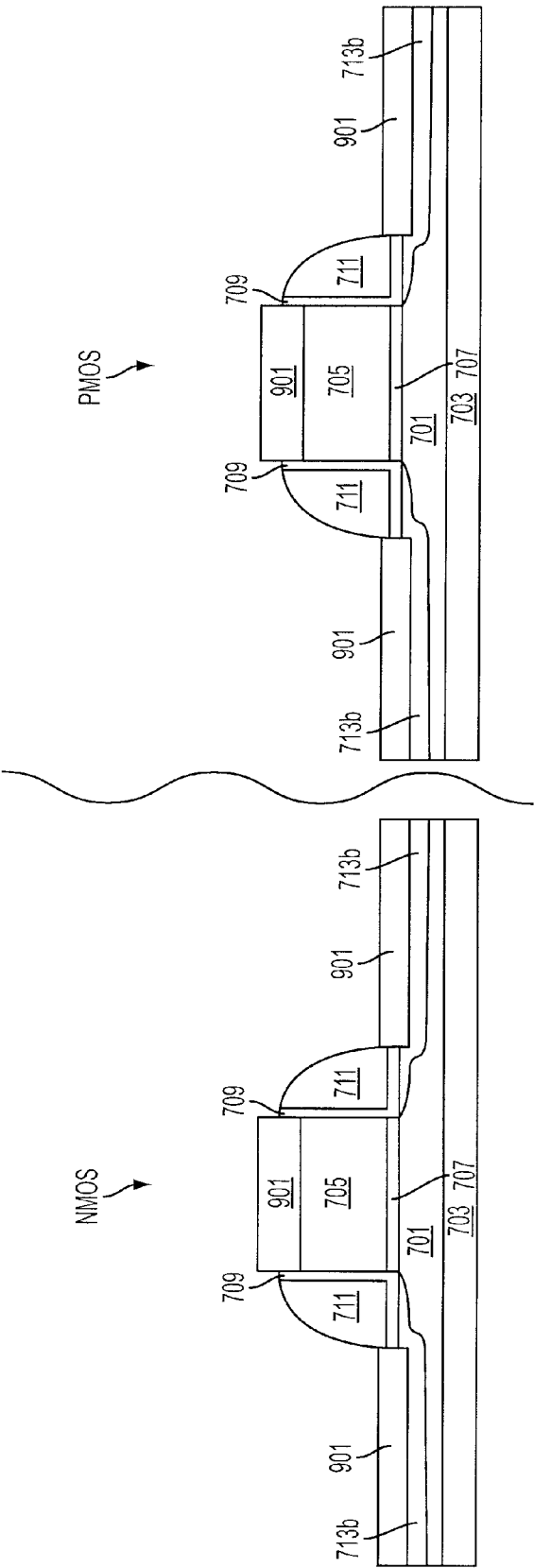


FIG. 9

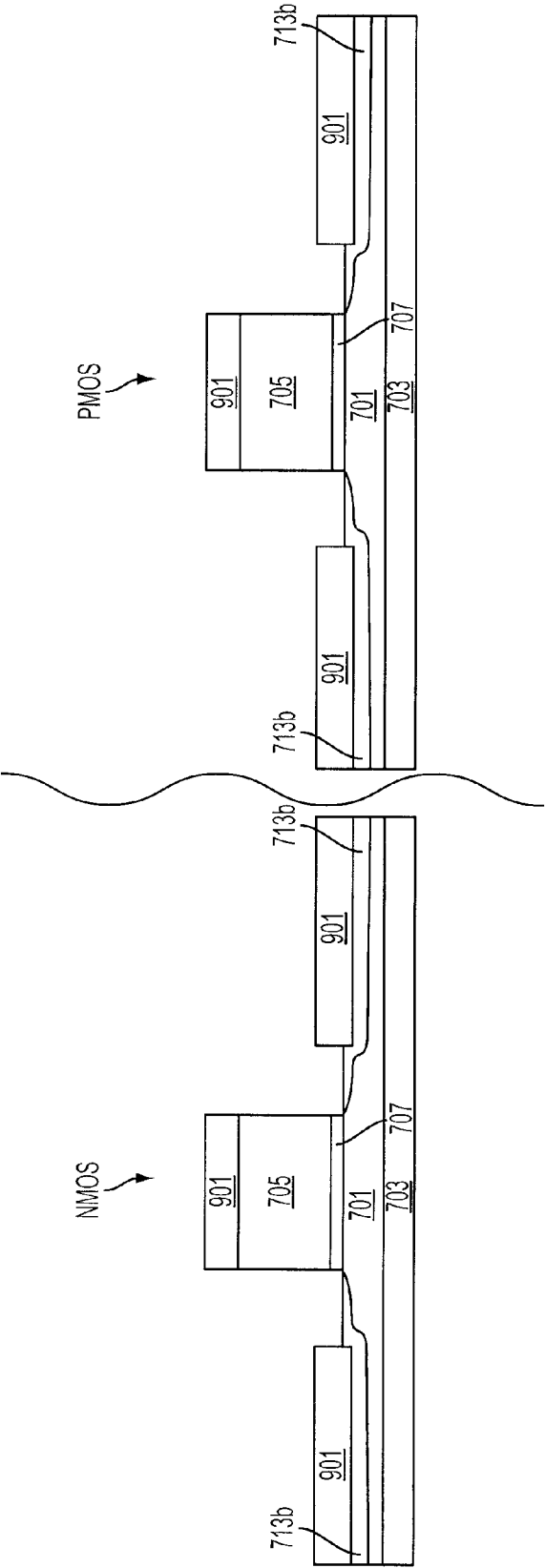
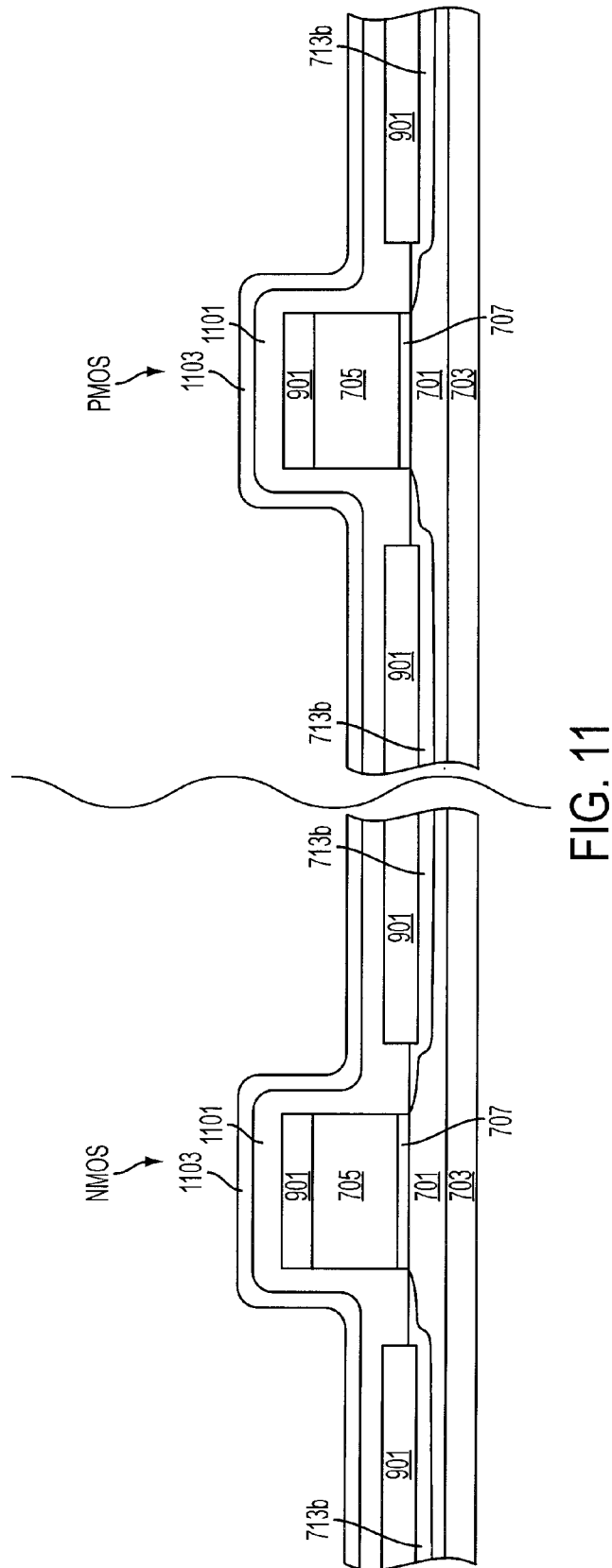


FIG. 10



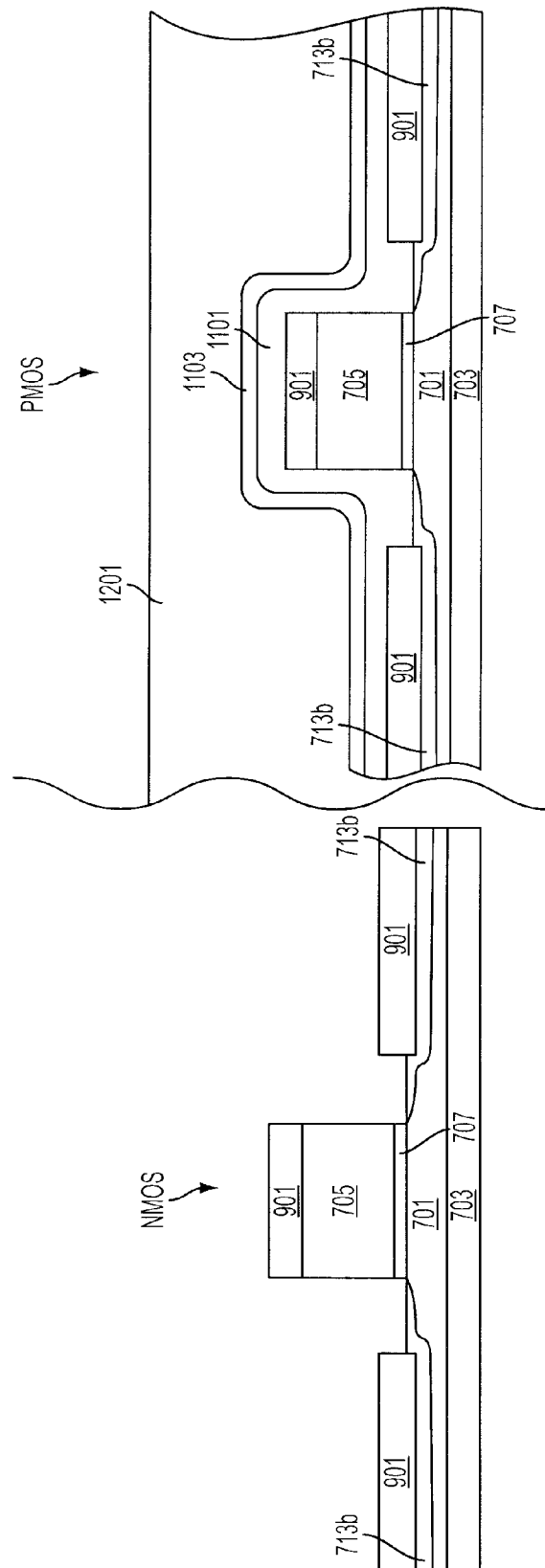


FIG. 12

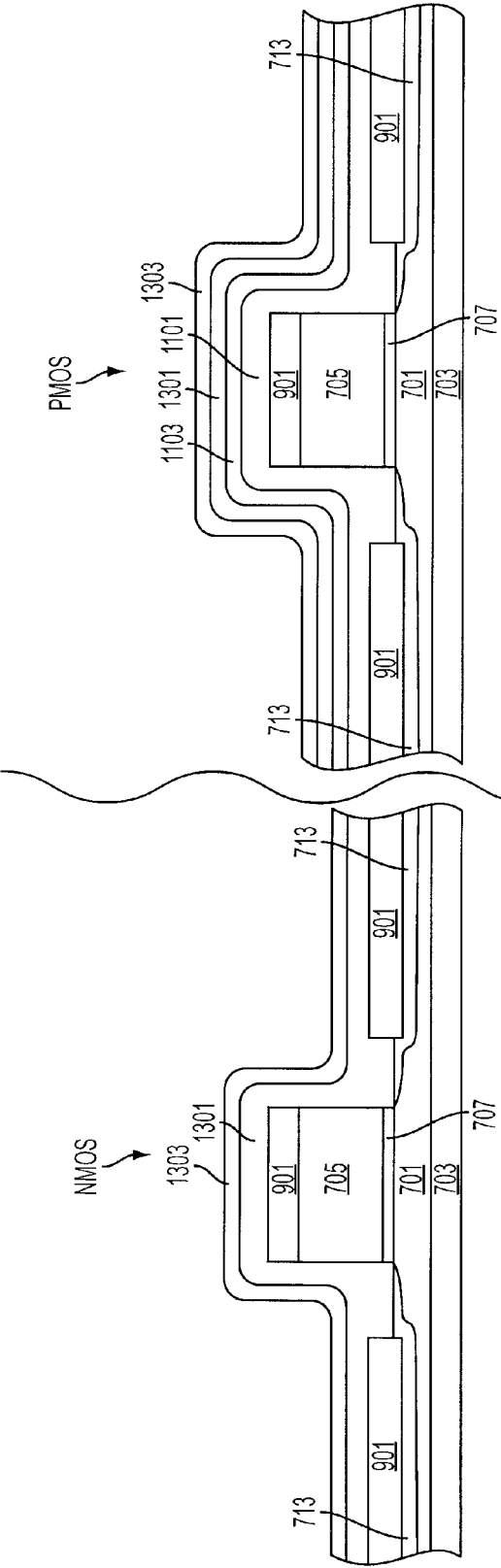


FIG. 13

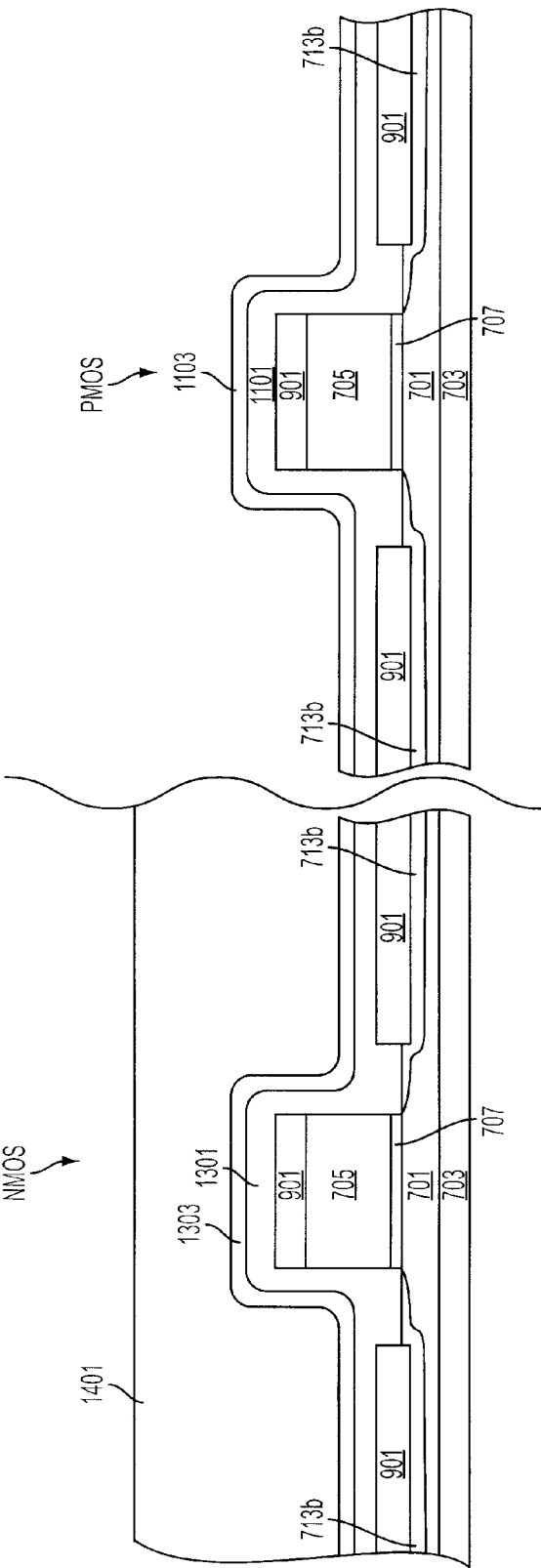


FIG. 14

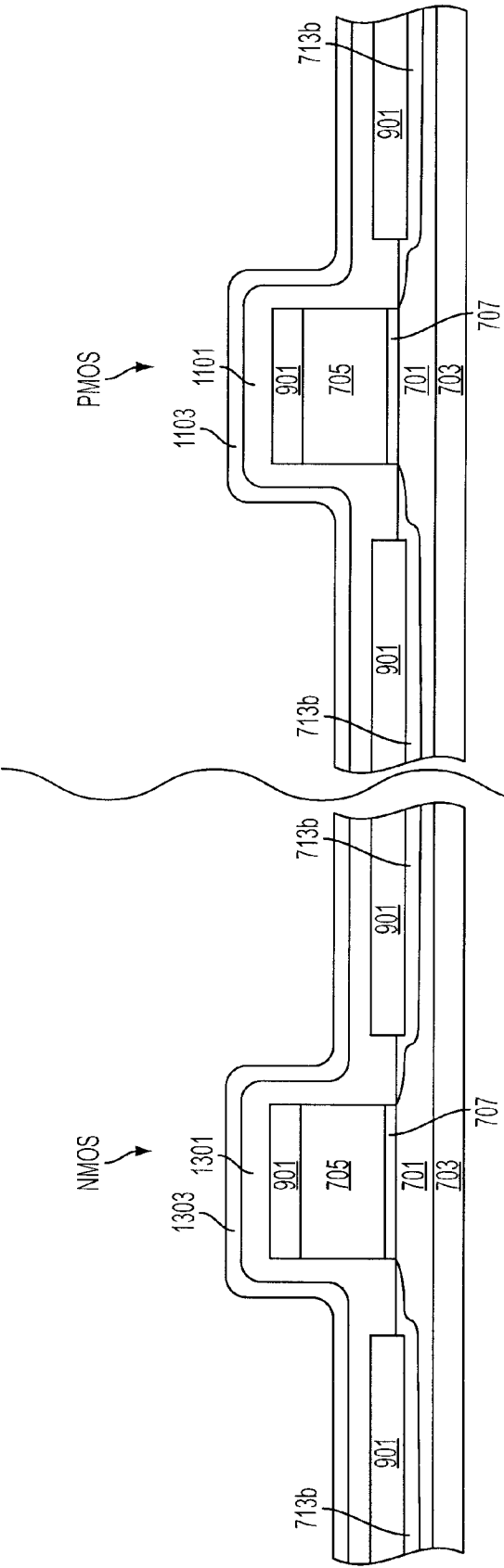


FIG. 15

1

METHOD TO REDUCE MOL DAMAGE ON NISI

RELATED APPLICATIONS

This application is a Divisional application of U.S. application Ser. No. 12/366,378 filed on Feb. 5, 2009.

TECHNICAL FIELD

The present disclosure relates to semiconductor devices with improved nickel silicide layers. The present disclosure is particularly applicable in fabricating transistors with improved device performance by avoiding nickel silicide degradation.

BACKGROUND

Conventional semiconductor devices, such as the one shown in FIG. 1, include a refractory metal deposited over all exposed upper surface features of a silicon substrate **101**, preferably by means of physical vapor deposition (PVD). The most common refractory metal employed is nickel, which forms a very low resistivity silicide with silicon and does not consume as much Si as other silicides commonly employed. NiSi has a great resistance to etchants which are used to etch silicon oxide, a common insulating material used in semiconductor devices.

The refractory metal can be sputtered by DC magnetron sputtering in an ultra-high vacuum, multi-chamber system. The formation of an MOS transistor requires the formation of a gate structure **103** and source/drain junctions **105**. The gate electrode **103** typically is formed by depositing a layer of heavily doped polysilicon **103a** on a metal oxide insulating layer **103b** and etching the layers to pattern the electrode. Sidewall spacers **107** are deposited on the opposing surfaces of the patterned gate electrode with oxide liner **109** between spacer **107** and both gate electrode **103** and substrate **101**. The sidewall spacers **107** comprise silicon oxide, silicon nitride or silicon oxynitride.

Generally, in forming the MOS transistor, the refractory metal is deposited after etching the gate electrode **103** and after forming the source/drain junction **105**, as illustrated in FIG. 1. After deposition, the refractory metal layer blankets the top surface of the gate electrode **103**, sidewall spacers **107**, oxide liners **109**, and source/drain regions **105**. As a result of thermal processing, e.g., a rapid thermal annealing (RTA) process, performed in an inert or reducing atmosphere, the refractory metal reacts with underlying silicon to form electrically conductive silicide layer portions **111** on the top surface of the polysilicon gate electrode **103** and on the exposed surfaces of the substrate where source and drain regions **105** are formed. Unreacted portions of the refractory metal layer, e.g., on the silicon oxide, nitride or oxynitride sidewall spacers **107** and the silicon oxide liner **109** are then removed, as by a wet etching process selective to the metal silicide portions **111**.

Substrates based on "strained silicon" have attracted interest as a semiconductor material which provides increased speeds of electron and hole flow therethrough, thereby permitting fabrication of semiconductor devices with higher operating speeds, enhanced performance characteristics, and lower power consumption. A layer **101a** of silicon-germanium (Si—Ge) is formed on a suitable crystalline substrate, e.g., a Si wafer or a silicon-on-insulator (SOI) wafer **101b**. The Si—Ge layer has a greater lattice constant (spacing of Si and Ge atoms therein) relative to the underlying Si. As a

2

consequence, the Si—Ge layer has a compressive strain. Electrons in such compressively strained layers have greater mobility than in conventional Si layers with smaller inter-atom spacings, i.e., there is less resistance to electron flow.

For example, electron flow may be up to about 70% faster compared to electron flow in conventional Si. Transistors and IC devices formed with such strained Si layers can exhibit operating speeds up to about 35% faster than those of equivalent devices formed with conventional Si, without necessity for reduction in transistor size. Conventional practices based on strained silicon technology also involve epitaxially growing a relaxed silicon layer on a tensilely stressed silicon layer which is subsequently doped to form relaxed source/drain regions in the relaxed silicon layer.

Conventional semiconductor devices typically comprise a plurality of active devices in or on a common semiconductor substrate, e.g., CMOS devices comprising at least a pair of PMOS and NMOS transistors in spaced adjacency. The mobility of electrons is faster than the mobility of holes in conventional bulk silicon substrates. Accordingly, in conventional CMOS transistors, the drive current of the PMOS transistor is less than the drive current of the NMOS transistor creating an imbalance. This imbalance is exacerbated in CMOS transistors fabricated on or within a tensilely stressed active device area formed in a strained lattice semiconductor substrate, e.g., strained Si—Ge on Si, because the increase in electron mobility is greater than the increase in hole mobility.

Channel carrier mobility of transistors based on strained Si substrates can be increased by applying a stress thereto. In forming P-channel transistors, channel carrier mobility is enhanced by applying a stressed dielectric layer exhibiting high compressive stress for increasing electron mobility. In N-channel transistors, channel carrier mobility is significantly increased by applying a stressed layer exhibiting high tensile stress for increasing hole mobility. Stressed dielectric layers, called stress liners, are applied to transistors wherein the source/drain regions are formed within the strained Si layer, and to transistors having relaxed source/drain regions formed on strained Si layers. The stress liner may comprise silicon carbide, silicon nitride or silicon oxynitride, and may be deposited by plasma enhanced chemical vapor deposition (PECVD) at a thickness of about 200 Å to about 1000 Å. Conventional PECVD conditions may be employed for deposition of a highly compressive layer or highly tensile dielectric layer. In depositing a stress liner exhibiting high compressive stress, both high frequency and low frequency power are applied. When depositing a stressed dielectric layer exhibiting high tensile stress, the low frequency power is significantly reduced.

Because the stress liners differ for P-channel and N-channel transistors, the stressed dielectric layers are typically patterned by reactive ion etching, which involves physically bombarding the surfaces to subtractively remove material. This etching process involves an "overetch" in order to clear the residue left behind. It is during this overetch process that the underlying NiSi layers are damaged, causing degradation of the NiSi. The degradation can be morphological, wherein mouse bites or missing NiSi is observed, as illustrated in FIG. 1 at **113a-113f**; or electrical, wherein the resistivity is increased. Either or both types of degradation ultimately result in poor device parameters such as higher sheet resistance and external resistance.

A need therefore exists for methodology enabling fabrication of semiconductor devices comprising transistors with degradation resistant NiSi layers and for the resulting improved semiconductor devices having reduced resistance and improved performance.

An aspect of the present disclosure is a semiconductor device comprising a transistor with reduced resistance.

Another aspect of the present disclosure is an efficient method of fabricating a semiconductor device comprising a transistor with reduced resistance.

Additional aspects and other features of the present disclosure will be set forth in the description which follows and in part will be apparent to those having ordinary skill in the art upon examination of the following or may be learned from the practice of the present disclosure. The advantages of the present disclosure may be realized and obtained as particularly pointed out in the appended claims.

According to the present disclosure, some technical effects may be achieved in part by a method comprising: forming a transistor on a substrate, the transistor including source/drain regions and a gate electrode; forming a first layer of nickel containing a first amount of platinum over the transistor; forming a second layer of nickel containing a second amount of platinum on the first layer of nickel, the second amount of platinum being greater than the first amount of platinum; and annealing the first and second layers of nickel to form a layer of NiSi over the source/drain regions and the gate electrode, wherein the NiSi layer has a platinum composition gradient with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode.

Aspects of the present disclosure include forming the first and second layers of nickel at a combined thickness of about 45 Å to about 100 Å with the first layer of nickel at a thickness of about 60% to about 80% of the combined thickness. Aspects also include annealing to form the layer of NiSi at a thickness of about 100 Å to about 220 Å. Further aspects include annealing by heating at a temperature of about 340° C. to about 450° C., such as at a temperature of about 360° C. to about 400° C. This technique can be used to effect annealing in a single step. Another aspect includes annealing by heating at a first stage at a first temperature of about 280° C. to about 350° C., for example about 300° C. to about 320° C., followed by heating at a second stage at a second temperature of about 360° C. to about 450° C., for example about 380° C. to about 420° C. Additional aspects include forming the first layer of nickel with a platinum content of about 8 weight % ("wt. %") to about 12 wt. %, for example about 10 wt. %, and the second layer of nickel with a platinum content of about 30 wt. % to about 70 wt. %, for example about 30 wt. % to about 50 wt. %. Aspects also include forming a capping layer of titanium nitride over the second layer before annealing, and removing the capping layer after annealing, followed by etching to remove unreacted nickel.

Another aspect of the present disclosure is a semiconductor device comprising: a silicon substrate; a transistor, comprising source/drain regions and a gate electrode, on the substrate; and a NiSi layer on the gate electrode and source/drain regions, the NiSi layer comprising platinum having a composition gradient, the platinum increasing in amount in a direction away from the source/drain regions and the gate electrode.

Aspects include devices with a P-channel transistor, an N-channel transistor, a first NiSi layer on the P-channel transistor, a second NiSi layer on the N-channel transistor, the first and second NiSi layers each comprising platinum having a concentration gradient with platinum increasing in amount toward the upper surface, i.e., increasing in amount in a direction away from the P-channel and N-channel transistors, a first stress liner on the P-channel transistor, and a second stress liner on the N-channel transistor. Further aspects

include the NiSi layers having a thickness of about 100 Å to about 220 Å, for example about 100 Å to about 150 Å.

Another aspect of the present disclosure is a method of fabricating a semiconductor device, the method comprising: forming a P-channel and an N-channel transistor on a silicon substrate, each transistor including a polysilicon gate electrode and source/drain regions formed in the silicon substrate; depositing a first layer of nickel containing a first amount of platinum on both transistors; depositing a second layer of nickel having a second amount of platinum on the first layer of nickel, the second amount of platinum being greater than the first amount of platinum; depositing a titanium nitride capping layer on the second layer of nickel, annealing the first and second layers of nickel to form a single layer of NiSi containing platinum having a concentration gradient with the platinum increasing in amount in a direction away from the gate electrode and source/drain regions; etching to remove the capping layer and unreacted nickel; forming a first stress liner exhibiting high compressive stress over both transistors; selectively removing the first stress liner from the N-channel transistor; forming a second stress liner exhibiting high tensile stress over both transistors; and selectively removing the second stress liner from the P-channel transistor.

Additional aspects and technical effects of the present disclosure will become readily apparent to those skilled in the art from the following detailed description wherein embodiments of the present disclosure are described simply by way of illustration of the best mode contemplated to carry out the present disclosure. As will be realized, the present disclosure is capable of other and different embodiments, and its several details are capable of modifications in various obvious respects, all without departing from the present disclosure. Accordingly, the drawings and description are to be regarded as illustrative in nature, and not as restrictive.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure is illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawing and in which like reference numerals refer to similar elements and in which:

FIG. 1 schematically illustrates a conventional transistor device with a degraded NiSi layer;

FIGS. 2 through 6 schematically illustrate sequential steps of a method in accordance with an embodiment of the present disclosure;

FIGS. 7 through 15 schematically illustrate sequential steps of a method in accordance with another embodiment of the present disclosure.

DETAILED DESCRIPTION

In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of exemplary embodiments. It should be apparent, however, that exemplary embodiments may be practiced without these specific details or with an equivalent arrangement. In other instances, well-known structures and devices are shown in block diagram form in order to avoid unnecessarily obscuring exemplary embodiments.

The present disclosure addresses and solves the problem of NiSi layer degradation during subsequent processing, as during middle of the line processing to remove overlying stress liners, which NiSi layer degradation causes resistance problems and adversely affects transistor performance. In accordance with embodiments of the present disclosure, a transis-

tor device is provided having a NiSi layer formulated to resist degradation, advantageously using methodology that does not impact process flow. In accordance with embodiments of the present disclosure, NiSi layer damage is prevented or significantly reduced by formulating the NiSi layer with a platinum composition gradient with the amount of platinum increasing in a direction toward its upper surface, i.e., in a direction away from the source/drain regions and the gate electrode, thereby shielding the NiSi layer from physical damage during post NiSi processing as when removing an overlying stress liner by etching.

Embodiments of the present disclosure include forming two layers of nickel on a silicon substrate and a silicon gate electrode, the first layer having a relatively low concentration of platinum and the second having a relatively high concentration of platinum, and annealing the two layers to form a single layer of NiSi. During annealing, platinum diffuses to form a composition gradient with the amount of platinum increasing in a direction away from the substrate and gate electrode. It was found that such a concentration gradient of platinum effectively shields the NiSi layer from physical damage during subsequent processing, e.g., etching, while concomitantly maintaining the integrity of the NiSi layer.

Methodology in accordance with embodiments of the present disclosure includes forming a transistor on a substrate, the transistor including source/drain regions and a gate electrode, depositing a first layer of nickel containing a relatively low amount of platinum, for example about 8 wt. % to about 12 wt. %, e.g., about 10 wt. %, over the transistor, forming a second layer of nickel containing a relatively high amount of platinum, such as about 30 wt. % to about 70 wt. %, e.g., about 30 wt. % to about 50 wt. %, on the first layer. The first and second layers of nickel are formed at a combined thickness of about 45 Å to about 100 Å, with the first layer of nickel at a thickness of about 60% to about 80% of the combined thickness. A capping layer of silicon nitride is formed over the second nickel layer. An RTA process is then implemented to convert the platinum containing nickel layers into a single layer of NiSi over the source/drain regions and the gate electrode, such that the NiSi layer has a platinum composition gradient with the amount of platinum increasing in a direction away from the source/drain regions and the gate electrode. The resulting platinum concentration gradient protects the NiSi film without increasing the resistivity of the layer.

The annealing process may be controlled to form a layer of NiSi at a thickness of about 100 Å to about 220 Å. Annealing may comprise a single heating stage at a temperature of about 340° C. to about 450° C., such as at a temperature of about 360° C. to about 400° C. Alternatively, a two stage annealing process can be performed, as by heating at a first temperature of about 280° C. to about 350° C., such as about 300° C. to about 320° C., followed by heating at a second temperature of about 360° C. to about 450° C., such as about 380° C. to about 420° C. In accordance with embodiments of the present disclosure, the capping layer is removed after annealing followed by etching to remove unreacted nickel, employing conventional methodology.

Embodiments of the present disclosure include forming two transistors, one P-channel and one N-channel transistor. A first stress liner exhibiting high compressive stress can be formed over both transistors, and the first stress liner is selectively removed, as by etching, from the N-channel transistor. A second stress liner exhibiting high tensile stress can be subsequently formed over both transistors, and the second stress liner is selectively removed, as by etching, from the P-channel transistor. During etching of the stress liners, the

thus formed platinum concentration gradient prevents the NiSi from being etched and damaged.

Still other aspects, features, and technical effects will be readily apparent to those skilled in this art from the following detailed description, wherein preferred embodiments are shown and described, simply by way of illustration of the best mode contemplated. The disclosure is capable of other and different embodiments, and its several details are capable of modifications in various obvious respects. Accordingly, the drawings and description are to be regarded as illustrative in nature, and not as restrictive.

A method in accordance with an embodiment of the present disclosure is schematically illustrated in FIGS. 2 through 5, wherein like reference characters denote like features. Adverting to FIG. 2, strained Ge—Si layer 201 is formed over Si layer 203. Ge—Si layer 201 may be globally strained or locally strained under the source/drain regions. Gate electrode 205 is formed over strained Ge—Si layer 201 with a gate dielectric 207 therebetween. Lightly doped source/drain regions 213a are formed in Ge—Si layer 201 using the gate electrode 205 as a mask in a conventional manner. A sidewall spacer comprising a silicon oxide liner 209, as at a thickness of about 60 Å to about 600 Å, is formed on side surfaces of the gate electrode 205 and on a portion of the upper surface of strained Ge—Si layer 201. Silicon nitride spacers 211 are then formed on the silicon oxide liner 209 as by employing PECVD followed by etching.

It should be appreciated that liner 209 may be deposited by ALD and may also comprise silicon nitride. Silicon oxide liner 209 advantageously prevents consumption of the gate electrode 205 by silicidation on the side surfaces thereof, and advantageously prevents a subsequently formed thin layer of NiSi on the silicon nitride sidewall spacers 211 from contacting a NiSi contact layer on the upper surface of the gate electrode 205 and/or from contacting NiSi contact layers on the upper surface of the strained Si layer 201, thereby preventing NiSi bridging along the silicon nitride sidewall spacers.

Subsequently, heavily doped source/drain regions 213b are formed in Ge—Si layer 201 using gate electrode 205 and sidewall spacers 209 and 211 as a mask, employing conventional implantation techniques, as by ion implanting boron, phosphorus, or arsenic.

Adverting to FIG. 3, a first layer of nickel 301 is deposited over the source/drain regions 213b, the spacers 211, the oxide liner 209, and the gate electrode 205. First layer of nickel 301 may be formed, for example, with a platinum content of about 8 wt. % to about 12 wt. %, such as about 10 wt. %. The first layer of nickel can be formed by sputtering using a target comprising Ni and 10 wt. % Pt. Over the first layer of nickel 301, a second layer of nickel 303 is deposited. The second layer of nickel 303 may be formed, for example, with a platinum content of about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %. The second layer of nickel can be sputtered using a target of Ni containing an amount of Pt within the range of 30 to 100%. The first nickel layer 301 and second nickel layer 303 may be formed to a combined thickness of about 45 Å to about 100 Å using a conventional deposition tool, for example on an AMAT Endura sputtering system, at a power of 1500 W to 2500 W, e.g., 2000 W, and using an Argon flow of 20 sccm to 40 sccm, e.g., 2000 sccm, or an Argon and Hydrogen mixture with a flow of 3 sccm to 10 sccm, e.g., 5 sccm. The first layer of nickel 301 may be formed at a thickness of about 60% to about 80% of the combined thickness, such as about 70% of the combined thickness.

A capping layer **305** may be formed over the second nickel layer **303** to prevent oxidation of the underlying metal. Capping layer **305** may be titanium nitride.

RTA and etch processes are performed to form a NiSi layer **401** over the source/drain regions **213b** and the gate electrode **205**, as illustrated in FIG. 4. The resulting NiSi layer **401** has a composition gradient of platinum from about 8 wt. % to about 12 wt. %, such as about 10 wt. %, at the silicon interface **403** to about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %, at the opposite surface. NiSi layer **401** may be formed at a thickness of about 100 Å to about 220 Å, such as at a thickness of about 100 Å to about 150 Å.

Nickel silicide is commonly doped with platinum to improve the material and electrical properties. Platinum has proven to be effective in preventing the nickel disilicide formation (NiSi₂). It was found that strategic platinum doping can also be used to effectively shield a NiSi layer from physical damage during MOL processing (post NiSi processing) while concomitantly maintaining the NiSi integrity. However, platinum also increases the resistivity of NiSi. In accordance with embodiments of the present disclosure, effective shielding of a NiSi layer is achieved while avoiding a problematic increase in resistivity by compositionally grading a platinum doped NiSi layer.

The RTA process may be a one-step or two-step process (with a clean in between the two steps) in an inert or reducing atmosphere, e.g., a nitrogen (N₂)-containing atmosphere. The single step anneal may be implemented at a temperature of about 340° C. to about 450° C., for example at a temperature of about 360° C. to about 400° C., for about 5 to 35 seconds. During the single step anneal the nickel layers are converted directly to a low resistivity monosilicide (NiSi).

The two-step anneal may be conducted by first heating at a temperature of about 280° C. to about 350° C., for example about 300° C. to about 320° C., e.g., 300° C., during which metal rich silicide, Ni₂Si, is formed. During the second step, the temperature may be raised to about 360° C. to about 450° C., such as about 380° C. to about 420° C., e.g., 420° C. During the second annealing step, the metal rich silicide is converted to monosilicide. The total time for the two step annealing can be the same as for the single step anneal process, for example 5 seconds for the first step and 30 seconds for the second step. The two-step anneal therefore reduces the time at which annealing occurs at a high temperature, thereby reducing the rough interfaces between the NiSi and the underlying Si layer.

After NiSi layer **401** is formed, the capping layer **305** and unreacted Ni on the wafer are then easily removed, as by wet chemical stripping. In an embodiment of the present disclosure, the capping layer and unreacted Ni are removed by an anisotropic sputter etching or by immersing the wafer into a solution of H₂SO₄, H₂O₂, and water (SPM) or a solution of NH₄OH, H₂O₂ and water (APM).

As illustrated in FIG. 5, the liner **209** and sidewall spacers **211** are then removed, exposing a portion of the upper surface of strained Si layer **201** between the silicide layers **401** and the side surfaces of gate electrode **205**, with a very thin oxide layer (not shown) thereon, e.g., less than 50 Å, serving as a buffer layer. A highly stressed dielectric layer **601**, such as a silicon nitride layer exhibiting high compressive stress, is then deposited by PECVD, as shown in FIG. 6. The highly stressed dielectric layer **601** serves to increase channel electron mobility, thereby increasing the drive current.

Another embodiment of the present disclosure is schematically illustrated in FIGS. 7 through 15. Adverting to FIG. 7, a CMOS device is schematically illustrated comprising an NMOS transistor portion at the left and a PMOS transistor

portion on the right, wherein similar features are denoted by similar reference characters. Strained Si—Ge layer **701** is formed over Si layer **703**. As in previous discussed embodiments, Si—Ge layer **701** may be globally strained or locally strained in the source/drain regions. Gate electrodes **705** are formed over strained Si—Ge layer **701** with gate dielectric **707** therebetween. Lightly doped source/drain regions **713a** are formed in Si—Ge layer **701** using gate electrodes **705** as a mask in a conventional manner. Sidewall spacers comprising oxide liner **709**, as at a thickness of about 60 Å to about 600 Å, are formed on side surfaces of gate electrodes **705** and on portions of the upper surface of strained Si—Ge layer **701**. Silicon oxide liner **709** can be formed in the same manner as discussed with respect to silicon oxide liner **209** in FIG. 2. Silicon nitride spacers **711** are then formed on silicon oxide liners **709** as by employing PECVD followed by etching.

Subsequently, heavily doped source/drain regions **713b** are formed in Si—Ge layer **701** using gate electrodes **705** and sidewall spacers **709** and **711** as a mask, employing conventional implantation techniques, as by ion implanting boron, phosphorus, or arsenic.

As illustrated in FIG. 8, a first layer of nickel **801** is deposited over the source/drain regions **713b**, spacers **711**, oxide liners **709**, and gate electrodes **705**. First layer of nickel **801** may be formed, for example, with a platinum content of about 8 wt. % to about 12 wt. %, such as about 10 wt. %. Over the first layer of nickel **801**, a second layer of nickel **803** is deposited. Second layer of nickel **803** may be formed, for example, with a platinum content of about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %. The first nickel layer **801** and second nickel layer **803** may be formed in a conventional deposition environment, for example on an AMAT Endura sputtering system, to a combined thickness of about 45 Å to about 100 Å. The first layer of nickel **801** may be formed at a thickness of about 60% to about 80% of the combined thickness, such as about 70% of the combined thickness.

A capping layer **805** is formed over the second nickel layer **803** to prevent oxidation of the underlying metal. Capping layer **805** may be titanium nitride.

RTA and etch processes are performed to form NiSi layers **901** over source/drain regions **713b** and gate electrodes **705**, as illustrated in FIG. 9. The resulting NiSi layers **901** have a composition gradient of platinum from about 8 wt. % to about 12 wt. %, such as about 10 wt. %, at the bottom surface that contacts the silicon gate electrodes **705** and source/drain regions **713b**, to about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %, at the opposite surface. NiSi layers **901** may be formed at a thickness of about 100 Å to about 220 Å, such as at a thickness of about 100 Å to about 150 Å. The RTA process may be a one-step or two-step process, as described above with respect to FIG. 4.

After NiSi layers **901** are formed, the capping layer **805** and unreacted Ni on the wafer are then easily removed, as by wet chemical stripping. In an embodiment of the present disclosure, the capping layer and unreacted Ni are removed by an anisotropic sputter etching or by immersing the wafer into a solution of H₂SO₄, H₂O₂, and water (SPM) or a solution of NH₄OH, H₂O₂ and water (APM).

As illustrated in FIG. 10, liners **709** and sidewall spacers **711** are then removed exposing portions of the upper surface of strained Si layer **701** between the silicide layers **901** and the side surfaces of gate electrodes **705**, with a very thin oxide layer (not shown) thereon, e.g., less than 50 Å, serving as a buffer layer.

A highly compressive stressed silicon nitride film **1101**, e.g., having a compressive stress greater than 1.5 GPa, is then

deposited over both the NMOS and PMOS transistors as shown in FIG. 11. Deposition of highly compressive stressed silicon nitride film 1101 may be implemented in a conventional manner. Multiple layer deposition and plasma treatment further increases compressive stress. Subsequently, a thin oxide or oxynitride film 1103 is deposited, as illustrated in FIG. 11, by a conventional CVD process. Typically oxide or oxynitride film 1103 is deposited at a thickness of about 30 Å to about 60 Å.

Subsequently, a mask 1201, e.g., photoresist or hard mask is applied over the PMOS transistor, as illustrated in FIG. 12, and the oxide or oxynitride film 1103 and high compressive stress silicon nitride film 1101 are removed from the NMOS transistor.

Adverting to FIG. 13, the mask 1201 is removed from the PMOS transistor. A highly tensile stressed silicon nitride film 1301, having a tensile stress of greater than 1.5 GPa, is then deposited over both the PMOS and NMOS transistors. Deposition of highly tensile stressed film 1301 can be implemented in a conventional manner. A thin oxide or oxynitride film 1303 is then deposited by a conventional CVD process, as at a thickness of about 30 Å to about 60 Å.

Adverting to FIG. 14, a mask 1401, such as a photoresist or hard mask, is then applied over the NMOS transistor, and the oxide or oxynitride film 1303 and the highly tensile stressed silicon nitride film 1301 are selectively removed from the PMOS transistor stopping on the oxide or oxynitride film 1103. The mask 1401 is then removed, and the resulting structure is illustrated in FIG. 15 which comprises an oxide or oxynitride film 1303 and a highly tensile stressed silicon nitride film 1301 over the NMOS transistor and an oxide or oxynitride film 1103 and a highly compressive stressed silicon nitride film 1101 over the PMOS transistor. The resulting CMOS device comprises both PMOS and NMOS transistors with increased channel carrier mobility and, hence, increased drive current, without increasing the resistivity of the transistors.

The embodiments of the present disclosure can achieve several technical effects, including resistance reduction, improved transistor and circuit performance, and reduced power requirements. The present disclosure enjoys industrial applicability in any of various types of highly integrated semiconductor devices.

In the preceding description, the present disclosure is described with reference to specifically exemplary embodiments thereof. It will, however, be evident that various modifications and changes may be made thereto without departing from the broader spirit and scope of the present disclosure, as set forth in the claims. The specification and drawings are, accordingly, to be regarded as illustrative and not as restrictive. It is understood that the present disclosure is capable of using various other combinations and embodiments and is capable of any changes or modifications within the scope of the inventive concept as expressed herein.

What is claimed is:

1. A semiconductor device comprising:

a silicon substrate,

a transistor, comprising source/drain regions and a gate electrode, on the substrate; and

a nickel silicide layer on the gate electrode and source/drain regions,

wherein the nickel silicide layer comprises platinum having a composition gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direc-

tion away from the source/drain regions and the gate electrode, from between about 8 wt. % and about 12 wt. % at the silicon interface.

2. The semiconductor device according to claim 1, wherein the concentration gradient of platinum in the nickel silicide layer ranges from between about 8 wt. % and about 12 wt. % to between about 30 wt. % and about 100 wt. %.

3. The semiconductor device according to claim 2, wherein the concentration gradient of platinum in the nickel silicide layer ranges from about 10 wt. % to between about 30 wt. % and about 50 wt. %.

4. The semiconductor device according to claim 1, wherein the nickel silicide layer has a thickness of about 100 Å to about 220 Å.

5. The semiconductor device according to claim 1, wherein the nickel silicide layer has a thickness of about 100 Å to about 150 Å.

6. The semiconductor device according to claim 1, wherein the silicon substrate comprises a strained silicon germanium (Si—Ge) layer on a silicon layer.

7. The semiconductor device according to claim 1, further comprising a stress liner on the nickel silicide layer.

8. The semiconductor device according to claim 7, further comprising an oxide or oxynitride layer on the stress liner.

9. The semiconductor device according to claim 8, wherein the stress liner comprises stressed silicon nitride.

10. A semiconductor device comprising:

a silicon substrate;

a P-channel transistor and an N-channel transistor, each comprising source/drain regions and a gate electrode, on the silicon substrate; and

a first nickel silicide layer on the gate electrode and source/drain regions of the P-channel transistor; and

a second nickel silicide layer on the gate electrode and source/drain regions of the N-channel transistor,

wherein each of the first and second nickel silicide layers comprises platinum having a concentration gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the P-channel and N-channel transistors, from between about 8 wt. % and about 12 wt. % at the silicon interface.

11. The semiconductor device according to claim 10, further comprising:

a first stress liner on the P-channel transistor; and

a second stress liner on the N-channel transistor.

12. The semiconductor device according to claim 11, wherein the first stress liner exhibits high compressive stress and the second stress liner exhibits high tensile stress.

13. The semiconductor device according to claim 12, wherein the concentration gradient of platinum in each nickel silicide layer ranges from between about 8 wt. % and about 12 wt. % to between about 30 wt. % and about 100 wt. %.

14. The semiconductor device according to claim 13, wherein the concentration gradient of platinum in each nickel silicide layer ranges from about 10 wt. % to between about 30 wt. % and about 50 wt. %.

15. The semiconductor device according to claim 12, wherein each nickel silicide layer has a thickness of about 100 Å to about 220 Å.

16. The semiconductor device according to claim 15, wherein each nickel silicide layer has a thickness of about 100 Å to about 150 Å.

17. The semiconductor device according to claim 12, wherein the silicon substrate comprises a strained silicon germanium (Si—Ge) layer on a silicon layer.

11

18. The semiconductor device according to claim 12, further comprising an oxide or oxynitride layer on the first and second stress liners.

19. The semiconductor device according to claim 18, wherein the first and second stress liners comprise stressed silicon nitride. 5

20. A semiconductor device comprising:

- a silicon substrate;
- a strained silicon germanium (Si—Ge) layer on the silicon substrate; 10
- a P-channel transistor comprising source/drain regions and a gate electrode on the strained Si—Ge layer;
- an N-channel transistor comprising source/drain regions and a gate electrode on the strained Si—Ge layer, spaced from the P-channel transistor; 15
- a first nickel silicide layer, having a thickness of about 100 Å to about 220 Å, on the gate electrode and source/drain regions of the P-channel transistor;

12

- a second nickel silicide layer, having a thickness of about 100 Å to about 220 Å, on the gate electrode and source/drain regions of the N-channel transistor;
- a compressively stressed silicon nitride layer on the first nickel silicide layer;
- a tensile stressed silicon nitride layer on the second nickel silicide layer;
- a first oxide or oxynitride layer on the compressively stressed silicon nitride layer; and
- a second oxide or oxynitride layer on the tensile stressed silicon nitride layer,

wherein each of the first and second nickel silicide layers comprises platinum having a concentration gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount from between about 8 wt. % and about 12 wt. % at the silicon interface to between about 30 wt. % and about 100 wt. % in a direction away from the P-channel and N-channel transistors.

* * * * *

EXHIBIT 5



(10) **Patent No.:** US 8,507,983 B2
(45) **Date of Patent:** Aug. 13, 2013

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,589,414	A	12/1996	Wann et al.	
6,759,675	B2	7/2004	Csutak et al.	
6,894,350	B2	5/2005	Shimizu et al.	
6,979,864	B2 *	12/2005	Negoro et al.	257/335
7,180,132	B2	2/2007	Cai et al.	
7,776,700	B2	8/2010	Yang et al.	
7,829,402	B2 *	11/2010	Matocha et al.	438/197
7,935,992	B2 *	5/2011	Chang	257/288
2009/0059111	A1	3/2009	Jang	

OTHER PUBLICATIONS

Leonard Rubin et al, Narrow n+/p+ isolation in retrograde well implants, Solid State Technology Magazine, Jul. 1, 2003, pp. 119-125, vol. 46 Issue 7, Solid State Technology, USA.

* cited by examiner

Primary Examiner — Quoc Hoang

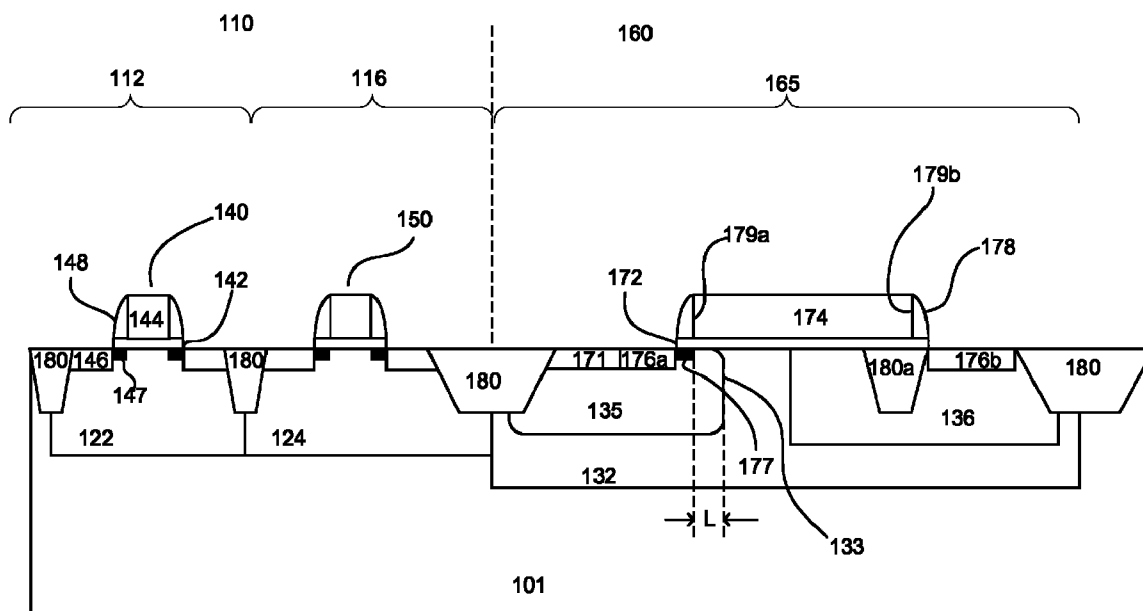
(74) *Attorney, Agent, or Firm* — Horizon IP Pte Ltd

(57) **ABSTRACT**

A device is disclosed. The device includes a substrate prepared with an active device region. The active device region includes a gate. The device also includes a doped channel well disposed in the substrate adjacent to a first edge of the gate. The first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate. The first edge of the gate and channel edge defines an effective channel length of the device. The effective channel length is self-aligned to the gate. A doped drift well adjacent to a second edge of the gate is also included.

20 Claims, 29 Drawing Sheets

(58) **Field of Classification Search**
USPC 257/335, 346, E29.256
See application file for complete search history.



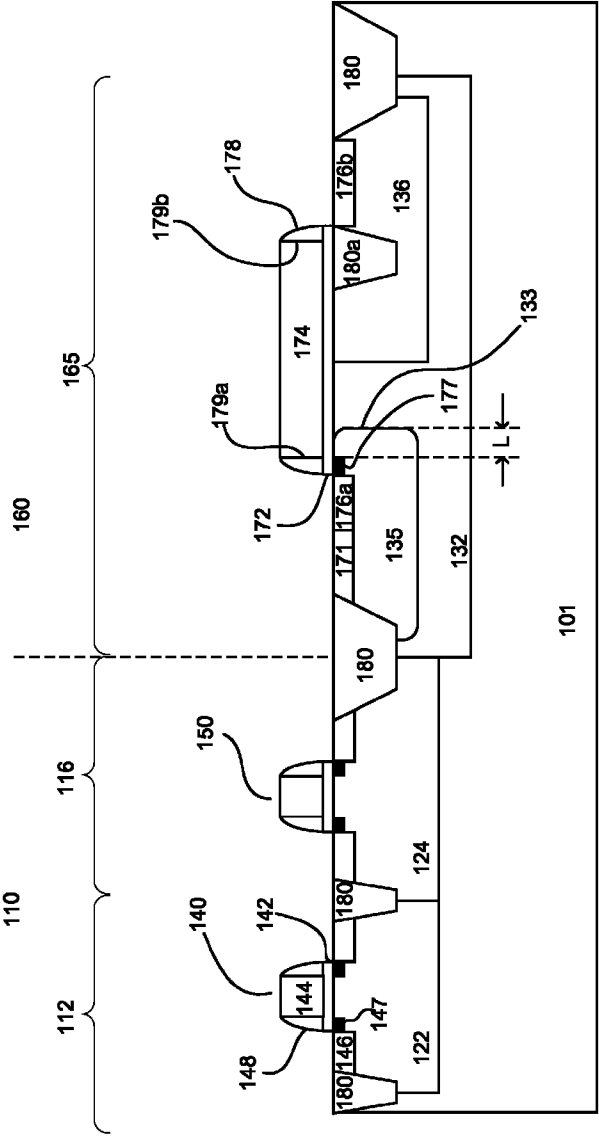


Fig. 1

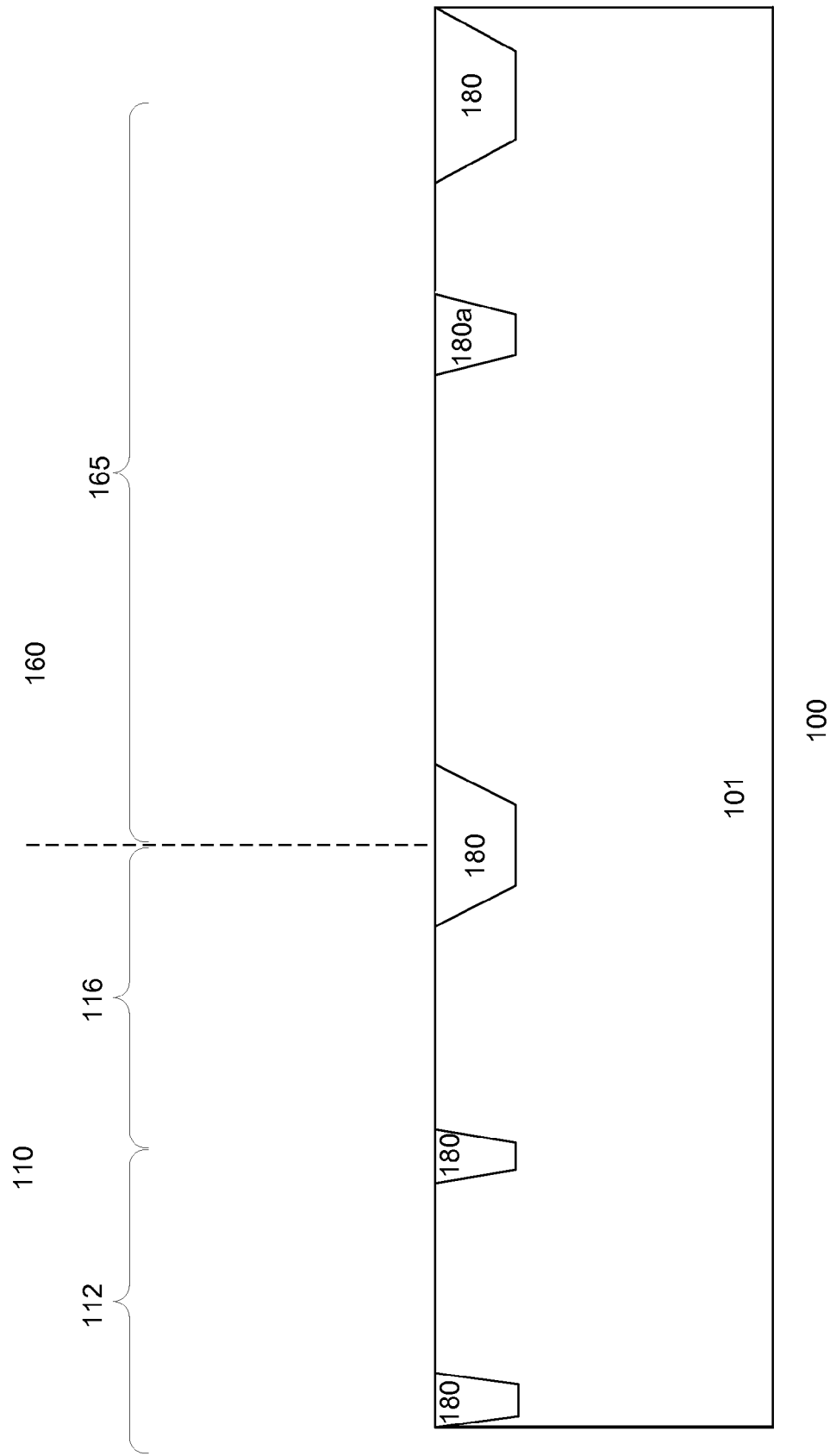


Fig. 2a

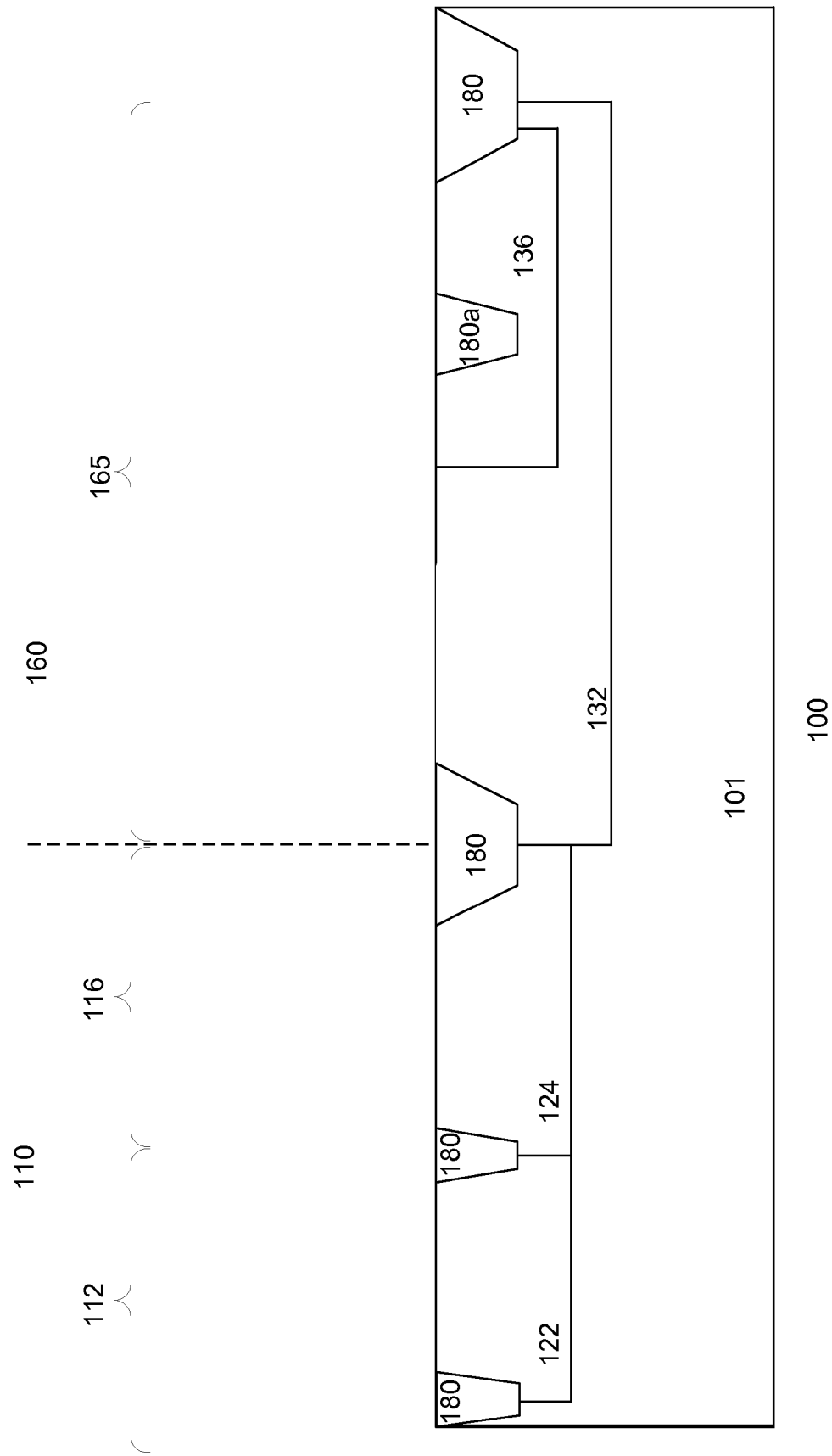


Fig. 2b

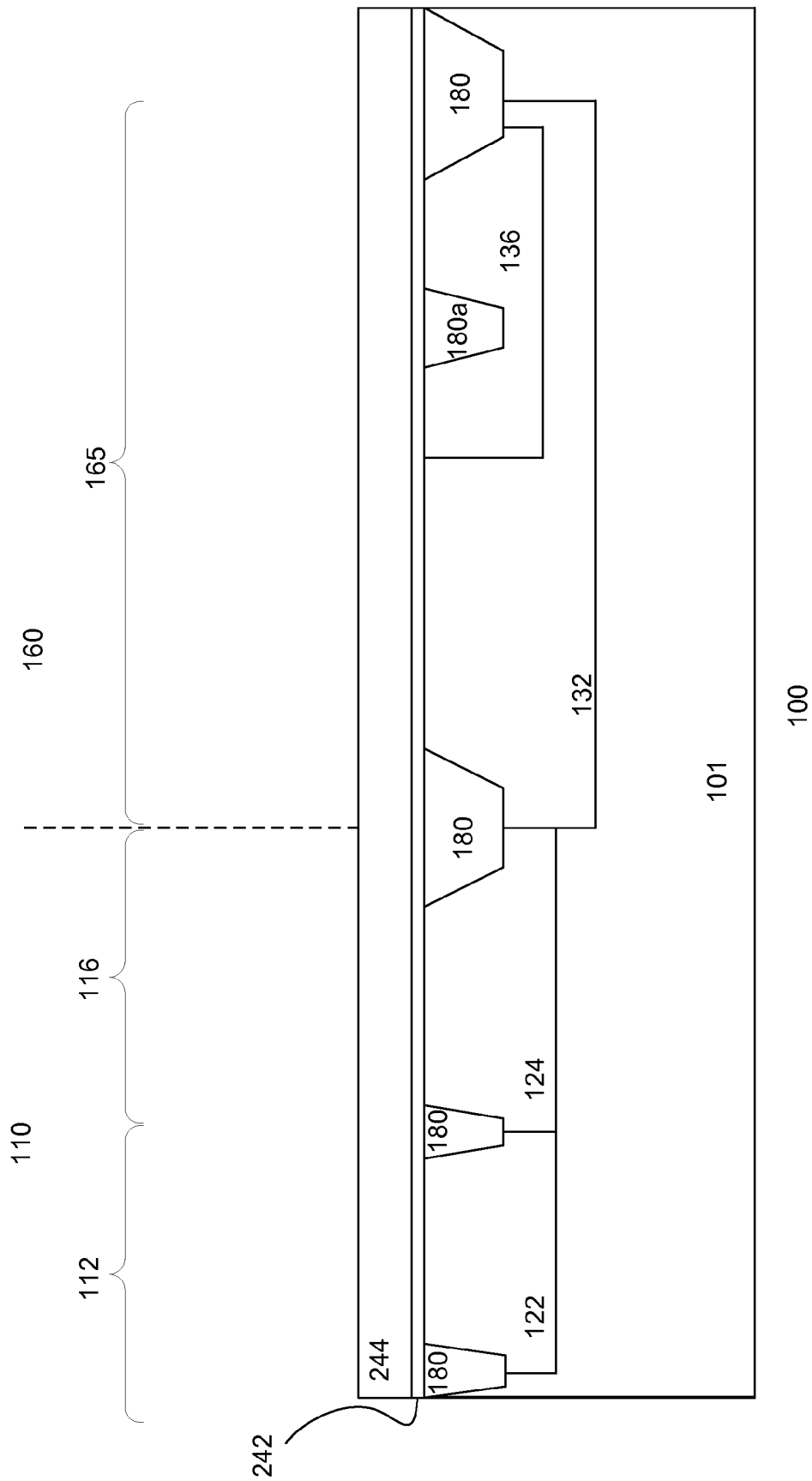


Fig. 2c

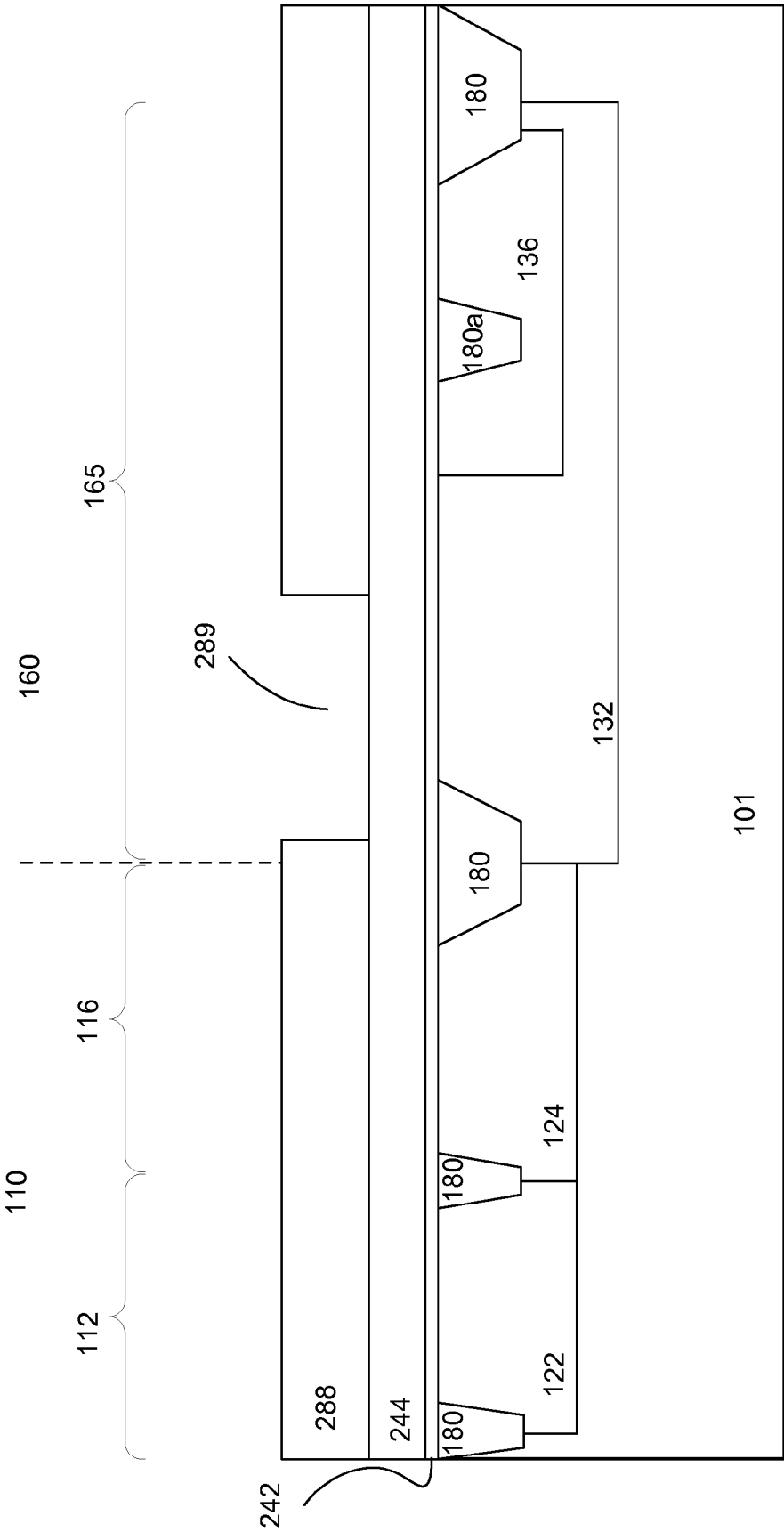


Fig. 2d

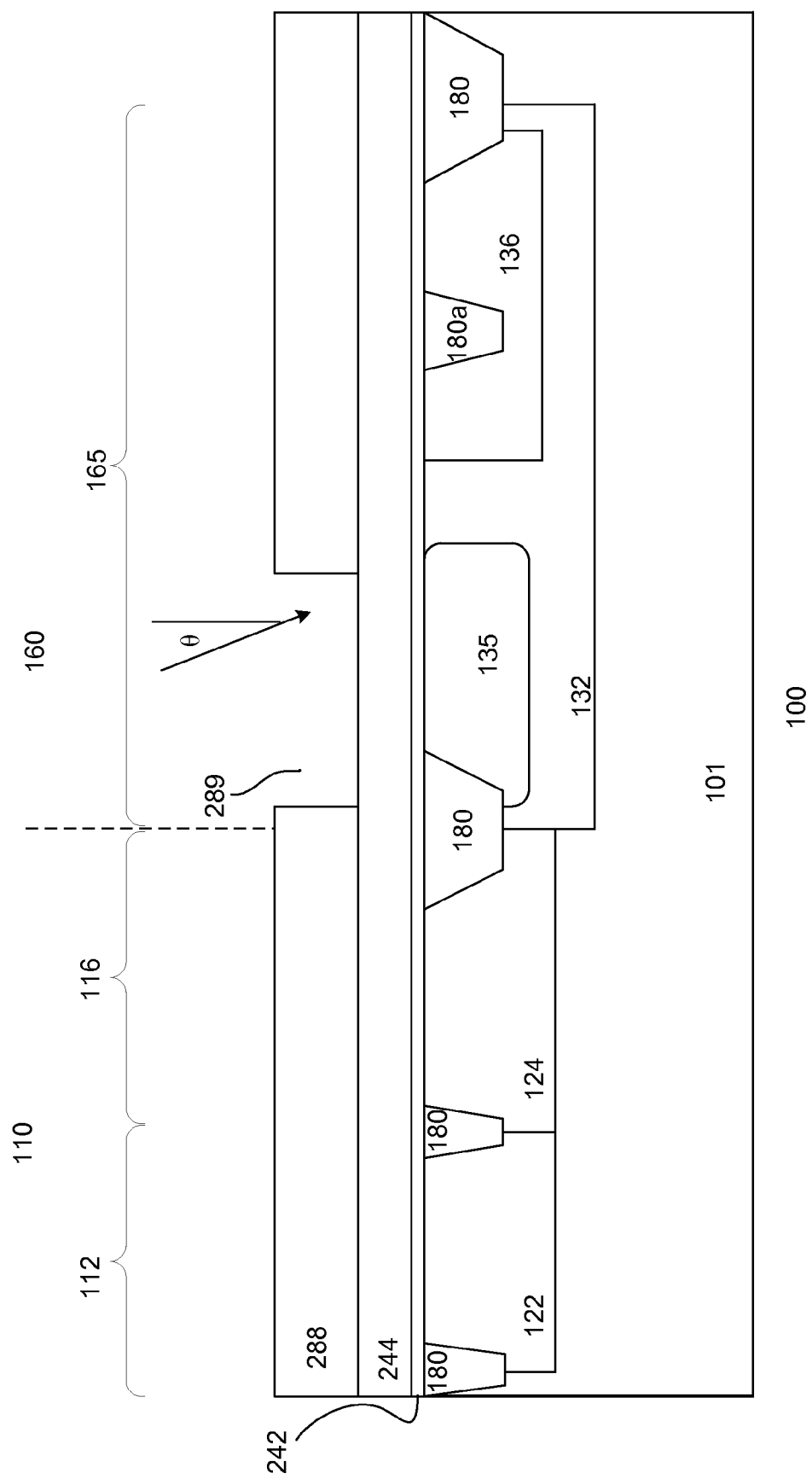


Fig. 2e

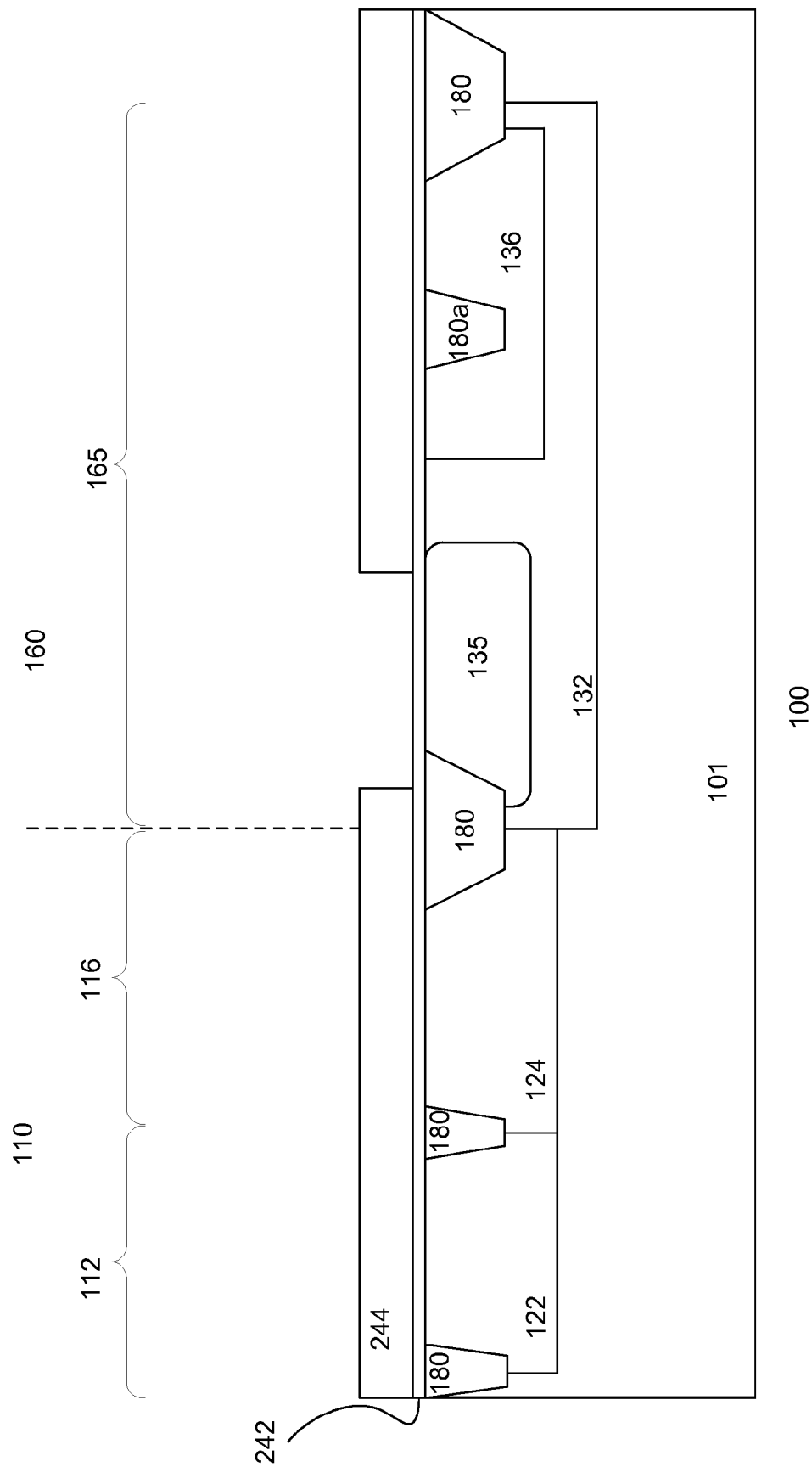


Fig. 2f

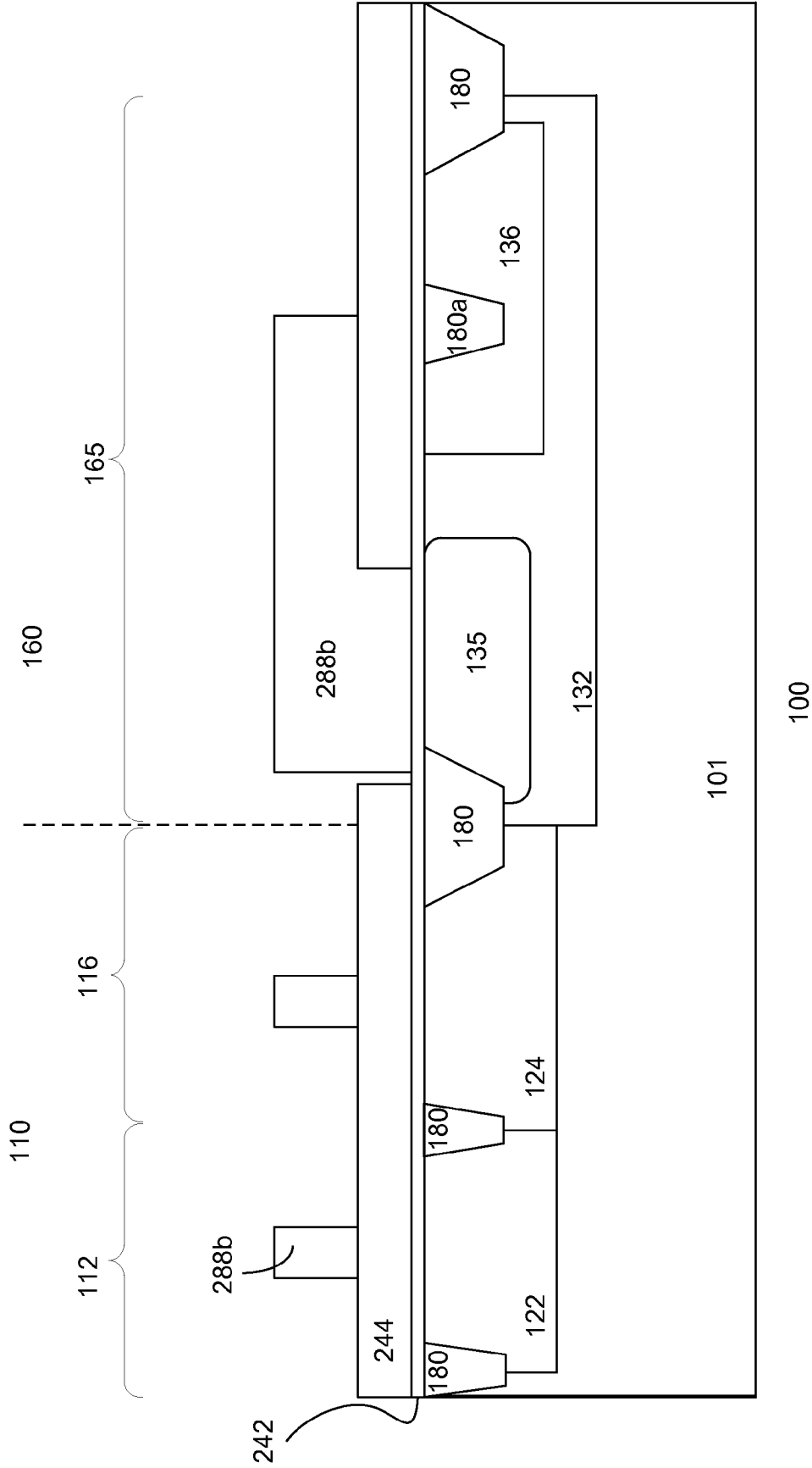


Fig. 2g

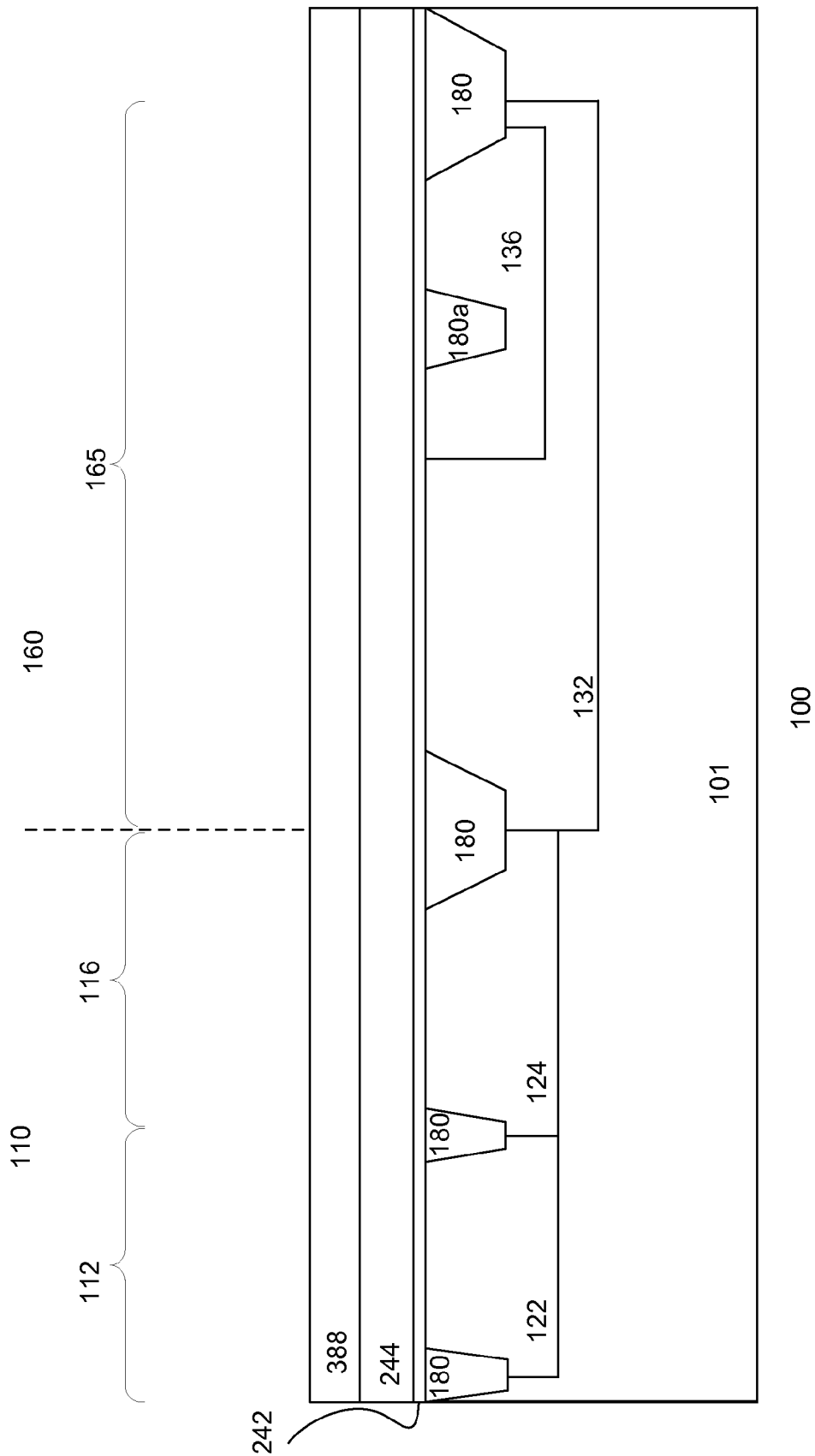


Fig. 3a

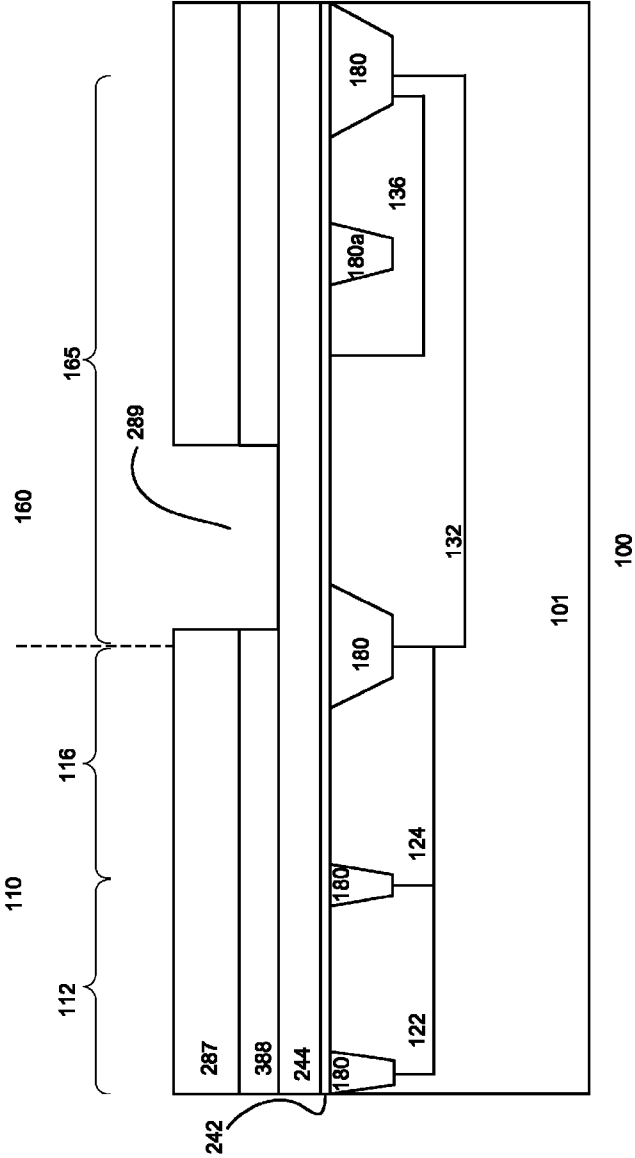


Fig. 3b

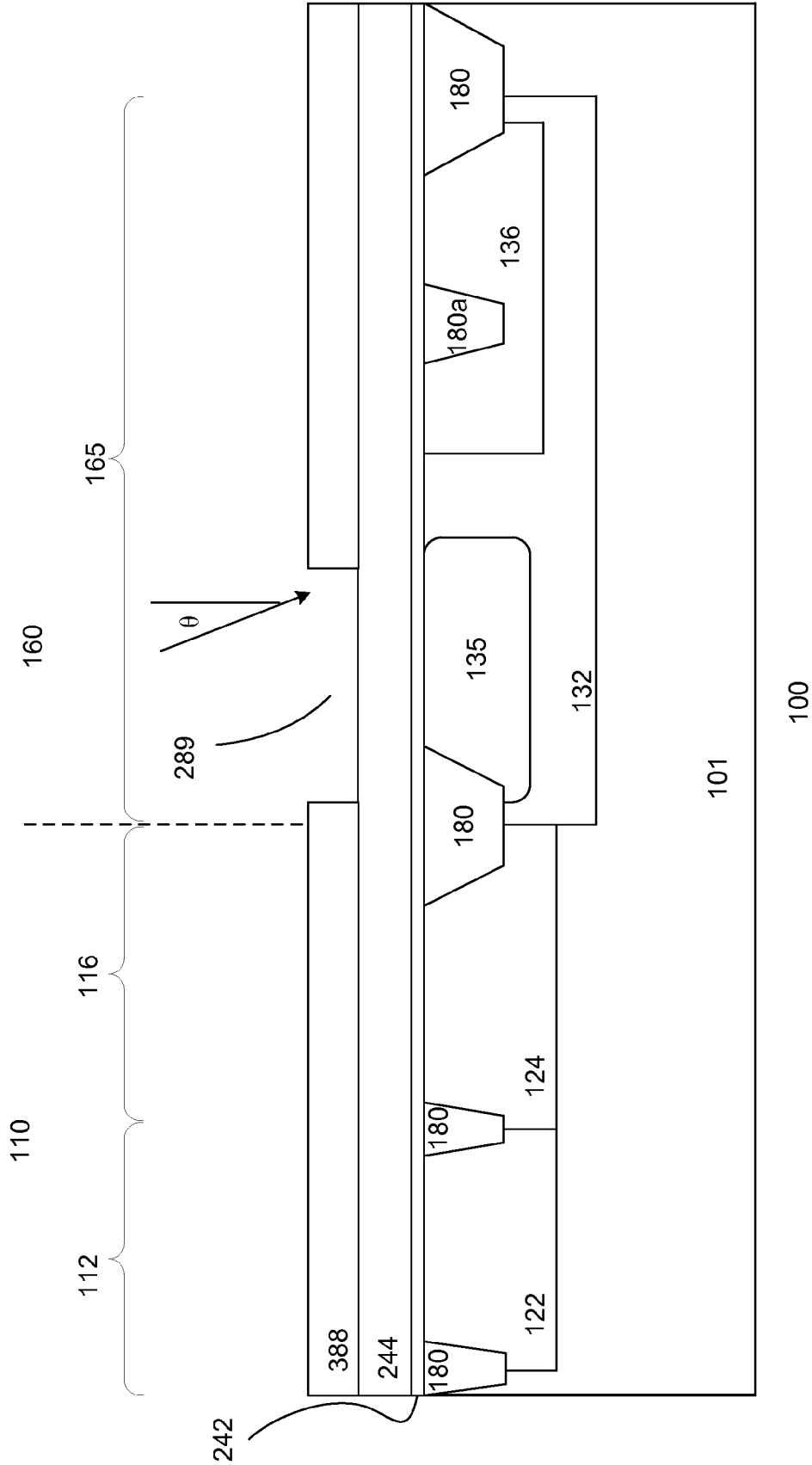


Fig. 3c

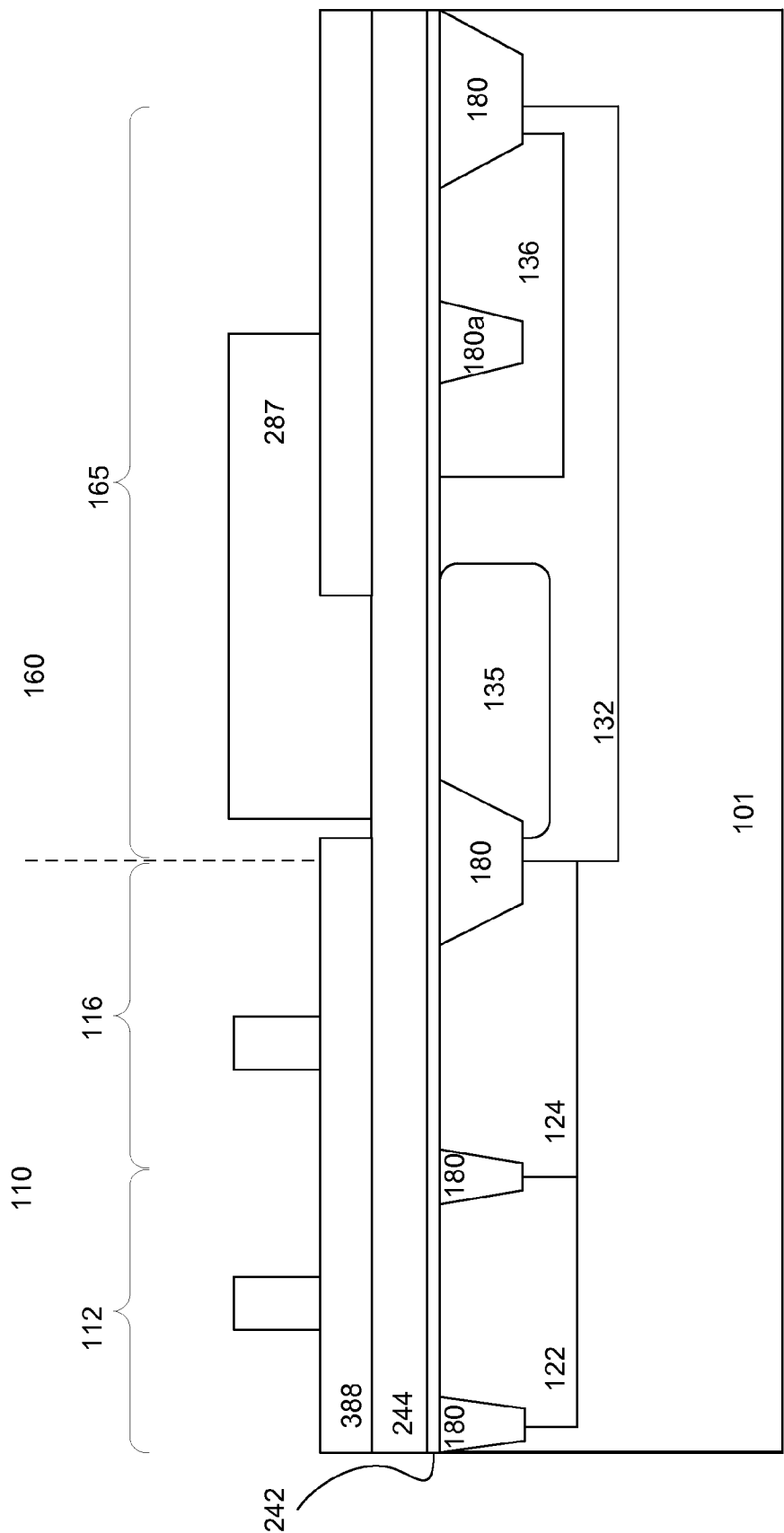


Fig. 3d

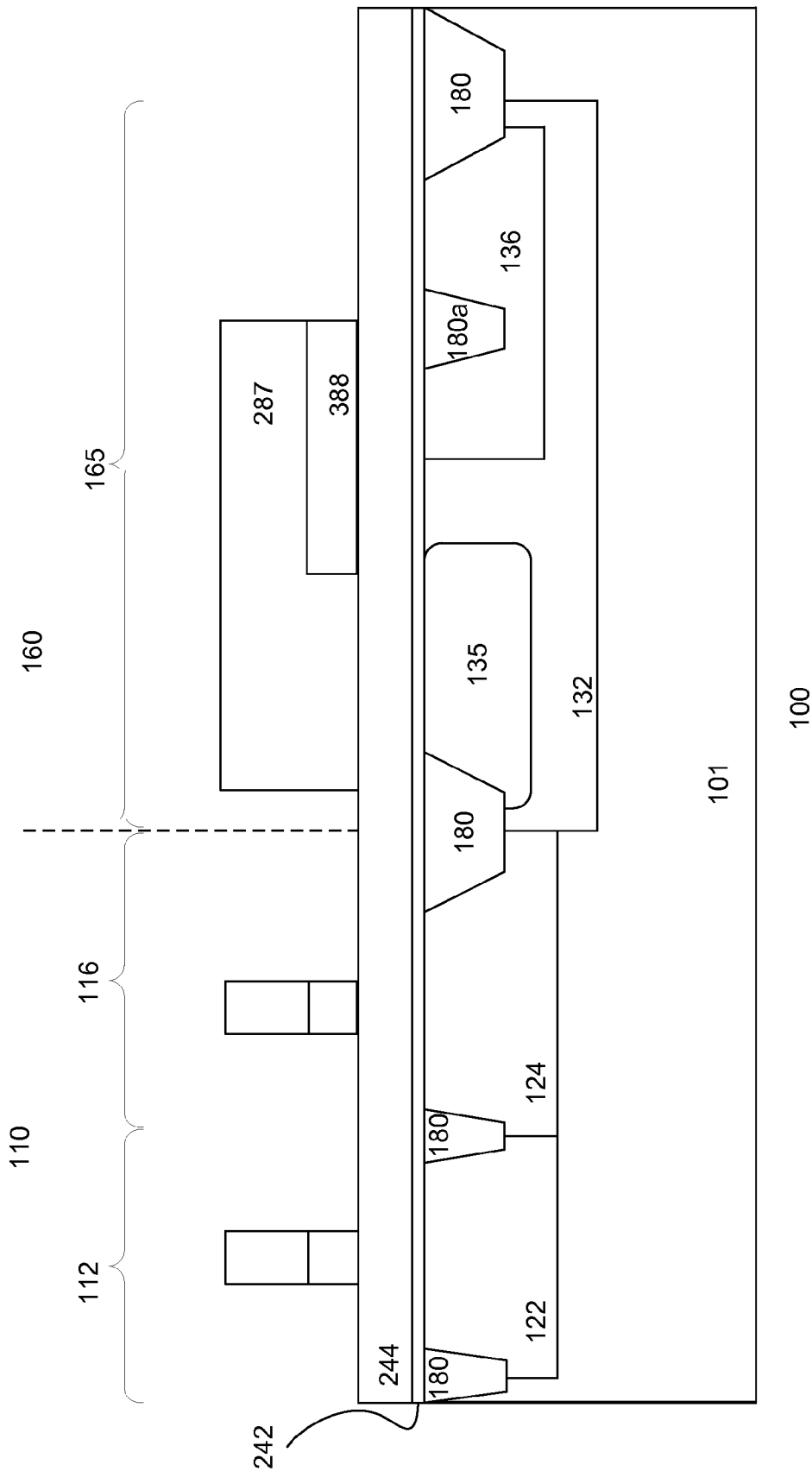


Fig. 3e

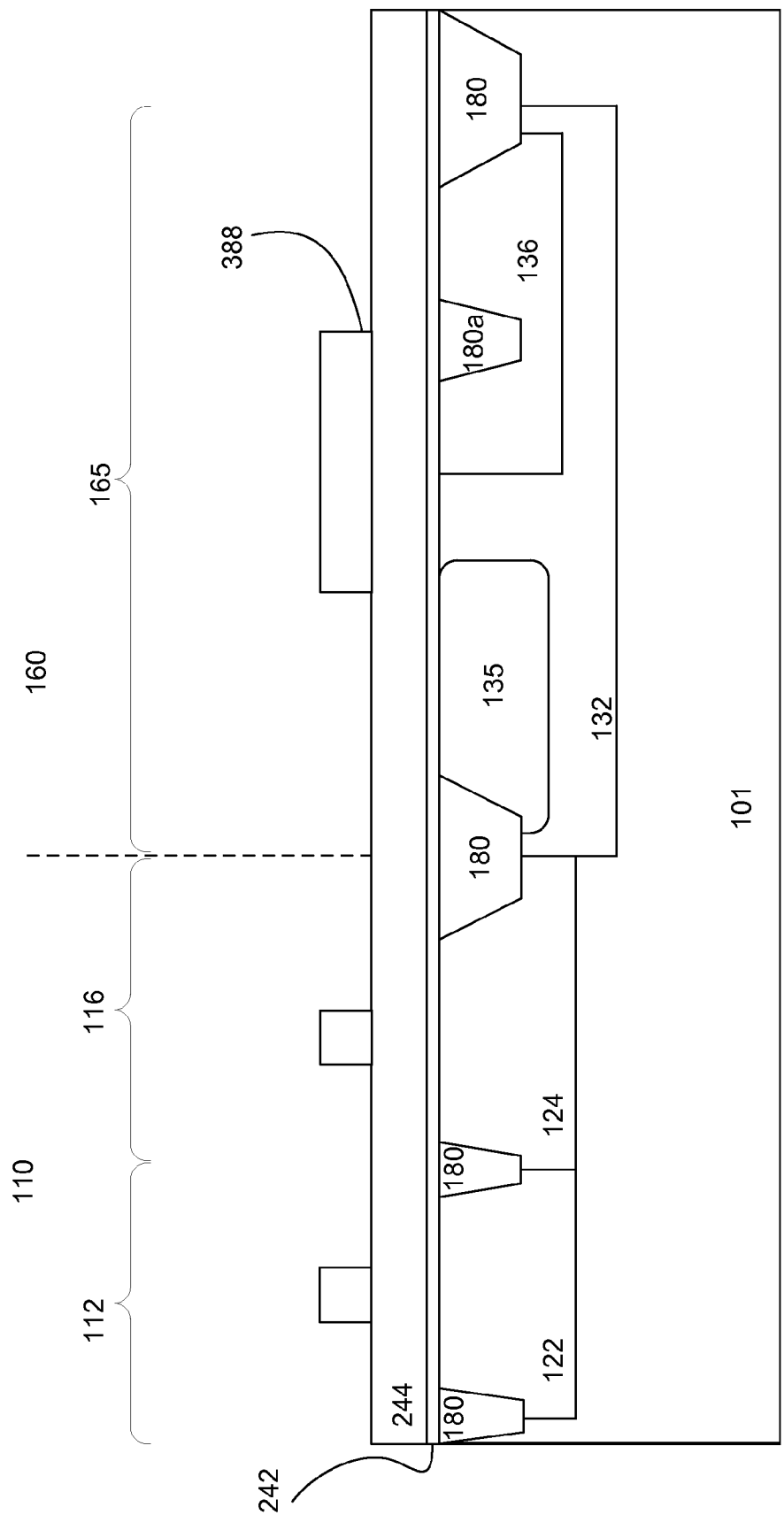
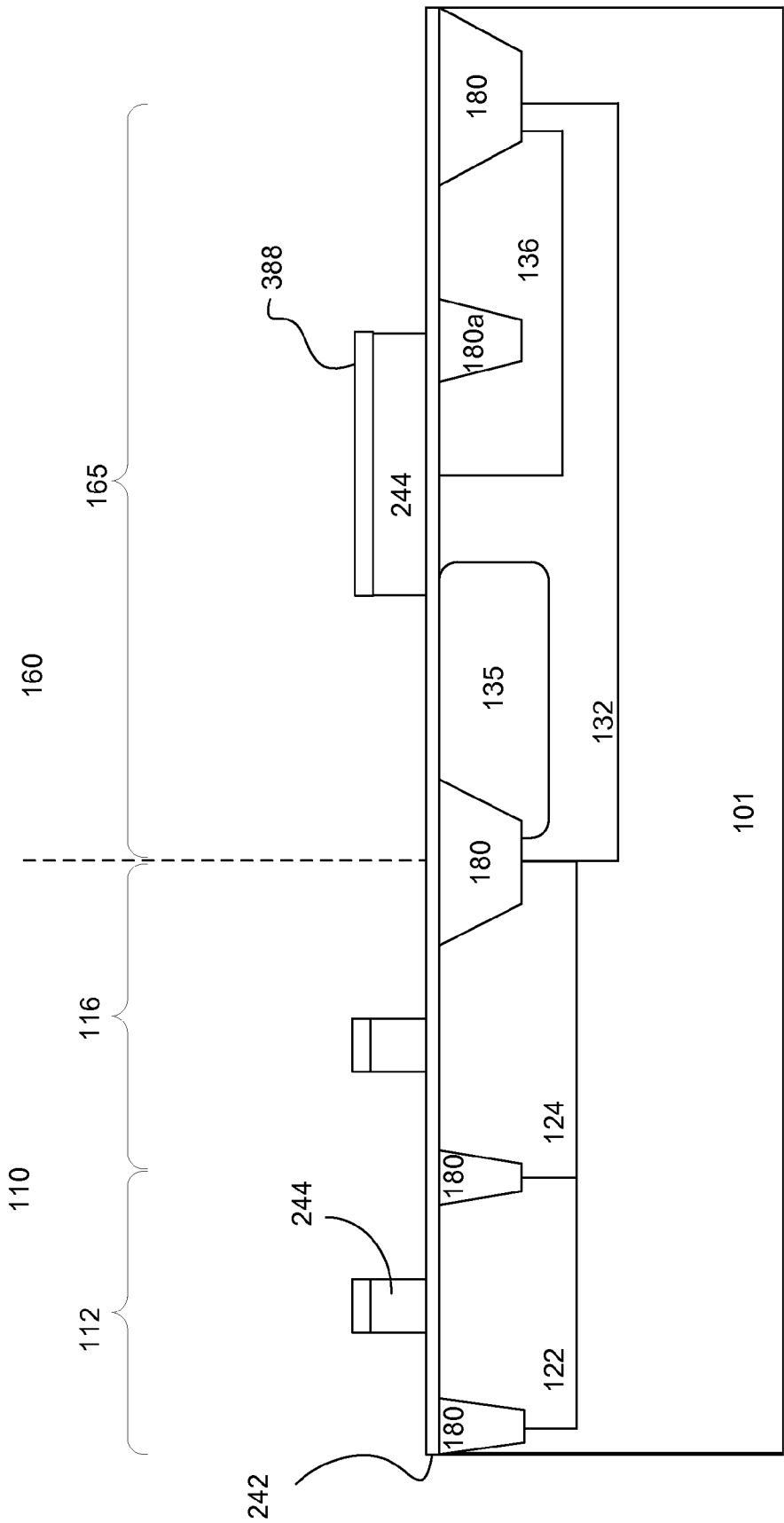


Fig. 3f



100

Fig. 3g

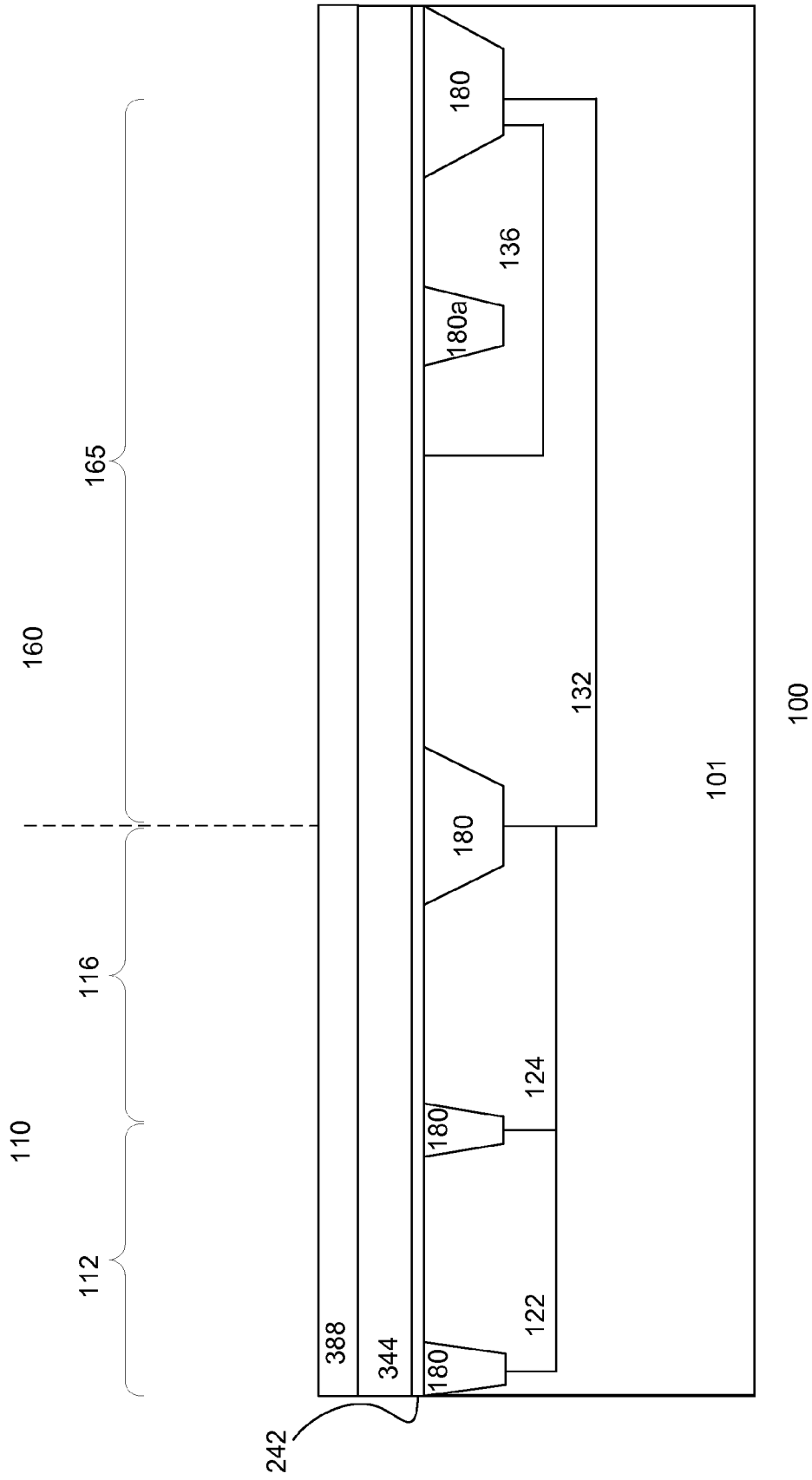


Fig. 4a

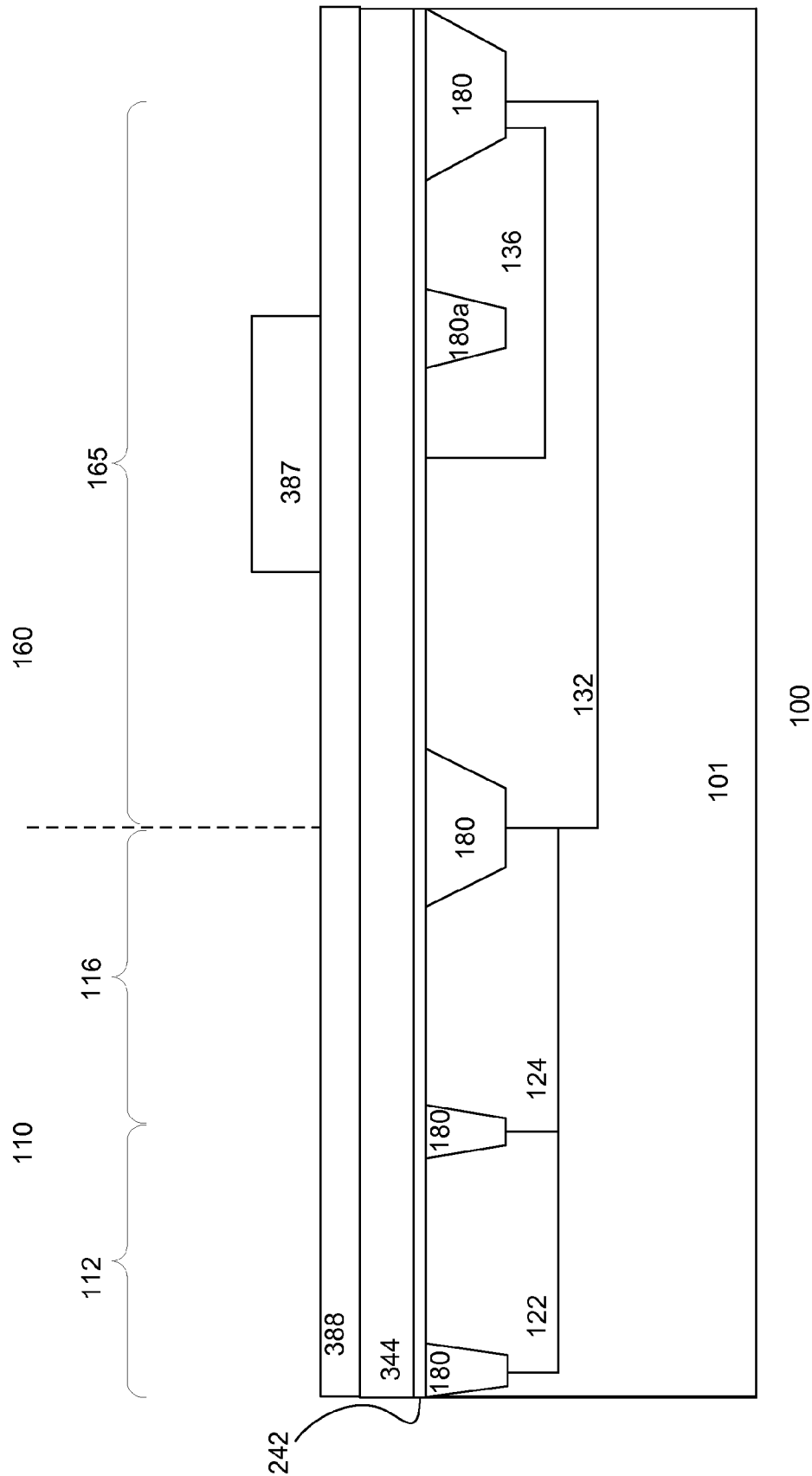


Fig. 4b

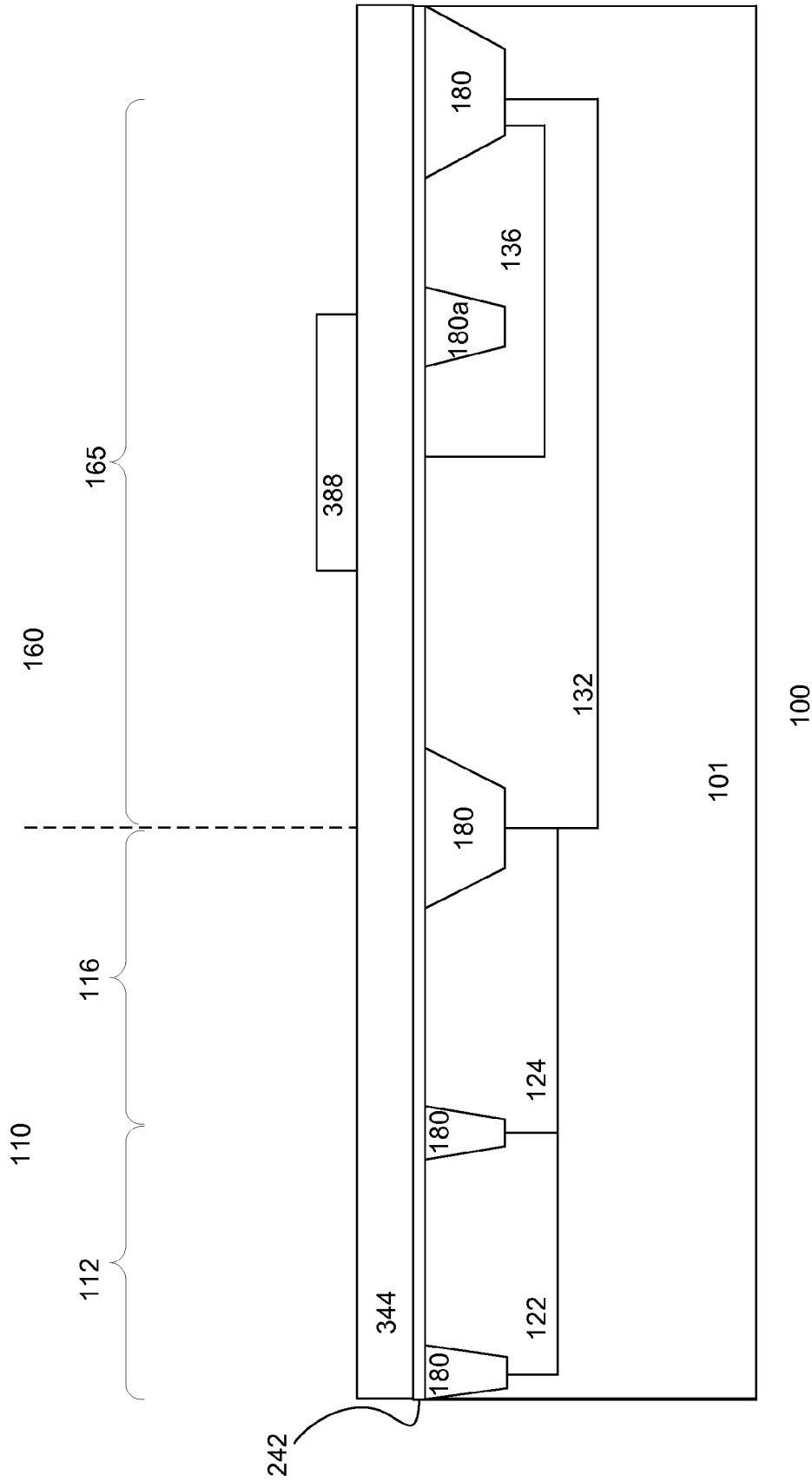


Fig. 4c

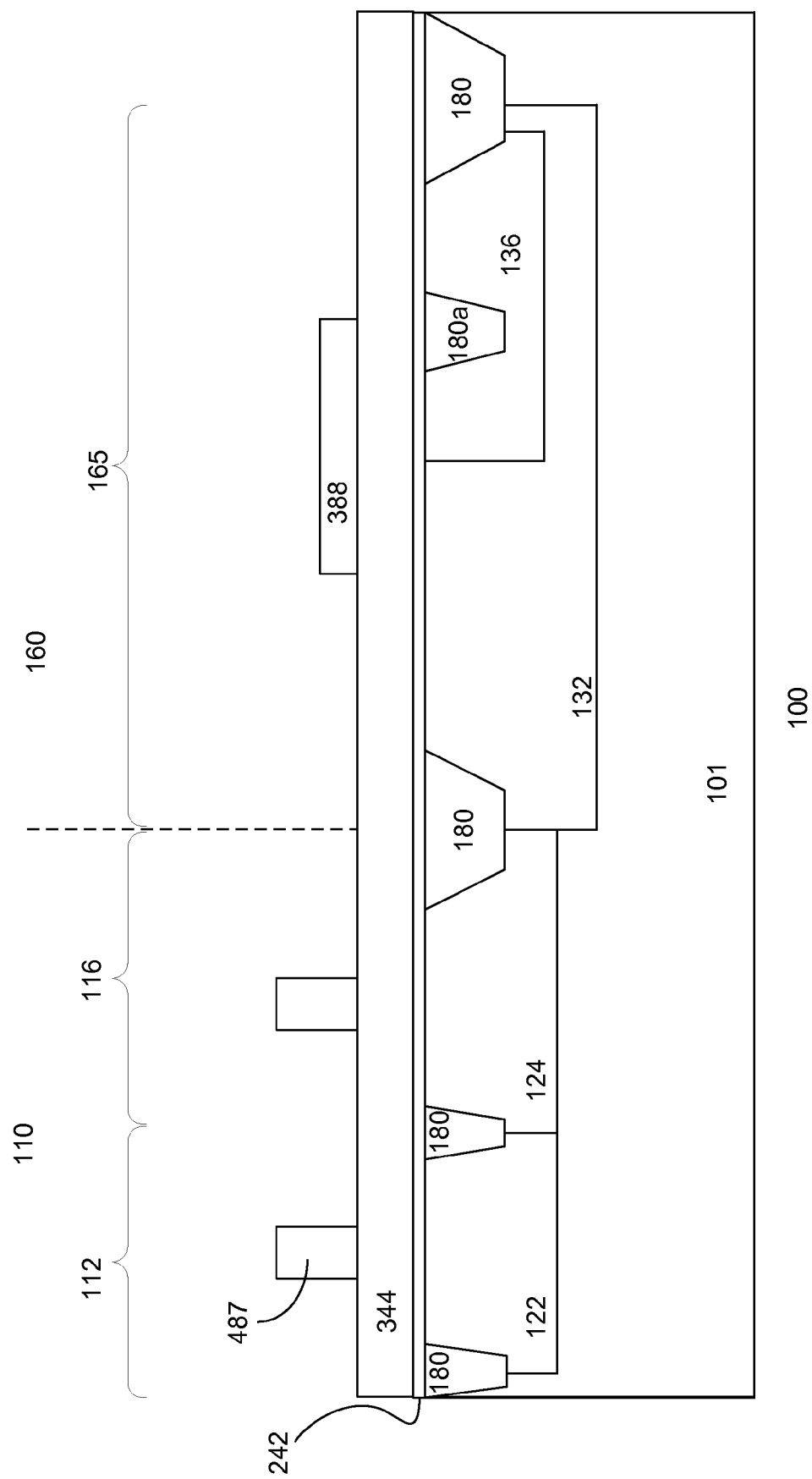


Fig. 4d

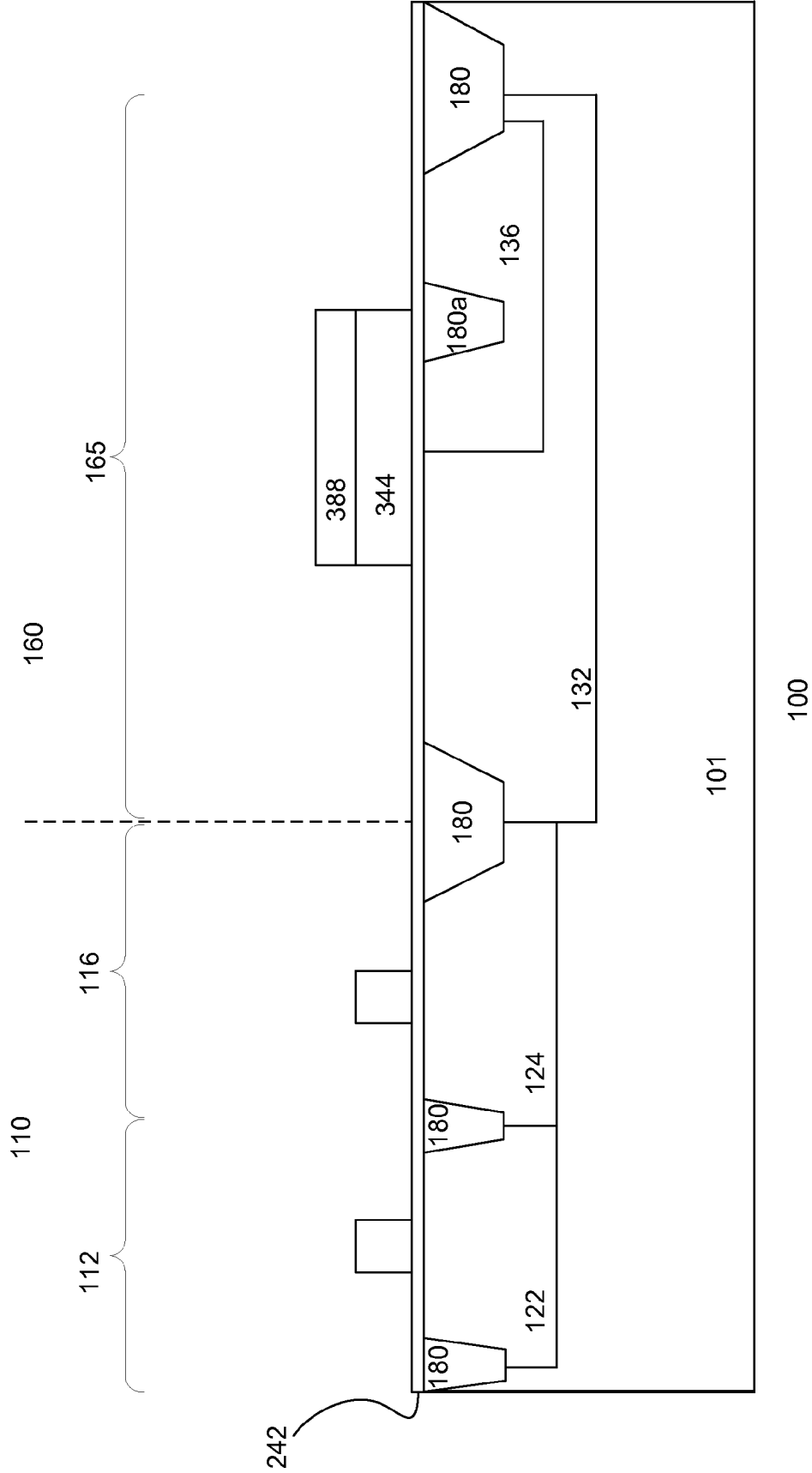


Fig. 4e

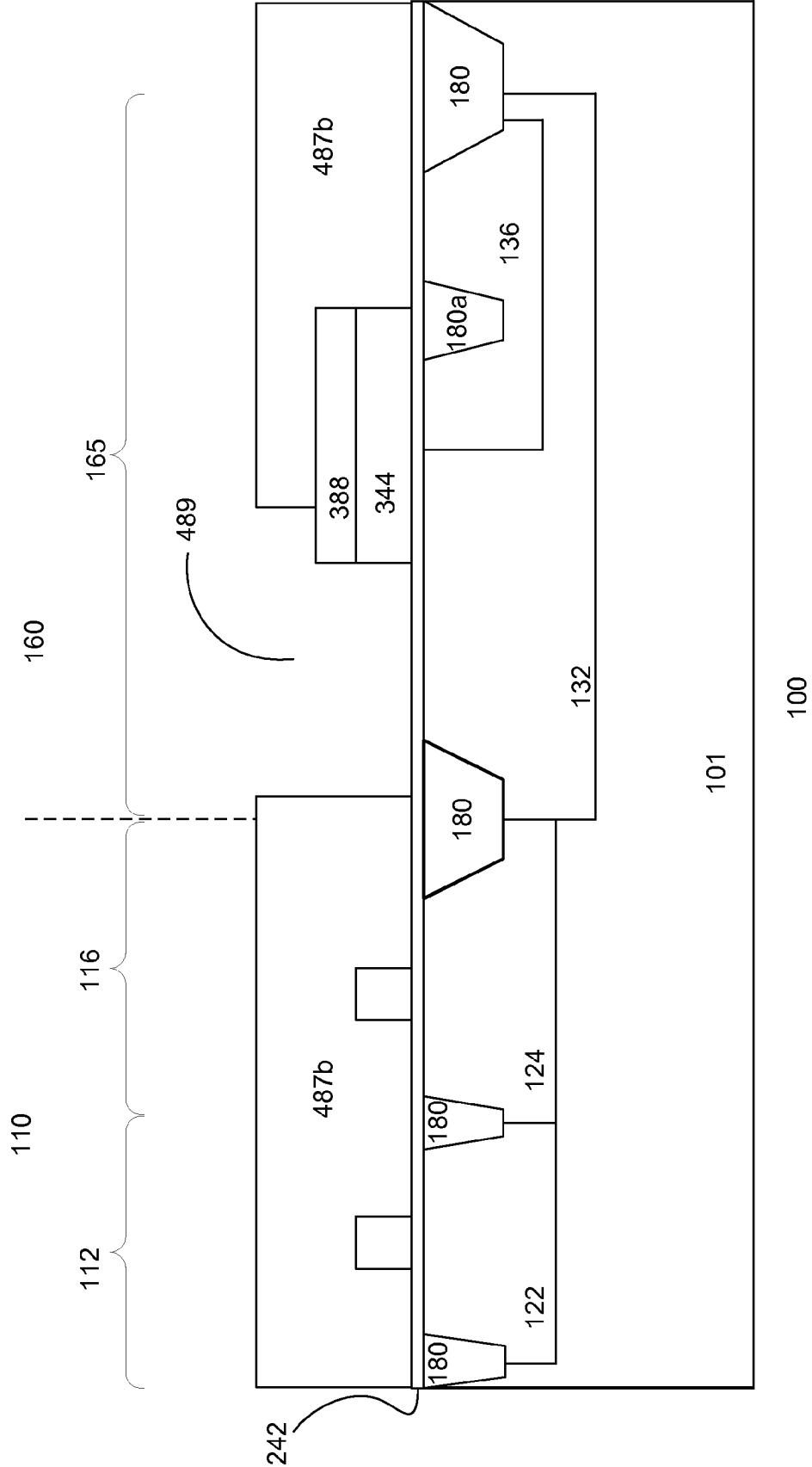


Fig. 4f

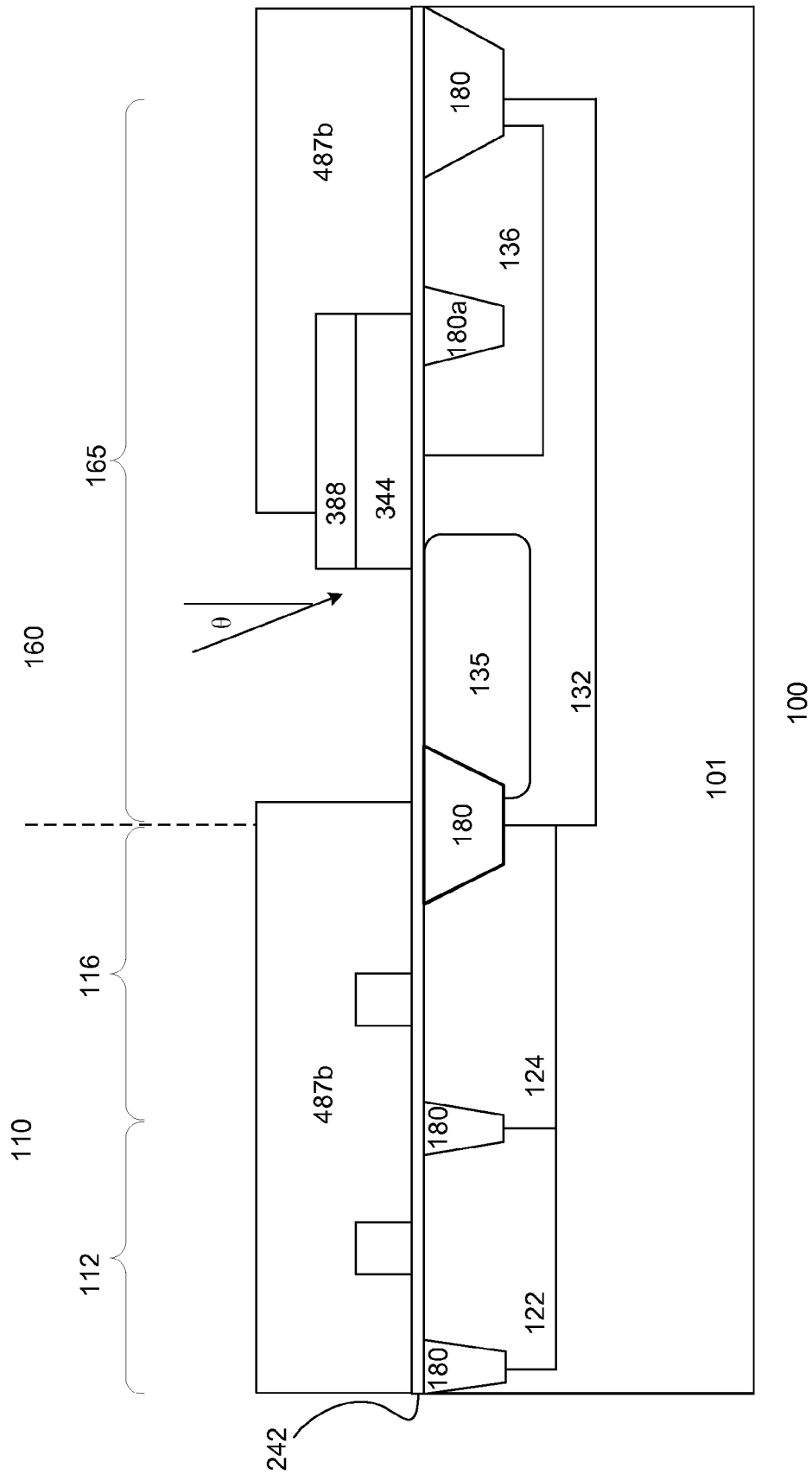


Fig. 49

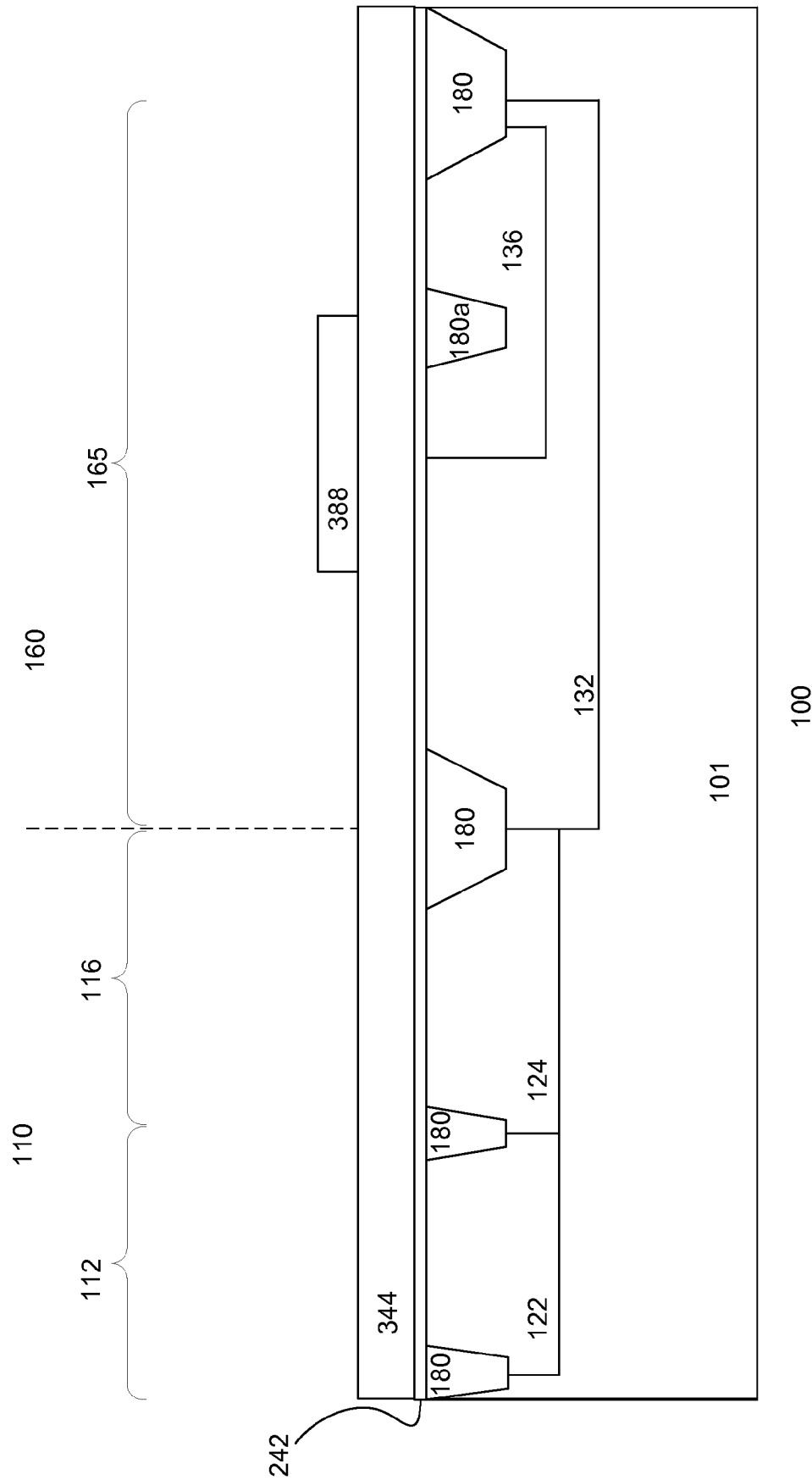


Fig. 5a

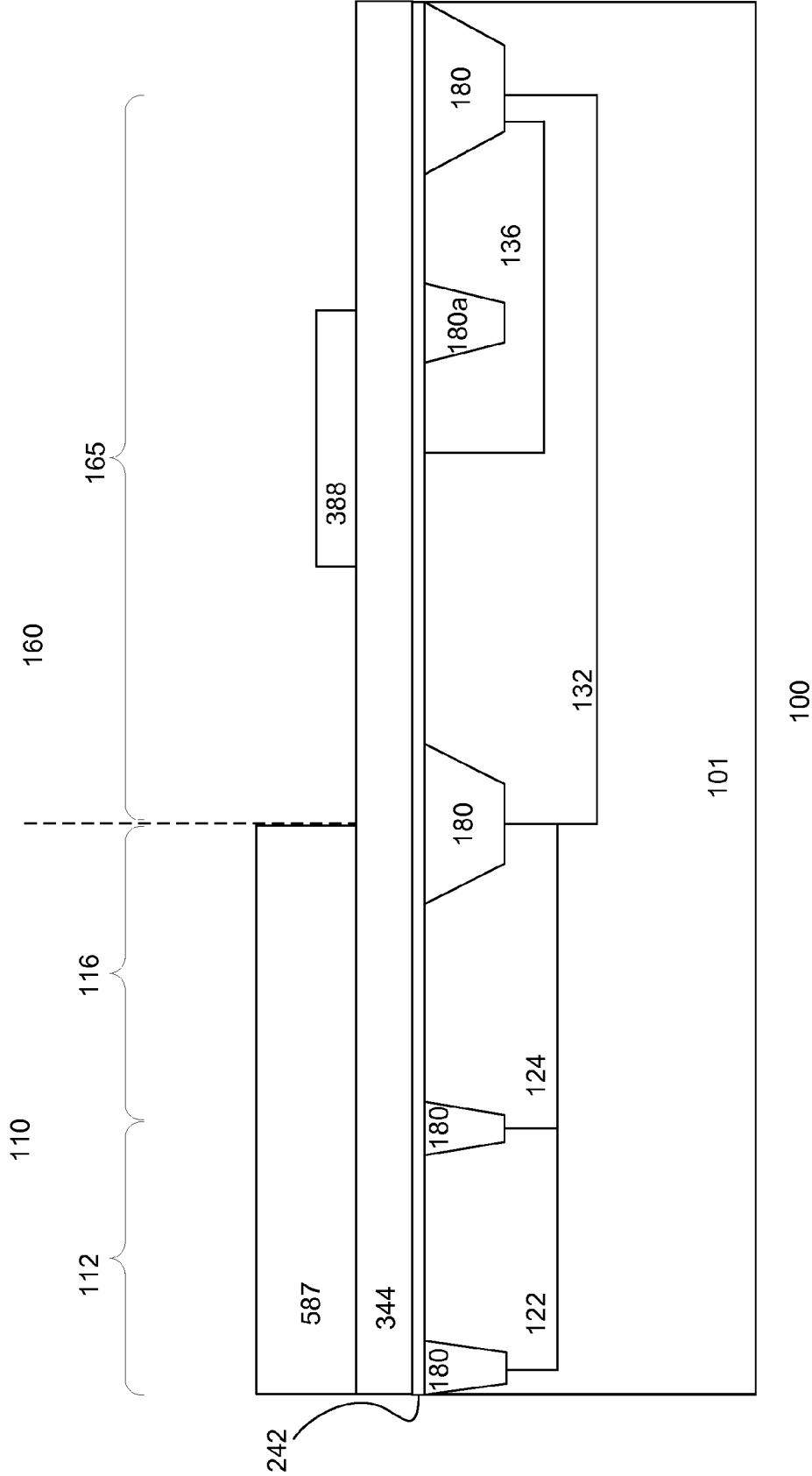


Fig. 5b

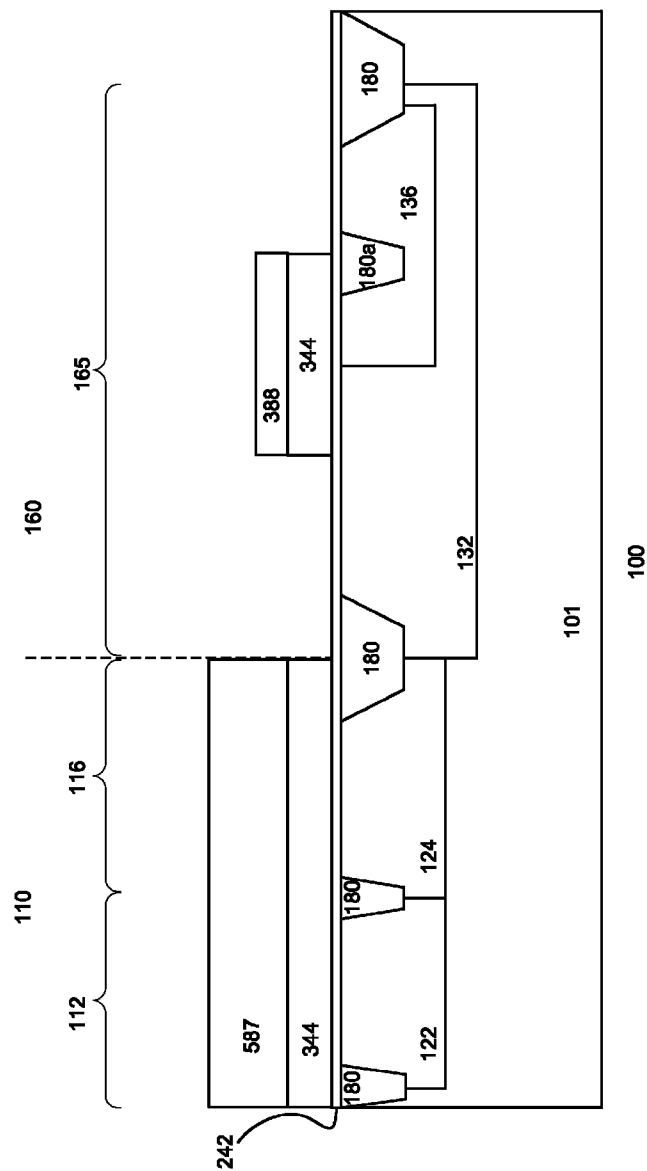


Fig. 5c

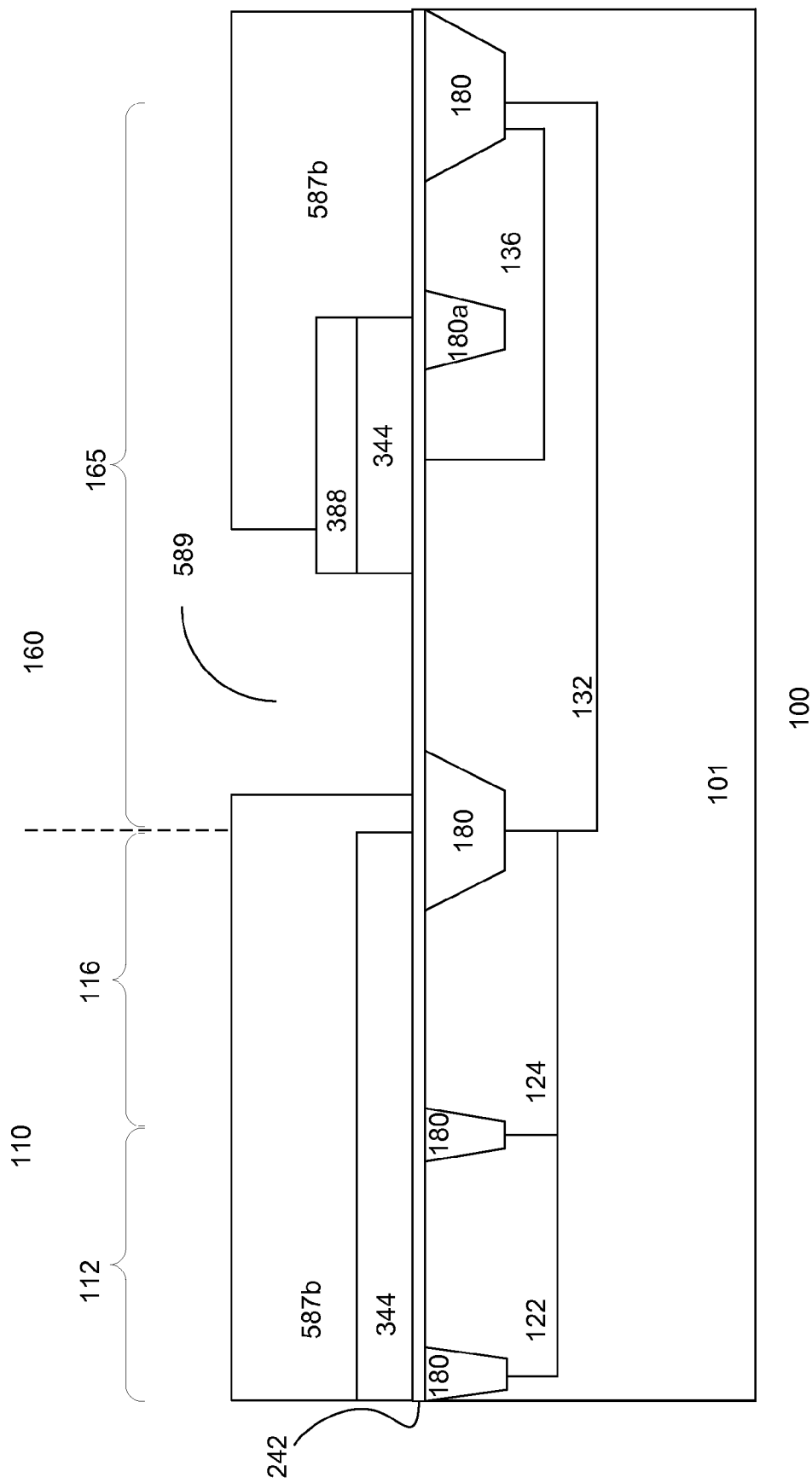


Fig. 5d

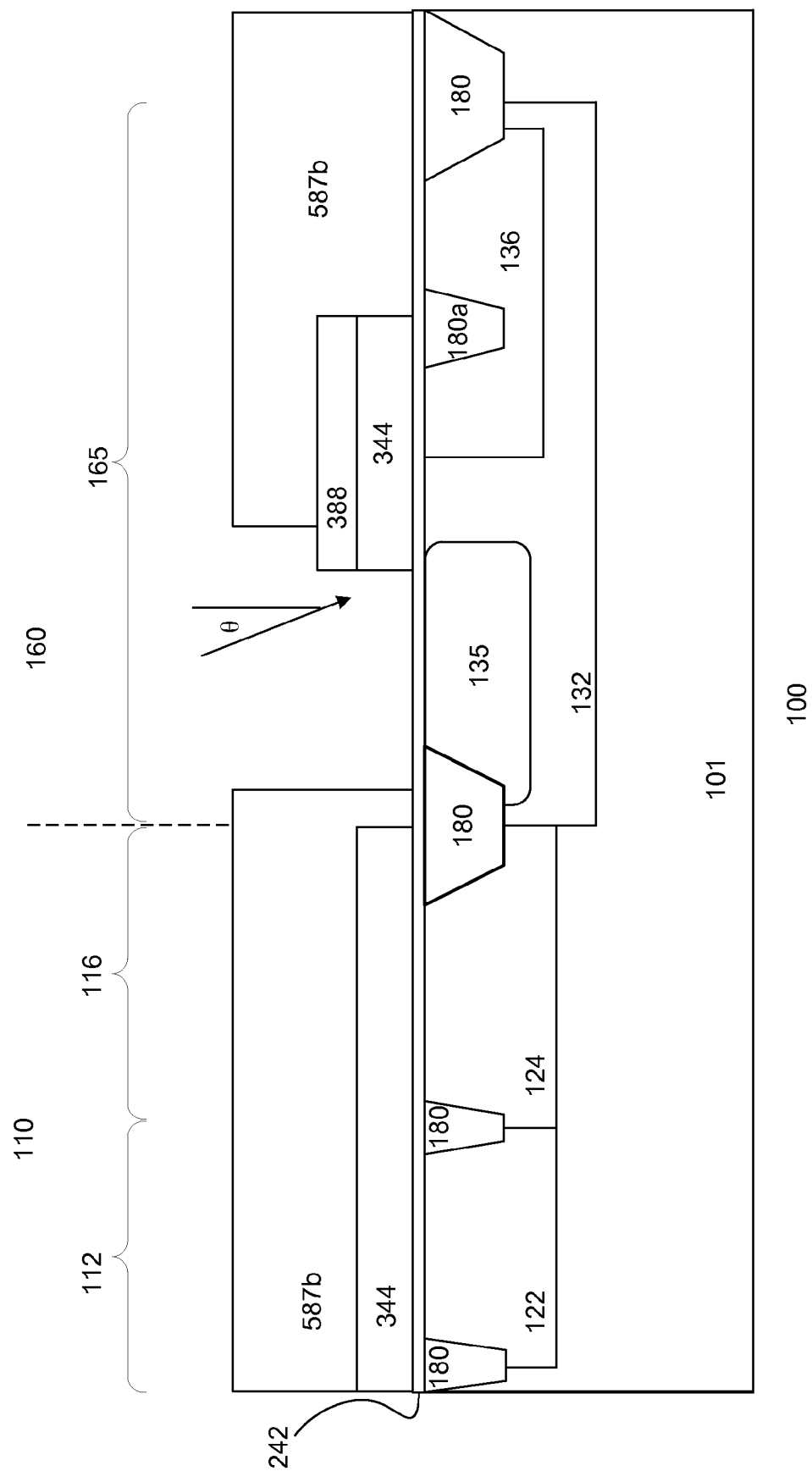


Fig. 5e

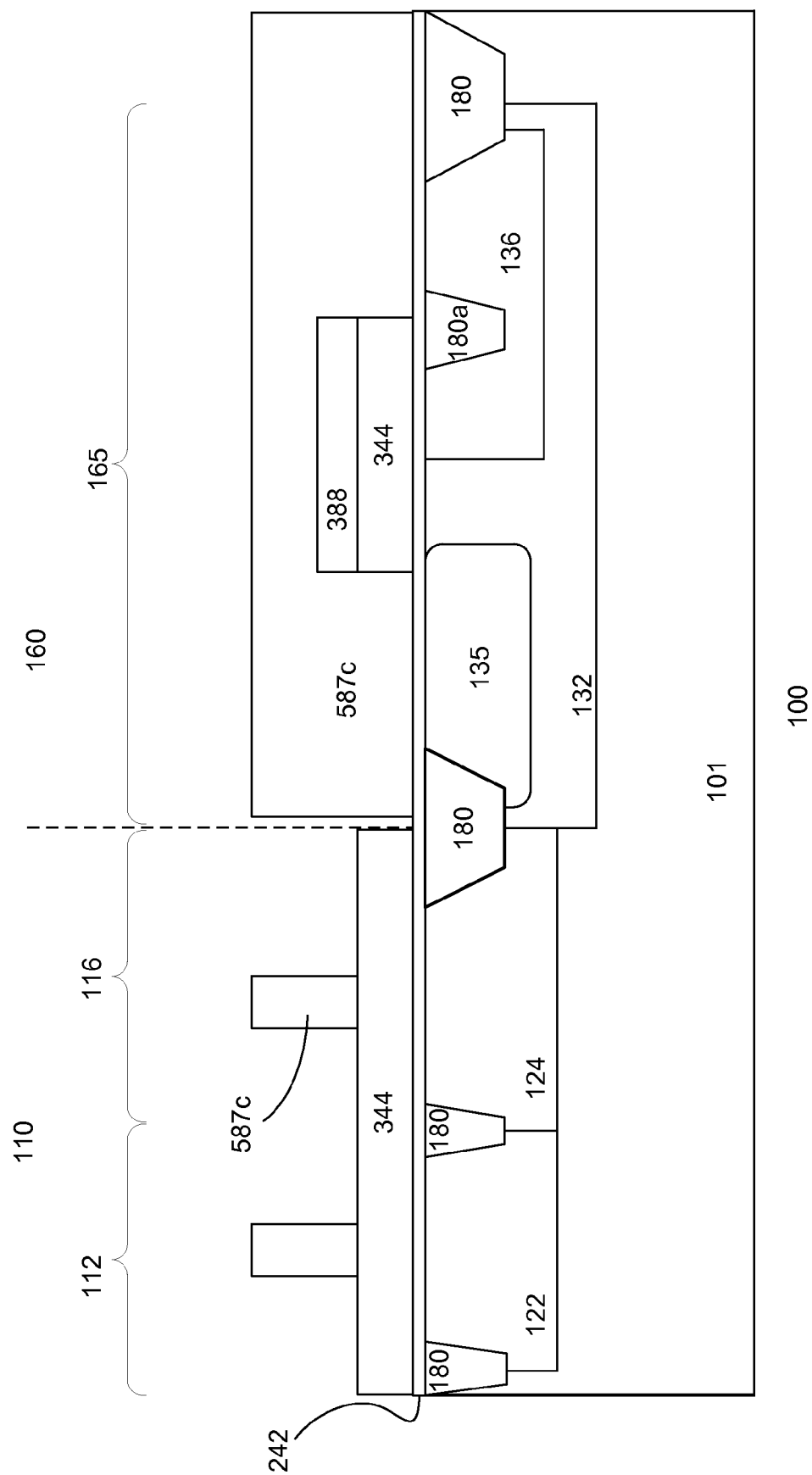


Fig. 5f

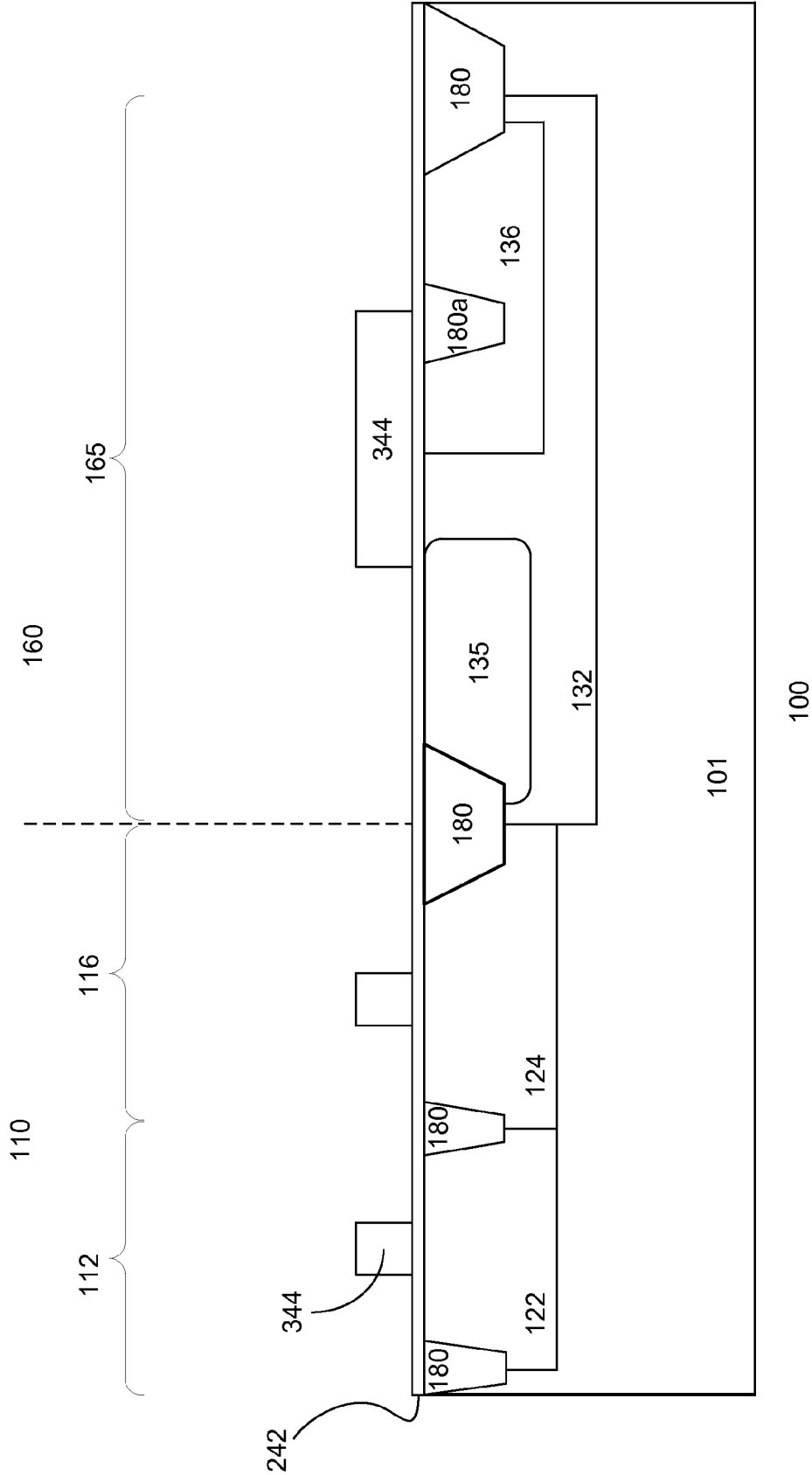


Fig. 5g

1

HIGH VOLTAGE DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

This application is a divisional application which claims benefits of co-pending U.S. patent application Ser. No. 12/500,620, filed on Jul. 10, 2009, which is a continuation-in-part application of U.S. patent application Ser. No. 12/390,509, filed on Feb. 23, 2009, now issued as U.S. Pat. No. 8,053,319. All disclosures are incorporated herewith by reference.

BACKGROUND

Lateral Double-Diffused (LD) transistors have been widely employed in high voltage applications. For high performance LD transistors, low drain-to-source on-resistance ($R_{ds_{on}}$) is desired to minimize its power dissipation when it is turned on, as well as high breakdown voltage to maximize its voltage capability. To achieve low $R_{ds_{on}}$, the channel of the high LD transistor should be as short as possible.

As technology evolves into era of deep sub-micron (e.g., beyond 0.35 μm) Very-Large-Scale Integration (VLSI), there is a desire for both high voltage (HV) LD transistors and low voltage (LV) transistors to be fabricated on the same substrate. Generally, the processes for forming LV devices, such as complementary metal-oxide-semiconductor (CMOS) processes, are used to form the HV devices. However, conventional CMOS processes are incompatible for forming HV transistors. For example, process overlay issues make it difficult for alignment of the HV channel well, which defines the channel length. Process variations in process overlay requires large process windows. This makes forming short channel lengths to achieve low $R_{ds_{on}}$ difficult if not impossible. Furthermore, the thin gate electrodes used prevent them serving as a hard mask, further exacerbating process control in forming the channel well.

From the foregoing discussion, it is desirable to provide reliable HV devices formed with short channel lengths to have low $R_{ds_{on}}$.

SUMMARY

A device is disclosed. In one embodiment, the device includes a substrate prepared with an active device region. The active device region includes a gate. The device also includes a doped channel well disposed in the substrate adjacent to a first edge of the gate. The first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate. The first edge of the gate and channel edge defines an effective channel length of the device. The effective channel length is self-aligned to the gate. The device further includes a doped drift well adjacent to a second edge of the gate.

In another embodiment, another device is disclosed. The device includes a substrate prepared with an active device region. The active device region includes a gate. The device also includes a doped channel well disposed in the substrate adjacent to a first edge of the gate. The first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate. The first edge of the gate and channel edge defines an effective channel length of the device. The effective channel length is self-aligned to the gate. The device also includes a doped drift well adjacent to a second edge of the gate. The device further includes a drift isolation region in the doped drift well.

2

In yet another embodiment, a semiconductor device is disclosed. The semiconductor device includes a substrate prepared with an active device region. The active device region includes a gate. The device also includes a doped channel well disposed in the substrate adjacent to a first edge of the gate. The first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate. The first edge of the gate and channel edge defines an effective channel length of the device. The effective channel length is self-aligned to the gate. The device also includes a doped drift well adjacent to a second edge of the gate. A drift isolation region is disposed in the doped drift well. The device further includes first and second diffusion regions adjacent to the gate.

These and other objects, along with advantages and features of the present invention herein disclosed, will become apparent through reference to the following description and the accompanying drawings. Furthermore, it is to be understood that the features of the various embodiments described herein are not mutually exclusive and can exist in various combinations and permutations.

BRIEF DESCRIPTION OF THE DRAWINGS

In the drawings, like reference characters generally refer to the same parts throughout the different views. Also, the drawings are not necessarily to scale, emphasis instead generally being placed upon illustrating the principles of the invention. In the following description, various embodiments of the present invention are described with reference to the following drawings, in which:

FIG. 1 shows an embodiment of a device;

FIGS. 2a-g show an embodiment of a process for forming a device;

FIGS. 3a-g show an alternative embodiment of a process for forming a device;

FIGS. 4a-g show another embodiment of a process for forming a device; and

FIGS. 5a-g show another embodiment of a process for forming a device.

DETAILED DESCRIPTION

Embodiments generally relate to semiconductor devices or integrated circuits. More particularly, some embodiments relate to high power devices. For example, high power devices include lateral double-diffused (LD) transistors, such as metal oxide transistors (MOS). The high power devices can be employed as switching voltage regulators for power management applications. The LD transistors can be easily integrated into devices or ICs. The devices or ICs can be incorporated into or used with, for example, consumer electronic products, particularly portable consumer products, such as cell phones, laptop computers and personal digital assistants (PDAs).

FIG. 1 shows a portion 100 of an embodiment of a device. As shown, the portion includes a substrate 101. The substrate, for example, comprises a silicon substrate. The substrate can be lightly doped with p-type dopants. N-type or other types of substrates, such as silicon germanium or silicon-on-insulator (SOI), are also useful.

The substrate may be prepared with first and second regions 110 and 160. In one embodiment, the first region comprises a low voltage (LV) device region while the second region comprises a high voltage (HV) device region. The regions may be other types of device regions or additional device regions may be provided. For example, a dual gate oxide (DGO) region may be provided for DGO devices. The

DGO devices may be used for I/O circuitries. Alternatively, the device may include only a HV region.

The first region, in one embodiment, comprises first and second type active sub-regions **112** and **116**. The first and second types are complementary types, forming a complementary type device. For example, the complementary type device comprises a complementary metal oxide semiconductor (CMOS) device. Forming non-complementary or other types of devices is also useful.

The first type active sub-region, for example, comprises a n-type active sub-region and the second type active sub-region comprises a p-type active region. The first type active sub-region comprises a second type doped well **122**; the second type active sub-region comprises a first type doped well **124**. For example, the first type active sub-region comprises a p-type doped well and the second type active sub-region comprises a n-type doped well. Other configurations of active sub-regions are also useful. P-type dopants can include boron (B), indium (In) or a combination thereof while n-type dopants can include phosphorus (P), arsenic (As), antimony (Sb) or a combination thereof.

A first type transistor **140** is provided in the first type active sub-region and a second type transistor **150** is provided in the second type active sub-region. For example, a n-type transistor is provided in the n-type active sub-region with a p-type doped well while a p-type transistor is provided in the p-type active sub-region with a n-type doped well. The transistors, in one embodiment, are MOS field effect transistors (MOS-FETs). Other types of transistors may also be useful.

A transistor, for example, comprises a gate having a gate electrode **144** over a gate dielectric **142**. The gate electrode, for example, comprises polysilicon. Other types of gate electrode materials, such as polysilicide or metal silicide, are also useful. Polysilicide or metal silicide gates can be formed by reacting the desired metal with polysilicon. Various types of metal, such as cobalt, titanium or nickel, can be used. Other types of gate electrode materials may also be useful. Furthermore, the gate electrode may be doped with dopants. Doping the gate electrode may depend on the technology. The gate electrode, for example, may be doped with the same dopant type of the transistor type. Doping the gate electrode with other dopant types is also useful. The gate dielectric can be silicon oxide. Other types of gate electrode or gate dielectric materials can also be useful. The material of the gate electrode and gate dielectric may depend on the technology. For example, cobalt silicide gate electrodes may be used for 0.18 μm and 0.13 μm technologies while nickel silicide may be used for 0.09 μm technologies. Other configurations of gate electrode and gate dielectric materials are also useful.

Sidewalls of the gate can be provided with sidewall spacers **148**. The sidewall spacers, for example, comprise a dielectric material, such as silicon oxide, silicon nitride or a combination thereof. Other types of materials can be used for the sidewall spacers.

Adjacent to the gate are diffusion regions **146**. The diffusion regions serve as, for example, source and drain of the transistor. Extension diffusion regions **147** can be provided. The extension diffusion regions, for example, are shallow diffusion regions which extend under the spacers. The diffusion regions (including the extension diffusion regions) of the first type transistor are doped with the first type dopants and the diffusion regions of the second type transistor are doped with the second type dopants. For example p-type diffusion regions are provided for p-type transistors and n-type diffusion regions are provided for n-type transistors. The depth and dopant concentration of the diffusion regions may depend, for example, on the application, such as voltage requirements.

The HV region includes a HV active region **165**. In one embodiment, the HV active region comprises a first type active region. For example, the HV active region comprises a n-type active region. Providing a p-type HV active region is also useful. Furthermore, it is understood that the HV region is depicted with only one active region for purpose of simplification and that the HV region can include numerous HV active regions or sub-regions. For example, the HV active sub-regions can be complementary types of HV sub-regions for complementary types of HV devices.

Isolation regions **180**, such as shallow trench isolation (STI) regions are provided to isolate the LV and HV regions as well as active sub-regions within the regions. Other types of isolation regions are also useful.

In one embodiment, the first type HV active region comprises at least first and second portions. The first portion comprises a second type HV channel (or body) well **135** and the second portion comprises a first type HV drift well **136**. The HV drift well, for example, defines the type of HV device. For example, a p-type HV active region would have a n-type HV channel well and a p-type HV drift well. In one embodiment, a third or middle portion separates the channel and drift wells. In other embodiments, the first and second portions are contiguous portions. For example, the HV channel and HV drift wells contact each other.

A deep well **132** may be provided in the HV active region, encompassing the HV channel and drift wells. Generally, the deep well comprises dopants of the first type or same as the HV device type. The deep well may be used to isolate the drift well from the substrate. For example, a p-type device will have a n-type deep well. In some cases, the deep well can be optional or the deep well can comprise dopants of the second or opposite type of the HV device type. For example, in the case of a n-type HV device, the deep well can be a p-type or n-type deep well. The different types of deep well selected may depend on the type of substrate. Generally, p-type doped substrates are used for fabricating devices. When p-type substrates are used, the n-type deep well may be optional. This is because the n-type drift well is already different from the p-type substrate. For a n-type substrate, a p-type deep well may be used to isolate the channel well from the substrate. Other configurations of deep wells may also be useful.

As shown, a first type HV device is provided in the first type HV active region. The first type HV device, in one embodiment, comprises a LD device, such as a LDMOS device. The HV device comprises a gate with first and second edges **179a-b** or sidewalls.

The gate includes a HV gate electrode **174** over a HV gate dielectric **172**. The HV gate electrode, for example, comprises polysilicon. Other types of HV gate electrode materials, such as polysilicide or metal silicide, are also useful. Various types of metal, such as cobalt, titanium or nickel, can be used. Other types of HV gate electrode materials may also be useful. Furthermore, the gate electrode may be doped with dopants. The HV gate electrode, for example, may be doped with the same dopant type of the transistor type. Doping the HV gate electrode with other dopant types is also useful. The HV gate dielectric can be silicon oxide. Other types of HV gate dielectric materials can also be useful.

In one embodiment, the material of the HV gate electrode and HV gate dielectric may depend on the CMOS process and/or technology. In one embodiment, the material of the HV gate electrode and HV gate dielectric may depend on the CMOS process for LV devices and/or technology. For example, cobalt silicide gate electrodes may be used for 0.18 μm and 0.13 μm technologies while nickel silicide may be

used for 0.09 μm technologies. Other configurations of gate electrode and gate dielectric materials are also useful.

Sidewalls of the gate can be provided with sidewall spacers **178**. The sidewall spacers, for example, comprise a dielectric material, such as silicon oxide, silicon nitride or a combination thereof. Other types of materials can be used for the sidewall spacers.

In one embodiment, the various gate elements of the HV gate, such as gate electrode, gate dielectric and spacers are of the same type of material as the LV devices in the LV region. This facilitates process compatibility for forming HV and LV devices.

The HV gate overlaps the channel and drift wells in the first and second portions of the HV active region. A drift isolation region **180a** may be provided within the drift well **136** on the high voltage side of the HV device. The drift isolation region **180a** can improve breakdown voltage of the HV device due to high voltage. For example, the drift isolation region is particularly useful to sustain high voltages for applications beyond about 10V. As shown, the drift isolation region **180a** is disposed in the drift well **136** under the gate. Disposing the drift well at other positions, such as partially underlying the gate may also be useful.

A first type drain diffusion region **176b** is provided between the drift isolation region **180a** and isolation region **180** which isolates the HV active region from other sub-active regions in the HV region. A first type source diffusion region **176a** is provided in the channel well **135** of the first type HV active region. A source extension region **177** can be provided which extends under the spacer on the channel side of the HV gate. The first type source and drain diffusion regions, for example, comprise a p-type diffusion regions for a p-type HV active region. Providing n-type diffusion regions for a n-type HV active region is also useful. The dopant depth and concentration of the diffusion regions may be selected depending on the application. In one embodiment, the dopant depth and concentration of diffusion regions may be the same as those of LV devices. Other dopant depths and concentrations are also useful.

A second type body contact region **171** may be provided in the first portion of the HV active region. As shown, the second type body contact region **171** is between the source diffusion region **176a** and the isolation region **180**. The second type body contact region comprises second type dopants. In one embodiment, dopant depth and concentration of the body contact region may be the same as the diffusion regions of the LV devices. Other dopant depths and concentrations are also useful. The body contact region **171** provides electrical coupling to the channel well **135**.

An edge **133** of the channel well **135** extends below the gate on the source side of the gate. The edge **133**, for example, can be referred to as the channel edge of the channel well. The channel edge of the channel well **135** and the first edge **179a** of the gate on the source side define a channel of the HV transistor having an effective channel length L .

The effective channel length, in one embodiment, is controlled by ion implantation which is self-aligned to the gate edge. For example, dopant ions are implanted using an implant mask through the gate electrode or a hard mask and the gate electrode. In one embodiment, the implant mask is also used to pattern the channel edge of the HV gate. The effective channel length may be controlled by channel or body implant conditions selected to satisfy threshold voltage and breakdown voltage requirements. For example, shorter effective channel length may be achieved by decreasing tilt angle, implant energy or dosage and vice-versa.

With the channel well being self-aligned to the gate edge of the implant, process overlay issues are avoided. Very short effective channel lengths with very low $R_{ds, on}$ can be achieved, enabling high current to pass through a small area. This is particularly useful for technologies beyond 0.35 μm , such as, for example, 0.25 μm , 0.18 μm and 0.13 μm technologies.

Smaller area requirements for power management applications with reduced power dissipation increases performance while reducing costs.

In one embodiment, the effective channel length L is less than 0.4 μm . In other embodiments, the effective channel length is less than 0.3 μm . In yet another embodiment, the effective channel length is less than 0.25 μm . Other effective channel lengths may also be useful, for example, depending on the application.

FIGS. 2a-g show cross-sectional views of an embodiment of a process for forming a device or IC **100**. Referring to FIG. 2a, a substrate **101** is provided. The substrate can comprise a silicon substrate, such as a lightly p-type doped substrate. Other types of substrates, including n-type doped substrates, silicon germanium or silicon-on-insulator (SOI), are also useful.

As shown, first and second device regions **110** and **160** are defined on the substrate. The first region, for example, comprises a LV device region while the second region comprises a HV device region. Other types of device regions or additional device regions may be provided. Alternatively, the device may include only a HV region.

The LV device region **110** can be defined with first and second type active sub-regions **112** and **116**. In one embodiment, the first and second type active sub-regions are complementary type active sub-regions to form a complementary device such as a CMOS device. Forming non-complementary or other types of devices is also useful.

As for the HV device region **160**, a HV active region **165** is defined therein. In one embodiment, the HV active region **165** can be a first or second type HV active region. Although only one HV active region is depicted, it is understood that the HV device region may include additional HV active regions or sub-regions. The HV active sub-regions can be complementary HV active sub-regions.

The substrate is also prepared with isolation regions **180** to separate the device regions as well as other active device sub-regions. In one embodiment, the isolation regions **180** comprise STIs. Various conventional processes can be employed to form the STI regions. For example, the substrate can be etched using conventional etch and mask techniques to form trenches which are then filled with dielectric materials such as silicon oxide. Chemical mechanical polishing (CMP) can be performed to remove the excess dielectric materials and provide a planar substrate top surface. Other processes or materials can also be used to form the STIs.

In FIG. 2b, the doped wells for the active regions of the device regions are formed. In one embodiment, a first type doped well **124** is formed in the second type LV active sub-region **116** and a second type doped well **122** is formed in the first type LV active sub-region **112**. For example, a n-type doped well is formed in the p-type active sub-region and a p-type doped well is formed in the n-type active sub-region.

In the HV active region **165**, it includes at least first and second portions. For example, the first portion corresponds to a drain portion and the second portion corresponds to a source portion. In some embodiments, a third or middle portion is provided to separate the first and second portions. Alternatively, the first and second portions are contiguous portions without a middle portion.

In one embodiment, a doped well **136** is formed in the drain portion of the HV active region. The doped well, for example, is a drift well of the HV device. The drift well comprises the same type as the HV active region. In the case of a first type HV active region, the drift well comprises a first type. For example, a n-type drift well is provided for a n-type HV active region.

A drift STI region **180a** may be optionally provided in the drift well **136**. The drift STI region is disposed in the drift well to improve reliability of high voltage devices. For example, the drift STI region improves breakdown reliability of the HV device. The drift STI region can be formed at the same time as the other STI regions.

A deep HV well **132** may be provided in the HV active region **165**. As shown, the deep HV well **132** is formed in the whole HV active region **165** and encompasses the first and second portions of the HV active region **165**. The deep HV well comprises an opposite type as the drift well and the HV active region. In the case of a first type HV active region, the deep well comprises a second type. For example, a p-type deep well is provided for a n-type HV active region.

The depth and concentration of dopants of the various wells may depend on, for example, application. For example, higher voltage applications may require lower doping concentration of dopants and greater depth of the wells. In one embodiment, the depth of the deep well **132** is about 3 μm with a dopant concentration of about $5 \times 10^{16} \text{ cm}^{-3}$. For the drift well **136**, it may have a depth of about 1.5 μm with a dopant concentration of about $5 \times 10^{17} \text{ cm}^{-3}$. Other depths and dopant concentrations are also useful for the wells.

As described, the STI regions are formed prior to forming the various doped wells. Other process schemes, such as forming the STI regions after the various doped wells are formed may also be useful.

The doped wells can be formed by ion implantation. P-type dopants can include boron (B), BF_2 (boron compounded with fluorine), indium (In) or a combination thereof while n-type dopants can include phosphorus (P), arsenic (As), antimony (Sb) or a combination thereof. Generally, the first and second types of the active regions are formed selectively in separate processes. For example, the n-type regions can be implanted with n-type dopants while an implant mask prevents doping of the p-type regions.

In one embodiment, the wells of the HV region are formed separately from the LV regions. For example, the LV wells can be formed first followed by the HV wells. Forming the HV wells prior to the LV wells is also useful. The different types of LV wells can be formed in separate processes and the different types of HV wells can also be formed in separate processes. For example, the first type and second type wells in the LV region are formed in separate implant processes. Similarly, the different types of wells in the HV region are formed in separate implant processes.

A doped well can be formed by a single implant process or multiple implant processes. In multiple implant processes, dopants can be implanted at different energy levels to achieve the desired doping profiles. In the case of a single implant process, a high temperature drive-in process is performed to achieve the desired doping profile.

A single implant process may be employed to, for example, form deep wells. For a n-type deep well, phosphorus may be implanted with a dopant concentration of about $6 \times 10^{12} \text{ cm}^{-2}$ at an energy level of about 2000 KeV followed by a two-hour drive-in at a temperature of about 1100° C. Single implant processes may also be used to form other types of wells.

Forming wells using multiple implant processes may also be useful. For example, LV wells may be formed by multiple implant processes.

In FIG. 2c, various layers of the gate are formed on the substrate. In one embodiment, a gate dielectric layer **242** is formed over the substrate. The gate dielectric layer, for example, comprises silicon oxide. Other types of dielectric materials, such as silicon oxynitride, may also be useful. Alternatively, high-k, low-k or composite of dielectric materials may be used. The thickness of the gate dielectric layer **242** may be about 30 Å for 1.8V gate voltage transistors. Other thicknesses may also be useful. The thickness, for example, may depend on the gate voltage applications. For example, higher gate voltages may require thicker gate dielectric layers. In one embodiment, the gate dielectric layer is formed by thermal oxidation. Other techniques, such as chemical vapor deposition (CVD), can also be used to form the gate dielectric layer.

A gate electrode layer **244** is deposited on the gate dielectric layer **242**. The gate electrode layer comprises, in one embodiment, polysilicon (poly). The gate electrode layer **244** can be formed as an amorphous or non-amorphous layer. For amorphous deposited layer, subsequent processing can be performed to crystallize it. Other types of gate electrode materials are also useful. For example, the polysilicon can be subsequently processed to form polysilicide or metal gates. The thickness of the gate electrode layer may be about 2000 Å or less. In another embodiment, the thickness of the gate electrode layer is about 2500 Å or less. In yet another embodiment, the thickness of the gate electrode layer can be about 4000 Å or less. For example, in the case of polysilicide gates, the polysilicon may be about 2000 Å while the tungsten is about 2000 Å. Other thicknesses are also useful. Various techniques can be used to form the gate electrode layer. For example, polysilicon can be deposited by CVD while metal can be deposited by sputtering. Other techniques, depending on the material, may also be useful.

Referring to FIG. 2d, a mask layer **288** is formed on the substrate, covering the gate electrode layer. In one embodiment, the mask layer comprises photoresist. The mask is patterned as desired. To pattern the mask layer, photolithography can be employed. For example, the mask can be selectively exposed to an exposure source through a lithographic mask. Depending on whether a positive or negative resist is used, the exposed or unexposed portions are removed by development. To enhance lithographic resolution, an ARC layer (not shown) may be provided beneath the mask layer.

The mask layer **288** is patterned to form an opening **289** to expose a portion of the gate electrode layer **244**. In one embodiment, the opening corresponds to the opening for the channel well implant. For example, the mask layer serves as the channel well implant mask. The ARC layer may be patterned with the mask layer. Subsequently patterning the ARC layer may also be useful.

In FIG. 2e, the substrate is implanted with dopants to form a channel well **135**. In one embodiment, dopants of the opposite type as the type of the HV active region are implanted. For example, p-type dopants are implanted into the p-type deep well of a n-type HV active region. In one embodiment, the dopants are implanted at an angle to form a channel from the gate edge to the edge of the channel well. The implant angle θ can be in a range of about 1 to 45°. Other implant angles may also be useful. The implant conditions can be varied depending on application requirements. For example, the angle, dose and energy can be selected to achieve the desired effective channel length.

In one embodiment, the channel well is formed by multiple implants. For example, the channel well can be formed by at least two tilt and rotate implants. In one embodiment, the implant comprises a quad implant. A quad implant comprises 4 tilted angled implants, each rotated by a rotation angle. For example, a p-type channel well can be formed with a quad implant at a tilt angle of about 30° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{ cm}^{-2}$ at an energy level of about 130 KeV. Other tilt angles, rotation angles, doses and energy levels are also useful. For example, a quad implant can comprise a tilt angle of about 7° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{ cm}^{-2}$ at an energy level of about 150 KeV.

After forming the channel doped well **135**, exposed portions of the gate electrode layer **244** are removed, as shown in FIG. 2f. In one embodiment, an anisotropic etch, such as a reactive ion etch (RIE) is performed to remove exposed portions of the gate electrode layer. The removal of the exposed portions of the gate electrode layer, in one embodiment, forms a first edge of a HV gate. The first edge, in one embodiment, corresponds to the source side of the HV gate. As shown, the gate dielectric layer **242** serves as an etch stop for etching the gate electrode layer. Leaving the gate dielectric layer can be advantageous since it can serve as an implant mask for subsequent ion implantation to protect the substrate from implant damage. Alternatively, the exposed portion of the gate dielectric layer can be removed. Removing the gate dielectric layer **242** to expose the substrate may also be useful. After etching the gate electrode layer, the mask layer **288** is removed.

As described, the implant mask, in one embodiment, serves as a channel well implant mask as well as a mask for patterning the first edge (e.g., source side) of the HV gate. By using the same mask to form the channel well and source side of the gate, the channel is self-aligned to the gate edge. This enables the channel length of the HV device to be well controlled to produce very short effective channel lengths to reduce $R_{ds, on}$ performance. In one embodiment, the effective channel length L is less than $0.4 \mu\text{m}$. In other embodiments, the effective channel length is less than $0.3 \mu\text{m}$. In yet another embodiment, the effective channel length is less than $0.25 \mu\text{m}$.

Referring to FIG. 2g, another mask layer **288b** is formed on the substrate and patterned. The mask layer is patterned to expose portions of the gate electrode layer **244** to be removed, protecting portions corresponding to gates of the LV devices. For example, the mask serves as a gate mask. Additionally, the mask layer **288b** protects the source portion of the HV region and the gate of the HV device. The exposed portions of the gate electrode layer **244** are removed to form gates of both the LV and HV devices.

After the gates are formed, the process continues to form the device, as illustrated in FIG. 1. The process, for example, includes forming extension regions **147** by ion implantation, followed by forming spacers **148** and **178** and diffusion regions **146** and **176**. The different types of extension and diffusion regions can be formed in different processes. For example, n-type diffusion regions are formed in one implant process and p-type diffusion regions are formed in another process. Furthermore, the diffusion and extension regions of the different device regions can be formed separately. Silicide gates and contacts can be formed after the diffusion regions are formed. For example, metals, such as cobalt are deposited on the substrate **101** and reacted to form silicide contacts and gates. The contacts and gates can be formed in the same or different processes. Unreacted metals are removed. A pre-metal and an inter-level dielectric layer can be formed in

which contacts and interconnects are formed. Additional interconnect levels can be formed by, for example, dual damascene techniques. Final passivation, dicing, assembly and testing may be performed to complete the IC.

An alternative embodiment of a process of forming a device **100** is shown in FIGS. 3a-g. Referring to FIG. 3a, a partially processed device is shown. The partially processed device is similar to that shown in FIG. 2c. A hard mask **388** is formed on the gate electrode layer **244**. The hard mask, for example, comprises silicon oxide. In one embodiment, the hard mask **388** is formed by CVD. The thickness of the hard mask, for example, is about 40 nm. Other materials, techniques or thicknesses may also be useful. For example, the hard mask may be formed from silicon nitride or other types of materials.

Referring to FIG. 3b, a soft mask **287**, such as photoresist, is formed on the hard mask **388**. The soft mask can be patterned by photolithographic techniques to form an opening **289** which can serve as a channel implant mask. An ARC layer can be provided between the soft mask **287** and the hard mask **388**. The pattern of the soft mask is transferred to the hard mask by, for example, RIE.

In one embodiment, the soft mask layer is removed after patterning of the hard mask, as shown in FIG. 3c. The substrate is implanted with dopants to form a channel doped well **135**. In one embodiment, dopants of the opposite type as the type of the HV active region are implanted. For example, p-type dopants are implanted into the p-type deep well of a n-type HV active region.

In one embodiment, the dopants are implanted at an angle to form a channel from the gate edge to the edge of the channel well. The implant angle θ can be in a range of about 1 to 45° . Other implant angles may also be useful. The implant conditions can be varied depending on application requirements. For example, the angle, dose and energy can be selected to achieve the desired effective channel length.

In one embodiment, the channel well is formed by multiple implants. For example, the channel well can be formed by at least two tilt and rotate implants. In one embodiment, the implant comprises a quad implant. A quad implant comprises 4 tilted angled implants, each rotated by a rotation angle. For example, a p-type channel well can be formed with a quad implant at a tilt angle of about 30° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{ cm}^{-2}$ at an energy level of about 130 KeV. Other tilt angles, rotation angles, doses and energy levels are also useful. For example, a quad implant can comprise a tilt angle of about 7° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{ cm}^{-2}$ at an energy level of about 150 KeV.

As shown in FIG. 3d, another soft mask layer **287** is deposited on the substrate, covering the hard mask layer and exposed gate electrode layer. The soft mask is patterned by lithography. The soft mask protects portions of the hard mask layer **388** corresponding to the transistor gates. A new ARC layer can be provided between the soft mask **287** and the hard mask **388**. In FIG. 3e, exposed portions of the hard mask are removed, exposing portions of the gate electrode layer. Patterning of the hard mask layer can be achieved by, for example, RIE. The patterned hard mask serves as a gate mask. The soft mask, as shown in FIG. 3f, is removed after patterning the hard mask.

Referring to FIG. 3g, the gate electrode layer **244** is patterned to form gates in the LV and HV region by, for example etch. Patterning of the gate electrode layer can be achieved using, for example, RIE. The patterning of the gate electrode layer also erodes the hard mask, reducing its thickness. The hard mask **388** can be removed by, for example, a clean step

11

after etching the gate electrode layer. The process continues as previously described to complete the device.

Another embodiment of a process for forming a device **100** is shown in FIGS. **4a-g**. Referring to FIG. **4a**, a partially processed device is shown. The partially processed device is similar to that shown in FIG. **3a**.

Referring to FIG. **4b**, a soft mask layer **387**, such as a photoresist layer, is formed on the hard mask **388**. An ARC layer can be provided between the soft mask and the hard mask. The soft mask layer is patterned to expose portions of the hard mask **388**. The remaining portion of the soft mask corresponds to the HV gate in the HV region **160**. Patterning of the soft mask layer can be achieved by photolithographic techniques. In one embodiment, the patterned soft mask layer also exposes the hard mask in the LV region **110**.

As shown in FIG. **4c**, the pattern of the soft mask is transferred to the hard mask by, for example, RIE. The RIE removes exposed portions of the hard mask to expose portions of the gate electrode layer **344** in the HV region which are to be removed. In one embodiment, the removal of the hard mask also exposes the gate electrode layer in the LV region. After patterning of the hard mask, the soft mask layer is removed.

In FIG. **4d**, another soft mask layer **487**, such as a photoresist layer, is formed over the substrate. An ARC layer may be provided beneath the soft mask layer. The soft mask layer can be patterned by photolithographic techniques to expose portions of the gate electrode layer **344** in the LV region which are to be removed, protecting portions corresponding to gates of the LV devices. For example, the patterned soft mask serves as a gate mask for the LV region. The soft mask layer is removed from the HV region.

The gate electrode layer is patterned in FIG. **4e**. In one embodiment, the gate electrode layer in the LV region is patterned using the soft mask while the hard mask is used in the HV region. The use of the soft mask advantageously facilitates forming gates of critical dimensions in the LV region.

Referring to FIG. **4f**, another soft mask layer **487b**, such as photoresist, is formed over the substrate. An ARC layer may be provided beneath the soft mask. The soft mask layer can be patterned by photolithographic techniques to form an opening **489**. The patterned soft mask layer serves as a channel implant mask.

As shown in FIG. **4g**, the substrate is implanted with dopants to form a channel doped well **135**. In one embodiment, dopants of the opposite type as the type of the HV active region are implanted. For example, p-type dopants are implanted into the p-type deep well of a n-type HV active region.

In one embodiment, the dopants are implanted at an angle to form a channel from the gate edge to the edge of the channel well. The implant angle θ can be in a range of about 1 to 45°. Other implant angles may also be useful. The implant conditions can be varied depending on application requirements. For example, the angle, dose and energy can be selected to achieve the desired effective channel length.

In one embodiment, the channel well is formed by multiple implants. For example, the channel well can be formed by at least two tilt and rotate implants. In one embodiment, the implant comprises a quad implant. A quad implant comprises 4 tilted angled implants, each rotated by a rotation angle. For example, a p-type channel well can be formed with a quad implant at a tilt angle of about 30° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{ cm}^{-2}$ at an energy level of about 130 KeV. Other tilt angles, rotation angles, doses and energy levels are also useful. For example, a quad implant can

12

comprise a tilt angle of about 45° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{ cm}^{-2}$ at an energy level of about 150 KeV.

In one embodiment, the implant mask and hard mask are removed after forming the channel well. For example, the implant mask is first removed followed by the removal of the hard mask. In some embodiments, the hard mask is remained. The process continues as previously described to complete the device after the formation of the gates.

An alternative embodiment of a process for forming a device **100** is shown in FIGS. **5a-g**. Referring to FIG. **5a**, a partially processed device is shown. The partially processed device is similar to that shown in FIG. **4c**.

As shown in FIG. **5b**, a soft mask layer **587**, such as a photoresist layer, is formed over the substrate. An ARC layer may be provided beneath the gate soft mask layer. The soft mask layer can be patterned by photolithographic techniques to expose the HV region, protecting the LV devices from being processed. As illustrated by FIG. **5c**, the gate electrode in the HV region is patterned by, for example, RIE, to form the gate using the hard mask as the gate mask. After patterning of the gate electrode in the HV region, the soft mask layer is removed.

Referring to FIG. **5d**, another soft mask layer **587b**, such as photoresist, is formed over the substrate. An ARC may be provided beneath the soft mask layer. The soft mask layer can be patterned by photolithographic techniques to form an opening **589**. The patterned soft mask layer serves as a channel implant mask.

As shown in FIG. **5e**, the substrate is implanted with dopants to form a channel doped well **135**. In one embodiment, dopants of the opposite type as the type of the HV active region are implanted. For example, p-type dopants are implanted into the p-type deep well of a n-type HV active region.

In one embodiment, the dopants are implanted at an angle to form a channel from the gate edge to the edge of the channel well. The implant angle θ can be in a range of about 1 to 45°. Other implant angles may also be useful. The implant conditions can be varied depending on application requirements. For example, the angle, dose and energy can be selected to achieve the desired effective channel length.

In one embodiment, the channel well is formed by multiple implants. For example, the channel well can be formed by at least two tilt and rotate implants. In one embodiment, the implant comprises a quad implant. A quad implant comprises 4 tilted angled implants, each rotated by a rotation angle. For example, a p-type channel well can be formed with a quad implant at a tilt angle of about 30° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{ cm}^{-2}$ at an energy level of about 130 KeV. Other tilt angles, rotation angles, doses and energy levels are also useful. For example, a quad implant can comprise a tilt angle of about 45° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{ cm}^{-2}$ at an energy level of about 150 KeV.

Referring to FIG. **5f**, another layer of soft mask layer **587c**, such as photoresist, is formed over the substrate. An ARC layer may be provided beneath the soft mask layer. The soft mask layer is patterned by photolithographic techniques to expose portions of the gate electrode layer **344** in the LV region to be removed. Additionally, the soft mask layer remains in the HV region to protect it from being processed. The exposed portions of the gate electrode layer **344** are removed to form the gates of the LV device, as illustrated in FIG. **5g**.

Similar to the embodiment of FIGS. **4a-g**, the LV gates are patterned using a soft mask while a hard mask is used to

13

pattern the HV gate. After forming the LV gates, the soft mask is removed followed by removal of the hard mask over the HV gate. In some embodiments, the hard mask can remain. The process continues to complete the device as described previously.

The invention may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. The foregoing embodiments, therefore, are to be considered in all respects illustrative rather than limiting the invention described herein. Scope of the invention is thus indicated by the appended claims, rather than by the foregoing description, and all changes that come within the meaning and range of equivalency of the claims are intended to be embraced therein.

What is claimed is:

1. A device comprising:
 - a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;
 - a doped channel well disposed in the substrate adjacent to a first edge of the gate, the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from a first heavily doped region adjacent to the first edge of the gate; and a doped drift well adjacent to a second edge of the gate.
2. The device of claim 1 further comprising spacer on the first edge of the gate, wherein the spacer defines the displacement of the channel from the first heavily doped region.
3. The device of claim 1 wherein the substrate comprises: a drift isolation region in the doped drift well; and a doped deep well in the substrate of the active device region encompassing the doped drift well.
4. The device of claim 3 wherein the drift isolation region is disposed under the gate.
5. The device of claim 4 wherein the first and a second heavily doped regions are provided adjacent to the gate, wherein:
 - the first heavily doped region is disposed in the doped channel well on a source side of the device; and
 - the second heavily doped region is disposed between the drift isolation region and a device isolation region on a drain side of the device.
6. The device of claim 1 wherein the first edge of the gate corresponds to a channel edge of the gate of the device and a second edge of the gate corresponds to a drain edge of the gate of the device.
7. The device of claim 1 wherein the channel having a channel length that is controlled by ion implantation.
8. The device of claim 7 wherein the ion implantation comprises a tilt angle implant.
9. The device of claim 8 wherein a tilt angle of the ion implantation is about 1-45°.
10. The device of claim 7 wherein the ion implantation comprises multiple tilt angle implants which are rotated about a plane of the substrate.
11. The device of claim 10 wherein a tilt angle of the implant is about 1-45°.

14

12. The device of claim 7 wherein the ion implantation comprises a quad tilt angle implant which is rotated about a plane of the substrate.

13. A device comprising:

- a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;
- a doped channel well disposed in the substrate adjacent to a first edge of the gate, wherein the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from a first heavily doped region adjacent to the first edge of the gate;
- a doped drift well adjacent to a second edge of the gate; and a drift isolation region in the doped drift well.

14. The device of claim 13 further comprising a spacer on the first edge of the gate, wherein the spacer defines the displacement of the channel from the first heavily doped region.

15. The device of claim 13 wherein the first edge of the gate corresponds to a channel edge of the gate of the device and a second edge of the gate corresponds to a drain edge of the gate of the device.

16. The device of claim 13 wherein the drift isolation region is disposed under the gate.

17. The device of claim 13 wherein the first and a second heavily doped regions are provided adjacent to the gate, wherein:

- the first heavily doped region is disposed in the doped channel well on a source side of the device; and
- the second heavily doped region is disposed between the drift isolation region and a device isolation region on a drain side of the device.

18. The device of claim 13 wherein the channel having a channel length that is controlled by ion implantation.

19. The device of claim 18 wherein the ion implantation comprises a tilt angle implant, multiple tilt angle implants or quad tilt angle implant.

20. A semiconductor device comprising:

- a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;
- first and second heavily doped regions adjacent to first and second edges of the gate;
- a doped channel well disposed in the substrate adjacent to the first edge of the gate, wherein the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from the first heavily doped region adjacent to the first edge of the gate; and
- a doped drift well adjacent to a second edge of the gate, wherein a drift isolation region is disposed in the doped drift well.

* * * * *

EXHIBIT 6

Technology ▼

RF-SOI & RF-CMOS Platforms

Contact Us

Back to RF & HPA

Low Ron-Coff, highly linear, feature-rich platforms to enable the highest performance 4G/LTE and 5G front-end module (FEM) products

The Tower Semiconductor RF-SOI process family combines a multi-metal layer CMOS process with options for 1.2V, 1.8V, 2.5V, and 5V transistors. The technology offering, with five generations in high volume manufacturing across both 200mm and 300mm fabs, is further enhanced by silicon proven accurate models and design libraries, and world-class design enablement.

These processes are well-suited for products requiring isolation such as cellular switches. Excellent channel isolation better than >40 dB, insertion loss of <0.35 dB, low harmonics of better than -110dBc at cellular power levels and inter-modulation distortion below -117 dBm have been demonstrated. Low noise amplifiers can also be integrated with specialized low-noise, high-gain, high-linearity devices, and low-loss inductors realized with thick Cu or Al layers. Specialized high voltage handling switch devices enable robust performance under high RF power and high VSWR conditions, ideal for applications such as antenna tuning and base stations. Selective shrinking of key design rules supports smaller switch and digital area. Our latest "Triple-play" platform enables designers to integrate high-performance RFLDMOS device for PA in addition to above switch and LNA devices: ideally suited for WiFi applications.

In addition to the active devices, process options include silicided and unsilicided poly resistors, RF metal-insulator-metal (MIM) capacitors, metal-fringe capacitors (MFCs), scalable geometry inductors, fixed geometry inductors, fixed geometry baluns, and transformers.

Substrate options include "thin-film" for the best Ron-Coff performance and "thick-film" for bulk-like behavior of the active MOSFETs, free of floating body effects.

Circuit designers new to the platform may draw upon a rich library of characterized demonstration IP and reference flows.

The platform is supported by our [Multi-Project Wafer](#) shuttle program for fast prototyping.

PLATFORM FEATURES:

- Sub-70fs Ron-Coff 2.5V NMOS
- Specialized devices for LNA integration
- High-performance RFLDMOS for monolithic Power Amplifier integration
- High density (4fF/μm²) and high voltage (>25V) MIM capacitors. High-linearity metal fringe capacitors
- Low value and high-value resistors, RF varactor, High-Q inductors
- High impedance, highly linear SOI substrate
- BSIM4 SOI, PSP and HiSIM SOI models
- Al or Cu low RC metallization
- Support for 1.2, 1.8, 2.5, 3.3, or 5V MOSFETs
- High utilization standard cell library
- High power handling devices for switch stack size reduction
- Highly accurate RF device models and fast parasitic extraction for wire bond or flipped die assembly
- Advanced substrate models

Phase Change Material (PCM) Switch:

A break-through PCM switch technology with record < 10 fs Ron x Coff is available for integration with Tower's RF-SOI and Silicon Germanium technologies for

Tower Semiconductor websites place cookies on your device to give you the best user experience. By using our websites, you agree to the placement of these cookies. Click to [Learn more](#)

Accept & Close

The Tower Semiconductor RF-SOI process family combines a multi-metal layer CMOS process with options for 1.2V, 1.8V, 2.5V, and 5V transistors. The technology offering, with five generations in high volume manufacturing across both 200mm and 300mm fabs, is further enhanced by silicon proven accurate models and design libraries, and world-class design enablement.

These processes are well-suited for products requiring isolation such as cellular switches. Excellent channel isolation better than >40 dB, insertion loss of <0.35 dB, low harmonics of better than 110dBc at cellular power levels and inter-modulation distortion below -117 dBm have been demonstrated. Low noise amplifiers can also be integrated with specialized low-noise, high-gain, high-linearity devices, and low-loss inductors realized with thick Cu or Al layers. Specialized high voltage handling switch devices enable robust performance under high RF power and high VSWR conditions, ideal for applications such as antenna tuning and base stations. Selective shrinking of key design rules supports smaller switch and digital area. Our latest "Triple-play" platform enables designers to integrate high-performance RFLDMOS device for PA in addition to above switch and LNA devices: ideally suited for WiFi applications.

In addition to the active devices, process options include silicided and unsilicided poly resistors, RF metal-insulator-metal (MIM) capacitors, metal-fringe capacitors (MFCs), scalable geometry inductors, fixed geometry inductors, fixed geometry baluns, and transformers.

Substrate options include "thin-film" for the best Ron-Coff performance and "thick-film" for bulk-like behavior of the active MOSFETs, free of floating body effects.

Circuit designers new to the platform may draw upon a rich library of characterized demonstration IP and reference flows.

The platform is supported by our [Multi-Project Wafer](#) shuttle program for fast prototyping.

- FEATURES:
- Sub-70fs Ron-Coff 2.5V NMOS
 - Specialized devices for LNA integration
 - High-performance RFLDMOS for monolithic Power Amplifier integration
 - High density (4fF/μm²) and high voltage (>25V) MIM capacitors. High-linearity metal fringe capacitors
 - Low value and high-value resistors, RF varactor, High-Q inductors
 - High impedance, highly linear SOI substrate
 - BSIM4 SOI, PSP and HiSIM SOI models
 - Al or Cu low RC metallization
 - Support for 1.2, 1.8, 2.5, 3.3, or 5V MOSFETs
 - High utilization standard cell library
 - High power handling devices for switch stack size reduction
 - Highly accurate RF device models and fast parasitic extraction for wire bond or flipped die assembly
 - Advanced substrate models

Phase Change Material (PCM) Switch:

A break-through PCM switch technology with record < 10 fs Ron x Coff is available for integration with Tower's RFSOI and Silicon Germanium technologies, for more efficient, advanced RF system architectures in applications including mobile, base-station and mmWave communications.

RF-SOI Markets Served

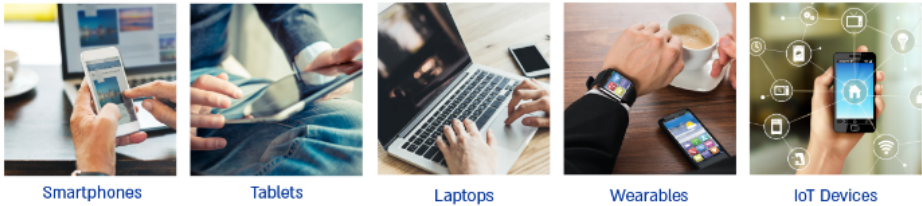


EXHIBIT 7

AMENDMENT AND PRESENTATION OF CLAIMS

Please replace all prior claims in the present application with the following claims.

1. - 13. (Canceled)

14. (Currently Amended) A semiconductor device comprising:

a silicon substrate,

a transistor, comprising source/drain regions and a gate electrode, on the substrate; and

a nickel silicide layer on the gate electrode and source/drain regions,

wherein the nickel silicide layer comprises platinum having a composition gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide,
with platinum increasing in amount in a direction away from the source/drain regions and
the gate electrode, from between about 8 wt. % and about 12 wt. % at the silicon interface.

15. (Canceled)

16. (Original) The semiconductor device according to claim 14, wherein the concentration gradient of platinum in the nickel silicide layer ranges from between about 8 wt. % and about 12 wt. % to between about 30 wt. % and about 100 wt. %.

17. (Original) The semiconductor device according to claim 16, wherein the concentration gradient of platinum in the nickel silicide layer ranges from about 10 wt. % to between about 30 wt. % and about 50 wt. %.

18. (Original) The semiconductor device according to claim 14, wherein the nickel silicide layer has a thickness of about 100 Å to about 220 Å.

19. (Original) The semiconductor device according to claim 14, wherein the nickel silicide layer has a thickness of about 100 Å to about 150 Å.

20. (Canceled)

21. (Previously Presented) The semiconductor device according to claim 14, wherein the silicon substrate comprises a strained silicon germanium (Si-Ge) layer on a silicon layer.

22. (Previously Presented) The semiconductor device according to claim 14, further comprising a stress liner on the nickel silicide layer.

23. (Previously Presented) The semiconductor device according to claim 22, further comprising an oxide or oxynitride layer on the stress liner.

24. (Previously Presented) The semiconductor device according to claim 23, wherein the stress liner comprises stressed silicon nitride.

25. (Currently Amended) A semiconductor device comprising:

a silicon ~~substrate~~, substrate;

a P-channel transistor and an N-channel transistor, each comprising source/drain regions and

a gate electrode, on the silicon substrate; and

a first nickel silicide layer on the gate electrode and source/drain regions of the P-channel transistor; and

a second nickel silicide layer on the gate electrode and source/drain regions of the N-channel transistor,

wherein each of the first and second nickel silicide layers comprises platinum having a concentration gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the P-channel and N-channel transistors, from between about 8 wt. % and about 12 wt. % at the silicon interface.

26. (Previously Presented) The semiconductor device according to claim 25, further comprising:

a first stress liner on the P-channel transistor; and

a second stress liner on the N-channel transistor.

27. (Previously Presented) The semiconductor device according to claim 26, wherein the first stress liner exhibits high compressive stress and the second stress liner exhibits high tensile stress.

28. (Previously Presented) The semiconductor device according to claim 27, wherein the concentration gradient of platinum in each nickel silicide layer ranges from between about 8 wt. % and about 12 wt. % to between about 30 wt. % and about 100 wt. %.

29. (Previously Presented) The semiconductor device according to claim 28, wherein the concentration gradient of platinum in each nickel silicide layer ranges from about 10 wt. % to between about 30 wt. % and about 50 wt. %.

30. (Previously Presented) The semiconductor device according to claim 27, wherein each nickel silicide layer has a thickness of about 100 Å to about 220 Å.

31. (Previously Presented) The semiconductor device according to claim 30, wherein each nickel silicide layer has a thickness of about 100 Å to about 150 Å.

32. (Previously Presented) The semiconductor device according to claim 27, wherein the silicon substrate comprises a strained silicon germanium (Si-Ge) layer on a silicon layer.

33. (Previously Presented) The semiconductor device according to claim 27, further comprising an oxide or oxynitride layer on the first and second stress liners.

34. (Previously Presented) The semiconductor device according to claim 33, wherein the first and second stress liners comprise stressed silicon nitride.

35. (Currently Amended) A semiconductor device comprising:

a silicon substrate;

a strained silicon germanium (Si-Ge) layer on the silicon substrate;

a P-channel transistor comprising source/drain regions and a gate ~~electrode~~, electrode on the strained Si-Ge layer;

an N-channel transistor comprising source/drain regions and a gate ~~electrode~~, electrode on the strained Si-Ge layer, spaced from the P-channel transistor;

a first nickel silicide layer, having a thickness of about 100 Å to about 220 Å, on the gate electrode and source/drain regions of the P-channel transistor;

a second nickel silicide layer, having a thickness of about 100 Å to about 220 Å, on the gate electrode and source/drain regions of the N-channel transistor;

a compressively stressed silicon nitride layer on the first nickel silicide layer;

a tensile stressed silicon nitride layer on the second nickel silicide layer;

a first oxide or oxynitride layer on the compressively stressed silicon nitride layer; and

a second oxide or oxynitride layer on the tensile stressed silicon nitride layer,

wherein each of the first and second nickel silicide layers comprises platinum having a concentration gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount from between about 8 wt. % and about 12 wt. % at the silicon interface to between about 30 wt. % and about 100 wt. % in a direction away from the P-channel and N-channel transistors.

REMARKS

Claims 14, 16 through 19, and 21 through 35 are pending in this Application. Claims 1 through 13, 15, and 20 were previously canceled. Claims 14, 25, and 35 are currently amended. Care has been exercised to avoid the introduction of new matter. Adequate descriptive support for the present Amendment should be apparent throughout the originally filed disclosure as, for example, in paragraph [0036] of the written description of the specification. Applicants submit that the present Amendment does not generate any new matter issue.

Telephonic Interview

Applicants express appreciation for Examiner Nicely's courtesy in granting and conducting a telephonic interview on July 24, 2012. During the interview, a proposed amendment which has been implemented herewith and the rejections under 35 U.S.C. § 102(b) and 103 were discussed. In addition during the interview, the Examiner courteously suggested reciting the platinum content at the silicon interface, which has also been incorporated in this amendment. The Examiner agreed during the interview that the proposed amendment, clarifying that the platinum composition gradient starts at a nonzero value at the silicon interface and spans the entire depth of the nickel silicide, appears to overcome the rejections relying upon *Okuno* for a platinum gradient. The Examiner, however, indicated that he still needed to update his search.

The specification was objected to as including informalities.

The Examiner stated on page 2 of the Office Action that in paragraph [0006] "compressive strain.." should be "compressive strain." Further, in paragraphs [0042] and

[0043], layer 701 should be a Si-Ge layer and layer 703 should be a Si layer. Last, in paragraph [0049], “e..g.” should be “e.g.”. The specification has been clarified accordingly.

Claims 25 and 35 were objected to as including informalities.

Specifically, on pages 2-3, the Examiner stated that in claim 25 “substrate,” should be “substrate;” and in claim 35, “electrode,” should be “electrode”. Claims 25 and 35 have been clarified accordingly.

Claim 14 is rejected under 35 U.S.C. 102(b) as being anticipated by *Okuno et al.* (U.S. Pub. 2007/0093047).

In the statement of the rejection, the Examiner asserted that *Okuno* discloses a method identically corresponding to that claimed. This rejection is traversed.

The factual determination of lack of novelty under 35 U.S.C. §102(b) requires the identical disclosure in a single reference of each element of a claimed invention, as those elements are set forth in the claims, such that the claimed invention is placed into the recognized possession of one having ordinary skill in the art. *Therasense, Inc. v. Beckton, Dickenson and Company*, 593 F.3d 1289 (Fed. Cir. 2010); *Praxair, Inc. v. ATMI, Inc.*, 543 F.3d 1308, (Fed. Cir. 2008); *Dayco Prods., Inc. v. Total Containment, Inc.*, 329 F.3d 1358 (Fed. Cir. 2003); *Crown Operations International Ltd. v. Solutia Inc.*, 289 F.3d 1367 (Fed. Cir. 2002); *Candt Tech Ltd. v. Resco Metal & Plastics Corp.*, 264 F.3d 1344 (Fed. Cir. 2001). Further, as a matter of procedural due process of law, the Examiner is required to specifically identify where in an applied reference is alleged to disclose each and every feature of a claimed invention, particularly when such is not apparent as in the present case. *In re Rijckaert*, 9 F.3d 1531 (Fed. Cir. 1993);

Lindemann Maschinenfabrik GMBH v. American Hoist & Derrick Co., 730 F.2d 1452 (Fed. Cir. 1984). Indeed, there are fundamental differences between the claimed semiconductor method and *Okuno*'s method that scotch the factual determination that *Okuno* discloses, or even remotely suggests, a method identically corresponding to that claimed.

Specifically, as courteously suggested by the Examiner during the interview, claims 14, 25, and 35 have been clarified by reciting that the gradient begins with an amount of 8 to 12 % platinum at the silicon surface and spans the entire depth of the NiSi layer. As agreed during the interview, *Okuno* fails to teach or suggest a platinum composition gradient that begins at the silicon surface with a nonzero amount and spans the entire depth of the silicide layer. Paragraphs [0130] and [0147] of *Okuno*, referenced by the Examiner at page 3 of the Office Action, do suggest a nickel silicide including platinum having a composition gradient with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode. However, paragraph [0130] further states that the gradient is formed as a platinum containing NiSi layer at the top surface of the NiSi. Therefore, the gradient does not begin at the silicon surface with a nonzero amount and increase across the entire NiSi layer from the silicon surface to the opposite surface. The gradient can only be across a top portion of the NiSi. Further, since the gradient is formed by first forming NiSi and then depositing a very thin layer of platinum over the NiSi and heating (see for example paragraphs [0135]-[0137]), the platinum could not extend to or close to the silicon surface to extend the gradient across the entire NiSi layer, nor is there any teaching or suggestion to do so.

Since *Okuno* does not disclose all of the limitations of independent claim 14 arranged as in the claim, Applicants, therefore, submit that the imposed rejection of claim 14 under 35 U.S.C.

§ 102 (b) for lack of novelty as evidenced by *Okuno* is not factually viable and, hence, solicit withdrawal thereof.

Claims 16 and 17 are rejected under 35 U.S.C. 103(a) as being unpatentable over *Okuno* in view of *Dussarrat et al.* (U.S. Pub. 2010/0256405) in view of *Lee et al.* (“Fully Silicided Ni_{1-x} PT_xSi Metal Gate Electrode for p-MOSFETs) in view of *Cabral et al.* (U.S. Pub. 2008/0274611).

Claims 18 and 19 are rejected under 35 U.S.C. 103(a) as being unpatentable over *Okuno* in view of *Wang et al.* (U.S. Pub. 2005/0095799).

Claim 21 is rejected under 35 U.S.C. 103(a) as being unpatentable over *Okuno* in view of Applicant’s Admitted Prior Art (*AAPA*).

Claims 22 through 24 are rejected under 35 U.S.C. 103(a) as being unpatentable over *Okuno* in view of *Sun et al.* (U.S. Pub. 2005/0247926).

The above-identified rejections are traversed.

Specifically, claims 16 through 19 and 21 through 24 depend ultimately from claim 14. Applicants incorporate herein the arguments previously advanced in traversing the imposed rejection of claim 14 under 35 U.S.C. §102(b) for lack of novelty as evidenced by *Okuno*.

In addition, as agreed during the interview, modifying *Okuno*’s gradient to span the entire depth of the NiSi would be contrary to the explicit teachings of *Okuno* to employ platinum only near the top surface of the NiSi. Accordingly, the skilled artisan would not have been motivated to include a platinum composition gradient in the NiSi starting at a nonzero amount at the silicon surface and spanning the entire depth of the NiSi layer.

Regarding claims 16 and 17, the Office Action (pages 4-6) states that *Dussarratt* teaches a range of 5-10 %, *Lee* discloses a range of 5-33 %, and *Okuno* teaches a gradient, and that the skilled artisan “would have had it within his or her ordinary capabilities through routine experimentation and optimization to arrive at platinum concentrations at the lower and upper ends of 505 that are within the respective claimed ranges as applicant has not disclosed that the limitations are for a particular unobvious purpose, produce an unexpected result, or are otherwise critical.” However, Applicants explain in paragraph [0037] of the specification that the particular gradient shields NiSi during etching or middle-of-the-line (MOL) processing without undesirably increasing the resistivity of the NiSi. Therefore, the ranges are critical.

Further, although the references may suggest that the platinum concentration affects the electrical resistance and heat-resistant temperature of the nickel silicide, the teachings apply to a uniform nickel platinum silicide, with no indication as to how forming a gradient of platinum concentration would affect the NiSi characteristics. For example, if a higher platinum percentage yields a better work function, the references would suggest employing a high percentage of platinum across the entire silicide layer, not merely at one surface. In addition, as agreed during the interview, since none of *Dussarratt*, *Lee*, and *Cabral* teaches or suggests a composition gradient for the platinum, they cannot, alone or in combination, teach or suggest a platinum composition gradient beginning with a nonzero value at the silicon surface and spanning the entire depth of the silicide layer, and, therefore, do not overcome the deficiencies of *Okuno*.

Similarly, *Wang*, *AAPA*, and *Sun* do not teach or suggest including platinum in the NiSi layer, no less including a composition gradient of platinum. Therefore, *Wang*, *AAPA*, and *Sun*, alone or in combination, do not cure the shortcomings of *Okuno*. Accordingly, even if *Okuno*’s

method were modified as proposed by the Examiner, and Applicants certainly do not agree that the requisite basis to support the asserted motivation has been established, the claimed invention would not result. *Uniroyal, Inc. v. Rudkin-Wiley Corp.*, 837 F.2d 1044 (Fed. Cir. 1988). Applicants therefore submit that the imposed rejections of claims 16 through 19 and 21 through 24, rejected under 35 U.S.C. §103(a) for obviousness predicated upon *Okuno* in view of *Wang*, *AAPA*, or *Sun*, are not factually or legally viable and, hence, solicit withdrawal thereof.

Claims 14, 22 through 27, 33, and 34 are rejected under 35 U.S.C. 103(a) as being unpatentable over *Sun* in view of *Okuno*.

Claims 16, 17, 28, and 29 are rejected under 35 U.S.C. 103(a) as being unpatentable over *Sun* in view of *Okuno* as applied to claims 14 and 27 above, and further in view of *Dussarrat* in view of *Lee* in view of *Cabral*.

Claims 18, 19, 30, and 31 are rejected under 35 U.S.C. 103(a) as being unpatentable over *Sun* in view of *Okuno* as applied to claims 14 and 27 above, and further in view of *Wang*.

Claims 21 and 32 are rejected under 35 U.S.C. 103(a) as being unpatentable over *Sun* in view of *Okuno* as applied to claims 14 and 27 above, and further in view of *AAPA*.

Claim 35 is rejected under 35 U.S.C. 103(a) as being unpatentable over *Sun* in view of *Okuno*, *AAPA*, *Wang*, *Dussarrat*, *Lee*, and *Cabral*.

The above-identified rejections are traversed.

Sun discloses a NiSi layer with no teaching or suggestion to include platinum or a platinum gradient. The Office Action (at pages 9-10) states that it would have been obvious to include a platinum concentration gradient in the NiSi of *Sun* as taught by *Okuno*. However, as

agreed during the interview, *Okuno* does not teach or suggest a platinum gradient across the entire NiSi layer beginning with a nonzero amount at the silicon surface. Further, modifying *Okuno*'s gradient to span the entire depth of the NiSi would be contrary to the explicit teachings of *Okuno* to employ platinum only near the top surface of the NiSi. Accordingly, the skilled artisan would not have been motivated to include a platinum composition gradient in *Sun*'s NiSi layer starting at a nonzero amount at the silicon surface and spanning the entire depth of the NiSi layer. In other words, even if *Sun*'s method were modified in view of *Okuno*, as proposed by the Examiner, and Applicants certainly do not agree that the requisite basis to support the asserted motivation has been established, the claimed invention would not result. *See Uniroyal*.

In addition, regarding claims 16, 17, 28, and 29, the Office Action (pages 11-13) states that *Dussarratt* teaches a range of 5-10 %, *Lee* discloses a range of 5-33 %, and *Okuno* teaches a gradient, and that the skilled artisan "would have had it within his or her ordinary capabilities through routine experimentation and optimization to arrive at platinum concentrations at the lower and upper ends of 505 that are within the respective claimed ranges as applicant has not disclosed that the limitations are for a particular unobvious purpose, produce an unexpected result, or are otherwise critical." However, as discussed *supra*, Applicants explain in paragraph [0037] of the specification that the particular gradient shields NiSi during etching or middle-of-the-line (MOL) processing without undesirably increasing the resistivity of the NiSi. Therefore, the ranges are critical.

Further, although the references may suggest that the platinum concentration affects the electrical resistance and heat-resistant temperature of the nickel silicide, the teachings apply to a uniform nickel platinum silicide, with no indication as to how forming a gradient of platinum concentration would affect the NiSi characteristics. For example, if a higher platinum

percentage yields a better work function, the references would suggest employing a high percentage of platinum across the entire silicide layer, not merely at one surface. In addition, as agreed during the interview, since none of *Dussarratt*, *Lee*, and *Cabral* teaches or suggests a composition gradient for the platinum, they cannot, alone or in combination, teach or suggest a platinum composition gradient beginning with a nonzero value at the silicon surface and spanning the entire depth of the silicide layer, and, therefore, do not overcome the deficiencies of *Sun* in view of *Okuno*.

Furthermore, *AAPA* and *Wang* do not teach or suggest a platinum gradient across a NiSi layer. Therefore, *AAPA*, *Wang*, *Dussarrat*, *Lee*, and *Cabral*, alone or in combination, cannot cure the deficiencies of *Sun* and *Okuno*. Accordingly, even if *Sun*'s method were modified in view of *Okuno* and *AAPA*, *Wang*, *Dussarrat*, *Lee*, and/or *Cabral*, as proposed by the Examiner, and Applicants certainly do not agree that the requisite basis to support the asserted motivation has been established, the claimed invention would not result. See *Uniroyal*. Applicants therefore submit that the imposed rejections of claims 14, 16 through 19 and 21 through 35, rejected under 35 U.S.C. §103(a) for obviousness predicated upon *Sun* in view of *Okuno*, *AAPA*, *Wang*, *Dussarrat*, *Lee*, and/or *Cabral*, are not factually or legally viable and, hence, solicit withdrawal thereof.

Therefore, the present application, as amended, overcomes the objections and rejections of record and is in condition for allowance. Favorable consideration is respectfully requested. If any unresolved issues remain, it is respectfully requested that the Examiner telephone the undersigned attorney at (703) 519-9954 so that such issues may be resolved as expeditiously as possible.

As Applicants' remarks with respect to the Examiner's rejections are sufficient to overcome these rejections, Applicants' silence as to assertions by the Examiner in the Office Action or certain requirements that may be applicable to such rejections (e.g., whether a reference constitutes prior art, ability to combine references, assertions as to patentability of dependent claims) is not a concession by Applicants that such assertions are accurate or such requirements have been met, and Applicants reserve the right to analyze and dispute such assertions in the future.

To the extent necessary, a petition for an extension of time under 37 C.F.R. § 1.136 is hereby made. Please charge any shortage in fees due in connection with the filing of this paper, including extension of time fees, to Deposit Account 504213 and please credit any excess fees to such deposit account.

Respectfully Submitted,

DITTHAVONG MORI & STEINER, P.C.

July 26, 2012
Date

/Anita Pellman Gross/
Anita Pellman Gross
Attorney/Agent for Applicants
Reg. No. 63325

Arthur J. Steiner
Attorney for Applicants
Reg. No. 26,106

918 Prince Street
Alexandria, VA 22314
Tel. (703) 519-9951
Fax (703) 519-9958



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

99963 7590 08/15/2012
DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314

EXAMINER

NICELY, JOSEPH C

ART UNIT

PAPER NUMBER

2813

DATE MAILED: 08/15/2012

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
-----------------	-------------	----------------------	---------------------	------------------

13/096,511

04/28/2011

Karthik Ramani

P2343US01

4709

TITLE OF INVENTION: METHOD TO REDUCE MOL DAMAGE ON NISI

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	NO	\$1740	\$300	\$0	\$2040	11/15/2012

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the SMALL ENTITY status shown above.

If the SMALL ENTITY is shown as YES, verify your current SMALL ENTITY status:

A. If the status is the same, pay the TOTAL FEE(S) DUE shown above.

B. If the status above is to be removed, check box 5b on Part B - Fee(s) Transmittal and pay the PUBLICATION FEE (if required) and twice the amount of the ISSUE FEE shown above, or

If the SMALL ENTITY is shown as NO:

A. Pay TOTAL FEE(S) DUE shown above, or

B. If applicant claimed SMALL ENTITY status before, or is now claiming SMALL ENTITY status, check box 5a on Part B - Fee(s) Transmittal and pay the PUBLICATION FEE (if required) and 1/2 the ISSUE FEE shown above.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

99963 7590 08/15/2012
DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
-----------------	-------------	----------------------	---------------------	------------------

13/096,511 04/28/2011 Karthik Ramani P2343US01 4709

TITLE OF INVENTION: METHOD TO REDUCE MOL DAMAGE ON NISI

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
-------------	--------------	---------------	---------------------	----------------------	------------------	----------

nonprovisional NO \$1740 \$300 \$0 \$2040 11/15/2012

EXAMINER	ART UNIT	CLASS-SUBCLASS
----------	----------	----------------

NICELY, JOSEPH C 2813 257-413000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) the names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____
(2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____
3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s); (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27. ☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature _____ Date _____

Typed or printed name _____ Registration No. _____

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/096,511	04/28/2011	Karthik Ramani	P2343US01	4709
99963 7590 08/15/2012 DITTHAVONG MORI & STEINER, P.C. 918 PRINCE STREET ALEXANDRIA, VA 22314			EXAMINER NICELY, JOSEPH C	
			ART UNIT 2813	PAPER NUMBER

DATE MAILED: 08/15/2012

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment to date is 0 day(s). If the issue fee is paid on the date that is three months after the mailing date of this notice and the patent issues on the Tuesday before the date that is 28 weeks (six and a half months) after the mailing date of this notice, the Patent Term Adjustment will be 0 day(s).

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No.	Applicant(s)	
	13/096,511	RAMANI ET AL.	
	Examiner	Art Unit	
	Joseph C. Nicely	2813	

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the amendment filed 07/26/2012.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 14 and 16-35.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
 - a) ☐ All b) ☐ Some* c) ☐ None of the:
 1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: ____.

Applicant has **THREE MONTHS FROM THE "MAILING DATE"** of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ A SUBSTITUTE OATH OR DECLARATION must be submitted. Note the attached EXAMINER'S AMENDMENT or NOTICE OF INFORMAL PATENT APPLICATION (PTO-152) which gives reason(s) why the oath or declaration is deficient.
 6. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
 - (a) ☐ including changes required by the Notice of Draftsperson's Patent Drawing Review (PTO-948) attached
 - 1) ☐ hereto or 2) ☐ to Paper No./Mail Date ____.
 - (b) ☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date ____.
- Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).**
7. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|---|
| 1. ☒ Notice of References Cited (PTO-892) | 5. ☐ Notice of Informal Patent Application |
| 2. ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948) | 6. ☐ Interview Summary (PTO-413),
Paper No./Mail Date ____. |
| 3. ☐ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date ____ | 7. ☐ Examiner's Amendment/Comment |
| 4. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | 8. ☒ Examiner's Statement of Reasons for Allowance |
| | 9. ☐ Other ____. |

/J. C. N./
Examiner, Art Unit 2813

/Matthew C. Landau/
Supervisory Patent Examiner, Art Unit 2813

DETAILED ACTION

1. This Office action is in response to the amendment filed 07/26/2012 in which claims 14, 25, and 35 were amended.
2. Claims 14, 16-19, and 21-35 remain pending.

REASONS FOR ALLOWANCE

3. Claims 14, 16-19, and 21-35 are allowed over the prior art of record.
4. The following is an examiner's statement of reasons for allowance:

As to claims 14, 25, and 35: the prior art of record fails to teach in combination a semiconductor device where the nickel silicide layer comprises platinum having a composition gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode (the P-channel and N-channel transistors), from between about 8 wt. % and about 12 wt. % at the silicon interface, and including all limitations.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

5. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.
 - a. Okuno et al. (US 2007/0093047) discloses that the platinum in the nickel silicide exists only at the top portion of the nickel silicide, thus, there is no

Art Unit: 2813

concentration of platinum at the silicon interface from between about 8 wt. % and about 12 wt. % and that increases in concentration a direction away from the gate electrode/(P or N) channel transistors.

b. Tsai et al (US 7,768,072) discloses fully silicided gates with a platinum concentration that is greatest near the gate dielectric, thus, the platinum concentration does not increase in a direction away from the source/drain regions/(P or N) channel transistors.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Joseph C. Nicely whose telephone number is (571)270-3834. The examiner can normally be reached on Monday through Friday 8:30AM-6:00PM EST.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Matthew Landau can be reached on (571) 272-1731. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Art Unit: 2813

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/J. C. N./
Examiner, Art Unit 2813

/Matthew C. Landau/
Supervisory Patent Examiner, Art Unit 2813

EXHIBIT 8

Based on Merriam-Webster's
Collegiate Dictionary, Eleventh Edition

The Merriam- Webster Dictionary

®

NEW EDITION

- 2,000 new words
- Clear and precise
- Over 75,000 definitions



A GENUINE MERRIAM-WEBSTER

The name *Webster* alone is no guarantee of excellence. It is used by a number of publishers and may serve mainly to mislead an unwary buyer.

*Merriam-Webster*TM is the name you should look for when you consider the purchase of dictionaries or other fine reference books. It carries the reputation of a company that has been publishing since 1831 and is your assurance of quality and authority.

Copyright © 2004 by Merriam-Webster, Incorporated

ISBN-13: 978-0-87779-930-6

ISBN-10: 0-87779-930-X

All rights reserved. No part of this work covered by the copyrights hereon may be reproduced or copied in any form or by any means—graphic, electronic, or mechanical, including photocopying, recording, taping, or information storage and retrieval systems—without written permission of the publisher.

MADE IN THE UNITED STATES OF AMERICA

080910 DFC:QP/B 1009080706

ad hoc • admonishingly

ad hoc *adj*: concerned with or formed for a particular purpose <an *ad hoc* committee> <*ad hoc* solutions>

adi-a-bat-ic \a-dē-ə-'ba-tik\ *adj*: occurring without loss or gain of heat — **adi-a-bat-i-cal-ly** \-ti-k(ə)-lē\ *adv*

adieu \ə-'dū, -'dyū\ *n, pl adieus or adieux* \ə-'düz, -'dyüz\ : FAREWELL — often used interjectionally

ad in-fi-ni-tum \ad-'in-fə-'nī-təm\ *adv or adj*: without end or limit

ad in-ter-im \ad-'in-tə-rəm, -'rim\ *adv*: for the intervening time — **ad interim** *adj*

adi-os \a-dē-'ōs, ä-\ *interj* [Sp *adiós*, lit., to God] — used to express farewell

ad-i-pose \a-də-'pōs\ *adj*: of or relating to animal fat : FATTY

adj *abbr* 1 adjective 2 adjutant

ad-ja-cent \ə-'jā-sənt\ *adj*: situated near or next ♦ **Synonyms** ADJOINING, CONTIGUOUS, ABUTTING, JUXTAPOSED, CONTERMINOUS — **ad-ja-cent-ly** *adv*

ad-jec-tive \a-jik-tiv\ *n*: a word that typically serves as a modifier of a noun —

ad-jec-ti-val \a-jik-'tī-vəl\ *adj* — **ad-jec-ti-val-ly** *adv*

ad-join \ə-'jōin\ *vb*: to be situated next to

ad-join-ing *adj*: touching or bounding at a point or line

ad-journ \ə-'jərn\ *vb* 1: to suspend indefinitely or until a stated time 2: to transfer to another place — **ad-journ-ment** *n*

ad-judge \ə-'jəj\ *vb* **ad-judged; ad-judging** 1: JUDGE, ADJUDICATE 2: to hold or pronounce to be : DEEM 3: to award by judicial decision

ad-ju-di-cate \ə-'jü-di-'kāt\ *vb* -cat-ed; -cat-ing : to settle judicially — **ad-ju-di-ca-tion** \ə-'jü-di-'kā-shən\ *n*

ad-junct \a-'jəŋkt\ *n*: something joined or added to another but not essentially a part of it ♦ **Synonyms** APPENDAGE, APPURTENANCE, ACCESSORY, APPENDIX —

adjunct *adj*

ad-jure \ə-'jür\ *vb* **ad-jured; ad-jur-ing** : to command solemnly : urge earnestly

♦ **Synonyms** BEG, BESEECH, IMPLORE — **ad-ju-ra-tion** \a-jə-'rā-shən\ *n*

ad-just \ə-'jəst\ *vb* 1: to bring to agreement : SETTLE 2: to cause to conform : ADAPT, FIT 3: REGULATE <~ a watch>

— **ad-just-able** *adj* — **ad-just-er** *also*

ad-jus-tor \ə-'jəs-tər\ *n* — **ad-just-ment** \ə-'jəst-mənt\ *n*

ad-ju-tant \a-jə-tənt\ *n*: one who assists; *esp*: an officer who assists a commanding officer by handling correspondence and keeping records

ad-ju-vant \a-jə-vənt\ *n*: one that helps or facilitates; *esp*: something that enhances the effectiveness of medical treatment — **adjuvant** *adj*

ad-lib \ad-'lib\ *vb* **ad-libbed; ad-lib-bing** : IMPROVISE — **ad-lib** *n*

ad-lib *adj*: spoken, composed, or performed without preparation

ad lib \ad-'lib\ *adv* [NL *ad libitum*] 1: at one's pleasure 2: without limit

ad-lib *adv*: spoken, composed, or performed without preparation

ad lib \ad-'lib\ *adv* [NL *ad libitum*] 1: at one's pleasure 2: without limit

ad lib \ad-'lib\ *adv* [NL *ad libitum*] 1: at one's pleasure 2: without limit

ad lib \ad-'lib\ *adv* [NL *ad libitum*] 1: at one's pleasure 2: without limit

ad lib \ad-'lib\ *adv* [NL *ad libitum*] 1: at one's pleasure 2: without limit

ADM *abbr* admiral

ad-man \ad-'man\ *n*: one who solicits, or places advertisements

admin *abbr* administration; time

ad-min-is-ter \əd-'mi-nə-st- : MANAGE, SUPERINTEND

out : DISPENSE <~ punish> give ritually or remedially <~ malaria> 4: to perform the

ministrator — **ad-min-is-trant** bəl\ *adj* — **ad-min-is-trant**

ad-min-is-tra-tion \əd-'mi-n- n 1: the act or process of

2: MANAGEMENT 3: the of ing the government of a cou

term of office of an adminis or body — **ad-min-is-tra-**

nə-,strā-tiv\ *adj* — **ad-min-** adv

ad-min-is-tra-tor \əd-'mi-n- : one that administers; *esp* :

tles an intestate estate

ad-mi-ra-ble \ad-m(ə)-rə-b- thy of admiration : EXCEL

mi-ra-bil-i-ty \ad-m(ə)-rə- ad-mi-ra-ble-ness *n* —

\-blē\ *adv*

ad-mi-ral \ad-m(ə)-rəl\ *n* [Ar *amīr-al-* commander o

amīr-al-bahr commander o commissioned officer in the

next below a fleet admiral

ad-mi-ral-ty \ad-m(ə)-rəl-tē British government departr

having authority over nav : the court having jurisdic-tions of maritime law

ad-mire \əd-'mī(-ə)r\ *vb* a mir-ing [MF *admirer*, fr. I

ad- to + *mirari* to wonder with high esteem — **ad-mi-**

mə-'rā-shən\ *n* — **ad-mir-e-** ing-ly \-'mī-rīŋ-lē\ *adv*

ad-mis-si-ble \əd-'mi-sə-b- can be or is worthy to be a

lowed : ALLOWABLE <~ ad-mis-si-bil-i-ty \-,mi-sə-

ad-mis-sion \əd-'mi-shən\ of admitting 2: the privi

admitted 3: a fee paid for : the granting of an argume

knowledge of a fact

ad-mit \əd-'mit\ *vb* **ad-mit-** ting 1: PERMIT, ALLOW

nize as genuine or valid 3 enter <*admitted* to the club

cept into a hospital as an in

ad-mit-tance \əd-'mi-təns\ process of admitting : perm

ad-mit-ted-ly \əd-'mi-təd-lē has been or must be admitte

be admitted

ad-mix \ad-'miks\ *vb* : to mi

ad-mix-ture \ad-'miks-chər- thing added in mixing 2: n

ad-mon-ish \ad-'mā-nish\ gently : reprove with a war

nyms CHIDE, REPROACH, RIMAND, REPROVE — **ad-**

ad-mon-ish-ing-ly *adv*